Ammar Khanfer

Measure Theory and Integration

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Preface

By the time students finish their upper-level courses in an undergraduate program of mathematics, they start to gain good mathematical knowledge about linear algebra and classical real analysis. Some major theories were assumed to be covered in detail: Riemann's theory of integration, the theory of vector spaces, and an introduction to the theory of ordinary and partial differential equations. The following three limitations are noticed:

1. Most vector spaces that have been studied are of finite dimensions, and their elements are merely vectors. Infinite-dimensional linear abstract spaces were not treated rigorously.
2. The Riemann theory of integration is used to integrate some, but not, all functions, and there is a need to improve the integration theory to allow integrating a larger class of functions.
3. In addition to these two limitations, derivative techniques are used to find classical solutions to some, but not all, differential equations, so a general theory of differentiation is needed to allow differentiating a larger class of functions.

One may start to wonder if it is possible to treat these obstacles in more general settings; i.e., extending the vector spaces to infinite-dimensional space consisting of functions or other type of elements instead of vectors. Also, generalizing the Riemann theory to yield a more flexible and reliable theory allowing for the integration of more functions, and generalizing the notion of derivative to allow differentiating functions that are not even continuous. This can be made possible when we extend the notions of linear transformations and functions to operators and distributions and vector spaces to function spaces.

This treatment of the theories mentioned above is the beginning of an "advanced mathematical analysis" that fills in perfectly for these gaps which eventually gives rise to some primary and central theories: the theory of measure and Lebesgue's integration, theory of Banach and Hilbert spaces, operator theory, and lastly, distribution theory. These theories lie at the heart of the advanced mathematical analysis with its three aspects: real analysis, functional analysis, and applied functional analysis, where the first aspect deals with the notion of measure and integration and the second

aspect deals with the notion of vector spaces of infinite dimensions. Meanwhile, the third aspect deals with the notion of differentiation and the idea of employing the preceding aspects to tackle many problems related to the theory of differential equations and its application. It can be said that real and functional analysis bridge the gap between elementary mathematics to advanced mathematics, where more maturity, knowledge, and skills are required to master the subject. This particular area of mathematics is rich in history and deep in treatment. Above all, it is substantially required in almost all graduate programs of mathematics. Usually, students study this subject only when they become familiar with the basic concepts of set theory (sets and their operations), linear algebra (vector spaces, norms, bases, linear transformations), and classical real analysis (Cauchy sequence, series, the convergence of sequences, differentiation and integration of functions and sequences of functions). A good understanding of these concepts not only facilitates the study of the subject but also provides motivation and a clear understanding of why one should get introduced to these concepts, which will cast some beauty onto the subject.

One pitfall arises when reading this area of mathematics from books. The complexity of the treatments of the books to this subject and the deepness of its concepts which may end up giving the subject a negative reputation among some weak and average students that the subject is too complicated and difficult to understand. Many excellent books are available about the subject. However, the exposition in most of them might not be simple enough for students to fully grasp, and thus, the gap between the student and this subject continues to exist and even widen. Therefore, a typical book for the student should discuss these areas of analysis with a comprehensive treatment and clear exposition and in a self-contained reader-friendly text.

With this goal in mind, we have started this project to write a series of books that is designed to serve this purpose for the field of mathematical analysis. This project has brought forth a series of books consisting of three volumes on real analysis, functional analysis, and applied functional analysis. This multi-volume series is intended to fulfill these guidelines and meet the expectations of students. It also aims at introducing this subject to the reader as a bridge from classical analysis to advanced analysis, wherein this aforementioned conceptually widening gap can be closed by providing a detailed explanation and clear exposition of the ideas and main results with a simple treatment. We present in this series the standard material that should be pursued and taught at this level of courses. The purpose of this series is to provide an excellent first step towards these goals and prepares the student for the next study stage. Efforts have been made to make each volume a self-study and reader-friendly text to allow the student to get acquainted with the basics of advanced analysis and provide the basis for further studies that can be carried more deeply.

About This Book

The present book is the first volume of the group of three books and is devoted to the area of real analysis. It consists of five chapters, with eight sections in each chapter.

Chapter 1 introduces measure theory and the notions of inner and measure of sets, and discusses Lebesgue's measurability.

Chapter 2 deals with measurable functions as a generalization of continuous functions by defining them in terms of measurable sets instead of intervals. It also covers the sequences of measurable functions and their convergence and establishes approximation theorems. Then we study some classes of measurable functions that will be carried forward in Chap. 3.

Chapter 3 introduces Lebesgue's theory of integration as a replacement for Riemann's theory of integration. The theory is based on measurable sets rather than intervals. The integral of a bounded function is defined, then the concept is extended to the integration of a general function. Convergence theorems are discussed for Lebesgue's integral. A comparison of the integrability of functions in terms of Riemann's and Lebesgue's theories is discussed. Then the concept of double integrals is extended to Lebesgue's integral.

Chapter 4 introduces the infinite-dimensional spaces with a rigorous treatment. These are normed vector spaces endowed with suitable norms. We investigate these spaces discussing some classes of them, in particular the Lebesgue spaces. We establish some fundamental inequalities and obtain some convergence and approximation results. Finally, we introduce linear operators and functionals on Lebesgue spaces, followed by a discussion on the idea of representing members of conjugate Lebesgue spaces, and the Riesz representation theorem.

Chapter 5 introduces measure theory and integration in more general settings. In particular, we introduce abstract measure theory and general measure spaces other than the Euclidean spaces, and with measures other than the Lebesgue measure. The signed measure and decomposition of measures are introduced, and the Radon–Nikodym theorem is presented and discussed thoroughly.

This textbook can be used as a reference for graduate courses in real analysis. The book provides more than 210 problems distributed at the end of each chapter. The questions aim to test whether the reader has absorbed the topic's basics and gained a complete understanding. Therefore, the levels of the questions range from straightforward to challenging, and some of them are standard in their subjects that can be found in many other books. Moreover, these questions help students prepare for the Ph.D. comprehensive exams. It should be noted that some questions are answered in detail, while others are left with some hints or key answers. Similarly, some details in the proofs throughout the book are left to the reader to complete independently.

It is important to know that mastering all the topics in this book is not easily possible without the time and effort needed to overcome all the difficulties that may arise in terms of grasping the material and mastering the subject. This will prepare students to think independently and test their maturity and background to accomplish

the task by filling in the gaps and unveiling all the hidden details whenever needed—a common practice when one enrolls in a graduate program and inevitable when one starts independent research.

Riyadh, Saudi Arabia Ammar Khanfer

Acknowledgments

I have to start thanking the God most of all, who gave me strength, health, knowledge, and most importantly, patience to endure and complete this work successfully.

Writing a book is more exhausting than one can imagine and more gratifying than one might expect. Before being rewarded, it is important in this regard to recall our teachers. I am immensely thankful to everyone who taught me mathematics at various stages of my life from childhood to the PhD level. Words cannot express gratitude to them. First and foremost, I'm forever thankful and grateful to Prof. Kirk Lancaster for being a great teacher, a PhD supervisor, a collaborator, and more importantly, an inspirer. This is a moment where I need to express my sincere gratitude and pay tribute to him. He is internationally recognized for his work on the analysis and geometry of elliptic partial differential equations. He is also well-known for proving the famous Concus–Finn conjecture. In 1996, together with Prof. David Siegel, they established the existence of the *central fans phenomena*, which is a cornerstone in understanding the behavior of capillary surfaces near convex corners. Recently, he, together with myself, were successful in investigating connections between geometry of hypersurfaces and generalized solutions of elliptic PDEs in higher dimensions.

Further, I would also like to express my deepest appreciation to all professors who taught me mathematics in the past. In particular, I'm indebted to the following (in chronological order starting from 1986): H.I. Karakas, Mashhoor Al-Refai, Bahaettin Cengiz (passed away), Fuad Kittaneh, Roshdi Khalil, Hasan Al-Ezeh (passed away), Yuri Ledyaev (the author of the book: *Nonsmooth Analysis and Control Theory*), John Martino, John Petrovic (the author of the book: *Advanced Calculus: Theory and Practice*), Qiji Jim Zhu (the author of the book: *Techniques of Variational Analysis*), Alan Elcrat (passed away, the author of the book: *Theory and Applications of Partial Differential Equations*), Alexander Bukhgeym (the author of the book: *Introduction to the Theory of Inverse Problems*), Victor Isakov (passed away, the author of the book: *Inverse Problems for Partial Differential Equations*, and the series editor of the book: *Sobolev Spaces in Mathematics*). I'm fortunate to have had the chance to be a student of those great mathematicians who influenced me the most.

I would like to express my sincere thanks to the staff at Springer who helped me in publishing the book to fruition. In particular, I'm grateful and thankful to Shamim

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Contents

1 Measure Theory 1
1.1 Notion of Measure 1
1.1.1 Motivation 1
1.1.2 Extended Real Numbers 2
1.1.3 Axiom of Measure 2
1.2 Outer Measure of a Set 3
1.2.1 Definition of Outer Measure 3
1.2.2 Basic Properties of Outer Measure 4
1.2.3 Additive Property of Outer Measure 6
1.3 Inner Measure 9
1.3.1 Definition of Inner Measure 9
1.3.2 Properties of Inner Measure 10
1.3.3 The Inner Measure Problem 10
1.3.4 Inner Measure As a Supremum 11
1.4 Lebesgue Measurability 13
1.4.1 Measurable Sets 13
1.4.2 Characterization of Measurable Sets 14
1.4.3 Measurable Sets as Sigma-Algebra 17
1.4.4 Borel Sets 19
1.4.5 Additivity of Lebesgue Measure 20
1.4.6 Continuity of Lebesgue Measure 21
1.5 Caratheodory Criterion 22
1.5.1 The Idea of Caratheodory's Approach 22
1.5.2 Characterization of Measurable Sets 23
1.5.3 Lebesgue Versus Caratheodory Approaches 24
1.5.4 Sigma-Algebra of Measurable Sets By Caratheodory Criterion 24
1.5.5 Additivity 28
1.6 Unusual Sets 28

1.6.1 Vitali Sets . . . 28
1.6.2 Nonmeasurable Sets . . . 30
1.6.3 Axiomatic Controversy . . . 31
1.6.4 Cantor Set . . . 31
1.7 Lebesgue Product Measure . . . 34
1.7.1 The Measure of Rectangles and Cubes . . . 34
1.7.2 Caratheodory Criterion in $\mathbb{R}^n$ . . . 36
1.7.3 Measure of Product Sets . . . 37
1.8 Problems . . . 39

2 Measurable Functions . . . 43
2.1 Introduction . . . 43
2.2 Measurable Functions . . . 43
2.2.1 Simple Function . . . 43
2.2.2 Simple Versus Step Functions . . . 44
2.2.3 Definition of Measurable Function . . . 45
2.2.4 Algebra of Measurable Functions . . . 46
2.2.5 Almost Everywhere Property . . . 48
2.3 Sequence of Measurable Functions . . . 48
2.3.1 Supremum and Infimum . . . 48
2.3.2 Limits of Sequences . . . 50
2.3.3 Modes of Convergence . . . 51
2.4 Approximation Theorems . . . 53
2.4.1 Nearly Versus A.E. . . . 53
2.4.2 First Littlewood Principle . . . 53
2.4.3 Approximation of Simple Functions . . . 54
2.4.4 Simple Approximation Theorem . . . 56
2.4.5 Egoroff Theorem . . . 57
2.4.6 Lusin Theorem . . . 58
2.5 Differentiability . . . 59
2.5.1 Dini Derivative . . . 59
2.5.2 Vitali Covering . . . 61
2.5.3 Derivative of Monotone Functions . . . 63
2.6 Functions of Bounded Variation . . . 66
2.6.1 Total Variation . . . 66
2.6.2 Bounded Variation . . . 66
2.6.3 Jordan Decomposition . . . 71
2.7 Absolute Continuity . . . 72
2.7.1 Absolute Continuous Functions . . . 72
2.7.2 Functions of Bounded Derivatives . . . 74
2.7.3 Derivative of Absolute Continuous Function . . . 75
2.8 Problems . . . 76

3 Lebesgue Integration 81
3.1 The Riemann Integral 81
3.1.1 Introduction 81
3.1.2 Riemann Integral of a Bounded Function 81
3.1.3 Deficiencies of Riemann Integral 83
3.2 Integral of Bounded Measurable Functions 85
3.2.1 The Idea of Lebesgue Integral 85
3.2.2 Integral of Simple Functions 85
3.2.3 Integral of Bounded Functions 89
3.3 Integral of General Measurable Functions 93
3.3.1 The Indeterminate Value Problem 93
3.3.2 Integral of Nonnegative Functions 94
3.3.3 The General Lebesgue Integral 97
3.4 Convergence Theorems 100
3.4.1 Uniform Convergence 100
3.4.2 Bounded Convergence Theorem 101
3.4.3 Monotone Convergence Theorem 102
3.4.4 Fatou's Lemma 103
3.4.5 Dominated Convergence Theorem 105
3.4.6 Historical Remark 107
3.5 Lebesgue Integrability 107
3.5.1 Riemann Integrability 107
3.5.2 Improper Riemann Integral 108
3.5.3 Lebesgue Integrability 109
3.5.4 Improper Riemann Integral Versus Lebesgue Integral 110
3.5.5 Riemann–Lebesgue Lemma 115
3.6 Lebesgue Fundamental Theorem of Calculus 116
3.6.1 The Recovering Problem 116
3.6.2 Integrability of The Derivative 117
3.6.3 Differentiation of Integral 118
3.6.4 Indefinite Integral of Derivative 122
3.7 Lebesgue Double Integral 123
3.7.1 The Double Integral 123
3.7.2 Sections of Sets and Functions 123
3.7.3 Fubini–Tonelli Theorems 125
3.7.4 Convolution 132
3.8 Problems 133

4 Lebesgue Spaces 139
4.1 Norms and Linear Spaces 139
4.1.1 Finite-Dimensional Linear Spaces 139

4.1.2 Definition of Norm . . . 140
4.1.3 The $p-$Norm . . . 141
4.2 Basic Theory of L^p Spaces . . . 142
4.2.1 The Space of Lebesgue Integrable Functions . . . 142
4.2.2 Definition of L^p Space . . . 143
4.2.3 L^∞ Spaces . . . 144
4.2.4 Inclusions of L^p Spaces . . . 145
4.3 Fundamental Inequalities . . . 147
4.3.1 Young's Inequality . . . 147
4.3.2 Holder Inequality . . . 147
4.3.3 Minkowski Inequality . . . 148
4.4 Further Spaces of Order p . . . 149
4.4.1 L^p Spaces, $0 < p < 1$ . . . 149
4.4.2 ℓ^p Spaces . . . 151
4.4.3 $\mathcal{R}^p$ Spaces . . . 154
4.5 Convergence in L^p . . . 155
4.5.1 Convergence in $p-$Norm . . . 155
4.5.2 Comparison Between Types of Convergence . . . 156
4.5.3 $p-$Norm Dominated Convergence Theorem . . . 157
4.5.4 Inclusion Relations . . . 158
4.5.5 Convergence in L^∞ . . . 159
4.6 Approximations in L^p . . . 159
4.6.1 Dense Subspaces . . . 159
4.6.2 Density Results in L^p . . . 160
4.6.3 Density Results in L^∞ . . . 163
4.7 Bounded Linear Functionals on L^p . . . 164
4.7.1 Notion of Functional . . . 164
4.7.2 Integral Functional on L^p . . . 166
4.7.3 Riesz Representation Theorem . . . 167
4.7.4 The Case $p = \infty$ . . . 170
4.8 Problems . . . 171

5 Abstract Measure Theory . . . 177
5.1 Generalization of Measure Theory . . . 177
5.1.1 Introduction . . . 177
5.1.2 Measurable and Measure Spaces . . . 178
5.1.3 Measurable Functions . . . 179
5.1.4 Integration Over Abstract Measurable Spaces . . . 179
5.2 Signed Measure . . . 182
5.2.1 Notion of Signed Measure . . . 182
5.2.2 Positive and Negative Sets . . . 183
5.2.3 Hahn's Lemma . . . 184
5.3 Decomposition of Measures . . . 186

5.3.1 Hahn Decomposition Theorem 186
5.3.2 Jordan Decomposition Theorem 187
5.4 Absolute Continuity of Measures 188
5.4.1 Motivating Problem 188
5.4.2 Definitions 189
5.5 Radon–Nikodym Theorem 191
5.5.1 Radon–Nikodym Theorem for Finite Measures 191
5.5.2 Extended Radon-Nikodym Theorem 193
5.6 Radon–Nikodym Derivative 195
5.6.1 Fundamental Theorem of Calculus for RN Derivatives 195
5.6.2 Calculus of RN Derivatives 195
5.6.3 Chain Rule of RN Derivative 197
5.7 Lebesgue Decomposition of Measures 198
5.7.1 Mutually Singular Measures 198
5.7.2 Lebesgue Decomposition Theorem 198
5.8 Problems 200

Answer Key 203

Chapter 1 203

Chapter 2 205

Chapter 3 209

Chapter 4 212

Chapter 5 217

Bibliography 221

Index 223

About the Author

Ammar Khanfer earned his Ph.D. from Wichita State University, USA. His area of interest is analysis and partial differential equations (PDEs), focusing on the interface and links between elliptic PDEs and hypergeometry. He has notably contributed to the field by providing prototypes studying the behavior of generalized solutions of elliptic PDEs in higher dimensions in connection to the behavior of hypersurfaces near nonsmooth boundaries. He also works on the qualitative theory of differential equations, and in the area of inverse problems of mathematical physics. He has published articles of high quality in reputable journals.

Ammar taught at several universities in the USA: Western Michigan University, Wichita State University, and Southwestern College in Winfield. He was a member of the Academy of Inquiry Based Learning (AIBL) in the USA. During the period 2008–2014, he participated in AIBL workshops and conferences on effective teaching methodologies and strategies of creative thinking, which made an impact on his engaging and motivational writing style. He then moved to Saudi Arabia to teach at Imam Mohammad Ibn Saud Islamic University, where he taught and supervised undergraduate and graduate students of mathematics. Furthermore, he was appointed as coordinator of the PhD program establishment committee in the department of mathematics. In 2020, he moved to Prince Sultan University in Riyadh, and has been teaching there since then.

Chapter 1
Measure Theory



1.1 Notion of Measure

1.1.1 Motivation

There is a motivation to generalize the notion of $length$ to another concept that could capture length in a more general setting. In principle, the measure of a set $A \subseteq \mathbb{R}$ should refer to the size of A, and it should agree with the natural properties of length if the settings were reduced to length. Recall that a bounded set is a set that is contained in a finite interval I, so it has an upper bound and a lower bound. Intuitively, the measure of an interval should be nothing but its length. However, not all sets in $\mathbb{R}$ are intervals, so two questions arise in this regard:

1. Does every set of $\mathbb{R}$ have a "sensible" size? In other words, can every subset of $\mathbb{R}$ be measured somehow by a certain number?
2. If yes, then how to measure it? What tool should we use to represent it?

We may need at the beginning to classify the subsets of $\mathbb{R}$ into admissible sets, which are sets that can be measured, and inadmissible sets, which are sets that cannot be measured. Keep in mind that there is no information yet on these classes and how big is each class, or even if one of them is empty. We need to develop a mathematical tool that enables us to measure the size of subsets, and this tool will allow us to determine if all sets are admissible (i.e., can be measured by that tool) or not.

Another observation led by intuition is that the measure of a set under the same measuring tool must be unique since it represents the size of the set according to a specific type of measure. This indicates that the measuring tool that we are considering should be a real-valued function, and it will be called a $measure$, and will be denoted by μ. The domain of μ is the collection Σ of all admissible sets of $\mathbb{R}$, which is indeed contained in the collection $\mathcal{P}(\mathbb{R})$ of all subsets of $\mathbb{R}$ (which is called $power\ set$). Later, we will investigate this collection and see how big it is and if $\Sigma = \mathcal{P}(\mathbb{R})$ or $\Sigma \subsetneq \mathcal{P}(\mathbb{R})$. Meanwhile, we will be dealing with Σ as a collection of subsets on the understanding that it consists of all admissible sets that can be measured by μ.

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1.1.2 Extended Real Numbers

A final observation is that, infinite intervals have infinite length. So it is customary to consider $+\infty$ as a possible outcome, and hence should be added to the set of real numbers. Therefore, it is convenient to extend the set of real numbers to include $+\infty$ and $-\infty$ as elements. The new enlarged set is called: "*extended real numbers*", and is expressed by

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}.$$

In $\overline{\mathbb{R}}$, the two elements $\pm\infty$ play the role of a number in the following sense:

1. For any number $x \in \mathbb{R}$, $x + \infty = \infty$ and $x - \infty = -\infty$.
2. $x \cdot \infty = \infty$ for $x \in \mathbb{R}^+$ and $x \cdot (\infty) = -\infty$ for $x \in \mathbb{R}^-$.
3. $\infty + \infty = \infty$, $-\infty - \infty = -\infty$.
4. $\infty \cdot \infty = \infty$ and $(-\infty)\infty = -\infty$.

The new extended system of real numbers simplifies the description of the behavior of functions. Under the new system, every subset of $\mathbb{R}$ has an upper bound and a lower bound, and we can say for instance that $\lim_{x\to\infty} x^2 = \infty$, and $\lim_{x\to 0^+} \ln x = -\infty$.

1.1.3 Axiom of Measure

We propose the following definition.

Definition 1.1.1 (*Measure*) Let Σ be a collection of subsets of $\mathbb{R}$ containing the empty set $\emptyset$. Then, a $measure\ \mu$ on $(\mathbb{R}, \Sigma)$ is a nonnegative function

$$\mu : \Sigma \to [0, \infty]$$

such that

1. Nullity Axiom: $\mu(\emptyset) = 0$.
2. Nonnegativity Axiom: $\mu(A) \geq 0$ for every set A. Moreover, if A is a bounded interval then $\mu(A) < \infty$, and if A is an unbounded interval then $\mu(A) = \infty$.
3. Monotonicity Axiom: If $A \subseteq B$ then $\mu(A) \leq \mu(B)$.
4. Additivity Axiom: If $A \cap B = \emptyset$ then

$$\mu(A \cup B) = \mu(A) + \mu(B).$$

In general, for a collection of pairwise disjoint sets $\{A_i\}_{i=1}^{\infty}$ we have:

$$\mu(\bigcup_{i}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i).$$

The above axioms are necessary to hold for any measure on the set of real numbers, and any set function μ on $(\mathbb{R}, \Sigma)$ satisfies the four axioms above is called a measure. These axioms represent the natural properties that any measure μ should intuitively satisfy, regardless of the size of the admissible sets in Σ measured by μ. The third axiom can be immediately deduced from the fourth axiom as

$$B = A \cup (B \setminus A),$$

so some authors ignore this axiom. We, however, prefer to mention it because it implies two important facts: If a set in $A \in \Sigma$ is contained in a bounded interval, then $\mu(A) < \infty$, and if A contains an unbounded interval, then $\mu(A) = \infty$. As said earlier, the collection Σ may be as big as the power set $\mathcal{P}(\mathbb{R})$, and may be strictly smaller. Later, we will see that the fourth axiom makes it hard for Σ to be as big as $\mathcal{P}(\mathbb{R})$.

There are several approaches to define a measure, but we will confine ourselves to Lebesgue's approach. There are two ways to measure a set using Lebesgue's approach: from outside the set and from inside. We will call the former: "*outer measure*" and will be denoted by $\mu^*(A)$, and we will call the latter: "*inner measure*" and will be denoted by $\mu_*(A)$. The main idea is to approximate the measure of a set by other standard sets whose size is known to us, and use them as a device to compute the measure of other sets. In Lebesgue's approach, the interval is the device to measure sets from outside, which is a good choice since we already know the measure (or the length) of an interval. However, we need to keep in mind that μ should define a nonnegative function, and each set in the domain of μ should have a unique measure $\mu(A)$. The inner measure $\mu_*(A)$ must be taken with extra care since we have no idea of the set's structure to be measured, and if the device to be used is the interval, we do not know if our set contains intervals. So it is safer at the beginning to use the outer measure to get a clue of what is going on.

1.2 Outer Measure of a Set

1.2.1 Definition of Outer Measure

Definition 1.2.1 (*Outer Measure*) Let $A \subseteq \mathbb{R}$. Then, the outer measure of A is a function

$$\mu^* : \mathcal{P}(\mathbb{R}) \to [0, \infty]$$

given by

$$\mu^*(A) = \inf_{I_n} \left(\sum \ell(I_n) \right), \ I_n \supseteq A,$$

where $\ell(I_n)$ is the length of the interval I_n, and the infimum is taken over all possible collections of $\{I_n\}$ that cover A.

The definition implies that for every $\epsilon > 0$ there exists a collection $\{I_n\}$ such that

$$\sum \ell(I_n) < \mu^*(A) + \epsilon.$$

Note that if a set A consists of disjoint subsets, we cannot obtain $\mu^*(A)$ by simply adding the measure of each subset using the additive property indicated in axiom 4 above. This observation justifies the importance of the additivity property and its validity to guarantee correct measuring for all sets since, otherwise, we can't measure all sets. We will see, however, that this property is, in fact, the source of many problems and difficulties the measure theory encountered and still encountering since it emerged at the beginning of the twentieth century.

1.2.2 Basic Properties of Outer Measure

The outer measure satisfies the following basic properties.

Proposition 1.2.2 (Basic Properties of Outer Measure) *Let μ^* be the Lebesgue outer measure defined in Definition 1.2.1. Then:*

1. If $A \subseteq B$, then $\mu^*(A) \leq \mu^*(B)$.
2. If A is a singleton set, then $\mu^*(A) = 0$. Consequently, let A be any countable infinite set. Then $\mu^*(A) = 0$.
3. If I is an interval then $\mu^*(I) = \ell(I)$, and $\mu^*([a, \infty)) = \infty$. That is, the outer measure of an interval is its length.
4. $\mu^*(A) = \mu^*(A + x)$ for any $x \in \mathbb{R}$.

Proof Part (1) can be readily seen from the definition. Indeed, every cover of B is a cover of A and so the infimum over all covers of A must be less than the infimum over all covers of B.

For (2), let $A = \{a\}$. Let

$$I_n = \left(a - \frac{1}{n}, a + \frac{1}{n}\right)$$

be a cover of $\{a\}$, then $\ell\{I_n\} = \dfrac{2}{n}$ for all n. Taking the infimum, we obtain: $\mu^*(\{a\}) = 0$. Let $A = \{a_n\}_{n=1}^{\infty}$, and consider the cover

$$I_n = \left(a_n - \frac{\epsilon}{2^{n+1}}, a_n + \frac{\epsilon}{2^{n+1}}\right).$$

Then

$$\ell(I_n) = \sum_n^\infty \frac{\epsilon}{2^n} = \epsilon.$$

Since ϵ is arbitrary, this proves (2).

Now we prove (3). Let $I = [a, b]$. Note that for $\epsilon > 0$, $I \subset (a - \epsilon, b + \epsilon)$. Hence

$$\mu^*(I) \leq b - a + 2\epsilon.$$

Since ϵ is arbitrary, we obtain

$$\mu^*(I) \leq b - a.$$

Let $\{I_n\}_{n=1}^\infty$ be collection of open intervals that cover $[a, b]$, which is compact, so by Heine–Borel Theorem, there exists a finite set of intervals $I_1, I_2, \ldots, I_m$ such that

$$[a, b] \subseteq \bigcup_{n=1}^m I_n.$$

There exists $I_1 = (a_1, b_1)$ containing a. If $b < b_1$ then $b \in I_1$. If not, then there exists $I_2 = (a_2, b_2)$ containing b_1. Repeat the argument and obtain $a_i < b_{i-1} < b_i$ for $i = 1, \ldots, m$. Hence

$$\begin{aligned}
\sum_{n=1}^\infty \ell(I_n) &\geq \sum_{n=1}^m \ell(I_n) \\
&= \sum_{n=1}^m (b_n - a_n) \\
&= (b_m - a_1) + (b_1 - a_2) + \cdots \\
&\geq b_m - a_1 \geq b - a.
\end{aligned}$$

Taking the infimum over all covers I_n, we obtain the other direction

$$\mu^*(I) \geq b - a.$$

Let $I = [a, \infty)$, then we can find a closed interval $[a, n] \subset I$ for all n. So

$$\mu^*([a, n]) = n - a < \mu^*(I), \ \forall n.$$

Letting $n \to \infty$, this proves (3).

To prove (4), note that $\ell(I) = \ell(I + x)$ for any interval I. Let $\epsilon > 0$, then from definition of infimum we can find $\{I_n\}$ such that $A \subseteq \cup I_i$

$$\sum \ell(I_i) < \mu^*(A) + \epsilon. \tag{1.2.1}$$

Since $A \subseteq \cup I_i$,

$$A + x \subseteq \cup(I_n + x),$$

so by (1.2.1) we have

$$\mu^*(A + x) \leq \sum \ell(I_n + x) = \sum \ell(I_n) < \mu^*(A) + \epsilon.$$

Since ϵ is arbitrary, we obtain $\mu^*(A + x) \leq \mu^*(A)$, and a similar argument using for $A + x - x$ gives the other direction, and this proves (4). □

The preceding properties are natural and should be satisfied by any function representing a measure to demonstrate that it extends the notion of length naturally and does not conflict with it.

1.2.3 Additive Property of Outer Measure

It remains to prove the additive property, which is problematic, as will be seen. If we can show that the outer measure defined in Definition 1.2.1 satisfies axiom (4) then it is a measure. Unfortunately, the following result shows that the outer measure is only countably subadditive.

Proposition 1.2.3 *Let $\{A_i\}_{i=1}^{\infty}$ be a collection of subsets of $\mathbb{R}$. Then*

$$\mu^*(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu^*(A_i).$$

Proof The result holds trivially if $\mu^*(A_i) = \infty$ for at least one A_i, so we may assume that $\mu^*(A_i) < \infty$ for all set A_i. For $\epsilon > 0$ and for each A_i there exists a cover $\{I_{i,n}\}$ such that

$$\sum_{n=1}^{\infty} \ell(I_{i,n}) < \mu^*(A_i) + \frac{\epsilon}{2^i}.$$

Since

$$\bigcup_{i=1}^{\infty} A_i \subseteq \bigcup_{i=1}^{\infty} I_{i,n},$$

we have

$$\begin{aligned}\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) &\leq \sum_i \sum_{n=1}^{\infty} \ell(I_n) \\ &\leq \sum_i \left(\mu^*(A_i) + \frac{\epsilon}{2^i}\right) \\ &= \sum_i \mu^*(A_i) + \epsilon.\end{aligned}$$

Note that ϵ is arbitrary. This completes the proof. □

Proving countable additivity is not an easy task. However, we can use a prominent property that will reduce the result to a more simple version. Recall that the distance between sets $A, B \in \mathbb{R}$ is given by

$$\begin{aligned}d(A, B) &= \inf\{d(a, b) \mid a \in A, b \in B\} \\ &= \inf\{|a - b| \mid a \in A, b \in B\}.\end{aligned}$$

The property $d(A, B) > 0$ means the two sets are not only disjoint, but positively separated. The positively separation property will be used in the following result to prove additivity of μ^*.

Proposition 1.2.4 *If A and B are two subsets of $\mathbb{R}$ such that $d(A, B) > 0$, then*

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B).$$

Proof The direction

$$\mu^*(A \cup B) \leq \mu^*(A) + \mu^*(B)$$

is already established by Proposition 1.2.3. To establish the other direction, let $\epsilon > 0$. Then we can find intervals $\{I_k\}$ such that $A \cup B \subseteq \cup I_k$ and

$$\sum \ell(I_k) \leq \mu^*(A \cup B) + \epsilon.$$

Now, let $d(A, B) = \rho$. Then subdivide the intervals into smaller intervals such that the length of each subinterval is no more than $\frac{\rho}{2}$. Let $\{I'_k\}$ be the collection of subintervals intersecting with A but not B, and $\{I''_k\}$ be the collection of subintervals intersecting with B but not A. Then $A \subseteq \cup I'_k$, $B \subseteq \cup I''_k$, and

$$\ell(I_k) = \ell(I'_k) + \ell(I''_k).$$

Hence

$$\mu^*(A) + \mu^*(B) \leq \sum \ell(I'_k) + \ell(I''_k) = \sum \ell(I_k) \leq \mu^*(A \cup B) + \epsilon.$$

The result follows since ϵ is arbitrary. □

It turns out that the condition $d(A, B) > 0$ can be used efficiently in proving additivity property. It is obvious from the definition that if $d(A, B) > 0$ then $A \cap B = \emptyset$, but the converse is not necessarily true. For example, take $A = (0, 1)$ and $B = (1, 2)$. It suffices to require that both sets are closed and at least one of them is compact. Let us take a simple case. Let $x \notin A$ where A is closed. Then, $d(x, A)$ gives the distance between x and the boundary of A, which is indeed positive since the boundary of A is included in A. In other words, $d(x, A) = 0$ if and only if $x \in \overline{A}$ (the closure of A). Now, let $x \in K$ where K is compact, and $A \cap K = \emptyset$ for some closed set A. Then, $d(x, A) = \rho_x > 0$ by the argument above. Moreover, by the compactness of K we can choose $O_1, O_2, \ldots, O_n$ covering K with $d(x_i, A) = \rho_{x_i}$ for each $x_i \in O_i$. So we have

$$d(x, A) = \rho = \min\{\rho_{x_i}\},\ i = 1, \ldots, n\,.$$

for every $x \in K$, which implies that $d(A, K) > 0$. Thus, we have the following fact:

Lemma 1.2.5 *Let A, B be closed sets of $\mathbb{R}$, and A is compact. If $A \cap B = \emptyset$, then $d(A, B) > 0$.*

The previous lemma will be used in the following result to prove the desired additivity property for μ^* on any collection of pairwise disjoint compact sets.

Proposition 1.2.6 *Let $\{A_i\}$ be a collection of pairwise disjoint compact sets. Then,*

$$\mu^*(\bigcup_i^{\infty} A_i) = \sum_{i=1}^{\infty} \mu^*(A_i).$$

Proof We use Proposition 1.2.4 and finite induction to get

$$\mu^*\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \mu^*(A_i). \tag{1.2.2}$$

It suffices to show that

$$\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \geq \sum_{i=1}^{\infty} \mu^*(A_i) \tag{1.2.3}$$

since the other direction is established by Proposition 1.2.4 and the previous Lemma. Taking into account the inclusion

$$\bigcup_{i=1}^{n} A_i \subseteq \bigcup_{i=1}^{\infty} A_i,$$

monotonicity of μ^* implies

$$\sum_{i=1}^{n} \mu^*(A_i) = \mu^*\left(\bigcup_{i=1}^{n} A_i\right) \leq \mu^*\left(\bigcup_{i=1}^{\infty} A_i\right).$$

Now take $n \longrightarrow \infty$. This proves (1.2.3). □

1.3 Inner Measure

1.3.1 Definition of Inner Measure

Proposition 1.2.3 proves subadditivity for general sets not necessarily disjoint. But what if the sets $\{A_i\}_{i=1}^{\infty}$ are pairwise disjoint? Does μ^* satisfy the countably additive property in this case? Well, taking into account that countable additivity is one of the axioms of measure, we need to establish this property. Unfortunately, there is no direct way to prove it for this measure, but the aid will come from the *inner* measure. Let us look at the additivity property from the following simple way: Let $X = A \cup (X \setminus A)$ for some $A \subseteq X$ and $\mu^*(X \setminus A) < \infty$. Proposition 1.2.3 implies that

$$\mu^*(X) \leq \mu^*(A) + \mu^*(X \setminus A),$$

that is,

$$\mu^*(X) - \mu^*(X \setminus A) \leq \mu^*(A). \tag{1.3.1}$$

This will not lead to any problem since we assume $\mu^*(X \setminus A) < \infty$. But $\mu^*(X \setminus A)$ is the outer measure of the complement of A under X, so the LHS of (1.3.1) can be thought of as the measure of A from inside, i.e., the inner measure of A. This motivates us[1] to introduce the following.

Definition 1.3.1 (*Inner Measure*) Let $X \subset \mathbb{R}$ with $\mu^*(X) < \infty$, and let $A \subseteq X$. Then, the inner measure of A, denoted $\mu_*(A)$, is defined as

$$\mu_*(A) = \mu^*(X) - \mu^*(X \setminus A).$$

The condition $\mu^*(X) < \infty$ is essential to avoid the indeterminate form $\infty - \infty$. Since the definition of inner measure holds for any superset, we need to find a superset $X \supseteq A$ such that $\mu^*(X \setminus A) < \infty$. But in this case, obtaining an inner measure for an unbounded set of the form $[a, \infty)$ or $(-\infty, a)$ becomes problematic. We also notice that the definition is based on the outer measure, which justifies why the outer measure is usually preferred.

[1] In fact, this is what motivated Lebesgue in the first place to define the inner measure.

1.3.2 Properties of Inner Measure

The definition implies the following properties.

Proposition 1.3.2 *Let $X \subset \mathbb{R}$ with $\mu^*(X) < \infty$, and let $A \subseteq X$. Let μ_* be the Lebesgue inner measure defined in Definition 1.3.1. Then*

1. $0 \leq \mu_*(A) \leq \mu^*(A)$.
2. If $B \subseteq A$, then $\mu_*(B) \leq \mu_*(A)$.
3. If A is countable then $\mu_*(A) = 0$. Consequently, $\mu_*(\emptyset) = 0$.
4. If A is a bounded interval, then $\mu_*(A) = \ell(A)$.
5. $\mu_*(A) = \mu_*(A + x)$ for any $x \in \mathbb{R}$.

Proof Statement (1) is immediate, and (2) follows from definition since $\mu^*(A^c) \subseteq \mu^*(B^c)$.

If A is countable then by (1) we have

$$0 \leq \mu_*(B) \leq \mu^*(B) = 0.$$

From (2) we obtain

$$0 \leq \mu_*(\emptyset) \leq \mu_*(\{a\}) = 0,$$

which proves (3).

For (4), choose X to be an interval containing A, and using Definition 1.3.1 and Proposition 1.2.2(3) we obtain (4).

For (5), we use Definition 1.3.1 and Proposition 1.2.2(4). □

1.3.3 The Inner Measure Problem

If the structure of the set is too bad in such a way that the inner measure differs from the outer measure, then the set will have two various measures, which is absurd. In this case, we shall say that this set cannot be measured (or *nonmeasurable*) in the sense of Lebesgue. If it happens that $\mu_*(A) = \mu^*(A)$, then we say that the set A has a unique Lebesgue measure

$$\mu_*(A) = \mu^*(A) = \mu(A).$$

We need to show in this case that μ is a measure. Another point to consider is that the definition of inner measure does not generally work for unbounded sets. So we need to restrict our attention to bounded sets, then we use it to define measurability for unbounded sets. Now, let us see what the measurability of the bounded set means. If $\mu_*(A) = \mu^*(A)$, then

$$\mu^*(A) = \mu^*(X) - \mu^*(X \setminus A)$$

for $X = A \cup (X \setminus A)$, or

$$\mu^*(X \setminus A) = \mu^*(X) - \mu^*(A). \tag{1.3.2}$$

The reader should realize now that we have some troubles in using the definition of inner measure. In view of (1.3.2), saying that the inner measure of a set agrees with its outer measure means countable additive property of outer measure is valid for subsets and their complements if both are inside a given set. This, however, might lead to a problem, as the availability of the superset X makes it hard to prove the measurability of sets. It is not clear if the complement set of a measurable set is measurable or not. If X is chosen to contain A, then $A^c = \mathbb{R} \setminus A$ is unbounded and the outer measure of any set containing A^c is infinite. We can get around this way by two approaches: 1. Generalizing (1.3.2). 2. Formulating another equivalent definition for the inner measure. We postpone the second approach until Sect. 1.5. Let us now explain the first approach.

Let E be a bounded set and let A be a set of finite outer measure such that $E \subseteq A$. Then

$$\begin{aligned} \mu_*(E) &= \mu^*(A) - \mu^*(A \cap E^c) \\ &= \mu^*(A) - \inf_{O \supseteq E^c} \mu^*(A \cap O). \end{aligned}$$

Since $E^c \subseteq O$, $F = O^c \subseteq E$, and note also that

$$(-\inf(X)) = \sup(-X).$$

Then

$$\mu_*(E) = \sup_{F \subseteq E} (\mu^*(A) - \mu^*(A \cap F^c)) = \sup_{F \subseteq E} \mu_*(F). \tag{1.3.3}$$

One can use a similar argument for the reverse direction (verify). The existence of the open set $O \supseteq E^c$ guarantees the existence of a closed set contained in E, and singleton sets and the empty set are closed sets; therefore, we can talk about the supremum of the closed sets contained in E.

1.3.4 Inner Measure As a Supremum

We will adopt this approach to formulate the following form of inner measure.

Definition 1.3.3 (*Inner Measure of a Set*) The inner measure of subsets of $\mathbb{R}$, denoted μ_*, is a nonnegative function

$$\mu_* : P(\mathbb{R}) \to [0, \infty],$$

defined by

$$\mu_*(A) = \sup\{\mu_*(F) : \text{ for all closed sets } F \subseteq A\}.$$

For Lebesgue's outer measure, Definition 1.3.3 is in fact equivalent to Definition 1.3.1. One advantage of this definition over Definition 1.3.1 is that it is no longer restricted to bounded sets since closed sets may be unbounded. However, a drawback is that it defines an inner measure as the supremum of all lower inner measures, so it is defined in terms of itself. Some authors get around this by taking the outer measure of all compact subsets in A. The definition in this case will take the form:

$$\mu_*(A) = \sup\{\mu^*(K) : \text{ for all compact sets } K \subseteq A\}.$$

While the definition may appear to have some advantages, it has two drawbacks:

1. The definition only serves bounded sets A, but not any arbitrary set in $\mathbb{R}$. This is because compact sets are bounded in $\mathbb{R}$.
2. For bounded sets, all closed subsets are compact, so replacing a closed set F with a compact set K is valid, but considering the outer measure of this set is not valid because, from (1.3.3), this would imply that $\mu^*(K) = \mu_*(K)$, which means that compact sets by definition have the same inner measure and outer measure. However, the definition should not automatically grant this ultimate result because the main purpose of defining a set's inner measure is to allow characterizing sets that have one measure.

These arguments led some mathematicians to prefer Definition 1.3.1, and others to search for alternative measurability options other than this approach (e.g., Caratheodory definition which will be discussed in Sect. 1.5).

Note that one can also use Definition 1.3.3 to prove Proposition 1.3.2(4) in a more general setting.

Proposition 1.3.4 *If J is any interval in $\mathbb{R}$, then $\mu_*(J) = \ell(J)$.*

Proof If J is bounded, then given that $\ell(\overline{J}) = \ell(J)$, it suffices to assume J to be open. So let $J = (a, b)$. Let $F \subseteq J$ be any closed set. Then

$$\mu_*(F) \leq \mu^*(F) \leq b - a.$$

Taking the supremum over all closed sets F gives

$$\mu_*(J) \leq b - a.$$

On the other hand, for any $\epsilon > 0$ the set $A = [a + \frac{\epsilon}{2}, b - \frac{\epsilon}{2}]$ is closed, and

$$\ell(A) = b - a - \epsilon \leq \mu_*(J)$$

since $A \subset J$. Taking $\epsilon \to 0$ and taking into account Proposition 1.2.2(3), we obtain

$$\mu_*(J) = b - a = \ell(J),$$

which proves Proposition 1.3.2(4) for J. Now let J be unbounded. It suffices to assume that $J = [a, \infty)$. Define the closed interval $J_n = [a, n] \subset [a, \infty)$. Then

$$\mu_*(J_n) = n - a \leq \mu_*(J),$$

and taking $n \to \infty$, we obtain $\mu_*([a, \infty)) = \infty$. □

1.4 Lebesgue Measurability

1.4.1 Measurable Sets

In the preceding two sections, we defined the measure of a set from outside and from inside. Combining the two concepts, we are ready to define measurability in the sense of "Lebesgue".

Definition 1.4.1 (*Lebesgue Measurability*) Let A be a bounded set in $\mathbb{R}$, (i.e., $\mu^*(A) < \infty$). If $\mu_*(A) = \mu^*(A)$, then we say that A is *measurable*, and that its measure is

$$\mu(A) = \mu_*(A) = \mu^*(A).$$

Otherwise, the set A is said to be "*nonmeasurable*". If A is unbounded set, that is, $\mu^*(A) = \infty$, then A is said to be measurable if $A \cap I$ is measurable for every bounded interval I.

The restriction on unbounded sets is imposed because A could be unbounded with infinite inner and outer measure, and this does not guarantee that A has the same inner and outer measure. The condition in the unboundedness case is to ensure that the structure of the set is good enough to produce unique measurability. One way to motivate the condition is the fact that for an unbounded set $A \subseteq \cup I_k$ where $I_k = (k, k + 1)$, we can write

$$A = \bigcup_k (A \cap I_k),$$

which implies that A is measurable provided that $A \cap I_k$ is measurable for each k and the countable union of measurable sets is measurable. It turns out that the treatment of measure for unbounded sets will not be complete until we determine whether the countable union of measurable sets is measurable. In lights of Definition 1.4.1, the following central questions arise:

1. What type of examples are there for measurable sets?
2. How large is the collection of measurable sets?
3. If $\mu^* = \mu_*$, does this define a measure satisfying countable additive property?

4. Are there examples of nonmeasurable sets?

We shall provide an answer for each one of the questions above. The following theorem gives a simple answer to the first question.

Proposition 1.4.2 *The following statements hold true.*

1. All sets of outer measure zero are measurable. That is, if $\mu^*(A) = 0$, then A is measurable.
2. All intervals are measurable.

Proof (1) follows from Proposition 1.3.2(1) and Definition 1.4.1. Proof of (2) follows from propositions: 1.2.2(3) and 1.3.4, noting that for any bounded interval I, the set $I \cap [a, \infty)$ will either be a singleton, I, a subinterval of I, or the empty set. All of them are measurable sets, hence by Definition 1.4.1 $[a, \infty)$ is measurable. Similarly, we can prove all unbounded intervals are measurable. □

1.4.2 Characterization of Measurable Sets

In view of Proposition 1.4.2, we see that all types of intervals and all finite and countable sets, in addition to the empty set, are measurable sets. Is this all? To be able to characterize measurable sets efficiently we need the following important result.

Proposition 1.4.3 *(Approximation Criterion) Let $E \subseteq \mathbb{R}$. Then:*

1. E is measurable if and only if for every $\epsilon > 0$, there exists an open set $O \supseteq E$ such that $\mu^*(O \setminus E) < \epsilon$.
2. E is measurable if and only if for every $\epsilon > 0$, there exists a closed set $F \subseteq E$ such that $\mu^*(E \setminus F) < \epsilon$.

Proof The proof of (1) is split into two cases: for the case when E is bounded and when E is unbounded. Let E be a bounded measurable set, so $\mu^*(E) < \infty$. For $\epsilon > 0$, there exists a collection of intervals $\{I_n\}$ such that $E \subseteq \bigcup_n I_n$ and

$$\sum \ell(I_n) < \mu^*(E) + \epsilon.$$

Let $O = \bigcup_n I_n$ then O is an open set and

$$\mu^*(O) \leq \sum \ell(I_n) < \mu^*(E) + \epsilon.$$

Since E is measurable, using (1.3.2)

$$\mu^*(O \setminus E) = \mu^*(O) - \mu^*(E) < \epsilon.$$

Now, let E be unbounded and $\mu^*(E) = \infty$. Let $\{I_n\}$ be a collection of disjoint finite intervals, and define $E_n = E \cap I_n$. Then $E = \bigcup E_n$, and by definition of measurability of unbounded sets, each E_n is measurable. Since E_n is bounded, by the first case, given $\epsilon > 0$ there exists open set O_n such that $E_n \subseteq O_n$ and

$$\mu^*(O_n \setminus E_n) < \frac{\epsilon}{2^n}.$$

Define $O = \bigcup O_n$. Then O is open set, and using set operations and monotonicity of outer measure, it is easy to see that

$$\mu^*(O \setminus E) \leq \sum \mu^*(O_n \setminus E_n) < \sum \frac{\epsilon}{2^n} = \epsilon.$$

For the reverse direction, let $\epsilon_n = \dfrac{1}{n} > 0$. Then for each n, there exists an open set O_n such that

$$\mu^*(O_n \setminus E) < \frac{1}{n}.$$

Define $G = \bigcap O_n$. Then $G \supseteq E$ and $G \setminus E \subseteq O_n \setminus E$, which implies

$$\mu^*(G \setminus E) \leq \mu^*(O_n \setminus E) < \frac{1}{n}.$$

Letting $n \to \infty$ gives

$$\mu^*(G \setminus E) = 0.$$

Since $E \subseteq G$, we write $G = (G \setminus E) \cup E$. So

$$\mu^*(G) \leq \mu^*(G \setminus E) + \mu^*(E) = \mu^*(E).$$

But monotonicity gives

$$\mu^*(E) \leq \mu^*(G).$$

Hence

$$\mu^*(E) = \mu^*(G). \tag{1.4.1}$$

On the other hand, using Definition 1.3.1 for inner measure,

$$\mu_*(E) = \mu^*(G) - \mu^*(G \setminus E) = \mu^*(G). \tag{1.4.2}$$

Hence E is measurable following (1.4.1) and (1.4.2).

For the reverse direction of (1), we need to prove that each $E \cap I$ is measurable for every bounded interval I because E is unbounded. From (2), for each n there exists an open set O_n such that $E \subseteq O_n$ and

$$\mu^*(O_n \setminus E) < \frac{1}{n}.$$

Define $G = \bigcap O_n$, and using the same argument in the first case we have

$$\mu^*(G \setminus E) = 0.$$

Now, let I be a bounded interval. Then $E \cap I \subseteq G \cap I$, so

$$\mu^* ((G \cap I) \setminus E \cap I)) = \mu^*(G \setminus E) = 0.$$

By monotonicity, $\mu^*(E \cap I) \leq \mu^*(G \cap I)$. On the other hand, by additivity of outer measure

$$\mu^*(G \cap I)) \leq \mu^*(E \cap I) + \mu^* ((G \cap I) \setminus E \cap I)) = \mu^*(E \cap I).$$

Hence

$$\mu^*(E \cap I) = \mu^*(G \cap I). \tag{1.4.3}$$

But using inner measure gives

$$\mu_*(E \cap I) = \mu^*(G \cap I)) - \mu^* ((G \cap I) \setminus E \cap I)) = \mu^*(G \cap I)).$$

Hence

$$\mu_*(E \cap I) = \mu^*(E \cap I). \tag{1.4.4}$$

From (1.4.3) and (1.4.4) we conclude that $E \cap I$ is measurable, and therefore E is measurable since I is arbitrary. This completes the proof of (1).

Proof of (2). This is a routine proof that can be concluded using result (1) above and the next theorem after we show that E^c is measurable (see Problem 13). We will, however, prove only the result for bounded sets. By definition of supremum, for $\epsilon > 0$ there exists closed set $F \subseteq E$ such that $\mu_*(E) < \mu_*(F) + \epsilon$, or

$$\mu_*(E) - \mu_*(F) < \epsilon. \tag{1.4.5}$$

Since E is bounded and measurable, $\mu_*(E) = \mu^*(E)$ and

$$\mu_*(F) = \mu^*(E) - \mu^*(E \setminus F),$$

which implies from (1.4.5) that

$$\mu^*(E \setminus F) = \mu_*(E) - \mu_*(F) < \epsilon.$$

For the reverse direction, let E be bounded set. Then for each n, there exists a closed set $F_n \subseteq E$ such that

$$\mu^*(E \setminus F_n) < \frac{1}{n}.$$

Define $K = \bigcup F_n$. Then using similar argument as above,

$$\mu^*(E \setminus K) = 0.$$

Since $K \subseteq E$, we have $\mu_*(K) \leq \mu_*(E)$. On the other hand,

$$\mu_*(K) = \mu^*(E) - \mu^*(E \setminus K) = \mu^*(E).$$

It follows that

$$\mu^*(E) \leq \mu_*(K) \leq \mu_*(E),$$

and therefore $\mu^*(E) = \mu_*(E)$, so E is measurable. □

1.4.3 Measurable Sets as Sigma-Algebra

According to the previous proposition, a set is measurable if and only if it can be approximated by an open set containing it or a closed set contained in it. This provides us with a powerful tool for characterizing measurable sets. The following theorem is a consequence of the previous theorem.

Theorem 1.4.4 *All countable unions, countable intersections, complements, and differences of measurable sets are measurable.*

Proof Let $\{E_i\}$ be a collection of measurable sets, and let $\epsilon > 0$. Then for every E_i there exists $O_i \supseteq E_i$ such that

$$\mu^*(O_i \setminus E_i) \leq \frac{\epsilon}{2^i}.$$

Let $E = \bigcup E_i$. Then

$$E \subseteq O = \bigcup O_i$$

which is an open set, and

$$O \setminus E \subseteq \bigcup (O_i \setminus E_i).$$

By monotonicity,

$$\mu^*(O \setminus E) \leq \mu^*(\bigcup (O_i \setminus E_i)) \leq \sum \mu^*(O_i \setminus E_i) = \sum \frac{\epsilon}{2^i} = \epsilon.$$

Then, by Proposition 1.4.3, E is measurable.

To prove complements are measurable, let E be measurable. Again, by approximation criterion, for every n there exists open set O_n such that

$$\mu^*(O_n \setminus E) < \frac{1}{n}.$$

Then $O_n^c = F_n$ is closed set, and $E^c \setminus F_n = O_n \setminus E$, and so

$$\mu^*(E^c \setminus F_n) < \frac{1}{n},$$

Let $F = \bigcup F_n$, then by previous argument it is easy to see that

$$\mu^*(E^c \setminus F) = 0.$$

Let I be an arbitrary bounded interval. Then $F \cap I \subseteq E^c \cap I$ and

$$\mu^*(E^c \cap I \setminus (F \cap I)) = 0.$$

By the same argument shown above, we can show that

$$\mu^*(F \cap I)) = \mu^*(E^c \cap I). \tag{1.4.6}$$

Using Definition 1.3.1,

$$\mu_*(F \cap I)) = \mu^*(E^c \cap I) - \mu^*(E^c \cap I \setminus (F \cap I) = \mu^*(E^c \cap I). \tag{1.4.7}$$

Now (1.4.6) and (1.4.7) imply

$$\mu_*(F \cap I)) = \mu^*(F \cap I)),$$

and thus $E^c \cap I$ is measurable. If E^c is unbounded, then it is measurable since I is arbitrary. If E^c is bounded, choose I such that $E^c \subseteq I$, so $E^c \cap I = E^c$.

Now, that we proved that countable unions and complements of bounded measurable sets are measurable, we can use set operations and the fact that

$$\left(\bigcap_n^\infty A_n\right) = \left(\bigcup_n^\infty A^c\right)^c \quad \& \quad A \setminus B = A \cap B^c$$

to conclude that the countable intersection of bounded measurable sets and differences between bounded measurable sets are measurable. □

One of the immediate consequences of the preceding theorem is that once open sets are proven to be measurable, we can immediately conclude that closed sets

(and hence all clopen sets) are measurable as well. Nevertheless, Proposition 1.4.3 automatically implies the measurability of open sets and closed sets. It characterizes measurable sets by approximating them to open and closed sets, which means that open sets and closed sets are themselves measurable sets. Hence, Proposition 1.4.3 and Theorem 1.4.4 lead to the following essential corollary:

Corollary 1.4.5 *Open sets, closed sets, and all countable intersections of open sets and countable unions of closed sets are measurable.*

Proof Let U be an open set in $\mathbb{R}$. Choose $O = U$ in Proposition 1.4.3(1) gives $\mu^*(O/U) = 0$. This implies that U is an open set. The proof for closed sets is similar, and the result now follows from the previous theorem. □

In light of the preceding results, all countable unions and intersections of intervals and measurable sets and their complements and differences are measurable, in addition to the set of irrational numbers and even the whole set $\mathbb{R}$. The collection of Lebesgue measurable subsets of $\mathbb{R}$ forms a special type of algebra of sets, known as: "σ-algebra".

1.4.4 Borel Sets

Definition 1.4.6 (*Algebra, σ-algebra, Borel σ-algebra*) Let X be a set. Then, a collection Σ of subsets of X that is closed under complements and finite unions is called: $algebra$. If the algebra Σ is also closed under countable unions, then it is called σ-*algebra*. If the σ-*algebra* is generated by open sets, then it is called: $Borel$ σ-*algebra*, and denoted by $\mathcal{B}$. A set that belongs to a Borel σ-algebra is called: "$Borel\ set$".

Borel σ-algebra is the smallest σ-algebra containing open sets. The intersection of open sets is a Borel set and is denoted by G_δ, and the union of closed sets is a Borel set and is denoted by F_σ. Here, the symbols σ and δ refer to unions and intersections, respectively, i.e., a set in $G_{\delta\sigma}$ can be written as a countable union of sets in G_δ. Similarly, a set in $F_{\sigma\delta}$ can be written as a countable intersection of sets in F_σ. In lights of the previous definition and all preceding results, we can say that the Lebesgue measurable sets forms σ-algebra of $\mathbb{R}$, and we have the following

$$F \subset F_\sigma \subset F_{\sigma\delta} \subset F_{\sigma\delta\sigma} \subset \cdots,$$

$$G \subset G_\delta \subset G_{\delta\sigma} \subset G_{\delta\sigma\delta} \subset \cdots.$$

That's quite large!

As an example, we can a singleton set $\{a\}$ as

$$\{a\} = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, a + \frac{1}{n}\right).$$

Hence $\{a\} \in \mathcal{B}$, i.e., a Borel set, and hence the set of rational numbers is a Borel set. Moreover,

$$[a, b) = \{a\} \cup (a, b),$$

and

$$[a, b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, a + \frac{1}{n}\right).$$

Hence all closed intervals and half-open intervals are Borel sets.

1.4.5 Additivity of Lebesgue Measure

The tools are handy now to answer the third question: Does $\mu = \mu^* = \mu_*$ define a measure on $\mathbb{R}$? The only axiom remaining is the countable additivity. We first prove the following result.

Proposition 1.4.7 *Let $\{A_n\}_{n=1}^{\infty}$ be a collection of subsets of $\mathbb{R}$ that are mutually disjoint. Then*

$$\sum_{n=1}^{\infty} \mu_*(A_n) \leq \mu_*(\bigcup_{n}^{\infty} A_n).$$

Proof The result is trivial if $\mu_*(A_n) = \infty$ for some n. Hence we may assume $\mu_*(A_n) < \infty$ for every $n \in \mathbb{N}$. For $\epsilon > 0$ and each n, there exists compact set K_n such that $K_n \subseteq A_n$ and

$$\mu(K_n) \geq \mu_*(A_n) - \frac{\epsilon}{2^n}.$$

But since A_n are disjoint, so are K_n and $d(K_i, K_j) > 0$ for each $i, j \in \mathbb{N}$. Fix a number $m \in \mathbb{N}$. Using monotonicity, Proposition 1.2.6, and Corollary 1.4.5 (since compact sets are closed) we obtain

$$\mu_*\left(\bigcup_{n=1}^{m} A_n\right) \geq \mu\left(\bigcup_{n=1}^{m} K_n\right) = \sum_{n=1}^{m} \mu(K_n) \geq \sum_{n=1}^{m} \mu_*(A_n) - \epsilon.$$

The result follows since ϵ is arbitrary. □

Now we are ready to prove the long-awaited result, which surprisingly has short and simple proof after all the preceding tools and results.

Theorem 1.4.8 *(*Countable Additivity) Let $\mu : (\mathbb{R}, \Sigma) \longrightarrow [0, \infty]$ be the Lebesgue measure as in Definition 1.4.1. If $\{E_n\}$ be a countable collection of mutually disjoint measurable sets in $(\mathbb{R}, \Sigma)$, then

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i).$$

Proof By Proposition 1.4.7, Proposition 1.3.2 (1), and Proposition 1.2.3, we have

$$\sum_{i=1}^{\infty} \mu_*(E_i) \le \mu_*\left(\bigcup_{i=1}^{\infty} E_i\right) \le \mu^*\left(\bigcup_{i=1}^{\infty} E_i\right) \le \sum_{i=1}^{\infty} \mu^*(E_i).$$

But for each measurable E_i we have

$$\mu_*(E_i) = \mu^*(E_i) = \mu(E_i). \qquad \square$$

1.4.6 Continuity of Lebesgue Measure

The following theorem discusses the continuity of the measure function in the following sense:

Theorem 1.4.9 *Let $\{E_i\}$ be a collection of measurable sets in $\mathbb{R}$. Then we have the following:*

1. **Continuity of measure from below**: If $\{E_i\}$ is ascending (i.e., $E_n \subseteq E_{n+1}$ for all n), then
$$\mu(\bigcup_{i}^{\infty} E_i) = \lim \mu(E_i).$$

2. **Continuity of measure from above**: If $\{E_i\}$ is descending (i.e., $E_n \supseteq E_{n+1}$ for all n), and $\mu(E_1) < \infty$, then
$$\mu(\bigcap_{i}^{\infty} E_i) = \lim \mu(E_i).$$

Proof Let $\{E_i\}$ be ascending, i.e., $E_n \subseteq E_{n+1}$. Note that the sets $E_{i+1} \setminus E_i$ are disjoint, hence

$$\begin{aligned}
\mu(\bigcup_{i=1}^{\infty} E_i) &= \mu(E_1 \bigcup (\bigcup_{i=1}^{\infty} E_{i+1} \setminus E_i) \\
&= \mu(E_1) + \sum_{i=1}^{\infty} \mu(E_{i+1} \setminus E_i) \\
&= \lim_{k\to\infty} [\mu(E_1) + \sum_{i=1}^{k} \mu(E_{i+1}) - \mu(E_i)] \\
&= \lim_{k\to\infty} \mu(E_k),
\end{aligned}$$

which proves (1).

Let $\{E_i\}$ be descending, i.e., $E_{n+1} \subseteq E_n$. Note that $E_1 \setminus E_i \subset E_1 \setminus E_{i+1}$. So by the preceding case

$$\mu\left(\bigcup_{i=1}^{\infty}(E_1 \setminus E_i)\right) = \lim \mu(E_1 \setminus E_i), \tag{1.4.8}$$

but

$$\lim \mu(E_1 \setminus E_i) = \lim \mu(E_1) - \mu(E_i) = \mu(E_1) - \lim \mu(E_i). \tag{1.4.9}$$

On the other hand,

$$\mu\left(\bigcup_{i=1}^{\infty}(E_1 \setminus E_i)\right) = \mu\left(E_1 \setminus \bigcap_{i=1}^{\infty} E_i\right) = \mu(E_1) - \mu\left(\bigcap_{i=1}^{\infty} E_i\right). \tag{1.4.10}$$

Combining (1.4.8), (1.4.9), and (1.4.10) establishes (2). □

1.5 Caratheodory Criterion

1.5.1 The Idea of Caratheodory's Approach

Earlier in the chapter, we illustrated two approaches to get around the problem of (1.3.2). Lebesgue measurability was explained as the first approach, and now it is time to explain the second approach. Let E be a given measurable set. Let I be bounded interval; then $E \cap I$ is measurable, i.e.,

$$\mu^*(E \cap I) = \mu_*(E \cap I) = \mu^*(I) - \mu^*(I \cap E^c),$$

or we can write it as

$$\mu^*(I) = \mu^*(E \cap I) + \mu^*(I \cap E^c). \tag{1.5.1}$$

In other words, every bounded interval should satisfy (1.5.1). Caratheodory's idea is to generalize the treatment from a bounded interval to any set A. It is known that

$$\mu^*(A) \leq \mu^*(A \cap E) + \mu^*(A \cap E^c). \tag{1.5.2}$$

1.5.2 Characterization of Measurable Sets

The following criterion provides another, but equivalent, way of characterizing measurable sets.

Proposition 1.5.1 (Caratheodory Criterion) *Let $E \subseteq \mathbb{R}$ be a set. Then E is measurable iff for any set A we have*

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c). \tag{1.5.3}$$

Proof Assume E to be measurable according to Definition 1.4.1. Choose an arbitrary set $A \subseteq \mathbb{R}$ and let O be open set such that $A \subseteq O$. Then by monotonicity,

$$\mu^*(A) \leq \mu^*(A \cap E) + \mu^*(A \setminus E) \leq \mu^*(O \cap E) + \mu^*(O \setminus E). \tag{1.5.4}$$

Since E and O are measurable sets, $O \cap E$ is measurable, and so

$$\mu^*(O \cap E) = \mu_*(O \cap E).$$

Substituting in (1.5.4), gives

$$\begin{aligned} \mu^*(A) &\leq \mu_*(O \cap E) + \mu^*(O \setminus E) \\ &\leq \mu^*(O) - \mu^*(O \setminus E) + \mu^*(O \setminus E), \end{aligned}$$

which implies

$$\mu^*(A) \leq \mu(O).$$

Now, since O is open and contains A, and knowing that any open set can be written as a union of countable intervals, we can take the infimum over all open set $O \supseteq A$ and substitute back in (1.5.4) to obtain the result.

For the reverse direction, assume E satisfies (1.5.3), and let I be any bounded interval. Then

$$\mu^*(I) = \mu^*(I \cap E) + \mu^*(I \setminus E),$$

or

$$\mu^*(I) - \mu^*(I \setminus E) = \mu^*(I \cap E). \tag{1.5.5}$$

But from Definition 1.2.1, the LHS of (1.5.5) is $\mu_*(I \cap E)$, and therefore $I \cap E$ is measurable, and since I is arbitrary bounded, measurability follows. □

Remark In (1.5.3), and since

$$A = (A \cap E) \cup (A \cap E^c),$$

by subadditivity of μ^*, we have

$$\mu^*(A) \leq \mu^*(A \cap E) + \mu^*(A \cap E^c).$$

So in order to prove measurability of E, it suffices to show that

$$\mu^*(A) \geq \mu^*(A \cap E) + \mu^*(A \cap E^c).$$

1.5.3 Lebesgue Versus Caratheodory Approaches

Under this clever criterion, a measurable set E can split any set A into two disjoint parts such that the measure of A is the total measure of both parts. This "split" condition characterizes measurable sets and gives the notion of measurability more flexibility in dealing with all sets. This condition is named after Caratheodory, who proposed his definition in 1914. Due to its popularity, many textbooks now introduce this condition as the definition of measurable sets, and the previous proposition proves the legitimacy of this adoption since they are equivalent. Both definitions are common and widespread in the literature, and they both treat the measurability of sets in the sense of Lebesgue theory. Still, the inner–outer approach was the idea of Lebesgue, while the splitting approach was the idea of Caratheodory. The former is more natural and appealing. It also gives credit to Lebesgue, who proposed this definition, and to appreciate his precursor work. However, the latter is more efficient and flexible in dealing with measurable sets and studying abstract spaces other than Euclidean spaces $\mathbb{R}^n$ because the topological structure of Euclidean spaces is clear and well understood and so it becomes simple to determine open and closed sets to perform outer and inner measures. This may be hard in general abstract spaces, if impossible, which explains why Caratheodory's criterion is favorable for researchers working on probability theory, for instance.

Another advantage of the Caratheodory condition is that it enables us to prove that the collection Σ of measurable sets is σ-algebra in less work and efforts that were needed using the inner–outer Lebesgue's approach. For example, it is immediate from (1.5.3) that the sets $\mathbb{R}$ and $\emptyset$ are measurable. Moreover, if $E \subseteq \mathbb{R}$ is measurable, then E^c is measurable.

1.5.4 Sigma-Algebra of Measurable Sets By Caratheodory Criterion

The next result will help us in establishing our main result about the collection of all measurable sets.

Lemma 1.5.2 *Let $\{E_k\}$ be a collection of mutually disjoint measurable sets. Then for every $A \subseteq \mathbb{R}$,*

$$\mu^*(A \cap (\bigcup^{\infty} E_k) = \sum^{\infty} \mu^*(A \cap E_k).$$

Proof We will establish the result in two steps. The first step is to prove

$$\mu^*\left(A \cap \left(\bigcup^{n} E_k\right)\right) = \sum^{n} \mu^*(A \cap E_k) \tag{1.5.6}$$

by induction. In the second step, we extend the result to infinity. Equation (1.5.6) is obviously true for $n = 1$. Assume the statement is true for $n - 1$, that is,

$$\mu^*\left(A \cap \left(\bigcup^{n-1} E_k\right)\right) = \sum^{n-1} \mu^*(A \cap E_k). \tag{1.5.7}$$

To prove it for n, consider the set

$$S = A \cap (\bigcup^{n} E_k).$$

Notice that E_n^c is measurable since E is so, and using Caratheodory criterion

$$\mu^*(S \cap E_n) + \mu^*(S \cap E_n^c) = \mu^*(S) = \mu^*\left(A \cap \left(\bigcup^{n} E_k\right)\right). \tag{1.5.8}$$

But

$$S \cap E_n = \left(A \cap \left(\bigcup^{n} E_k\right)\right) \cap E_n = A \cap E_n$$
$$S \cap E_n^c = \left(A \cap \left(\bigcup^{n} E_k\right)\right) \cap E_n^c = A \cap \bigcup^{n-1} E_k.$$

Substituting back in (1.5.8) and then using (1.5.7) we obtain

$$\mu^*(A \cap E_n) + \mu^*\left(A \cap \bigcup^{n-1} E_k\right) = \mu^*(A \cap E_n) + \sum^{n-1} \mu^*(A \cap E_k).$$
$$= \sum^{n} \mu^*(A \cap E_k).$$

Comparing to (1.5.8) the result follows.

Now we extend (1.5.6) to infinity. Notice that by monotonicity we have

$$\sum^{n} \mu^*(A \cap E_k) = \mu^*\left(A \cap \left(\bigcup^{n} E_k\right)\right) \leq \mu^*\left(A \cap \left(\bigcup^{\infty} E_k\right)\right).$$

Taking $n \to \infty$ gives

$$\sum^{\infty} \mu^*(A \cap E_k) \leq \mu^*(A \cap (\bigcup^{\infty} E_k)).$$

On the other hand, subadditivity of outer measure gives

$$\mu^*\left(A \cap \left(\bigcup^{\infty} E_k\right)\right) = \mu^*\left(\bigcup^{\infty}(A \cap E_k)\right) \leq \sum^{\infty} \mu^*(A \cap E_k).$$

It follows that

$$\mu^*\left(A \cap \left(\bigcup^{\infty} E_k\right)\right) = \sum^{\infty} \mu^*(A \cap E_k). \qquad \square$$

We'll use the preceding lemma to prove the following theorem which establishes the same result of Theorem 1.4.4, using the Caratheodory condition.

Theorem 1.5.3 *All differences, complements, countable unions, and countable intersections of measurable sets are measurable.*

Proof Let E_1 and E_2 be two measurable sets. Let A be arbitrary set in $\mathbb{R}$. First, we show $E_1 \setminus E_2$ is measurable. The idea of the proof is to split A by E_1 then split $A \cap E_1$ by E_2. So we have

$$\begin{aligned}\mu^*(A) &= \mu^*(A \cap E_1) + \mu^*(A \cap E_1^c) \\ &= \mu^*(A \cap E_1^c) + \mu^*(A \cap E_1 \cap E_2) + \mu^*(A \cap E_1 \cap E_2^c).\end{aligned}$$

By subadditivity of μ^*, we write

$$\mu^*(A) \geq \mu^*\left((A \cap E_1^c) \cup (A \cap E_1 \cap E_2)\right) + \mu^*(A \cap E_1 \cap E_2^c). \tag{1.5.9}$$

But it is readily seen that

$$(A \cap E_1^c) \cup (A \cap E_1 \cap E_2) = A \cap (E_1 \cap E_2^c)^c.$$

Substituting back in (1.5.9) we conclude by Caratheodory criterion that $E_1 \cap E_2^c = E_1 \setminus E_2$ is measurable. Complements can be directly derived from (1.5.3), or immediately concluded from the previous result with $E_1 = \mathbb{R}$. Next we show $E_1 \cup E_2$, hence any finite union, is measurable. For convience, let $B = E_1 \cup E_2$. Since

$$(A \cap B) = (A \cap E_1) \cup (A \cap (E_2 \setminus E_1)),$$

we obtain

$$\mu^*(A \cap B) \leq \mu^*(A \cap E_1) + \mu^*\left(A \cap (E_2 \cap E_1^c)\right) \tag{1.5.10}$$

Add $\mu^*(A \cap (E_1 \cup E_2)^c)$ to both sides of (1.5.10) and taking into account that E_2 is measurable and so it splits $A \cap E_1^c$,

$$\mu^*(A \cap B) + \mu^*\left(A \cap B^c\right) \leq \mu^*(A \cap E_1) + \mu^*(A \cap E_1^c). \tag{1.5.11}$$

But writing A as

$$A = A \cap \mathbb{R} = A \cap (B \cup B^c) = (A \cap B) \cup (A \cap B^c)$$

and therefore, we obtain

$$\mu^*(A) \leq \mu^*(A \cap (E_1 \cup E_2)) + \mu^*(A \cap (E_1 \cup E_2)^c). \tag{1.5.12}$$

Equations (1.5.11) and (1.5.12) imply that $E_1 \cup E_2$, hence by induction and the fact that

$$\bigcup_{i=1}^{n} E_i = \left(\bigcup_{i=1}^{n-1} E_i\right) \cup E_n$$

we conclude that $\bigcup_{i=1}^{n} E_i$ is measurable.

Now, we prove the result for a countable union. Construct the following sequence of sets:

$$S_1 = E_1 \ \& \ S_n = E_n \setminus \left(\bigcup_{i=1}^{n-1} E_i\right).$$

Then it is easy to see that $\{S_n\}$ is a collection of mutually disjoint sets with the fact that $S_n \subseteq E_n$ and

$$\bigcup_{i=1}^{\infty} S_i = \bigcup_{i=1}^{\infty} E_i = E,$$

which implies that

$$E^c \subseteq \left(\bigcup_{i=1}^{n} S_i\right)^c.$$

Moreover, by the previous argument about the difference of measurable sets, we see that the sets $\{S_i\}$ are measurable so, by measurability of finite union proved up, $\bigcup_{i=1}^{n} S_i$ is measurable. Hence by Caratheodory criterion,

$$\mu^*(A) \geq \mu^*\left(A \cap \bigcup_{i=1}^{n} S_i\right) + \mu^*\left(A \cap \left(\bigcup_{i=1}^{n} S_i\right)^c\right)$$
$$\geq \mu^*\left(A \cap \bigcup_{i=1}^{n} S_i\right) + \mu^*(A \cap E^c).$$

Using Lemma 1.5.2, and taking the limit of the sum as $n \to \infty$, then using Lemma 1.5.2 again, we can extend the above inequality to

$$\mu^*(A) \geq \mu^*(A \cap \bigcup_{i=1}^{\infty} S_i) + \mu^*(A \cap E^c)$$
$$= \mu^*(A \cap E) + \mu^*(A \cap E^c)\,.$$

Therefore, we conclude that $E = \bigcup_{i=1}^{\infty} E_i$ is measurable.

Now, intersections can be deduced directly from unions and complements. □

The fact that the collection of all measurable sets forms σ-algebra can be concluded now. One can begin with proving that all intervals of the form $(-\infty, a)$, $(-\infty, a]$, (a, ∞), and $[a, \infty)$ are measurable (see Problems 24, 25), then using intersections to show that all intervals are measurable, then the previous theorem to show that all Borel sets are measurable as predicted in Sect. 1.4.

1.5.5 Additivity

An immediate but important conclusion is the following:

Corollary 1.5.4 (Countable Additivity of Measure) *Let $\{E_k\}$ be a collection of mutually disjoint measurable sets. Then*

$$\mu(\bigcup_{\infty} E_k) = \sum_{\infty} \mu(E_k).$$

Proof Let $A = \mathbb{R}$ in Lemma 1.5.2. Then from the hypothesis and the previous theorem, $\bigcup_{\infty} E_k$ is also measurable; hence the result follows. □

1.6 Unusual Sets

1.6.1 Vitali Sets

We've seen that the collection of measurable sets contains a multiple type of sets, such as finite sets, countable sets, open sets, closed sets, intersections of open sets,

unions of closed sets, and any combinations of unions and intersections of these sets, in addition to the empty set (which can be expressed as the intersection of two disjoint open sets). In fact, it is hard (mostly impossible) to visualize a nonmeasurable set, and this gives birth to the fourth question: "Does there exist such sets that are nonmeasurable in Lebesgue sense?" In other words, does Σ equals $P(\mathbb{R})$, the power set of $\mathbb{R}$?

In 1905, B. Vitali, an Italian mathematician, proposed a set that cannot be measured by Lebesgue measure. Before we start defining the set, it is well known that a coset of $\mathbb{Q}$ in $\mathbb{R}$ is any set of the form

$$x + \mathbb{Q} = \{x + q : \ q \in \mathbb{Q}, \ x \in \mathbb{R}\}.$$

Similarly, a coset of $\mathbb{Q}$ in $[0, 1]$ is any set of the form

$$\{x + q : \ q \in \mathbb{Q}, \ \ x \in [0, 1]\}.$$

It is easy to see that the collection of cosets of $\mathbb{Q}$ in $[0, 1]$ form a partition of $[0, 1]$. That is, if $x - y \in \mathbb{Q}$ then

$$x + \mathbb{Q} = y + \mathbb{Q}$$

and if $x - y \notin \mathbb{Q}$ then

$$(x + \mathbb{Q}) \cap (y + \mathbb{Q}) = \emptyset.$$

This partition can be formed by the following equivalence relation:

$$x \sim y \ \text{ in } [0, 1] \Longleftrightarrow x - y \in \mathbb{Q} \cap [0, 1].$$

This equivalence relation produces equivalence classes of the form of cosets of $\mathbb{Q}$ in $[0, 1]$. So under the equivalence relation $\sim$, any two numbers belonging to the same coset are equivalent. Vitali's idea was to form a set by selecting a single point of each of the equivalent classes, i.e., cosets $x + \mathbb{Q}$. Let us denote the formed set by V. Then, we observe the following facts about V:

1. $V \subseteq [0, 1]$. This is obvious since each point in V belongs to some equivalence class of the coset $x + \mathbb{Q}$ form.
2. V is uncountable. Indeed, if $x \in V$ then x is in some coset. Adding x to any rational number q gives another number in the same coset, so each coset is countable; therefore, we must have an uncountable number of cosets because $[0, 1]$ is uncountable.

Now let $q \in C = \mathbb{Q} \cap [-1, 1]$, and consider the set $V_i = V + q_i$ for $q_i \in C$. Define the following set

$$N = \biguplus_{C} V_i. \tag{1.6.1}$$

We observe the following facts about N:

1. The sets V_i are pairwise disjoint. To see this, let $x \in V_i \cap V_j$. Then
$$x = x_i + q_i = x_j + q_j$$
for $x_i \in V_i$ and $x_j \in V_j$. But this implies $x_i - x_j \in C$ so $x_i \sim x_j$, hence they cannot be both in V.
2. $[0, 1] \subseteq N \subseteq [-1, 2]$. If $x \in [0, 1]$ then $x = v + q$ for some $v \in V \subseteq [0, 1]$. So $q = v - x$, and since $x, v \in [0, 1]$ then $q \in C$. Hence $x \in N$. This proves the first inclusion. The second inclusion follows from the fact that $q \in [-1, 1]$.

1.6.2 Nonmeasurable Sets

Now we come to our assertion:

Proposition 1.6.1 *The set N defined in (1.6.1) is nonmeasurable.*

Proof Suppose it is measurable, then $\mu(V_i) = \mu(V)$ for all i since μ is translation-invariant. Moreover, since all v_i are disjoint measurable sets, by countable additivity we have

$$\mu(N) = \mu\left(\bigcup V_i\right) = \sum^{\infty} \mu(V_i) = \sum^{\infty} \mu(V).$$

If $\mu(V) = 0$ then $\mu(N) = 0$. If $\mu(V) > 0$ then $\mu(N) = \infty$. On the other hand, since

$$[0, 1] \subseteq N \subseteq [-1, 2],$$

monotonicity implies

$$1 \leq \mu(N) \leq 3.$$

This is a contradiction, and the proof is complete. □

Let us look at the set N in a different way. Since $[0, 1] \subseteq N$,

$$1 \leq \mu^*\left(\bigcup V_i\right) \leq \sum \mu^*(V_i)\,.$$

Hence

$$\mu^*(V) = \mu^*(V_i) > 0.$$

On the other hand,

$$\mu_*\left(\bigcup V_i\right) \geq \sum \mu_*(V_i),$$

which implies that $\mu_*(V) = 0$. So we have

$$\mu_*(V) < \mu^*(V)$$

and this explains why V is nonmeasurable.

1.6.3 Axiomatic Controversy

But what is the structure of V? We don't know. Is there a concrete example of such a set? Well, the answer is NO. Can we construct such a set? NO. Does this set really exist? YES and NO. This seems a little surprising. It all depends on an axiom that we have used in the proof of Proposition 1.6.1. It is when it was stated that "*we select a single point from each coset*". The whole secret lies in this step, as it's not clear how to select uncountably infinite times without using a function. Thus, to proceed, we need to take it for granted and use an axiom known as "*axiom of choice*", denoted by AC, which asserts that it is possible to make such selection. We don't wish to go into deep details since this will pull us away from the scope of this book. In brief, the axiom of choice has been one of the most controversial ideas in mathematics. The mystery surrounding its validity is still around till today since 1904 when Ernst Zermelo proposed the principle. It is not a theorem, so there is no proof of it. Some consider it as an axiom, a hypothesis while others oppose it. However, both agree that it always leads to non-constructive proofs because AC tells us we "*can*" make a choice, but it doesn't show us "*how*" to make that choice. This justifies why we can't form such a set. It is worth noting that Solovay proved in 1970 that it is impossible to prove the existence of nonmeasurable sets without using AC. Consequently, one can conveniently assert that all subsets of $\mathbb{R}$ are measurable, but only after denying the AC, which won't be simple to consider, especially knowing that most mathematicians today accept AC and use it without hesitation.[2]

1.6.4 Cantor Set

The second unusual set is the set proposed by the German mathematician Georg Cantor in 1883. The construction of the Cantor set is nice but weird. We begin the procedure as follows: First, we consider the interval $I = [0, 1]$. Call this set as C_0. Remove the middle third of the interval C_0without the endpoints. So

$$C_1 = \left[0, \frac{1}{3}\right] \bigcup \left[\frac{2}{3}, 1\right].$$

[2] We have to admit that accepting AC will make our mathematical life easier and proofs and techniques become more efficient, because it has numerous applications. Still, on the other hand, mathematics will not be miserable without AC. Paradoxes and disasters will come out whether we accept AC or deny it. Whether we should accept or deny the AC is entirely beyond this book's scope, and the interested reader may consult books on Set Theory and Mathematical Philosophy.

The second step is to remove the middle third of each of the intervals above to obtain

$$C_2 = \left[0, \frac{1}{9}\right] \bigcup \left[\frac{2}{9}, \frac{3}{9}\right] \bigcup \left[\frac{6}{9}, \frac{7}{9}\right] \bigcup \left[\frac{8}{9}, 1\right].$$

We observe the following:

1. Each C_n is the union of 2^n closed intervals.
2. C_n is closed.
3. The length of each interval above is $\frac{1}{3^n}$. Hence

$$\ell(C_n) = 2^n . \frac{1}{3^n} = \left(\frac{2}{3}\right)^n .$$

4. C_n is measurable, being the union of disjoint measurable sets, hence

$$\mu(C_n) \leq \left(\frac{2}{3}\right)^n .$$

We continue the process infinitely many times, and we define the Cantor set

$$C = C_\infty = \bigcap_{n=1}^{\infty} C_n .$$

Then we see that $\mu(C) = \lim \mu(C_n) = 0$. Let us look at it differently. At the nth step, we remove 2^{n-1} intervals, each of which is of length 3^{-n}. The total length of the removed intervals is

$$\sum 2^{n-1} 3^{-n} = \frac{1}{3} \sum \left(\frac{2}{3}\right)^n = \frac{1}{3} \left(\frac{1}{1 - \frac{2}{3}}\right) = 1.$$

But this is in fact the measure of the complement of C. Hence

$$\mu(C) = 1 - \mu([0, 1]) = 1 - 1 = 0.$$

We observe the following facts about the Cantor set C.

1. C is compact. (Verify).
2. C is measurable. This is due to the previous observation and the fact that it is of measure zero.
3. C has no intervals inside. To see this, for every $x, y \in C$ we can find n sufficiently large such that

$$|x - y| > 3^{-n}.$$

This means x and y belong to two disjoint intervals. Choose a between x and y such that $a \notin C_n$, so $a \notin C$. We conclude that C is totally disconnected, so its interior is empty.

4. C is uncountable. Due to the construction of C, it is best to write all numbers of $[0, 1]$ in ternary (i.e., base 3) representation. So every number in $[0, 1]$ can be written as $\sum a_n 3^{-n}$, where $a_n \in \{0, 1, 2\}$. In the process of constructing C, we remove the middle third. This means that we remove numbers with ternary representation in step n of the process, including 1 in their n^{th} decimal place. So we left with 0 and 2. Hence every $x \in C$ can be expressed as $\sum a_n 3^{-n}$, with $a_n \in \{0, 2\}$. Suppose C is countable, i.e., $C = \{x^1, x^2, \ldots\}$. Then for each $x^j \in C$ we write

$$x^j = 0.r_1^j r_2^j r_3^j \ldots$$

i.e.

$$x^1 = 0.r_1^1 r_2^1 r_3^1 \ldots$$

$$x^2 = 0.r_1^2 r_2^2 r_3^2 \ldots$$

$$x^3 = 0.r_1^3 r_2^3 r_3^3 \ldots$$

Consider the diagonal sequence $0.\, r_1^1 r_2^2 r_3^3 \ldots$. Then the representation

$$x = 0.r_1 r_2 r_3 \ldots$$

defined as

$$r_j = \begin{cases} 2 & : r_j^j = 0 \\ 0 & : r_j^j = 2 \end{cases}$$

is not listed above although $x \in C$. This is a contradiction. This argument is known as the *Cantor diagonal argument*.

The fact that a totally disconnected and uncountable set with zero measure seems stunning. The Cantor set is another set type of set that is hard to interpret and visualize but bears exciting features. Due to its weird construction, it is a great tool to provide counterexamples that explain profound ideas in many topics of analysis.

1.7 Lebesgue Product Measure

1.7.1 The Measure of Rectangles and Cubes

The concept of measure can be extended to $\mathbb{R}^n$ for $n > 1$. This seems plausible, knowing that the notion of length is extended to the notion of the area in two dimensions and to volume in three dimensions. Therefore, the intervals $\{I_n\}$ which are the primary tool of computing measure on $\mathbb{R}$ are extended to rectangles $\{R\}$ in $\mathbb{R}^2$ and to cubes $\{Q\}$ in $\mathbb{R}^n$ for all $n \geq 3$. The $n-$dimensional cubes can be written as

$$Q = [a_1, b_1] \times \cdots \times [a_n, b_n].$$

Then we define the nth-dimensional measure

$$\mu_n(Q) = (b_1 - a_1) \cdot (b_2 - a_2) \cdots (b_n - a_n).$$

All results of this chapter apply to measure on $\mathbb{R}^n$. The outer measure of a set $A \subseteq \mathbb{R}^n$ is

$$\mu_n^* : \mathcal{P}(\mathbb{R}^n) \longrightarrow [0, \infty]$$

defined by

$$\mu_n^*(A) = \inf\{\sum Vol(Q)\},$$

where the infimum is taken over all

$$\bigcup_{\infty} Q_n \supseteq A.$$

If $n = 2$, then the cubes Q reduce to rectangles R, and an open set can be written as a countable union of almost disjoint rectangles. Recall from Sect. 1.3 that the symbols σ and δ refer to unions and intersections, respectively. In this regard, we write $O \in R_\sigma$ to denote a countable union of almost disjoint rectangles. So a countable intersection of open sets belongs to $R_{\sigma\delta}$. We also have the following:

1. $\mu_n^*(\emptyset) = 0$.
2. If $A \subseteq B$ then $\mu_n^*(A) \leq \mu_n^*(B)$.
3. $\mu_n^*(\bigcup A_k) \leq \sum \mu_n^*(A_k)$.

Let us establish (3). We may assume that $\mu_n^*(A_k) < \infty$ for all k, since there is nothing to prove otherwise. Let $\epsilon > 0$. Then there exists a countable cover of rectangles $\{R_{ij}\}$ such that $E_i \subset \bigcup_j R_{ij}$, and

$$\sum_j \mu_n(R_{ij}) \leq \mu_n^*(E_i) + \frac{\epsilon}{2^i}.$$

So

$$\begin{aligned}\mu_n^*(E) &\leq \sum_i \sum_j \mu_n(R_{ij}) \\ &\leq \sum_i (\mu_n^*\left(E_i\right) + \frac{\epsilon}{2^i}\Big) \\ &= \sum_i \mu_n^*(E_i) + \epsilon.\end{aligned}$$

Since ϵ is arbitrary, we obtain (3).

It is well known that any open set in $\mathbb{R}$ can be written as a countable union of disjoint open intervals. However, this is not the case in $\mathbb{R}^n$ for $n > 1$. We instead have the following: An open set $O \in \mathbb{R}^n$ can be written as a countable union of almost disjoint closed sets. The property "almost disjoint" means the interior of the set is disjoint, but the sets can be intersected at the boundaries. In particular, we have the following fundamental result:

Proposition 1.7.1 *Let $Q \in \mathbb{R}^n$ be a cube. Then $\mu_n^*(Q) = Vol(Q)$.*

Proof It suffices to prove the result for compact cubes, and the general cubes are left to the reader. Using the definition, we have

$$\mu_n^*(Q) \leq Vol(Q). \tag{1.7.1}$$

Let $\{Q_k\}$ be a cover of Q, and let $\epsilon > 0$. For each Q_k, let $\hat{Q}_k$ be cubes such that $Q_k \subseteq (\hat{Q}_k)^\circ$ the interior of $\hat{Q}_k$, and

$$\bigcup \hat{Q}_k \leq (1+\epsilon) Vol(\hat{Q}_k).$$

So $\{(\hat{Q}_k)^\circ\}$ is an open cover of Q. Then, by Heine–Borel theorem, there exists $n \in \mathbb{N}$ such that

$$Q \subseteq \bigcup_{i=1}^{n} (\hat{Q}_k)^\circ \subseteq \bigcup_{i=1}^{n} \hat{Q}_k.$$

Hence

$$Vol(Q) \leq \bigcup_{i=1}^{n} \hat{Q}_k \leq \mu_n^*(Q). \tag{1.7.2}$$

The result follows from (1.7.1) and (1.7.2). □

Moreover, we define the inner measure as

$$\mu_{n*}(A) = \sup\left\{\sum \mu_n(K)\right\},$$

where supremum is taken over all compact $K \subseteq A$. So $A \in \mathbb{R}^n$ is measurable iff

$$\mu_{n*}(A) = \mu_n^*(A).$$

1.7.2 Caratheodory Criterion in $\mathbb{R}^n$

We also have the Caratheodory Criterion.

Proposition 1.7.2 *Let $E \in \mathbb{R}^n$. Then E is measurable if for every $A \in \mathbb{R}^n$ we have*

$$\mu_n^*(A) = \mu_n^*(A \cap E) + \mu_n^*(A \setminus E).$$

According to the proposition, all sets of measure zero are measurable in $\mathbb{R}^n$. Let $\mu_n^*(E) = 0$ for some $E \subset \mathbb{R}^n$. For any $A \subset \mathbb{R}^n$ we have

$$0 \leq \mu_n^*(E \cap A) \leq \mu_n^*(E) = 0.$$

Then

$$\mu_n^*(A) \geq \mu_n^*(E \cap A) + \mu_n^*(E^c \cap A).$$

The other direction is clear, and hence the set E is measurable.

Proposition 1.7.2 can be used to show that the collection of measurable sets forms a σ-algebra in $\mathbb{R}^n$. Using the last two propositions, one can show that every open set in $\mathbb{R}^n$ is measurable, and since the complement of a measurable set is measurable, all closed sets are measurable. We can therefore show that all Borel sets in $\mathbb{R}^n$ are Lebesgue measurable. We also have the following approximation property.

Proposition 1.7.3 *The following are equivalent.*

1. E is measurable in $\mathbb{R}^n$.
2. For every $\epsilon > 0$, there exists an open set $O \in R_\sigma$ such that $O \supseteq E$ and $\mu_n^*(O \setminus E) < \epsilon$.
3. There exists $G \in R_{\sigma\delta}$ such that $\mu_n^*(G \setminus E) = 0$.
4. For every $\epsilon > 0$, there exists closed $F \supseteq E$ such that $\mu_n^*(E \setminus F) < \epsilon$.
5. There exists $J \in R_{\delta\sigma}$ such that $\mu_n^*(E \setminus J) = 0$.

Proof The proof is similar to Proposition 1.4.3. For $1 \Rightarrow 2$ let $\epsilon > 0$. Then there exists $\{Q_k\}$ be a cover of E such that

$$\mu_n^*(E) \leq \sum Vol(Q_k) \leq \mu_n^*(E) + \frac{\epsilon}{2}.$$

Find the extended cubes $\hat{Q}_k$ such that $Q_k \subseteq (\hat{Q}_k)^\circ$ and

$$Vol(\hat{Q}_k) \leq Vol(Q_k) + \frac{\epsilon}{2^{k+1}}.$$

Let $O = \bigcup(\hat{Q}_k)^\circ$, then O is open and

$$\mu_n^*(O) \leq \sum Vol(\hat{Q}_k)^\circ \leq \sum \left(Vol(Q_k) + \frac{\epsilon}{2^{k+1}} \right) \leq \mu_n^*(E) + \epsilon.$$

So

$$\mu_n^*(O \setminus E) = \mu_n^*(O) - \mu_n^*(E) \leq \epsilon.$$

The implication $2 \Rightarrow 4$ can easily be established using the fact that if $E \subseteq O$ then $F = O^c \subseteq E^c$, and $O \setminus E^c = E \setminus F$. The implications $2 \Rightarrow 3$ and $4 \Rightarrow 5$ can be established similar to the argument for Proposition 1.4.3 by replacing ϵ with $\frac{1}{n}$ to obtain $F_n \subset E \subset O_n$, then defining $G = \bigcap O_n$ and $J = \bigcup F_n$. The proof of the implications $3 \Rightarrow 1$ (and $5 \Rightarrow 1$) is similar to that for Proposition 1.4.3, or using Proposition 1.7.2. The details are left to the reader as an exercise (see Problem 34). □

1.7.3 Measure of Product Sets

One of the main results of this section is the following:

Theorem 1.7.4 *Let $A, B \in \mathbb{R}^n$. Then*

$$\mu_n^*(A \times B) \leq \mu_n^*(A) \cdot \mu_n^*(B),$$

with equality

$$\mu_n^*(A \times B) = \mu_n^*(A) \cdot \mu_n^*(B)$$

if A and B are measurable sets.

Proof Let $\epsilon_1, \epsilon_2 > 0$. Then there exist $\{Q'_k\}$ and $\{Q''_k\}$ in $\mathbb{R}^n$ such that $E_1 \subseteq \bigcup Q'_k$ and $E_2 \subseteq \bigcup Q''_k$ where

$$\sum Vol(Q'_k) \leq \mu_n^*(A) + \epsilon_1$$

and

$$\sum Vol(Q''_k) \leq \mu_n^*(B) + \epsilon_2.$$

But $A \times B \subseteq \bigcup Q'_k \times Q''_k$, so

$$\begin{aligned} \mu_n^*(A \times B) &\leq \sum Vol(Q'_k \times Q''_k) \\ &= (\sum Vol(Q'_k))(\sum Vol(Q''_k) \\ &\leq (\mu_n^*(A) + \epsilon)(\mu_n^*(B) + \epsilon), \end{aligned}$$

where $\epsilon = \max\{\epsilon_1, \epsilon_2\}$. Letting $\epsilon \to 0$ gives

$$\mu_n^*(A \times B) \leq \mu_n^*(A) \cdot \mu_n^*(B). \tag{1.7.3}$$

If A and B are measurable, then using Proposition 1.7.3 there exist Borel sets $G, H \in R_{\sigma\delta}$, where $A \subseteq G$, $B \subseteq H$, such that

$$\mu_n(G \setminus A) = \mu_n(H \setminus B) = 0.$$

Clearly $G \times H$ is measurable in $\mathbb{R}^{2n}$ (why?). We have

$$(G \times H) \setminus (A \times B)) \subseteq [(G \setminus A) \times B] \cup [G \times (H \setminus B)].$$

So

$$\mu_n\left((G \times H) \setminus (A \times B)\right) = 0.$$

Hence $A \times B$ is measurable, and

$$\mu_n(G \times H) \leq \mu_n(A \times B) + \epsilon.$$

But

$$\mu_n(G \times H) = \mu_n(G) \cdot \mu_n(H) = \mu_n(A) \cdot \mu_n(B).$$

Combining with (1.7.3), the result follows. □

Under the previous theorem, lines in $\mathbb{R}^2$ are of measure zero since they can be written as $\{(x, a) : x \in \mathbb{R}\}$ for vertical lines or $\{(a, y) : y \in \mathbb{R}\}$ for horizontal lines. Observe that the measure used here is μ_2 (the two-dimensional measure) not the standard measure discussed in the preceding sections.

Continuity of the measure from below and from above extends to μ_n. In particular, we have the following.

Theorem 1.7.5 *Let μ_2 be the Lebesgue product measure in $\mathbb{R}^2$. Let $\{E_n\}$ be a countable collection of measurable sets in $\mathbb{R}^2$. Then the following statements hold:*

1. **Countable additivity**. If $E_1, E_2, \ldots$ are pairwise disjoint sets, then

$$\mu_2\left(\bigcup_i^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu_2(E_i).$$

2. **Continuity of measure from below**. If $\{E_i\}$ increases (i.e., $E_n \subseteq E_{n+1}$ for all n) then

$$\mu_2\left(\bigcup_i^{\infty} E_i\right) = \lim \mu_2(E_i).$$

3. **Continuity of measure from above**. If $\{E_i\}$ decreases, (i.e., $E_n \supseteq E_{n+1}$ for all n) and $\mu_2(E_1) < \infty$, then

$$\mu_2(\bigcap_i^{\infty} E_i) = \lim \mu_2(E_i).$$

1.8 Problems

1. Let $A = \mathbb{Q} \cap [0, 1]$. Let $\{I_n\}$ be a finite open cover of A. Prove $\sum \ell(I_i) \geq 1$.
2. Let A, B be sets in $\mathbb{R}$. Show that if $A \subset B$ and $\mu^*(A) = \mu^*(B)$, then $\overline{A} = B$.
3. Let $E \subseteq A \subseteq \mathbb{R}$ and $\mu(E) = 0$. Show that $E^c \cap A$ is dense in A.
4. **Counting Measure**. The counting measure is defined as $\mu(A) = |A|$ if A is finite, and $\mu(A) = \infty$ if A is infinite. This measure is used to measure discrete sets in terms of their cardinality, i.e., the measure of a set is the number of its elements. Show that μ is a measure.
5. **Dirac Measure**. Consider the following Dirac measure δ_x defined as: $\delta_x(A) = 1$ if $x \in A$ and $\delta_x(A) = 0$ if $x \notin A$. Show that for a fixed x, δ_x is a measure.
6. Let $\{\lambda_n^*\}$ be a sequence of outer measures, and define

$$\lambda^*(A) = \sup_n \lambda_n^*(A)$$

   for all $A \subseteq \mathbb{R}$. Show that λ^* is an outer measure.
7. Let $\{\mu_n\}$ be a sequence of measures on $\mathbb{R}$, and define

$$\lambda(A) = \sup\{\mu_n(A)\}.$$

   (a) Determine whether λ is a measure or not.
   (b) If $\mu_n \leq \mu_{n+1}$ in the construction of $\{\mu_n\}$ above, show that λ is a measure.
8. Show that if $E \subseteq \mathbb{R}$ is measurable and $x \in \mathbb{R}$, then $E + x$ is measurable.
9. (a) Show that E is measurable if and only if there exists $G \supseteq E$ such that $\mu^*(G) = \mu(E)$.
   (b) Show that G found in (a) is measurable.
10. Let $E_1 \subset A \subset E_2 \subset \mathbb{R}$ and E_1, E_2 are measurable, with $\mu(E_1) = \mu(E_2) < \infty$. Show that A is measurable.
11. Let $\{A_n\}$ be a sequence of measurable sets. Define

$$\liminf A_n = \bigcup_{n=1}^{\infty} \bigcap_{i=n}^{\infty} A_i.$$

    Show that

$$\mu(\liminf A_n) \leq \liminf \mu(A_n).$$

12. **Borel–Cantelli Theorem**. Let $\{E_n\}$ be a set of measurable sets of $\mathbb{R}$, and $\sum \mu(E_n) < \infty$. Let

$$E = \bigcap_{n=1}^{\infty} \bigcup_{i=n}^{\infty} A_i = \limsup(E_k).$$

(a) Show that E is measurable.
(b) Prove that $\mu(E) = 0$.
13. Prove Proposition 1.4.3(2) for the case when E is unbounded.
14. Prove that A is a measurable set if and only if for every $\epsilon > 0$, there exists a compact set F such that $\mu^*(A \setminus F) < \epsilon$.
15. Use Proposition 1.4.3 directly to show that compact sets are measurable. (Do not use Theorem 1.4.4).
16. **Open–Closed Characterization**. Prove that A is a measurable set if and only if for every $\epsilon > 0$, there exists closed set F and open set O such that $F \subseteq E \subseteq O$ and $\mu^*(O \setminus F) < \epsilon$.
17. Use the open–closed characterization in the previous problem to deduce the Caratheodory condition.
18. Let $A \subseteq \mathbb{R}$. Suppose that for every $\epsilon > 0$, there exists measurable E such that $\mu(A \triangle E) < \epsilon$. Show that A is measurable.
19. Give an example of a Borel set that is neither G_δ nor F_σ.
20. Show that the set of all irrationals in [0, 1] is a G_δ set.
21. **Complete Measure**. A measure μ is called "*complete*" if for every measurable E, with $\mu(E) = 0$, if $U \subseteq E$, then U is measurable.
(a) Show that the Lebesgue measure is complete.) Show that the counting measure (defined in problem 4) is complete.) If ν and λ are two complete measures, show that $\lambda + \nu$ is a complete measure.
22. If E_1, E_2 are measurable, prove that

$$\mu(E_1 \cup E_2) + \mu(E_1 \cap E_2) = \mu(E_1) + \mu(E_2).$$

23. Show that E is measurable if and only if $\mu(A \cup B) = \mu(A) + \mu(B)$ for all $A \subseteq E$ and $B \subseteq \overline{E}$.
24. Use the Caratheodory criterion to show that (a, ∞) is measurable. Deduce that every open, closed, clopen interval is measurable.
25. Use the Caratheodory criterion to prove the approximation result (Proposition 1.4.3).
26. Let A and B be two disjoint subsets of $\mathbb{R}$. Show that if either A or B is measurable, then $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$.
27. Show that for each $\epsilon > 0$, there exists a set A_ϵ that is dense in [0, 1] and such that $\mu(A_\epsilon) \leq \epsilon$.
28. Show that the measurability of sets is necessary for the additivity of measure.
29. Let $E \subseteq V \subseteq \mathbb{R}$, for some Vitali set V. If E is measurable, show that $\mu(E) = 0$.
30. Give an example of a set A with $\mu^*(A) = \infty$ but no intervals are inside A.

31. Give an example of
    a) a nonmeasurable set A with $\mu_*(A) = \mu^*(A) = \infty$.
    b) a set $A \times B$ which is measurable in $\mathbb{R}^n$ but A is nonmeasurable in $\mathbb{R}$. c) two sets A and B such that $\mu(A) = \mu(B) = 0$ but $\mu(A + B) > 0$.
32. Show that every set of positive measure has nonmeasurable subsets.
33. Show that the Cantor set can be mapped surjectively onto $[0, 1]$. Deduce C is uncountable.
34. Prove the implications $(3 \Rightarrow 1)$ and $(5 \Rightarrow 1)$ in Proposition 1.7.3.
35. Prove Theorem 1.7.5.

Chapter 2
Measurable Functions



2.1 Introduction

We will use Lebesgue's measure theory to define Measurable functions. This class of functions is a generalization of the class of continuous functions. It plays a significant role in many important applications, and they can replace continuous functions when they cannot interpret the problem under consideration. It is worth noting that the main motivation for Lebesgue to introduce measurable sets was to strengthen the Riemann integral, which suffers some deficiencies that we shall see later. It is well known from Calculus that the concept of definite Riemann integral is based on the idea continuity of functions, which turns out to be, in many cases, too strong to produce well-defined integrals. It was the reason to think about a wider class of functions that could generalize the class of continuous functions in more general (but weaker) settings. This means that every continuous function should fall in this new generalized class, but the converse is not true.

2.2 Measurable Functions

2.2.1 Simple Function

Recall that a function f is continuous if and only if $f^{-1}(O)$ is open for every open set O. To weaken the definition, we remove the requirement that $f^{-1}(O)$ is open. We may consider it to be merely measurable, and from the previous section we know that every open set is measurable but the converse is not necessarily true. But what property shall we give to f to produce measurable sets of the form $f^{-1}(O)$? The following function may give some motivation.

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Definition 2.2.1 (*Simple Function*) A function φ is called *simple function* if it is defined on a set A, and is written as

$$\varphi(x) = \sum_{i=1}^{n} c_i \chi_{A_i}(x),$$

where $\bigcup A_i = A$, and

$$\chi_{A_i}(x) = \begin{cases} 1 & x \in A_i \\ 0 & x \notin A_i \end{cases}.$$

The function χ_{A_i} is called "*characteristic function*", or the "*indicator function*".

2.2.2 Simple Versus Step Functions

As suggested by its name, its simplicity is because it takes only a finite number of values. In order for these functions to be well defined, we require $c_i = c_j$ on $A_i \cap A_j$ provided that the intersection is nonempty. If the sets $\{A_i\}$ are pairwise disjoint, and the values c_i are distinct, then φ is written in a standard form or canonical form. The collection $\{A_i\}$ is called: a $partition$ of A, meaning that the sets are pairwise disjoint whose union is A. But why are simple functions important? Recall in Riemann integration theory, the primary tool to define upper and lower Riemann sums was the step function given by

$$s(x) = \sum_{i=1}^{n} c_i \chi_{[a_i,b_i]}(x). \tag{2.2.1}$$

This is a function with finite values too, and since intervals are measurable sets, every step function is a simple function. The converse is not necessarily true, however, since not every measurable set is an interval. An example of a simple function that is not a step function is the characteristic function of the rationals $f(x) = \chi_{\mathbb{Q}}(x)$. To give the upper and lower sums more freedom and flexibility in calculating integrals, we can define a wider class of functions containing step functions. Considering that every interval is a measurable set but not the converse, the function defined in (2.2.1) could be the typical function to replace the step function, provided that the sets $\{E_1, E_2, \ldots, E_n\}$ are measurable. Therefore, to satisfy this condition we introduce $measurable\ function$ as a new class of functions which will require the set $f^{-1}(O)$ to be measurable.

2.2.3 Definition of Measurable Function

Definition 2.2.2 (*Measurable Function*) A function $f: \Omega \subseteq \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ is called $measurable$ if given any open set $O \subseteq \mathbb{R}$, $f^{-1}(O)$ is measurable set.

Here, we consider the extended real system

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{\infty, -\infty\}$$

as discussed before. A function having the range of $\overline{\mathbb{R}}$ is called: "$extended\ real\text{-}valued\ function$". Observe that if $f^{-1}(O)$ is open (hence f is continuous), then $f^{-1}(O)$ is measurable, but the converse is not true. So an immediate result that can be drawn from the definition is the following.

Proposition 2.2.3 *Every continuous function is measurable.*

As illustrated earlier in the section, this significant result is the main reason for introducing this new class of functions. The following theorem gives a simpler way to characterize measurable functions.

Proposition 2.2.4 *Let f be an extended real-valued function. Then the following are equivalent*

1. f is measurable.
2. $f^{-1}(F)$ is measurable set for every closed set F.
3. $f^{-1}([-\infty, r)) = \{x \in \Omega : f(x) < r\}$ is measurable set for every $r \in \mathbb{R}$.
4. $f^{-1}([-\infty, r]) = \{x \in \Omega : f(x) \leq r\}$ is measurable set for every $r \in \mathbb{R}$.
5. $f^{-1}((r, \infty]) = \{x \in \Omega : f(x) > r\}$ is measurable set for every $r \in \mathbb{R}$.
6. $f^{-1}([r, \infty]) = \{x \in \Omega : f(x) \geq r\}$ is measurable set for every $r \in \mathbb{R}$.

Proof (1) and (2) are equivalent by complement measurability property and the fact that $f^{-1}(A^c) = \left(f^{-1}(A)\right)^c$. Since $[-\infty, r)$ and $(r, \infty]$ are open sets, (1) implies (3) and (5). To prove (3 $\Rightarrow$ 6), note that

$$f^{-1}([r, \infty)) = f^{-1}((-\infty, r))^c = \left(f^{-1}((-\infty, r))\right)^c,$$

and complement of measurable set is measurable. This also proves (5 $\Rightarrow$ 4). Moreover, note that

$$(-\infty, r) = \bigcup_{n=1}^{\infty} \left[-\infty, r - \frac{1}{n}\right] \ \&\ (r, \infty) = \bigcup_{n=1}^{\infty} \left[r + \frac{1}{n}, \infty\right],$$

and we also have

$$f^{-1}\left(\bigcup E_n\right) = \bigcup f^{-1}(E_n),$$

which proves (6 $\Rightarrow$ 5) and (4 $\Rightarrow$ 3). Note also that

$$(a, b) = (a, \infty) \cap (-\infty, b),$$

and it is well known in elementary analysis that every open set is a countable union of open intervals and every closed set is a countable intersection of open sets. This implies (1) and (2). The details are left to the reader. □

The most basic measurable function is the characteristic function

$$\chi_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}.$$

The characteristic function is measurable if and only if the set A is measurable. The finite combination of characteristic functions gives rise to the simple function

$$\varphi(x) = \sum_{i=1}^{n} c_i \chi_{A_i}(x),$$

where $\chi_{A_i}(x)$ is the characteristic function and $A_i = \{x : \varphi(x) = c_i\}$. Thus, the simple function is measurable if and only if all sets $\{A_i\}$ are measurable.

2.2.4 Algebra of Measurable Functions

One advantage for the class of measurable functions is that it is closed under algebraic operations. We have the following proposition.

Proposition 2.2.5 *If f and g be extended real-valued measurable functions, then:*

1. *If f and g be measurable functions and $c \in \mathbb{R}$, then so is $cf + g$.*
2. *If f and g be measurable functions, then so is $f \cdot g$.*
3. *If f and g be measurable functions, then so is $\dfrac{f}{g}$.*
4. *If f and g be measurable functions, then $f \vee g = \max\{f, g\}$ and $f \wedge g = \min\{f, g\}$ are measurable.*

Proof Note that

$$\{x : cf < r\} = \{x : f < \frac{r}{c}\}$$

if $c > 0$ and

$$\{x : cf < r\} = \{x : f > \frac{r}{c}\},$$

if $c < 0$. In any case, cf is measurable. To prove $f + g$ with $c = 1$, let $r \in \mathbb{R}$, then

$$\{x : f(x) + g(x) < r\} = \{x : f(x) < r - g(x)\}.$$

Density of $\mathbb{Q}$ implies there exists $q \in \mathbb{Q}$ such that $f(x) < q < r - g(x)$. Hence

$$\{x : f(x) + g(x) < r\} = \bigcup_q (\{x : f(x) < r\} \cap \{x : g(x) < q - r\}.$$

The first and second sets are measurable, and countable union of measurable sets is measurable. This proves (1).

Now we prove that if f is measurable then so is f^2. If $r < 0$, then

$$\{x : f^2 > r\} = \mathbb{R},$$

which is measurable. So we may assume $r > 0$. Then

$$\{x : f^2 > r\} = \{x : f > \sqrt{r}\} \cup \{x : f < -\sqrt{r}\}.$$

Both sets are measurable, and this implies f^2 is measurable. Hence if f and g are measurable, then $(f + g)^2$ and $(f - g)^2$ are measurable functions. Now, note that

$$f \cdot g = \frac{1}{4}[(f + g)^2 - (f - g)^2].$$

To prove (3), it suffices to show that $\frac{1}{g}$ is measurable since $\frac{f}{g} = f \cdot \frac{1}{g}$. Let $r > 0$. Then

$$\left\{x : \frac{1}{g} > r\right\} = \left\{x : g < \frac{1}{r}\right\} \setminus \{x : g \leq 0\},$$

which is measurable. If $r \leq 0$ then

$$\left\{x : \frac{1}{g} > r\right\} = \left\{x : g < \frac{1}{r}\right\} \cup \{x : g > 0\},$$

which is measurable. Hence $\frac{1}{g}$ is measurable. This proves (3). Note that

$$\{x : \max\{f, g\} < r\} = \{x : f < r\} \cap \{x : g < r\}$$

and

$$\{x : \min\{f, g\} < r\} = \{x : f < r\} \cup \{x : g < r\}.$$

This proves (4). □

2.2.5 Almost Everywhere Property

The word "almost" in our daily life simply means something which holds about all but not exactly all, and that's why we don't have an exact definition for it. Measure theory provides us with a definition to characterize the word "almost", and when and where would it occurs.

Definition 2.2.6 (*Almost Everywhere Property*) If a property or a process holds everywhere in $\mathbb{R}$ except at a set of measure zero, we say that this property (or operation) holds almost everywhere. This is abbreviated as $a.e.$

According to the definition, if we say $f = g$ a.e. on $[a, b]$, then we mean that the set $\{x \in [a, b] : f(x) \neq g(x)\}$ has measure zero. We'll see in the next chapter that Lebesgue's theory of integration, which was the primary motivation for Lebesgue's Measure Theory, does not consider these sets of measure zero. Hence the a.e. property is vital in this regard as it removes difficulties that may arise from these sets by ignoring them. One of the crucial consequences of this property is the following result.

Proposition 2.2.7 *Let f be a measurable function. If $f = g$ a.e., then g is measurable.*

Proof Let D be the domain of g, and the set $E = \{x : f(x) \neq g(x)\}$. Then $\mu(E) = 0$, hence E is measurable. Let $r \in \mathbb{R}$. Then

$$\begin{aligned}\{x \in D : g < r\} &= \{x \in D \setminus E : g > r\} \cup \{x \in E : g > r\} \\ &= [\{x \in D : g > r\} \cap (\{x \in E : g > r\})^c] \cup \{x \in E : g > r\} \\ &= [\{x \in D : f(x) < r\} \cap \{x \in E : g \geq r\}] \cup \{x \in E : g > r\}.\end{aligned}$$

One can easily see that the first set is measurable because f is measurable and the last two sets are subsets of E, hence measurable. The proof is complete. □

2.3 Sequence of Measurable Functions

2.3.1 Supremum and Infimum

In order to establish convergence results more effectively using the extended system of real numbers, we generalize the notion of limit as follows: Consider

$$\sup_k \{f_k(x) : k \geq n\} = (\sup_k f_k)(x)$$

$$\inf\{f_k : k \geq n\} = (\inf_k f_k)(x),$$

where the supremum is known to be the smallest possible upper bound for the sequence, whereas the infimum is the largest possible lower bound for the sequence. We observe that the former sequence decreases and the latter sequence increases. By taking the limit of both sequences, we define the limit superior of the sequence $\{f_n\}$ as

$$\inf_n \sup_{k\geq n} f_k = \limsup f_n, \tag{2.3.1}$$

and the limit inferior of f_n as

$$\sup_n \inf_{k\geq n} f_k = \liminf f_n. \tag{2.3.2}$$

These are denoted by $\overline{\lim} f_n$ and $\underline{\lim} f_n$, respectively. To explain this, let

$$\overline{\lim} f_n = \limsup f_n = f.$$

This means for $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have

$$|\sup\{f_n, f_{n+1}, \ldots\} - f| < \epsilon.$$

This implies for all $n \geq N$ we have

$$f_n < f + \epsilon. \tag{2.3.3}$$

Similarly, let

$$\underline{\lim} f_n = \liminf f_n = f.$$

Then for all $n \geq N$ we have

$$f - \epsilon < f_n. \tag{2.3.4}$$

The following properties can be derived:

Proposition 2.3.1 *Let $\{f_n\}$ and $\{g_n\}$ be sequences of extended real-valued functions. Then:*

1. $\underline{\lim} f_n \leq \overline{\lim} f_n$.
2. $\overline{\lim}(-f_n) = -\underline{\lim} f_n$.
3. $\overline{\lim}(f_n + g_n) \leq \overline{\lim} f_n + \overline{\lim} g_n$.
4. $\underline{\lim} f_n + \underline{\lim} g_n \leq \underline{\lim}(f_n + g_n)$.

Proof For (1), this follows from the fact that $\inf_{m\geq n} f_k \leq \sup_{m\geq n} f_k$.

For (2) we use (2.3.3) and (2.3.4). Indeed, if $\overline{\lim}(-f_n) = f$ then for $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have

$$-f_n < f + \epsilon,$$

or in other words

$$f_n > -f - \epsilon$$

for all $n \geq N$. Hence $\underline{\lim} f_n = -f$. This proves (2).

For (3), let $m \geq k$ then

$$x_m + y_m \leq \sup_{n \geq k} x_n + \sup_{n \geq k} y_n.$$

Taking supremum over m then letting $k \to \infty$ gives (3).

For (4), note that

$$\begin{aligned} \liminf f_n + \liminf g_n &= -(\limsup -f_n + \limsup -g_n) \\ &\leq \liminf (f_n + g_n). \end{aligned}$$

This proves (4). □

2.3.2 Limits of Sequences

The following proposition gives another characterization of convergence in terms of supremum and infimum concepts. Throughout, we always deal with sequences of extended real-valued functions.

Proposition 2.3.2 *A sequence $\{f_n\}$ converges to f p.w. if and only if*

$$\limsup f_n = \liminf f_n = f.$$

Proof ($\Rightarrow$) Let $f_n \longrightarrow f$. Let $\epsilon > 0$. Then there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have

$$f - \epsilon < f_n < f + \epsilon.$$

Hence, $f - \epsilon$ is a lower bound of f_n, which implies that

$$f - \epsilon \leq \inf_{k \geq n} f_k \leq f_n.$$

So

$$f - \epsilon < \inf_{k \geq n} f_n < f + \epsilon.$$

Since ϵ is arbitrary, this implies

$$\inf_{k \geq n} f_k = f.$$

Similar argument shows that

$$\sup_{k\geq n} f_k = f.$$

($\Leftarrow$) Suppose

$$\limsup f_n = \liminf f_n = f.$$

Let $\epsilon > 0$. There exists $N_1, N_2 \in \mathbb{N}$ such that for all $n \geq N_1$ we have

$$f - \epsilon < \inf_{k\geq n} f_k \leq f_n\,, \tag{2.3.5}$$

and for all $n \geq N_2$ we have

$$f_n \leq \sup_{k\geq n} f_k < f + \epsilon. \tag{2.3.6}$$

Combining (2.3.5) and (2.3.6), and letting $N = \max\{N_1, N_2\}$, then for all $n \geq N$ we have

$$f - \epsilon < f_n < f + \epsilon.$$

This proves the other direction. □

The advantage of using limsup and liminf is that they always exist for any bounded sequence even if the limit does not exist, so they can be helpful in studying approximations and asymptotic behavior of sequences and finding upper and lower bounds for the sequence.

Proposition 2.3.3 *Let $\{f_n\}$ be a sequence of measurable functions, $f_n : X \to \overline{\mathbb{R}}$. Then $\sup_n f_n$, $\inf_n f_n$, $\limsup_{n\to\infty} f_n$, and $\liminf_{n\to\infty} f_n$ are measurable.*

Proof Note that for $r \in \mathbb{R}$,

$$\{\sup_n f_n \leq r\} = \bigcap_{n=1}^{\infty}\{f_n \leq r\} \;\;\&\;\; \{\inf_n f_n \leq r\} = \bigcup_{n=1}^{\infty}\{f_n \leq r\}.$$

So $\sup_n f_n$ and $\inf_n f_n$ are measurable. But from (2.3.1) and (2.3.2) we also conclude $\limsup_{n\to\infty} f_n$, and $\liminf_{n\to\infty} f_n$ are measurable. □

2.3.3 *Modes of Convergence*

Generally speaking, when we say a sequence $\{f_n\}$ converges to f, we mean that the functions f_n are getting closer to f as n increases. But in what sense does this closeness occur? In elementary analysis, we studied two types of convergence: a convergence in which each point converges individually and a convergence where all points

converge simultaneously at once. The former is called: "*pointwise convergence*", and the latter is called: "*uniform convergence*". Here are the definitions:

Definition 2.3.4 (*Pointwise Convergence*) The sequence $\{f_n\}$ converges to f pointwise (denoted by $f_n \xrightarrow{p.w.} f$) if for every $\epsilon > 0$ and $x \in Dom(f)$ there exists $N \in \mathbb{R}$ such that $|f_n(x) - f(x)| < \epsilon$ for every $n > N$.

Here, the number N depends on the choice of x and ϵ. Eventually, $\lim_{n\to\infty} f_n(x) = f(x)$ for every x in the domain of f.

The stronger type of convergence is the uniform one.

Definition 2.3.5 (*Uniform Convergence*) The sequence $\{f_n\}$ converges to f uniformly (denoted by $f_n \xrightarrow{u} f$) if given any $\epsilon > 0$, there exists $N \in \mathbb{R}$ such that $|f_n(x) - f(x)| < \epsilon$ for every $n \geq N$ and for every $x \in Dom(f)$.

The number N in this definition depends only on ϵ, which justifies the name: "uniform", which means that all points converge together. Another way to formulate it is to say:

$$\sup_x |f_n(x) - f(x)| < \epsilon, \quad \forall n \geq N.$$

The idea of the convergence is to have the tail of the sequence inside some open set. In pointwise convergence, fixing ϵ will give different lengths of tail sequences for each x, but in uniform convergence, there will be the same length of the tail for all x. An immediate consequence from the definitions is that uniform convergence implies pointwise convergence, but the converse need not be true. A fundamental result in analysis states that if $f_n \xrightarrow{u} f$ on Ω and $\{f_n\}$ is a sequence of continuous functions on Ω then f is continuous on Ω, and it is well known that a sequence of continuous functions does not necessarily converge to a continuous function unless the convergence is uniform. Take for instance $f_n(x) = x^n$ on $[0, 1]$. Then f_n is continuous for all n . However, $f_n \longrightarrow f(x) = 0$ for $0 \leq x < 1$ and $f(x) = 1$ for $x = 1$ and f is not continuous. So continuity is not preserved under limiting operations. This is not the case for measurability property.

Theorem 2.3.6 *If $\{f_n\}$ is a sequence of measurable functions that converges to f either p.w., or a.e., then f is measurable.*

Proof The p.w. limit follows from Propositions 2.3.2 and 2.3.3. For the a.e. limit, since $\{f_n\}$ are measurable functions, lim sup f_n is measurable by the previous proposition. If $f_n \to f$ a.e. then lim sup $f_n = f$ a.e. Proposition 2.2.7 implies that f is measurable. □

This demonstrates that measurability is closed under the limiting operations, which justifies why the class of measurable functions is an indispensable tool in analysis.

2.4 Approximation Theorems

2.4.1 Nearly Versus A.E.

It is important to discuss the following approximation theorems: Simple Approximation Theorem, Egoroff Theorem, and Lusin Theorem. These theorems provide fundamental results that play a crucial role in the theory of integration. Other important results about approximations are commonly known as "*Littlewood Three Principles*", in honor of the British mathematician John E. Littlewood in 1944.[1] Researches were motivated by these principles, and the principles quickly started to be widely known as *Littlewood Principles*. Littlewood proposed his principles as follows:

1. Every measurable set is *nearly* a finite union of intervals.
2. Every measurable function is *nearly* continuous.
3. Every convergence of a sequence of functions is *nearly* uniform.

The word "*nearly*" here means that the indicated property holds except for a set of measure ϵ. This is the best we can hope when using this word, and this implies that it is broader than the "a.e. property", which gives an exception for sets of measure zero. In other words, an exception of the type (a.e.) gives a countable set, where an exception of the type (nearly) gives a small but uncountable set of measure ϵ. This difference between countability and uncountability plays an essential role in some results. Since this has been a common misunderstanding by many readers and students of measure theory, we need to emphasize here that the word "nearly" in these principles means: except a set of measure ϵ, not a countable set as many may mistakenly think.

The results represented in this section will help us build Lebesgue's theory of integration as a better replacement of Riemann's theory.

2.4.2 First Littlewood Principle

Theorem 2.4.1 (First Littlewood Principle) *Let E be a measurable set of finite measure. Then for every $\epsilon > 0$, there exist bounded open intervals $I_1, I_2, \ldots, I_n$, $U = \bigcup_{j=1}^{n} I_j$ such that*

$$\mu(E \triangle U) = \mu((E \setminus U) \cup (U \setminus E)) < \epsilon.$$

Proof Let $\mu(E) < \infty$. Then, there exists $\{I_n\}$ such that $E \subseteq \bigcup_{j=1}^{\infty} I_j$ and

$$\sum \mu(I_j) \leq \mu(E) + \frac{\epsilon}{2}.$$

[1] J. E. Littlewood (1885–1977) stated these approximation results as principles in his classic book: "Lectures on the Theory of Functions".

So there exists $m \in \mathbb{N}$ such that

$$\sum_{j=m}^{\infty} \mu(I_j) \leq \frac{\epsilon}{2}.$$

Let $U = \cup_{j=1}^{m} I_j$. Then

$$\mu((E \setminus U) \leq \mu\left(\left(\bigcup_{j=1}^{\infty} I_j\right) \setminus E\right) = \mu\left(\bigcup_{j=m+1}^{\infty} I_j\right) \leq \sum_{j=m}^{\infty} \mu(I_j) \leq \frac{\epsilon}{2}. \tag{2.4.1}$$

Moreover,

$$\mu(U \setminus E) \leq \mu\left(\left(\bigcup_{j=1}^{\infty} I_j\right) \setminus E\right) \leq \sum \mu(I_j) - \mu(E) \leq \frac{\epsilon}{2}. \tag{2.4.2}$$

Combining (2.4.1) and (2.4.2), we obtain the result. □

2.4.3 Approximation of Simple Functions

An obvious consequence of the previous principle is the following:

Proposition 2.4.2 *Let f be a simple function defined on a set E of finite measure. Then for every $\epsilon > 0$, there exists a step function $s(x)$ such that*

$$\mu(\{x : f(x) \neq s(x)\}) < \epsilon.$$

Proof Let f be a simple function defined on a measurable set E. Then by the previous result, there exist intervals $\{I_1, I_2, \ldots, I_n\}$ such that $K = \bigcup_{j=1}^{n} I_j$ and

$$\mu(E \triangle U) = \mu((E \setminus U) \cup (U \setminus E)) < \epsilon. \tag{2.4.3}$$

Define the function

$$s(x) = \sum_{i=1}^{n} c_i \chi_{I_i}(x)$$

and

$$f(x) = \sum_{i=1}^{n} c_i \chi_{E_i}(x)$$

where $\{E_1, E_2, \ldots, E_n\}$ is a partition of a measurable set E and f is defined on E. Notice the value of the two functions is the same on the sets $\{E_i\}$ and $\{I_i\}$, hence the only way $f(x) \neq g(x)$ is that $x \in E \setminus K$ or $x \in K \setminus E$. Hence

$$\{x : f(x) \neq s(x)\} = (U \setminus E) \cup (E \setminus U).$$

Since the sets are measurable, we apply (2.4.3) to obtain the result. □

The previous result says that any simple function defined on a set of finite measure can be approximated by a step function. The following result discusses approximation of simple functions by continuous functions.

Theorem 2.4.3 *Let f be a simple function on a measurable set E. Then for each $\epsilon > 0$, there exists a continuous function g and a closed set F such that $\mu(E \setminus F) < \epsilon$ and $f = g$ on F.*

Proof We write f as

$$f(x) = \sum_{k=1}^{n} c_k \chi_{E_k},$$

where $E_1, E_2, \ldots E_n$ is a collection of disjoint measurable sets, and

$$\bigcup_{k=1}^{n} E_k = E$$

for some $n \in \mathbb{N}$. By Proposition 1.4.3(2), for every $\epsilon > 0$ there exist closed sets $F_k \subseteq E_k$ such that

$$\mu(E_k \setminus F_k) < \frac{\epsilon}{n}.$$

Then

$$F = \bigcup_{k=1}^{n} F_k$$

is a closed set, and this gives

$$\mu(E \setminus F) = \sum_{k=1}^{n} \mu(E_k \setminus F_k) < \epsilon.$$

Note that $f\,|_F$ (the restriction of f to F) is continuous, so we define the following function:

$$g(x) = \sum_{k=1}^{n} c_k \chi_{F_k}.$$

Then g is continuous on F and we clearly have $g = f$ on F. To make g continuous on $E \setminus F$, we write

$$E \setminus F = \bigcup_{k=1} (a_k, b_k),$$

where we assumed that $E \setminus F$ is open. Now, extend g continuously to E by connecting lines on each (a_k, b_k) so that $g(a_k) = f(a_k)$ and $g(b_k) = f(b_k)$, for example, let

$$g(x) = \frac{f(a_k)}{b_k - a_k}(b_k - x) + \frac{f(b_k)}{b_k - a_k}(x - a_k).$$

Then, we see that the extended g is continuous on E and therefore our proof is complete. □

2.4.4 Simple Approximation Theorem

The following theorem is one of the main results of measure theory and plays a major role in obtaining approximation results.

Theorem 2.4.4 (Simple Approximation Theorem (SAT)) *Let f be a measurable function on a measurable set E. Then there exists a sequence of simple functions $\{\varphi_n\}$ such that $|\varphi_n| \leq |f|$ for all n and $\varphi_n \longrightarrow f$ p.w. on E. If f is bounded on E then $\varphi_n \longrightarrow f$ uniformly.*

Proof Let $f \geq 0$. Consider the following collections of intervals

$$E_{n_k} = \left\{x : \frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n}\right\}$$

with n fixed and $k = 1, 2, \ldots, n2^n$, and

$$F_n = \{x : f(x) \geq n\}.$$

Then it is clear that sets E_{n_k} are disjoint, and

$$\bigcup_{k=1}^{n2^n} E_{n_k} = [0, n].$$

Now, define the following sequence of simple functions:

$$\varphi_1(x) = \begin{cases} 0 & 0 \leq f(x) < 1 \\ 1 & f(x) \geq 1 \end{cases}.$$

$$\varphi_2(x) = \begin{cases} 0 & 0 \leq f(x) < 1 \\ \frac{1}{2} & \frac{1}{2} \leq f(x) < 1 \\ 1 & 1 \leq f(x) < \frac{3}{2} \\ \frac{3}{2} & \frac{3}{2} \leq f(x) < 2 \\ 2 & f(x) > 2 \end{cases}.$$

In general, we write

$$\varphi_n(x) = \sum_{k=1}^{n2^n} \left(\frac{k-1}{2^n}\right) \chi_{E_{n_k}} + n\chi_{F_n}.$$

It follows that the sequence can be written as

$$\varphi_n(x) = \begin{cases} \frac{k-1}{2^n} & : \frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n},\ k = 1, 2, \ldots, n2^n \\ n & : f(x) \geq n \end{cases}.$$

Then it is clear that $0 \leq \varphi_n \leq f$ for all n. Note that $\{\varphi_n\}$ is measurable for all n since the sets

$$\{x : f(x) \geq n\}$$

and

$$\left\{x : \frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n}\right\}$$

are measurable. Moreover, $\varphi_n \leq \varphi_{n+1}$.

If $f(x) = \infty$ then $\varphi_n = n$. Hence we always have $\varphi_n \to f$. So the result is proved for the case $f \geq 0$. For arbitrary f, write $f = f^+ - f^-$. Then by the preceding argument we can obtain two sequences of simple functions $\{\psi_n\}$ and $\{\phi_n\}$ such that $\psi_n \nearrow f^+$ and $\phi_n \nearrow f^-$, hence we choose the following sequence $\varphi_n = \psi_n - \phi_n$. Therefore $\varphi_n \to f$.

If f is bounded above, then $f(x) < N$ for some $N \in \mathbb{N}$, which implies that

$$|\varphi_n - f| < \frac{1}{2^n}$$

for all $n \geq N$. This implies that $\varphi_n \to f$ uniformly. □

2.4.5 Egoroff Theorem

Theorem 2.4.5 (Egoroff Theorem (Third Littlewood Principle)) *Let $\{f_n\}$ be a sequence of measurable functions and $f_n \to f$ a.e. on a set E of finite measure. Then for every $\epsilon > 0$, there exists a set A, $\mu(A) < \epsilon$ such that $f_n \longrightarrow f$ uniformly on $E \setminus A$.*

Proof Let

$$E_{n,k} = \bigcup_{m=n}^{\infty} \left\{x \in E : |f_m(x) - f(x)| \geq \frac{1}{k}\right\}.$$

We see that for a fixed k, we have $E_{n+1,k} \subset E_{n,k}$ and

$$\bigcap_{n=1}^{\infty} E_{n,k} = \emptyset.$$

If $x \in E_{n,k}$, then $\lim f_n(x) \neq f(x)$. Hence

$$\mu\left(\bigcap_{n=1}^{\infty} E_n\right) = 0 = \lim \mu(E_n).$$

So for $\epsilon > 0$ and k, there exists $n_0 \in \mathbb{N}$ such that for $n \geq n_0$ we have

$$\mu(E_{n,k}) < \frac{\epsilon}{2^{k+1}}.$$

If $x \in A_k = E_{n_{0,k}}$, then

$$|f_n(x) - f(x)| < \frac{1}{k}.$$

Let $A = \overset{\infty}{\cup} A_k$. Then

$$\mu(E) \leq \sum \frac{\epsilon}{2^{k+1}} < \epsilon.$$

Hence, there exists $n_0 \in \mathbb{N}$ and E such that if $x \in E \setminus A$ then

$$|f_n - f| < \frac{1}{n}$$

for every $n > n_0$. This implies $f_n \longrightarrow f$ uniformly on $E \setminus A$. □

2.4.6 *Lusin Theorem*

Theorem 2.4.6 (Lusin[2] Theorem—(Second Littlewood Principle)) *Let f be measurable on a set E of finite measure. Then for every $\epsilon > 0$, there exists a closed set $F \subset E$ with $\mu(E \setminus F) < \epsilon$ such that $f\,|_F$ is continuous on F. Moreover, there exists a function g continuous on E such that $g = f$ on F.*

Proof By the Simple Approximation Theorem, there exists a sequence $\{f_n\}$ of simple functions such that $f_n \longrightarrow f$ p.w. on E. By Theorem 2.4.3 for each $n \in \mathbb{N}$ there exists a continuous function g_n and a closed set F_n such that $f_n = g_n$ on F_n and

$$\mu(E \setminus F_n) < \frac{\epsilon}{2^{n+1}}.$$

By Egoroff Theorem there exists a compact set $K \subseteq E$ such that

$$\mu(E \setminus K) < \frac{\epsilon}{2}$$

[2] Lusin (or in fact Luzin) published his result in 1912, and he was Egoroff's PhD student. Both are Russian mathematicians.

and $\{f_n\}$ converges uniformly to f on K. Now, take

$$F = \left(\bigcap_{n=1}^{\infty} F_n\right) \cap K.$$

Then it is easy to show that F is closed and $\mu(E \setminus F)) < \epsilon$ (verify). Since the convergence of f_n to f on F is uniform, and since all g_n are continuous functions, the restriction of f to F is continuous, being the limit of a uniformly convergent sequence $f_n \mid_F$. To prove the second part, g can be constructed similar to that for Theorem 2.4.3. □

We have the following remarks about Lusin Theorem.

1. The Lusin Theorem does not assert that f is continuous on all points of F, but only the restriction of f to F is continuous on F. The reader should realize the difference between the two assertions. The function $f(x) = \chi_{\mathbb{Q}}(x)$ should be a good example to demonstrate the difference.
2. The result is still valid if E is an unbounded set (see Problem 21).

2.5 Differentiability

2.5.1 Dini Derivative

In preceding sections, we saw that measurable functions are not necessarily continuous, and we know that a function cannot be differentiable at a point if it is discontinuous there. On the other hand, the second Littlewood principle states that every measurable set is nearly continuous, so this should enable us to discuss the differentiability of measurable functions in some sense, even if they are not continuous. Since we are dealing with the extended real line, one important concept that is needed to adjust is the range of the function as it must include the values $\pm\infty$. This motivates the idea of defining a generalized form of derivatives in terms of the supremum and infimum. One advantage of this approach is that it allows us to define derivatives for almost all functions, whether they are differentiable or not. This idea was first studied by the Italian mathematician Ulisse Dini in 1892, where he introduced the following derivatives.

Definition 2.5.1 (*Dini Derivative*) Let f be a real-valued function defined on an interval $[a, b]$, and let $x_0 \in [a, b]$. Then

1. The upper right Dini derivative of f at x_0 is defined as

$$D^+ f(x_0) = \limsup_{x \to x_0^+} \frac{f(x) - f(x_0)}{x - x_0}.$$

2. The lower right Dini derivative of f at x_0 is defined as

$$D_+ f(x_0) = \liminf_{x \to x_0^+} \frac{f(x) - f(x_0)}{x - x_0}.$$

3. The upper left Dini derivative of f at x_0 is defined as

$$D^- f(x_0) = \limsup_{x \to x_0^-} \frac{f(x) - f(x_0)}{x - x_0}.$$

4. The lower left Dini derivative of f at x_0 is defined as

$$D_- f(x_0) = \liminf_{x \to x_0^-} \frac{f(x) - f(x_0)}{x - x_0}.$$

To illustrate the differentiation process, let $f(x) = |x|$. Then using the above formulas, it is easy to see that

$$D^+ f(0) = \limsup_{x \to 0^+} \frac{|x|}{x} = \inf_{h \geq 0} \sup_{0 < x < h} \frac{|x|}{x} = \inf_{h \geq 0} 1 = 1,$$

$$D_+ f(0) = \liminf_{x \to 0^+} \frac{|x|}{x} = \sup_{h \geq 0} \inf_{0 < x < h} \frac{|x|}{x} = \sup_{h \geq 0} 1 = 1,$$

$$D^- f(0) = \limsup_{x \to 0^-} \frac{|x|}{x} = \inf_{h \leq 0} \sup_{h < x < 0} \frac{|x|}{x} = \inf_{h \leq 0} (-1) = -1,$$

$$D_- f(0) = \liminf_{x \to 0^-} \frac{|x|}{x} = \sup_{h \leq 0} \inf_{h < x < 0} \frac{|x|}{x} = \sup_{h \leq 0} (-1) = -1.$$

Remark Notice the following observations from the derivative definitions.

1. If we define the sequence

$$f_n = n\left(f(x + \frac{1}{n}) - f(x)\right),$$

   then by Proposition 2.3.3 each of the four Dini derivatives is measurable.
2. Due to supremum and infimum notions, any of the four Dini derivatives may be infinite.

However, we define differentiability as follows.

Definition 2.5.2 (*Differentiability*) A real-valued function f is said to be $differentiable$ at x_0 if all four Dini derivatives at x_0 are finite and equal, i.e.,

$$D^+ f(x_0) = D_+ f(x_0) = D^- f(x_0) = D_- f(x_0) = f'(x_0) < \infty.$$

The definition implies that a function may not be differentiable at a point, even if the four Dini derivatives exist at that point. In general, we have

$$D_+ f \leq D^+ f \ \ \& \ D_- f \leq D^- f.$$

If

$$D^+ f(x_0) = D_+ f(x_0) = f'_+(x_0) < \infty$$

then we say that the function is differentiable at x_0 from the right. Similar case holds for the left derivative.

2.5.2 Vitali Covering

One particularly important class of measurable functions that we need to differentiate is monotone measurable functions. First, we need to define an essential type of cover that helps establish our result about the differentiability of monotone measurable functions.

Definition 2.5.3 (*Vitali Cover*) Let $E \subset \mathbb{R}$ be a measurable set. A cover V is said to be a $Vitali\ covering$ of E if V is a collection of intervals such that for every $x \in E$ and for every $\epsilon > 0$ there exists $I \in V$ where $x \in I$ and $\mu(I) < \epsilon$.

The following theorem is known as: $Vitali\ covering\ theorem$.

Theorem 2.5.4 (Vitali Covering Theorem) *Let $E \subset \mathbb{R}$ be a measurable set of finite measure, and $V = \{I_\alpha : \alpha \in \Lambda\}$ be a Vitali covering of E. Then for every $\epsilon > 0$ there exist finite disjoint intervals $I_1, \ldots, I_n$ of V such that*

$$\mu\left(E \setminus \bigcup_{j=1}^{n} I_j\right) < \epsilon.$$

Proof Since the endpoints of any interval have measure zero, we can assume all intervals in V to be closed with no loss of generality. Let O to be any open set containing E. Then we can also assume V covers O by removing all intervals not contained in O. So $\bigcup I_n \subseteq O$ and

$$\sum \ell(I_n) \leq \mu(O) < \infty. \tag{2.5.1}$$

Construct the following collection of intervals in V: Choose $I_1 \in V$ and define the numbers $M_1, M_2, \ldots$ by

$$M_1 = \sup_{I \in V} \{\ell(I) : \ I \cap I_1 = \emptyset\},$$

$$M_n = \sup\{\ell(I) : \ I \in V, \ \ I \cap \bigcup_{k=1}^{n} I_k = \emptyset\}, \ \text{for } n \geq 2.$$

For each n, choose I_{n+1} to be such that

$$\ell(I_{n+1}) > \frac{1}{2} M_n.$$

Then we have

$$\ell(I) \leq M_n < \ell(I_{n+1}). \tag{2.5.2}$$

If there exists a number $n \in \mathbb{N}$ such that $\{I_1, I_2, \ldots, I_n\}$ covers E then the proof is complete. Otherwise, this will inductively construct a sequence $\{I_k\}_{k=1}^{\infty}$ of disjoint intervals in V such that for each n

$$E \setminus \bigcup_{k=1}^{n} I_k \neq \emptyset. \tag{2.5.3}$$

Let $\epsilon > 0$. Then by (2.5.1) we can find an integer number $N \in \mathbb{N}$ such that

$$\sum_{k=N+1}^{\infty} \ell(I_n) < \frac{\epsilon}{5}. \tag{2.5.4}$$

This gives

$$\lim \ell(I_n) = 0. \tag{2.5.5}$$

From (2.5.3), let

$$x \in E \setminus \bigcup_{k=1}^{N} I_k. \tag{2.5.6}$$

Then there exists $I \in V$ such that $x \in I$ and

$$I \cap \bigcup_{k=1}^{N} I_k = \emptyset.$$

But from (2.5.2) and (2.5.5), I must intersect one of the intervals. Let m be the first integer number such that

$$I \cap I_m \neq \emptyset. \tag{2.5.7}$$

Then $m > N$ and

$$I \cap \bigcup_{k=1}^{m-1} I_k = \emptyset.$$

But this implies that

$$\ell(I_m) > \ell(I)/2. \tag{2.5.8}$$

It follows from (2.5.7) and (2.5.8) that the distance from x to the middle point of I_m cannot exceed

$$\ell(I) + \frac{1}{2}\ell(I_m) < \frac{5}{2}\ell(I_m).$$

Let $J_m = 5I_m$ be an interval having the same midpoint, then from (2.5.4) and (2.5.6) we obtain

$$\mu(E \setminus \bigcup_{k=1}^{n} I_k) \leq \sum_{k=N+1}^{\infty} \ell(J_k) < \epsilon.$$

□

2.5.3 *Derivative of Monotone Functions*

Our goal now is to show that all monotone measurable functions are almost everywhere differentiable. Before proving this result, the following lemma is useful.

Lemma 2.5.5 *If f is nondecreasing on $[a, b]$, then*

$$\mu\{x \in [a, b] : \ D^+ f(x) = \infty\} = 0.$$

Proof Define the set

$$E = \{x \in [a, b] : \ D^+ f(x) = \infty\}.$$

Then we can write $E = \bigcap_{r>0} E_r$, where

$$E_r = \{x \in [a, b] : \ D^+ f(x) \geq r\}.$$

By remark (2), it can be readily seen that the set E_r is measurable, which implies that E is measurable. Consider the collection of closed intervals $J_k = [a_k, b_k]$ contained in $[a, b]$ for which

$$f(b_k) - f(a_k) \geq r(b_k - a_k). \tag{2.5.9}$$

This implies that $\{J_k\}$ is a Vitali covering of E_r (why?), and $E_r \subseteq \bigcup J_k$. By Vitali Covering Theorem, for every $\epsilon > 0$, there exists a finite disjoint intervals $J_1, \ldots J_n$ of V such that

$$\mu\left(E_r \setminus \bigcup_{k=1}^{n} J_k\right) < \epsilon. \tag{2.5.10}$$

Since

$$E_r = \left(E_r \setminus \bigcup_{k=1}^{n} J_k\right) \cup \left(E_r \cap \left(\bigcup_{k=1}^{n} J_k\right)\right),$$

using (2.5.9) and (2.5.10) we obtain

$$\begin{aligned}\mu(E_r) &\leq \epsilon + \sum_{k=1}^{n} \mu((a_k, b_k)) \\ &= \epsilon + \sum_{k=1}^{n} (b_k - a_k). \\ &\leq \epsilon + \frac{1}{r} \sum_{k=1}^{n} f(b_k) - f(a_k).\end{aligned}$$

The fact that f is increasing and ϵ is arbitrary gives

$$\mu(E_r) \leq \frac{1}{r}(f(b - f(a)).$$

Taking $r \to \infty$, then $\mu(E_r) \to 0$. This establishes the lemma. □

The lemma simply says that the set of all points with infinite right upper Dini derivative has measure zero for any measurable increasing function. Now we come to the main result of the section, which Lebesgue establishes.

Theorem 2.5.6 (Lebesgue Differentiation Theorem) *If a function f is monotone on an open interval (a, b), then f' exists a.e.*

Proof Let f be monotone, say nondecreasing. Then from the definitions of Dini derivatives we have

$$D_- f(x) \leq D^- f(x) \leq D_+ f(x) \leq D^+ f(x).$$

Thus, it suffices to show that $D^+ f(x) \leq D_- f(x)$ for all except a set of measure zero. So consider the following set

$$A = \{x : D_- f(x) < D^+ f(x)\}.$$

Then we can find two rational numbers, say r and R such that

$$E = A_{r,R} = \{x : D_- f(x) < r < R < D^+ f(x)\}.$$

Being the intersection of two measurable sets, it is easy to show that E is measurable. Assume $\mu(E) = M > 0$. Let $\epsilon > 0$. Then we can find an open set O such that $\mu(O) < M + \epsilon$. For each $x \in E \subseteq O$, we can find $h > 0$ such that $x \in I = [x - h, x]$. This implies that

$$f(x) - f(x-h) < rh.$$

By Vitali Covering Theorem, we can find a finite collection of disjoint intervals $\{I_j = [x_j - h_j, x_j]\}_{j=1}^n$. Hence

$$\sum_{j=1}^{n} f(x_j) - f(x_j - h_j) < r \sum_{j=1}^{n} h_j < r\mu(O) < r(M+\epsilon). \tag{2.5.11}$$

For each $z \in E \cap (\bigcup_{i=1}^n J_i)$, we can find $k > 0$ such that the interval $(z, z+k) \subset I_j$ for some j. This implies that

$$f(z+k) - f(z) > kM.$$

By Vitali Covering Theorem we can find a finite collection of disjoint intervals $\{I_i = [z_i + k_i, z_i]\}_{i=1}^m$ such that

$$\mu\left(E \bigcap \bigcup_{i=1}^{m} J_i\right) > \mu\left(E \bigcap \bigcup_{i=1}^{n} J_i\right) - \epsilon > M - 2\epsilon.$$

This gives

$$\sum_{i=1}^{m} f(z_i + k_i) - f(z_i) > R \sum_{i=1}^{m} k_i > R(M - 2\epsilon). \tag{2.5.12}$$

Note that every I_i is contained in some I_j. From (2.5.11) and (2.5.12) and the fact that f is increasing, we get

$$R(M-2\epsilon) < \sum_{i=1}^{m} f(z_i + k_i) - f(z_i) \leq \sum_{j=1}^{n} f(x_j) - f(x_j - h_j) < r(M+\epsilon).$$

But ϵ is arbitrary and $M > 0$. This gives $R \leq r$, which is a contradiction. It follows that $\mu(E) = 0$, which implies

$$D^+ f(x) \leq D_- f(x) \ a.e. \tag{2.5.13}$$

Now, replacing $f(x)$ by $-f(-x)$ in (2.5.13) we obtain (verify)

$$D^- f(x) \leq D_+ f(x).$$

Therefore, we obtain

$$D^+ f(x) \leq D_- f(x) \leq D^- f(x) \leq D_+ f(x) \leq D^+ f(x).$$

This establishes the result. □

2.6 Functions of Bounded Variation

2.6.1 Total Variation

The previous section shows that monotone measurable functions are differentiable almost everywhere. The difference between two monotone functions is thus differentiable. It turns out that this property characterizes another important and large class of functions that contains many other classes. Let $f : I \to \mathbb{R}$ be a function defined on $I = [a, b]$. Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$ such that $x_0 = a$ and $x_n = b$. Then we have the following definition.

Definition 2.6.1 (*Variation*) The $variation$ of f with respect to the partition P on an interval $[a, b]$ is denoted by $v_a^b(f; P)$ and is given by

$$v_a^b(f; P) = \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| .$$

The $total\ variation$ of f on $[a, b]$ is given by

$$V_a^b(f) = \sup_P v_a^b(f; P),$$

by taking the supremum over all partitions of $[a, b]$.

2.6.2 Bounded Variation

It is clear from the definition that the variation $v(f; P)$ depends on the partition P, but the total variation $V(f)$ is a fixed number. If this number is finite in the extended real line, then we have the following definition.

Definition 2.6.2 (*Bounded Variation*) Let f be a real-valued function. Then, we say that f is of $bounded\ variation$ on $[a, b]$ if $V(f) < \infty$. We denote this by $f \in BV[a, b]$.

The definition gives rise to boundedness, because every function of bounded variation is indeed bounded as the following proposition asserts.

Proposition 2.6.3 *Let f be real-valued function on $[a, b]$. Then*

1. *If f is monotone on $[a, b]$ then $f \in BV[a, b]$ and*

$$V_a^b(f) = f(b) - f(a).$$

2. *If $f \in BV[a, b]$ then f is bounded on $[a, b]$.*

Proof For (1), let f be increasing. Let

$$P = \{x_0 = a, x_1, \ldots, x_n = b\}$$

be a partition for $[a, b]$. Then

$$\sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| = \sum_{i=1}^{n} f(x_i) - f(x_{i-1}) = f(b) - f(a).$$

By taking the supremum over all partitions P, we obtain (1).

For (2), let $x \in [a, b]$. Then

$$\begin{aligned} |f(x)| &\leq |f(x) - f(a)| + |f(a)| \\ &\leq V_a^b(f) + |f(a)|. \end{aligned}$$

Taking supremum over all x in $[a, b]$ gives (2). □

The reverse direction of the result in the previous proposition is not necessarily valid. The following example illustrates the case for (2).

Example 2.6.4 Let f be real-valued function defined on $[0, 1]$ as follows:

$$f(x) = \begin{cases} x \sin(\frac{\pi}{x}) & 0 < x \leq 1 \\ 0 & x = 0. \end{cases}$$

The function is clearly bounded since $|f| \leq 1$. To show that f is not of bounded variation, let

$$P = \left\{0, \ldots, \frac{2}{2n+1}, \ldots, \frac{2}{5}, \frac{2}{3}, 1\right\}$$

be a partition of $[0, 1]$. Then

$$\begin{aligned} \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| &= \left|\frac{2}{3}\right| + \left|-\frac{2}{3} - \frac{2}{5}\right| + \cdots \\ &= \frac{2}{3} + \frac{2}{3} + \frac{2}{5} + \frac{2}{5} + \frac{2}{7} + \frac{2}{7} + \cdots \\ &= 4 \sum_{i=1}^{n} \frac{1}{2i+2}. \end{aligned}$$

Letting $n \to \infty$ gives

$$v_p(f, [0, 1]) \to \infty.$$

Hence $f \notin BV[0, 1]$.

In view of the previous example, we see that the condition of bounded variation is stronger than boundedness. If the function is bounded but possesses rapid oscillations at some point, its total variation grows infinitely large; hence, functions of bounded variations are bounded on an interval and behave well on that interval. Other examples are given in Problem 31. It should be noted that this strong condition does not mean these functions are rare or difficult to obtain. The following properties show how large the class of $BV[I]$ is.

Proposition 2.6.5 *Let f and g be two real-valued functions such that $f, g \in BV[a, b]$. Then*

1. *For any $r, t \in \mathbb{R}$ we have $rf + tg \in BV[a, b]$ with*

$$V_a^b(rf + tg) \leq |r|\, V_a^b(f) + |t|\, V_a^b(g).$$

2. $f \cdot g \in BV[a, b]$.
3. *If $\dfrac{1}{g}$ is bounded in $[a, b]$ then $\dfrac{f}{g} \in BV[a, b]$.*
4. *For any $c \in (a, b)$,*

$$V_a^c(f) + V_c^b(f) = V_a^b(f).$$

5. *If $f_n \to f$ p.w. on $[a, b]$, then*

$$V_a^b(f) \leq \underline{\lim} V_a^b(f_n).$$

Proof Let $P = \{x_0 = a, x_1, \ldots, x_n = b\}$ be a partition for $[a, b]$. Then for $r \in \mathbb{R}$,

$$\begin{aligned}\sum_{i=1}^{n} |rf(x_i) - rf(x_{i-1})| &= |r| \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| \\ &\leq |r|\, V_a^b(f).\end{aligned}$$

Similarly for g. For the summation we have

$$\begin{aligned}\sum_{i=1}^{n} |f(x_i) + g(x_i) - f(x_{i-1}) - g(x_{i-1})| &\leq \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| \\ &\quad + \sum_{i=1}^{n} |g(x_i) - g(x_{i-1})| \\ &\leq V_a^b(f) + V_a^b(g).\end{aligned}$$

This proves (1).

For (2), we have

$$\begin{aligned}\sum_{i=1}^{n} |fg(x_i) - fg(x_{i-1})| &= \sum_{i=1}^{n} |fg(x_i) - f(x_i)g(x_{i-1}) + f(x_i)g(x_{i-1}) - fg(x_{i-1})| \\ &\leq \sum_{i=1}^{n} |f(x_i)|\,|g(x_i) - g(x_{i-1})| \\ &\quad + \sum_{i=1}^{n} |g(x_{i-1})|\,|f(x_i) - f(x_{i-1})|\end{aligned}$$

But from Proposition 2.6.3(2) f and g are bounded, so $|f| \leq M_1$ and $|g| \leq M_2$. Hence

$$\sum_{i=1}^{n} |fg(x_i) - fg(x_{i-1})| \leq M_1 V_a^b(f) + M_2 V_a^b(g).$$

This proves (2).

For (3), let $\left|\frac{1}{g}\right| \leq M$. Then we have

$$\begin{aligned}\sum_{i=1}^{n} \left|\frac{1}{g(x_i)} - \frac{1}{g(x_{i-1})}\right| &= \sum_{i=1}^{n} \left|\frac{g(x_{i-1}) - g(x_i)}{g(x_i)g(x_{i-1})}\right| \\ &\leq M^2 V_a^b(g).\end{aligned}$$

For (4), let $c \in (a, b)$, and let P_1 and P_2 be two partitions for $[a, c]$ and $[c, b]$, respectively. Then $P = P_1 \cup P_2$ is a partition for $[a, b]$.

$$v_a^c(f; P_1) + v_c^b(f; P_2) = v_a^b(f; P) \leq V_a^b(f).$$

Taking supremum over all partitions of $[a, c]$ and $[c, b]$, gives

$$V_a^c(f) + V_c^b(f) \leq V_a^b(f). \tag{2.6.1}$$

Conversely, let P be partition for $[a, b]$ such that $c \notin P$. Let P_1 and P_2 be two partitions for $[a, c]$ and $[c, b]$, respectively. Then $P_1 \cup P_2 = \tilde{P}$ containing c, so it is a refinement of P. Choose $x_i \in P_2$ and $x_{i-1} \in P_1$. By triangle inequality we have

$$|f(x_i) - f(x_{i-1})| \leq |f(x_i) - f(c)| + |f(c) - f(x_{i-1})|\,.$$

This gives

$$\begin{aligned}\sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| &\le \sum_{P_1} |f(x_i) - f(x_{i-1})| + \sum_{P_2} |f(x_i) - f(x_{i-1})| \\ &= v_a^c(f; P_1) + v_c^b(f; P_2) \\ &\le V_a^c(f) + V_c^b(f).\end{aligned}$$

Taking supremum over all partitions P for $[a, b]$ gives

$$V_a^b(f) \le V_a^c(f) + V_c^b(f). \tag{2.6.2}$$

Now the result follows from (2.6.1) and (2.6.2).

For (5), for every $\epsilon > 0$, there exists $\delta > 0$, $N \in \mathbb{N}$ such that

$$\sum_{i=1}^{n} |f_n(x) - f(x)| < \frac{\epsilon}{2}$$

for all $n \ge N$. Let $P = \{x_0 = a, x_1, \ldots, x_{k-1}, x_k = b\}$ be a partition for $[a, b]$. Then we have

$$\begin{aligned}\sum_{i=1}^{n} |f(x_k) - f(x_{k-1})| &= \sum_{i=1}^{n} |f(x_k) - f(x_{k-1}) \\ &\quad + f_n(x_k) - f_n(x_k) + f_n(x_{k-1}) - f_n(x_{k-1})| \\ &\le \sum_{i=1}^{n} |f(x_k) - f_n(x_k)| + \sum_{i=1}^{n} |f_n(x_{k-1}) - f(x_{k-1})| \\ &\quad + \sum_{i=1}^{n} |f_n(x_k) - f_n(x_{k-1})| \\ &\le \frac{\epsilon}{2} + \frac{\epsilon}{2} + |f_n(x_k) - f_n(x_{k-1})| \\ &\le \epsilon + V_a^b(f_n).\end{aligned}$$

Now, we take the supremum over P, and then taking the infimum limit, knowing that ϵ is arbitrary, we obtain (5). □

The next proposition is useful in our main result of the section.

Proposition 2.6.6 *If $f \in BV[a, b]$, then the total variation of f is increasing on $[a, b]$.*

Proof Let $g(x) = V_a^x(f)$, and let $x_1 < x_2$ in $[a, b]$. By Proposition 2.6.5(4)

$$V_a^{x_2}(f) = V_a^{x_1}(f) + V_{x_1}^{x_2}(f).$$

So

$$g(x_2) - g(x_1) = V_{x_1}^{x_2}(f) \geq 0.$$

This establishes the result. □

2.6.3 *Jordan Decomposition*

The following theorem was proposed by C. Jordan in 1881, and it states that every function of bounded variation can be expressed as the difference of two increasing functions. This representation is called: $Jordan\ decomposition$.

Theorem 2.6.7 (Jordan Decomposition) *Every real-valued function $f \in BV[A, B]$ can be written as $f = f_1 - f_2$ where f_1 and f_2 are two increasing functions on $[a, b]$.*

Proof Let $f_1(x) = V_a^x(f)$. Then from previous proposition f_1 is increasing on $[a, b]$. Let $f_2(x) = f_1(x) - f(x)$. To show f_2 is increasing, let $x < y$. Since $f_1(a) = 0$ and f_1 increasing, using Proposition 2.6.3(1) we have

$$f(y) - f(x) \leq V_x^y(f) = f_1(y) - f_1(x). \tag{2.6.3}$$

On the other hand, by Proposition 2.6.5(4)

$$V_a^y(f) = V_a^x(f) + V_x^y(f).$$

This implies

$$f_1(y) = f_1(x) + V_x^y(f). \tag{2.6.4}$$

From (2.6.3) and (2.6.4) we obtain

$$f_1(y) - f_1(x) \geq f(y) - f(x),$$

which can also be written as

$$f_2(y) = f_1(y) - f(y) \geq f_1(x) - f(x) = f_2(x).$$

Hence f_2 is increasing. □

The previous result together with Lebesgue differentiation theorem implies the following result.

Corollary 2.6.8 *If $f \in BV[a, b]$, then f is differentiable a.e.*

2.7 Absolute Continuity

We conclude the study of measurable functions by an important class of functions called absolutely continuous functions. These are, in fact, continuous functions and of bounded variations. The reader probably wonders why to study continuous functions if the purpose of the chapter is to investigate measurable functions that are not necessarily continuous. As we shall see in the next chapter, this class of functions plays a dominant role in integrating measurable functions, and it has deep connections with functions of bounded variations. Therefore, it is important to study these functions in this accordance since this will further develop the machinery used to establish Lebesgue's theory of integration.

2.7.1 Absolute Continuous Functions

Definition 2.7.1 (*Absolute Continuity*) Let f be a real-valued function defined on $[a, b]$. Then f is said to be *absolutely continuous* on $[a, b]$, denoted by $f \in AC[a, b]$, if given $\epsilon > 0$, there exists $\delta > 0$ such that if $\{[x_i, y_i] : \ i = 1, 2, \ldots, n\}$ is a finite collection of pairwise disjoint subintervals in $[a, b]$ and

$$\sum_{i=1}^{n} |y_i - x_i| < \delta$$

then

$$\sum_{i=1}^{n} |f(y_i) - f(x_i)| < \epsilon.$$

One can see easily from the definition that an absolutely continuous function is necessarily continuous. We also expect the following basic properties hold.

Proposition 2.7.2 *Let $f, g \in AC[a, b]$. Then the following functions are absolutely continuous on $[a, b]$:*

1. $f \in AC[a, x]$ *on every* $a \leq x \leq b$.
2. $|f|$.
3. $f + g$.
4. $f - g$.
5. $f \cdot g$.
6. $\frac{f}{g}$, *provided that* $|g| \geq c > 0$ *on* $[a, b]$ *for some* c.

Proof The proof is similar to that of Proposition 2.6.5. The proof of (1) is clear, and (2) follows from the inequality

$$||f(x)| - |f(y)|| \leq |f(x) - f(y)|.$$

To prove (3), if $f, g \in AC[a, b]$, then for $\epsilon > 0$ there exists $\delta > 0$ such that for every disjoint (x_i, y_i), where $\sum_{i=1}^{n}(y_i - x_i) < \delta$ we have

$$\sum_{i=1}^{n} |f(x_i) - f(y_i)| < \frac{\epsilon}{2},$$

$$\sum_{i=1}^{n} |g(x_i) - g(y_i)| < \frac{\epsilon}{2}.$$

Then we have

$$\begin{aligned}\sum_{i=1}^{n} |f(x_i) + g(x_i) - f(y_i) - g(y_i)| &\leq \sum_{i=1}^{n} |f(x_i) - f(y_i)| + \sum_{i=1}^{n} |g(x_i) - g(y_i)| \\ &\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.\end{aligned}$$

Hence $f + g \in AC[a, b]$. Moreover, $f - g = f = (-g)$, so $f - g \in AC[a, b]$. For (5), we have

$$\sum_{i=1}^{n} |fg(x_i) - fg(y_i)| \leq \sum_{i=1}^{n} |f(x_i)|\,|g(x_i) - g(y_i)| + \sum_{i=1}^{n} |g(y_i)|\,|f(x_i) - f(y_i)|.$$

But f and g are bounded (why?), so $|f| \leq M_1$ and $|g| \leq M_2$. Hence

$$\sum_{i=1}^{n} |fg(x_i) - fg(x_{i-1})| \leq M_1 \frac{\epsilon}{2M_1} + M_2 \frac{\epsilon}{2M_2} = \epsilon.$$

So $f \cdot g \in AC[a, b]$.

We leave (6) for the reader as an exercise (see Problem 38). □

The following proposition shows how this property is strong for functions.

Proposition 2.7.3 *Let $f \in AC[a, b]$. Then*

1. $f \in BV[a, b]$.
2. *f is uniformly continuous on $[a, b]$.*

Proof Proof of (1). Let $\epsilon = 1$ in the definition of absolute continuity. Then, there exists $\delta > 0$ such that if $\{[x_i, y_i] : i = 1, 2, \ldots, n\}$ is a finite collection of pairwise disjoint subintervals in $[a, b]$ and $\sum_{i=1}^{n}(y_i - x_i) < \delta$ then

$$\sum_{i=1}^{n} |f(y_i) - f(x_i)| < 1.$$

Let M be the least integer such that $M > \dfrac{b-a}{\delta}$. Construct the following partition of $[a, b]$

$$P = \{a = x_0, x_1, \ldots, x_M = b\},$$

where

$$x_i = a + i\left(\frac{b-a}{M}\right),$$

which gives the pairwise disjoint collection $\{[x_{i-1}, x_i],\ i = 1, \ldots M\}$. It is easy to see that each subinterval $[x_{i-1}, x_i]$ is of length less than δ. Then

$$v_{x_{i-1}}^{x_i}(f; P) = \sum_{i=1}^{M} |f(x_i) - f(x_{i-1})| < 1,$$

which implies

$$v_a^b(f; P) = \sum_{i=1}^{M} v_{x_{i-1}}^{x_i}(f; P) < M.$$

Taking the supremum and using Proposition 2.6.5(4) gives

$$V_a^b(f) < M.1 = M.$$

Proof of (2). Since $f \in AC[a, b]$, for $\epsilon > 0$, there exists $\delta > 0$ such that for any finite collection of disjoint subintervals $\{[x_i, y_i] : i = 1, 2, \ldots, n\}$ with $\sum_{i=1}^{n}(y_i - x_i) < \delta$ we have

$$\sum_{i=1}^{n} |f(y_i) - f(x_i)| < \epsilon.$$

Consider the single subinterval $[x, y]$ of $[a, b]$, with $|y - x| \leq \delta$. This gives

$$|f(y) - f(x)| < \epsilon. \qquad \square$$

2.7.2 *Functions of Bounded Derivatives*

The converse of the statements of the previous proposition is not necessarily true. For example, the step function $f(x) = \lfloor x \rfloor$ is an example of a bounded variation but not continuous function. Furthermore, the function

$$f(x) = \begin{cases} x \sin(\frac{\pi}{x}) & 0 < x \leq 1 \\ 0 & x = 0 \end{cases}$$

was shown in Example 2.6.4 to be not of bounded variation on [0, 1], so by Proposition 2.7.3(1) it is not absolute continuous on [0, 1]. However, it is easy to see that the function is continuous on [0, 1], so it is uniformly continuous on [0, 1]. One sufficient condition to guarantee that a uniformly continuous function is absolutely continuous is the boundedness of the derivative.

Proposition 2.7.4 *Let f be continuous on $[a, b]$. If $\left|f'\right| \leq M < \infty$ on $[a, b]$, then $f \in AC[a, b]$.*

Proof For every $\epsilon > 0$, let $\delta = \dfrac{\epsilon}{M}$. Consider the collection of pairwise disjoint intervals $\{[x_i, y_i]\}, i = 1, \ldots n\}$ such that $\sum_{i=1}^{n} |y_i - x_i| < \delta$. Then

$$\sum_{i=1}^{n} |f(y_i) - f(x_i)| = \sum_{i=1}^{n} \frac{|f(y_i) - f(x_i)|}{|y_i - x_i|} . |y_i - x_i| .$$

Using the hypothesis, we use the Mean Value Theorem to obtain

$$\sum_{i=1}^{n} |f(y_i) - f(x_i)| \leq \sum_{i=1}^{n} M |y_i - x_i| < \delta M = \epsilon. \qquad \square$$

2.7.3 *Derivative of Absolute Continuous Function*

Another fundamental property of absolutely continuous functions is the following.

Proposition 2.7.5 *Let $f \in AC[a, b]$. Then*

1. *$f'(x)$ exists a.e. on $[a, b]$.*
2. *If $f'(x) = 0$ a.e. on $[a, b]$, then f is constant on $[a, b]$.*

Proof The proof of (1) follows immediately by Proposition 2.7.3(1), then Corollary 2.6.8.
To prove (2), let $f'(x) = 0$ a.e. on $[a, b]$. Then there exists a set N of measure zero such that $f'(x) = 0$ on $[a, b] \setminus N$. Let $E = [a, c] \setminus N$. There exists $h > 0$ such that

$$[x, x + h] \subseteq [a, c].$$

Let $\epsilon > 0$. Since $f'(x) = 0$ on $[a, b]$ and $f \in AC[a, b]$, there exists $\delta > 0$ such that if $|h| < \delta$ then

$$|f(x + h) - f(x)| < h\epsilon. \tag{2.7.1}$$

So the cover

$$\{[x, y] \subseteq [a, c] : \ |f(y) - f(x)| < \epsilon(y - x)\}$$

is a Vitali cover for E. Hence by Vitali Covering Theorem, there exists a finite collection of disjoint intervals $[x_i, y_i]$, $i = 1, \ldots, n$ in $[a, c]$ such that

$$\mu\left(E \setminus \bigcup_{i=1}^{n} [x_i, y_i]\right) < \delta. \tag{2.7.2}$$

Now, let

$$a = x_1 < y_1 < x_2 < y_2 < \ldots < x_n < y_n < x_{n+1} = c.$$

Then by (2.7.2) we have

$$\sum_{k=1}^{n} |x_{i+1} - y_i| = \mu\left(E \setminus \bigcup_{i=1}^{n} [x_i, y_i]\right) < \delta.$$

By absolute continuity, this implies that

$$\sum_{k=1}^{n} |f(x_{i+1}) - f(y_i)| < \epsilon, \tag{2.7.3}$$

and from (2.7.1) we have

$$\sum_{k=1}^{n} |f(y_i) - f(x_i)| \leq \sum_{k=1}^{n} \epsilon(y_i - x_i) \leq \epsilon(c - a). \tag{2.7.4}$$

It follows from (2.7.3) and (2.7.4) that

$$\begin{aligned} |f(c) - f(a)| &= \left| \sum_{k=1}^{n} (f(x_{i+1}) - f(y_i)) + \sum_{k=1}^{n} (f(y_i) - f(x_i)) \right| \\ &\leq \sum_{k=1}^{n} |f(x_{i+1}) - f(y_i)| + \sum_{k=1}^{n} |f(y_i) - f(x_i)| \\ &< \epsilon + \epsilon(c - a). \end{aligned}$$

Since $\epsilon > 0$ is arbitrary, $f(c) = f(a)$, and thus, f is constant on $[a, b]$. □

2.8 Problems

1. Give an example of a sequence of functions f_n such that lim inf f_n and lim sup f_n both exist but lim f_n does not exist.
2. If $f_n \leq g_n$ for all n show that

$$\limsup f_n \leq \limsup g_n.$$

3. Show that

$$\underline{\lim}(f_n + g_n) \leq \overline{\lim}(f_n) + \underline{\lim}(g_n) \leq \overline{\lim}(f_n + g_n).$$

4. (a) Show that

$$\overline{\lim}(f_n g_n) \leq (\overline{\lim}(f_n)).(\overline{\lim}(g_n)).$$

(b) If $\overline{\lim}(f_n)$ and $\overline{\lim}(g_n)$ both exist, show that

$$\overline{\lim}(f_n g_n) = (\overline{\lim}(f_n)).(\overline{\lim}(g_n)).$$

5. Show that f is measurable if and only if $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$ are measurable.
6. Show that if f is piecewise continuous, then f is measurable.
7. Show that f is measurable if and only if $|f|$ is measurable.
8. (a) If f is measurable, show that f^n is measurable for every $n \in \mathbb{N}$.
(b) If f is measurable show that $|f|^p$ is measurable for $p > 0$.
9. Show that if f is measurable and $b \in \mathbb{R}$, then the set $f^{-1}(b)$ is measurable.
10. If f is an extended real-valued measurable function, show that the set $f^{-1}\{\infty\}$ is measurable.
11. If f and g are measurable, show that $\{x : f = g\}$ is measurable.
12. Let $A \subseteq \mathbb{R}$ be a dense set. If the set $\{x : f(x) \geq r\}$ is measurable for all $r \in \mathbb{R}$, show that f is measurable.
13. Give an example of a nonmeasurable function whose square is measurable.
14. Show that if $\{f_n\}$ is a sequence of measurable functions and

$$\liminf f_n = \limsup f_n = \infty$$

then $f_n \longrightarrow \infty$.
15. Let g be continuous, and f be measurable and finite. Show that $g \circ f$ is measurable.
16. Show that the following functions are measurable:
(a) $f(x) = e^{\sqrt[3]{x}}$.
(b) $f(x) = |\sin x|$.
(c) $f(x) = x^3 \chi_{\mathbb{Q}}(x)$.
17. Let $f. \mathbb{R} \longrightarrow \overline{\mathbb{R}} = [-\infty, \infty]$ be a measurable function. Show that there exists a sequence of simple functions $\{\varphi_n\}$ converging to f a.e. and $|\varphi_n(x)| \leq f(x)$ for all n and x.
18. Let $f. \mathbb{R} \longrightarrow \overline{\mathbb{R}} = [-\infty, \infty]$ be a measurable function. Show that there exists a sequence of step functions converging almost everywhere to f.
19. If f is measurable and differentiable a.e., then show that f' is measurable.
20. Show that Egoroff Theorem does not hold if $\mu(E) = \infty$.
21. Extend the Lusin Theorem for the unbounded case, i.e., if $\mu(E) = \infty$.

22. Prove the reverse direction of Lusin Theorem.
23. Find the Dini derivatives for

$$f(x) = \begin{cases} x \sin(\frac{1}{x}) & 0 < x \leq 1 \\ 0 & x = 0. \end{cases}$$

at $x = 0$.
24. Prove the following:
(a) $D_+ f(x) = -D^+(-f(x))$.
(b) $D^+(f+g) \leq D^+(f) + D^+(g)$.
25. Show that the characteristic function $f(x) = \chi_{\mathbb{Q}}(x)$ is not of bounded variation.
26. If f is continuous on $[a, b]$ and possesses a local minimum value $f(c)$ for $c \in (a, b)$, show that

$$D^- f(c) \leq 0 \leq D_+ f(c).$$

27. Show that $V_a^b(f) = 0$ if and only if $f(x) = c$ for some constant c.
28. Show that if $f \geq 0$ on $[a, b]$ and $f \in BV[a, b]$ then

$$f^\alpha \in BV[a, b]$$

for every $\alpha > 1$.
29. Show that if $f, g \in BV[a, b]$ then

$$f \vee g = \max\{f, g\} \in BV[a, b].$$

30. Determine whether the following functions are of bounded variation on $[0, 1]$.

(1) $f(x) = \begin{cases} \sin(\frac{\pi}{x}) & 0 < x \leq 1 \\ 0 & x = 0. \end{cases}$

(2) $f(x) = \begin{cases} x \cos(\frac{\pi}{x}) & 0 < x \leq 1 \\ 0 & x = 0. \end{cases}$

(3) $f(x) = \begin{cases} \sqrt{x} \sin(\frac{\pi}{x}) & 0 < x \leq 1 \\ 0 & x = 0. \end{cases}$

(4) $f(x) = \begin{cases} x^2 \sin(\frac{\pi}{x}) & 0 < x \leq 1 \\ 0 & x = 0. \end{cases}$

31. Show that if $f \in BV[a, b]$ then $|f| \in BV[a, b]$ but the converse is not true.
32. Let $f \in BV[a, b]$. f is continuous at $c \in (a, b)$ iff $g(x) = V_a^x(f)$ is continuous at c.
33. If $f \in BV[a, b]$. Show that f can be written as the difference between two strictly increasing functions.
34. Show that if f is Lipschitz, and $g \in BV[a, b]$ then $f \circ g \in BV[a, b]$.

35. Show that if f is monotone on $[a, b]$ then the set of points of discontinuities in $[a, b]$ is at most countable.
36. Give an example of $f, g \in AC[a, b]$ but $f \circ g \notin AC[a, b]$.
37. Let $f, g \in AC[a, b]$. Show that if g is monotone then $f \circ g \in AC[a, b]$.
38. Prove part (6) of Proposition 2.7.2.
39. Consider the following function

$$f(x) = \begin{cases} x \sin \dfrac{1}{x} & 0 < x \leq 1 \\ 0 & x = 0. \end{cases}$$

(a) Show that $f \in AC[\epsilon, 1]$ for every $\epsilon > 0$.
(b) Show that the AC property breaks when $\epsilon \to 0$.

40. A function is called $Lipschitz$ if there exists a constant number $K > 0$ such that

$$|f(y) - f(x)| \leq K |y - x| .$$

(a) Show that if $\left|f'\right|$ is bounded then f is Lipschitz.
(b) Show that if f Lipschitz and absolutely continuous then $\left|f'\right|$ is bounded.
(c) Show that if a function is Lipschitz on $[a, b]$ then it is absolutely continuous on $[a, b]$.
(d) Show that $f(x) = \sqrt{x}$ is absolutely continuous on $[0, 1]$ but not Lipschitz.

41. Consider the function

$$f(x) = \begin{cases} x^{\alpha} \sin \dfrac{1}{x^{\beta}} & 0 < x \leq 1 \\ 0 & x = 0. \end{cases}$$

(a) Show that if $\alpha \leq \beta$ the f is uniformly continuous on $[0, 1]$ but not absolutely continuous on $[0, 1]$.
(b) Show that if $\alpha > \beta$ then $f \in AC[0, 1]$.

42. Let $f \in AC[a, b]$. Let $N \subset [a, b]$ with $\mu(N) = 0$. Prove that $\mu(f(N)) = 0$.
43. **Riemann–Stieltjes integral.** Let f and g be bounded functions on $[a, b]$, and consider the integral $\int_a^b f dg$. This integral is defined as follows:
Let

$$P_n = a = t_0 < t_1 < \ldots < t_n = b$$

be any partition of $[a, b]$, and consider the sum

$$S(f, g; P) = \sum_{j=1}^{n} f(\tau_j)[g(t_j) - g(t_{j-1})],$$

for $t_{j-1} \le \tau_j \le t_j$. Then we say that f is *Riemann–Stieltjes integrable with respect to* g if there exists a number I such that for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$|S(f, g; P) - I| < \epsilon$$

whenever

$$\max_{1 \le j \le n} \{t_j - t_{j-1}\} < \delta.$$

The number I is the Riemann–Stieltjes integral of f with respect to g and is written as

$$I = \int_a^b f dg = \lim \sum_{j=1}^{n} f(\tau_j)[g(t_j) - g(t_{j-1})].$$

(a) Show that the Riemann–Stieltjes integral is linear with respect to f and to g.
(b) Show that if f is continuous on $[a, b]$ and $g \in BV[a, b]$ then the integral $\int_a^b f dg$ exists.

44. Let f be Riemann–Stieltjes integrable with respect to a monotone increasing g on $[a, b]$, and $m \le f(x) \le M$ on $[a, b]$.
(a) Show that there exists c between m and M such that

$$\int_a^b f dg = c[g(b) - g(a)].$$

(b) If $g \in BV[a, b]$ show that

$$F(x) = \int_a^x f dg,$$

for $a \le x \le b$ is of bounded variation on $[a, b]$ (i.e., $F \in BV[a, b]$).)
Show that if g is continuous on (a, b) then F is continuous on (a, b).

Chapter 3
Lebesgue Integration



3.1 The Riemann Integral

3.1.1 Introduction

Riemann introduced his theory of integration in 1953 during his work on the theory of Fourier series. The theory proposed a rigorous definition of the integral, and it allows integrating any piecewise continuous function on an interval. As a consequence, any function that can be integrated "in the sense of Riemann" can also be represented by a Fourier series. The Riemann integral was soon considered among the most important mathematical tool that was discovered in the nineteenth century, due to its contributions to the theory of functions and Fourier series, and the simplicity of its definition and its geometric interpretation allows the idea of approximating the integral numerically. Therefore, it was used immensely and efficiently in physics and other areas of science until today.

3.1.2 Riemann Integral of a Bounded Function

Recall that the Riemann integral of a bounded function on a compact interval is defined by means of step functions. Let

$$P = \{a = x_0, x_1, \ldots x_i, x_{n-1}, x_n = b\}$$

be a partition of $[a, b]$. Consider the following function.

$$g(x) = \sum_{i=1}^{n} c_i \chi_{E_i}(x),$$

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where $E_i = (x_i, x_{i-1})$. This is clearly a step function, bounded and defined on $[a, b]$, and its Riemann integral is

$$\int_a^b g(x)dx = \sum_{i=1}^{n} \int_a^b c_i \chi_{E_i}(x) = \sum_{i=1}^{n} c_i (x_i - x_{i-1}).$$

The integral of the step function here gives the total areas of the rectangles formed by the graph of the step function over $[a, b]$. This can be used to evaluate the integral of any bounded function on $[a, b]$ as follows: Let $f : [a, b] \to \mathbb{R}$ be bounded function. That is, there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in [a, b]$. Then we define the lower integral of f as

$$\underline{\int_a^b} f(x)dx = L(P; f) = \sup_s \left\{ \int_a^b s(x) : \ s(x) \text{ is a step function}, s(x) \leq f(x) \right\}.$$

Similarly, the upper integral of f is defined as

$$\overline{\int_a^b} f(x)dx = U(P; f) = \inf_t \left\{ \int_a^b t(x) : \ t(x) \text{ is a step function}, t(x) \geq f(x) \right\}.$$

It is clear that we always have $L(P; f) \leq U(P; f)$, but f is called Riemann integrable on $[a, b]$ if $L(P; f) = U(P; f)$, i.e.,

$$\int_a^b f(x)dx = \underline{\int_a^b} f(x)dx = \overline{\int_a^b} f(x)dx. \tag{3.1.1}$$

Equivalently, we say that

$$\int_a^b f(x)dx = \sup_{s(x)} \left\{ \int_a^b s(x) : \ s(x) \text{ is a step function}, s(x) \leq f(x) \right\}, \tag{3.1.2}$$

or,

$$\int_a^b f(x)dx = \inf_t \left\{ \int_a^b t(x) : \ t(x) \text{ is a step function}, t(x) \geq f(x) \right\}. \tag{3.1.3}$$

A simple criterion for integrability is to say that the function f is Riemann integrable on $[a, b]$ if for every $\epsilon > 0$ there exist two step functions, say $s(x)$ and $t(x)$, such

that $s(x) \leq f(x) \leq t(x)$ on $[a, b]$ and

$$\left| \int_a^b t(x)dx - \int_a^b s(x)dx \right| < \epsilon.$$

If equality cannot be achieved in (3.1.1) then we say that the function f is not Riemann integrable on $[a, b]$.

3.1.3 *Deficiencies of Riemann Integral*

The Riemann integral suffers three critical deficiencies: The first is that some functions cannot be integrated using Riemann integral. The most famous example is the Dirichlet function

$$f(x) = \begin{cases} 1 & x \text{ is rational} \\ 0 & x \text{ is irrational} \end{cases}. \tag{3.1.4}$$

Here, $L(P; f) = 0$ and $U(P; f) = 1$, so f is not Riemann integrable on $[a, b]$. Piecewise continuous functions can be handled using Riemann integral. Still, Dirichlet proposed his function as an example of a function that cannot be integrated, hence failed to have Fourier series representation. The problem in this function lies in the fact that it has uncountable number of discontinuities. In fact, we have the following theorem, proved by Lebesgue, which gives a characterization of Riemann integrable functions.

Theorem (Lebesgue's Criterion for Riemann integrability) *A bounded function defined on a bounded set is Riemann integrable if and only if it has at most countable number of discontinuities.*[1]

In the language of measure theory, we can say that a bounded function defined on a bounded set E is Riemann integrable if and only if the set of discontinuities over E is of measure zero. Equivalently, a bounded function defined on a bounded set E is Riemann integrable if and only if it is continuous almost everywhere on E.

The second type of deficiencies is that the interchanging between the limit and the integral signs in the Riemann integral is hard to achieve because it requires uniform convergence in bounded domains. We have the following theorem: If $\{f_n\}$ is a sequence of Riemann integrable on a bounded set Ω and $f_n \to f$ uniformly on Ω, then f is integrable on Ω and

$$\lim \int_\Omega f_n = \int_\Omega f.$$

[1] Consult a book on real analysis for a proof of the theorem.

If the convergence is pointwise, then the result is invalid in general, and f may not even be integrable. As an example, let us consider the following sequence:

$$f_n(x) = \begin{cases} 1 & x = q_1, q_2, \ldots, q_n \\ 0 & \text{otherwise,} \end{cases}$$

where $q_1, q_2, \ldots, q_n$ is the enumeration of $[0, 1]$. Then it is clear that

$$\int_0^1 f_n(x) = 0.$$

However, $f_n(x)$ converges to the Dirichlet function in (3.1.4), and this function is non-Riemann integrable on $[0, 1]$, so

$$\lim \int_0^1 f_n(x) \neq \int_0^1 \lim f_n(x).$$

Another example for an integrable function f is to let

$$f_n(x) = \begin{cases} n & (0, \frac{1}{n}) \\ 0 & \text{elsewhere} \end{cases}.$$

Then $f_n \to 0 = f$. Nevertheless,

$$\lim \int_{\mathbb{R}} f_n = 1 \neq 0 = \int_{\mathbb{R}} f.$$

The uniform convergence condition is strong and can rarely be satisfied in many applied problems because the function can usually be discontinuous at some points, and this breaks the uniform convergence. The source of the problem originated from the Fourier series and the problem of calculating Fourier coefficients of the Fourier Series of an arbitrary function f.

The third type of deficiencies is that the Riemann integral has limited abilities to deal with unbounded functions. For example, the function $f(x) = \dfrac{1}{\sqrt{x}}$ is unbounded on $(0, 1)$, and the improper Riemann integral

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \infty.$$

These deficiencies cause problem to the theory of Riemann integration and motivate mathematicians to formulate another theory that can improve the integration process and surpass these shortcomings.

3.2 Integral of Bounded Measurable Functions

3.2.1 The Idea of Lebesgue Integral

The two aforementioned shortcomings of the Riemann integral gave enough motivation for researchers to look for possible ways to improve the definition of the definite integral. This problem captured Henri Lebesgue's interests, who investigated the idea of generalizing (or, to say, improving) the definition of integral to overcome these shortcomings. Riemann partitioned the domain of the function into subintervals and built his step functions over them from which the lower and upper sums can be evaluated. Lebesgue noticed that this partitioning was the reason for the crisis. It could not handle functions with infinitely many discontinuities (like the Dirichlet function) defined on sets that are not intervals. Step functions work effectively in approximating piecewise continuous functions as they possess regular connected graphs, except, possibly, at finitely many points. The Dirichlet function is discontinuous at countably infinitely many points over a dense subset (the rationals), so it cannot be approximated by step functions defined over intervals (i.e., connected sets). The idea was to look for another function that could approximate this Dirichlet-type functions, and is defined in a better generic settings than intervals. An interval is a connected set with a specific "length", and Lebesgue generalized the notion to "measure" and introduced measurable sets. As a result, the range of the function was partitioned, and the values of the function were taken over measurable sets.[2]

3.2.2 Integral of Simple Functions

We introduced measurable sets to generalize the intervals. Let us elaborate the idea of the integral for which the range of the function is partitioned (instead of the domain as in Riemann integral). Let

$$f : [a, b] \longrightarrow [c, d],$$

and consider the following partition:

[2] Splitting the range of the function enables us to deal with functions on domains other than $\mathbb{R}$. This fits perfectly with probability theory which deals with measuring the chances and likelihood of an event's occurrence. The probability of that event is the measure of the corresponding set.

$$P = \{c \leq y_0 < y_1 < \cdots < y_n \leq d\}.$$

Consider the sets

$$f^{-1}([y_i, y_{i+1})) = \{x \in [a, b] : \ y_i \leq f(x) \leq y_{i+1}\} = E_i. \tag{3.2.1}$$

Then choose $\tilde{y}_i$ from each interval $[y_i, y_{i+1})$. According to that choice of P, we get the following simple function

$$\varphi(x) = \sum_{i=1}^{n} \tilde{y}_i \chi_{E_i}(x),$$

where the characteristic function χ is given by

$$\chi_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}.$$

Recall that the step functions form the building stone of Riemann's integral. Now that we have proposed a broader class of functions that form a finer building stone, we shall use these functions to establish the new integral, which should be more general than the Riemann integral; in the sense that some bad-shaped functions can be approximated (hence integrated) using simple functions but not step functions. This, theoretically, should make more sense because of the Simple Approximation Theorem and taking it into account that all continuous functions are measurable but not vice versa in general. It is worth noting that one of the requirements of a "generalized" version of a theory is to agree with the basic theory if the conditions are reduced to the original conditions of that theory. In particular, let $f(x) = \chi_I(x)$ and $g(x) = \chi_E(x)$, where I is an interval and E is a measurable set. We know that $\int f$ represents the area over I. If I approximates E, then $\int g$ must be close to $\int f$, and they should be equal once we have $E = I$. This suggests the following definition for the integral of simple functions.

Definition 3.2.1 (*Integral of Simple Function*) Let $\varphi(x) = \sum_{i=1}^{n} c_i \chi_{A_i}$ be a simple function where $\{A_i\}$ is a partition of a measurable set E of finite measure, (i.e. $\mu(E) < \infty$). Define

$$\int_E \varphi(x)dx = \sum_{i=1}^{n} c_i \mu(A_i).$$

Since the simple function can be written in different forms, we need to make sure the integral is independent of the function's representation, or the integral will not be well defined. So assume

$$E = \bigcup_{i=1}^{n} A_i = \bigcup_{j=1}^{m} B_j$$

and write

$$\varphi(x) = \sum_{i=1}^{n} a_i \chi_{A_i} = \sum_{j=1}^{m} b_j \chi_{B_j}.$$

Then by Definition 3.2.1

$$\int_E \varphi = \sum_{i=1}^{n} a_i \mu(A_i). \tag{3.2.2}$$

Since $A_i = \cup_{j=1}^{m} A_i \cap B_j$, (3.2.2) can be written as

$$\int_E \varphi = \sum_{i=1}^{n} a_i \mu\left(\bigcup_{j=1}^{m} A_i \cap B_j\right) = \sum_{i=1}^{n} a_i \sum_{j=1}^{m} \mu(A_i \cap B_j). \tag{3.2.3}$$

But if φ is well defined, then $a_i = b_j$ on $A_i \cap B_j$. Hence (3.2.3) can be written as

$$\begin{aligned}\int_E \varphi &= \sum_{j=1}^{m} b_j \sum_{i=1}^{n} \mu(A_i \cap B_j) \\ &= \sum_{j=1}^{m} b_j \mu\left(\bigcup_{i=1}^{n} A_i \cap B_j\right) \\ &= \sum_{j=1}^{m} b_j \mu(B_j).\end{aligned}$$

So

$$\int_E \sum_{i=1}^{n} a_i \chi_{A_i} = \int_E \sum_{j=1}^{m} b_j \chi_{B_j} = \sum_{i=1}^{n} a_i \mu(A_i) = \sum_{j=1}^{m} b_j \mu(B_j).$$

The integral is therefore well defined.

The following proposition shows that the integral defined in Definition 3.2.1 satisfies the basic properties of the classical definite integral.

Proposition 3.2.2 *Let $\{A_i\}_{i=1}^{n}$ and $\{B_j\}_{j=1}^{m}$ be two partitions of E, that is,*

$$E = \bigcup_{i=1}^{n} A_i = \bigcup_{j=1}^{m} B_j,$$

and let

$$\varphi(x) = \sum_{i=1}^{n} a_i \chi_{A_i}$$

and

$$\psi(x) = \sum_{j=1}^{m} b_j \chi_{B_j}$$

be two simple functions. Then

(1) Monotonicity: If $0 \le \varphi(x) \le \psi(x)$ *on* Ω *then*

$$\int_E \varphi \le \int_E \psi.$$

(2) Linearity:

$$\int_E c\varphi + \psi = c \int_E \varphi + \int_E \psi.$$

(3) Additivity:

$$\int_{\bigcup A_i} \varphi = \sum_{i=1}^{n} \int_{A_i} \varphi.$$

(4) Nullity: If $\mu(E) = 0$, *then*

$$\int_E \varphi = 0.$$

Proof For (1), we have $0 \le a_i \le b_j$ on $A_i \cap B_j = C_{ij}$. Note that $\bigcup_{j=1}^{m} C_{ij} = A_i$ and $\bigcup_{i=1}^{n} C_{ij} = B_j$. So

$$\begin{aligned}
\int_E \varphi &= \sum_{i=1}^{n} a_i \mu(A_i) \\
&= \sum_{i=1}^{n} a_i \sum_{j=1}^{m} \mu(C_{ij}) \\
&\le \sum_{j=1}^{m} b_j \sum_{i=1}^{n} \mu(C_{ij}) \\
&= \sum_{j=1}^{m} b_j \mu(B_j) = \int_E \psi.
\end{aligned}$$

For (2), the case $c \neq 1$ with $\int_E c\varphi = c \int_E \varphi$ is straightforward and is left to the reader. So let $c = 1$. Then

$$\begin{aligned}\int_E \varphi + \psi &= \sum_i \sum_j (a_i + b_j)\mu(C_{ij}) \\ &= \sum_{i=1}^{n} a_i \sum_{j=1}^{m} \mu(C_{ij}) + \sum_{j=1}^{m} b_j \sum_{i=1}^{n} \mu(C_{ij}) \\ &= \sum_{i=1}^{n} a_i \mu(A_i) + \sum_{j=1}^{m} b_j \mu(B_j) \\ &= \int_E \varphi + \int_E \psi\,.\end{aligned}$$

(3) follows immediately from the additivity of measure for disjoint subsets, and (4) is immediate from the definition. □

3.2.3 *Integral of Bounded Functions*

After defining the integration of simple functions, the next step is to define the integral for a bounded and measurable function. These are the same conditions under which continuous (or piecewise) functions are Riemann integrable. Our goal is to extend the integration of simple functions to integral of bounded functions over sets of finite measure and compare the Riemann integral and the newly established Lebesgue integral to see if we benefit from this. Recall the Simple Approximation Theorem states that if a measurable function f is bounded on a measurable set then there exists a sequence of simple functions that converges uniformly to f.

Let $|f| \leq M$ on a measurable set E of finite measure, for some $M > 0$. In sets E_i in (3.2.1), let

$$y_i = \frac{(i-1)M}{n}, \ \& \ y_{i+1} = \frac{iM}{n},$$

for $-n \leq i \leq n$ and $n = 1, 2, 3, \ldots$. Then the sets $\{E_i\}$ are disjoint and

$$\bigcup_{i=-n}^{n} E_i = E.$$

We obtain the following two sequences of simple functions:

$$\varphi_n(x) = \frac{M}{n} \sum_{-n}^{n} (i-1)\chi_{E_i}(x),$$

$$\psi_n(x) = \frac{M}{n} \sum_{-n}^{n} i\,\chi_{E_i}(x).$$

Then

$$\varphi_n(x) \le f(x) \le \psi_n(x). \tag{3.2.4}$$

Integrating (3.2.4) over E, then taking the infimum over all possible $\psi \ge f$ and the supremum over all possible $\varphi \le f$, gives

$$0 \le \inf_{f \le \psi} \int \psi - \sup_{\varphi \le f} \int \varphi \le \int_E \psi_n - \varphi_n \le \frac{M}{n}\mu(E), \tag{3.2.5}$$

where $\mu(E) < \infty$. Taking $n \to \infty$, we see that the two sup and inf integrals are equal. From (3.2.5) we conclude that the integral of f over E exists and equal to the sup-integral and the inf-integral. From the above argument, we come to the following definition:

Definition 3.2.3 (*Integral of Bounded Function*) Let f be a bounded measurable function that is defined on a measurable set E with finite measure (i.e. $\mu(E) < \infty$). Then, we define the integral of f over E as follows:

$$\int_E f = \sup_{\varphi \le f} \int_E \varphi, \quad \text{or equivalently} \quad \int_E f = \inf_{\psi \ge f} \int \psi.$$

The first thing we should do is to make sure the newly defined integral satisfies the basic properties of integrals, i.e., monotonicity, linearity, and additivity, in addition to the a.e. property.

Proposition 3.2.4 *Let f and g be two bounded measurable functions on a set E with $\mu(E) < \infty$. Then*

(1) Monotonicity: If $f \le g$ on E then

$$\int_E f \le \int_E g.$$

(2) Linearity:

$$\int_E c(f+g) = c\int_E f + \int_E g$$

for any $c \in \mathbb{R}$.

(3) Additivity: If E_1 and $\{E_1, E_2\}$ is a partition of E then

$$\int_E f = \int_{E_1} f + \int_{E_2} f.$$

(4) Almost Everywhere property: If $f = g$ a.e. on E then

$$\int_E f = \int_E g.$$

Proof (1) can be proved directly from Definition 3.2.3. Since $f \leq g$, the collection of all simple functions less than or equal to f is a subset of the collection of all simple functions less than or equal to g, hence the supremum of the latter set is greater than or equal to the supremum of the former set. This proves (1).

For (2), let $c = 1$, and let $\varphi \leq f$ and $\psi \leq g$ be simple functions. Then

$$\varphi + \psi \leq f + g.$$

By monotonicity and Proposition 3.2.2(2)

$$\int_E f + g \geq \int_E \varphi + \psi = \int_E \varphi + \int_E \psi. \tag{3.2.6}$$

Taking the supremum of (3.2.6) over all simple functions $\varphi \leq f$ and then taking the supremum over all simple functions $\psi \leq g$ gives

$$\int_E f + g \geq \int_E f + \int_E g.$$

To prove the other direction, let φ and ψ be two simple functions such that $f \leq \varphi$ and $g \leq \psi$. Then

$$\int_E f + g \leq \int_E \varphi + \psi = \int_E \varphi + \int_E \psi.$$

Taking the infimum over all φ and ψ gives the desired direction. Now let $c \neq 1$. The case $c = 0$ is trivial. If $c > 0$ then

$$\int_E c.f = \sup_{\varphi \leq cf} \int_E \varphi = c \sup_{\frac{\varphi}{c} \leq f} \int_E \frac{\varphi}{c} = c \int_E f.$$

If $c < 0$ then

$$\int_E c.f = \inf_{\varphi \geq cf} \int_E \varphi = c \sup_{\frac{\varphi}{c} \leq f} \int_E \frac{\varphi}{c} = c \int_E f.$$

This proves (2).

For (3) we have

$$\int_E f = \int_E f.\chi_{E_1 \cup E_2} = \int_E f.\chi_{E_1} + \int_E f.\chi_{E_2} = \int_{E_1} f + \int_{E_2} f.$$

For (4), let $A = \{x : f(x) \neq g(x)\}$. Then $\mu(A) = 0$, and by Proposition 3.2.2 (4) and taking the supremum over all $\varphi \leq f$ gives

$$\int_A f = 0.$$

Hence we have

$$\int_E f = \int_{E\setminus A} f + \int_A f = \int_E g + 0 = \int_E g.$$

□

Note that Property (4) shows that the Lebesgue Theory of Integration has ignored sets of measure zero as if they are negligible. In other words, they don't affect operations and properties, such as integration, convergence, differentiation, continuity, and monotonicity. Now it remains to compare it with the Riemann integral. The simple functions are defined over measurable sets, so they are used to approximate measurable functions in much the same way the step functions approximate continuous functions. Since simple functions are defined over sets with more general settings than intervals, this provides more flexibility and convenience in evaluating the integrals. So we have

$$\overline{\int_E} f \leq \sup_{\varphi\leq f} \int \varphi \leq \int_E f \leq \inf_{\psi\geq f} \int \psi \leq \overline{\int_E} f, \tag{3.2.7}$$

where φ, ψ are simple functions, and $\int_E^- f$ is the lower Riemann integral defined by

$$\overline{\int_E} f = \sup_{s\leq f} \int_E s(x)dx,$$

for all possible step functions s, and $\int_E^+ f$ is the upper Riemann integral given by

$$\overline{\int_E} f = \sup_{s\geq f} \int_E s(x)dx,$$

for all possible step functions s. It should be noted that (3.2.7) explains why Lebesgue integral can exist when the Riemann integral fails to exist and how the two integrals coincide if both exist.

Theorem 3.2.5 *Let f be bounded on a set E of finite measure. If f is Riemann integrable on E, then it is Lebesgue integrable, and the two integrals are equal.*

Proof If f is Riemann integrable, then the two upper and lower Riemann integrals agree, i.e.,

$$\overline{\int_E} f = \underline{\int_E} f.$$

But from (3.2.7) this means

$$\sup_{\varphi \le f} \int \varphi = \int_E f = \inf_{\psi \ge f} \int \psi.$$

□

Remark To distinguish between the two integrals, we can write R-integrable and L-integrable to mean Riemann integrable and Lebesgue integrable, respectively. Moreover, since the Riemann theory is based on partitions of intervals in the domain, we usually use the notation $\int_a^b f(x)dx$ to denote the Riemann integral, and since this partition is not necessarily valid in the Lebesgue case, we don't refer to the Lebesgue's integration process by dx but rather $d\mu$, so we denote it by $\int_{[a,b]} f\,d\mu$. We can simply write $\int f$ when it is clear from the context.

Theorem 3.2.5 shows that every R-integrable function is L-integrable. What about the converse? Can we find functions that are L-integrable but not R-integrable? Well, the answer to this question is yes, and it is a must. There would not be any purpose in defining a new integral if it did not give us more flexibility in the integration process. The key is to look at simple and step functions. If we think of a simple function that is not a step function, we can easily get the answer. The famous Dirichlet function gives the answer. Let

$$f(x) = \chi_{\mathbb{Q}}(x) = \begin{cases} 1 & x \in \mathbb{Q} \cap [0,1] \\ 0 & x \in [0,1] \setminus \mathbb{Q} \end{cases}.$$

Let $A_1 = \mathbb{Q} \cap [0,1]$ and $A_2 = [0,1] \setminus \mathbb{Q}$. Then $\{A_1, A_2\}$ is a partition of $[0,1]$ and so

$$\begin{aligned} \int_{[0,1]} f\,d\mu &= \int_{A_1} f\,d\mu + \int_{A_2} f\,d\mu \\ &= 1.\mu(A_1) + 0.\mu(A_2) \\ &= 0. \end{aligned}$$

It worked so easily! This shows that the Lebesgue integral is a generalization of the Riemann integral.

3.3 Integral of General Measurable Functions

3.3.1 The Indeterminate Value Problem

We will generalize Definition 3.2.3 to any function f defined on any interval. Our primary motivation is always the Simple Approximation Theorem (SAT). But we

may encounter a problem of indeterminate quantities of the form $\infty - \infty$ when dealing with some integrals of simple functions, such as

$$\varphi(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}.$$

This problem does not appear when constructing the Riemann integral because it is defined over a bounded set. To avoid this unfortunate result, we can write any measurable function f as $f(x) = f^+(x) - f^-(x)$, where

$$f^+ = \max(f, 0) = \frac{1}{2}(|f| + f)$$

$$f^- = \max(-f, 0) = \frac{1}{2}(|f| - f)$$

and f^+ and f^- are measurable. So we can construct the integral of both parts of the function, then use them to build the integral for the desired function. But there is another problem that may (in fact will) arise in this case: since ∞ is an admitted value for f in the extended real number, we may encounter integrals of functions like $\infty \cdot \chi_A(x)$ where $\mu(A) = 0$. On the other hand, since the integral may be taken over an unbounded set, we may encounter functions like $0 \cdot \chi_A(x)$ where $\mu(A) = \infty$. If we are trying to construct the integral of a possibly unbounded function on a bounded set, then encountering the indeterminate form $0 \cdot \infty$ is inevitable. To get around it we use the convention

$$0 \cdot \infty = \infty \cdot 0 = 0.$$

Under this convention, we agree to say the following: Let f be a measurable function defined on a measurable set E. Then:

(1) If $\mu(E) = \infty$ and $f = 0$ on E, then $\int_E f = 0$.
(2) If $\mu(E) = 0$ and $f = \infty$ on E, then $\int_E f = 0$.

3.3.2 *Integral of Nonnegative Functions*

Let us assume that f is nonnegative measurable function that is not necessarily bounded on a measurable set E. We adopt Definition 3.2.3 and write

$$\int_E f = \sup_{0 \leq \varphi \leq f} \int_E \varphi \tag{3.3.1}$$

for all possible simple measurable function φ. Since the supremum always exist in the extended real system, this integral may approach ∞. The integral still exists and equals ∞ in this case, but we will say that f is *integrable* over E only if

$$\int_E f < \infty.$$

Definition 3.3.1 (*Integral of Nonnegative Function*) Let $f : E \longrightarrow [0, \infty]$ be a nonnegative measurable function defined on a measurable set E. Then the integral of f over E is defined as

$$\int_E f = \sup \int_E \varphi,$$

where the supremum is taken over all possible simple functions φ with $0 \leq \varphi \leq f$. The function f is said to be *integrable* over E if $\int_E f < \infty$.

It turns out that the definition of a nonnegative measurable function is the same for bounded and unbounded functions, except the fact that simple functions are bounded in the former case and unbounded in the latter case. Note that we allowed the values of f or/and the measure of E to attain ∞ in the definition above; so f can be an unbounded function and defined over an unbounded set.

The definition above is natural but may not be practical in application. The following theorem, motivated by SAT, might be more practical and useful, and some authors use it as an alternative definition of the integral of a nondecreasing function. Since $f \geq 0$, there exists a sequence of nonnegative and increasing simple functions $\varphi_n \nearrow f$. But since

$$0 \leq \varphi_n \leq \varphi_{n+1},$$

the value of the integral $\int \varphi_n$ increases by monotonicity and gets closer to $\int f$, and the supremum of the integrals $\int \varphi_n$ is simply the limit.

Theorem 3.3.2 *Let $f \geq 0$ be a measurable function defined on a measurable set E. If $\{\varphi_n\}$ is an increasing sequence of nonnegative simple functions and $\varphi_n \nearrow f$, then*

$$\int_E f = \lim \int_E \varphi_n.$$

Proof Since $\varphi_n \leq f$ for all n, we have

$$\int_E \varphi_n \leq \sup_{0 \leq \varphi \leq f} \int_E \varphi = \int_E f.$$

Taking the limit, we obtain

$$\lim \int_E \varphi_n \leq \int_E f. \tag{3.3.2}$$

Conversely, let ϕ be a simple function, $0 \leq \phi \leq f$. Then

$$f = \lim \varphi_n \geq \phi,$$

and

$$\int_E \phi = \sum_{k=1}^{m} c_k \mu(E_k), \tag{3.3.3}$$

where $\{E_k\}_{k=1}^m$ is a partition of E (i.e., $E_i \cap E_j = \emptyset$ and $\bigcup E_k = E$). Let $\epsilon > 0$ and define the following sequence of sets:

$$A_{n,k} = \{x \in E_k : \varphi_n(x) \geq (1-\epsilon)c_k\}.$$

Then, it can be readily seen that $A_{n,k} \subseteq A_{n+1,k}$ and $\bigcup_n A_{n,k} = E_k$, which implies

$$\mu\left(\bigcup A_{n,k}\right) = \lim_n \mu(A_{n,k}) = \mu(E_k).$$

But

$$\int_{E_k} \varphi_n \geq \int_{A_{n,k}} \varphi_n \geq (1-\epsilon)c_k \mu(A_{n,k}).$$

Using additivity of simple functions

$$\int_E \varphi_n = \int_{\bigcup E_k} \varphi_n \geq \sum_{k=1}^{m} (1-\epsilon)c_k \mu(A_{n,k}).$$

Taking the limit gives

$$\lim \int_E \varphi_n \geq \sum_{k=1}^{m} (1-\epsilon)c_k \mu(E_k).$$

Since $\epsilon > 0$ is arbitrary, letting $\epsilon \to 0$, and in view of (3.3.3), we obtain

$$\lim \int_E \varphi_n \geq \int_E \phi. \tag{3.3.4}$$

Taking the supremum over all $\phi \leq f$, this gives the reverse direction of (3.3.2) and thus completes the proof. □

Because of the previous result, many authors prefer to define the integral of a nonnegative measurable function in the following way:

$$\int_E f = \lim \int_E \varphi_n\,, \quad 0 \leq \varphi_n \leq f, \quad \varphi_n \text{ is a simple function.} \tag{3.3.5}$$

This is because SAT guarantees the existence of such sequence $\{\varphi_n\}$, and the previous result, together with the extended system of real numbers, guarantee the existence of the limit in (3.3.5). The reader can use the techniques of the proof above to show that the definition is well defined. Note that the supremum in Definition 3.3.1 is taken over all possible simple functions $\varphi \leq f$, while the limit definition (3.3.5) is taken over the sequence $\{\varphi_n\}$ that is converging to f. They may appear to differ, but in reality, they are equivalent, and it is a good exercise for the reader to prove it (see Problem 7).

3.3.3 The General Lebesgue Integral

After defining the integral of a nonnegative function, we can now define the integral of a general function using the decomposition for any function $f(x) = f^+(x) - f^-(x)$. Both functions f^+ and f^- are nonnegative; hence, we can apply the definition obtained in the preceding step. If $\int f^+ < \infty$ and $\int f^- < \infty$, then we say that f is integrable in Lebesgue sense.

Definition 3.3.3 (*Lebesgue Integrable Function*) Let f be a measurable function defined on a measurable set E. If f^+ and f^- both integrable over E, then f is *integrable*.

Note here that if, at least, one of the functions f^+ and f^- is integrable, then the integral $\int_E f$ exists and may equal ∞ in case the other function is not integrable over E. If neither is integrable over E, then f is not integrable, and the integral $\int_E f$ is undefined in this case. In the Riemann theory, the function is integrable if and only if its integral exists. This is one of the advantages of the Lebesgue integral over the Riemann integral because it gives the Lebesgue integral more power and flexibility in dealing with integrands of infinite integrals. Moreover, integrals considered improper in the Riemann case can be performed in the Lebesgue theory and might be well defined.

Recall that the idea of Lebesgue integral was to improve the integration process's performance to overcome all deficiencies of the Riemann integral. If the Riemann integral exists, then the Lebesgue integral exists, and they are equal, but the converse

is not necessarily true as seen and observed in the famous Dirichlet's function. We, therefore, expect that the two integrals possess the same basic properties for integrals. They will not coincide if they don't share the same basic properties, i.e., linearity, monotonicity, and additivity.

The following properties hold also in the most general form of Lebesgue integral. The importance of such result stems from the fact that it confirms that Lebesgue theory naturally extends the Riemann theory with no possible contradictions or conflicts in their values.

Proposition 3.3.4 *Let f and g be integrable over E. Then the following properties hold:*

(1) Monotonicity: If $f \leq g$ a.e. then

$$\int_E f \leq \int_E g.$$

(2) Linearity:

$$\int_E af + g = a \int_E f + \int_E g$$

for $a, b \in \mathbb{R}$.

(3) Additivity: If $E = A \cup B$ where $A \cap B = \emptyset$, then

$$\int_E f = \int_A f + \int_B f.$$

(4) Almost Everywhere Property: If $f = g$ a.e. on E and f is integrable on E, then g is integrable on E and

$$\int_E f = \int_E g.$$

(5) If $f \geq 0$ and $\int_E f = 0$, then $f(x) = 0$ a.e. on E.

Proof Proving the result when f and g are nonnegative functions is similar to the preceding results using Theorem 3.3.2. Proving the results for general functions f and g requires the decompositions $f = f^+ - f^-$ and $g = g^+ - g^-$ and applying the results above for each f^+, f^-, g^+, and g^- since all are nonnegative, the linearity property is valid for all of them. We leave (1) and (3) for the reader as an exercise. To prove (2), we have

$$\begin{aligned}\int_E f+g &= \int_E f^+ - f^- + g^+ - g^- \\ &= \int_E f^+ + g^+ - (f^- + g^-) \\ &= \int_E f^+ - \int_E f^- + \int_E g^+ - \int_E g^- \\ &= \int_E f^+ - f^- + \int_E g^+ - g^- \\ &= \int_E f + \int_E g.\end{aligned}$$

The proof of (4) is immediate from (3) with $E = (E \setminus A) \cup A$ where $\mu(A) = 0$. To prove (5), let

$$E_n = \left\{x \in E : f(x) > \frac{1}{n}\right\}.$$

For every n, we have

$$0 = \int_E f \geq \int_{E_n} f \geq \frac{1}{n}(\mu(E_n)).$$

So $\mu(E_n) = 0$ for every n, which implies that

$$\mu\left(\bigcup_n E_n\right) = 0 = \mu(\{x \in E : f(x) > 0\}).$$

□

One of the most important consequences is the following result which provides an approximation of integrable functions in terms of continuous functions.

Proposition 3.3.5 *Let f be L-integrable on a set E. Given $\epsilon > 0$, there exists a continuous function g such that*

$$\int_E |f - g| < \epsilon.$$

Proof Let $f = f^+ - f^-$, where $f^+, f^- \geq 0$. By SAT, there exist sequences of simple functions $\phi_n \nearrow f^+$ and $\psi_n \nearrow f^-$. So there exists a number (why?) $N \in \mathbb{N}$ such that

$$\int_E |f^+ - \phi_N| < \frac{\epsilon}{4}, \quad \int_E |f^- - \psi_N| < \frac{\epsilon}{4}.$$

Define $\varphi_N = \phi_N - \psi_N$. Then,

$$\int_E |f - \varphi_N| < \frac{\epsilon}{2}. \tag{3.3.6}$$

Now, using Theorem 2.4.3 we can find a continuous function g such that

$$\int_E |\varphi_N - g| < \frac{\epsilon}{2}. \tag{3.3.7}$$

Now, the result follows from (3.3.6) and (3.3.7). The details are left to the reader as an exercise. □

3.4 Convergence Theorems

As stated in the preceding sections, one of the main advantages of Lebesgue integral over the Riemann integral is the efficiency of the convergence of integrals. This is because the Lebesgue integral is more flexible in dealing with integral-limit interchanging problems. The Riemann integral cannot do much without uniform convergence. On the other hand, Lebesgue integral forgoes the uniform convergence, as we shall see soon.

3.4.1 Uniform Convergence

Interchanging limits and integral signs need uniform convergence if we deal with the Riemann integral. The following theorem explains this:

Theorem 3.4.1 (Uniform Convergence Theorem for R-Integrals) *Let* $\{f_n\}$ *be a sequence of Riemann integrable functions on* $[a, b]$. *If* $f_n \longrightarrow f$ *uniformly on* $[a, b]$ *then* f *is Riemann integrable on* $[a, b]$ *and*

$$\lim \int_a^b f_n = \int_a^b \lim f_n = \int_a^b f.$$

The theorem asserts that passage of limit inside integral sign is allowed whenever the convergence is uniform. It should be noted that some authors used this property to define the Lebesgue integral for a bounded measurable function to be $\int f = \lim \int \varphi_n$ for a sequence of simple functions $\varphi_n \longrightarrow f$ uniformly. We do not guarantee the functions $\{\varphi_n\}$ are R-integrable in order to perform integration and apply the theorem above. Thus, we need an analog theorem for the Lebesgue integrals.

Theorem 3.4.2 (Uniform Convergence Theorem for L-Integrals) *Let $\{f_n\}$ be a sequence of bounded measurable functions defined on a set E of finite measure. If $f_n \longrightarrow f$ uniformly, then*

$$\lim \int_E f_n = \int_E f.$$

Proof The uniform convergence of measurable and bounded functions $\{f_n\}$ to f implies f is measurable and bounded. Let $\epsilon > 0$. Then there exists $N \in \mathbb{N}$ such that

$$|f_n - f| \leq \epsilon \text{ for } n \geq N.$$

Then

$$\left| \int_E f_n - \int_E f \right| = \left| \int_E f_n - f \right| \leq \int_E |f_n - f| \leq \mu(E) \cdot \epsilon.$$

The result follows since $\mu(E) < \infty$. □

Notice that we used the linearity property of the Lebesgue integral of bounded functions to prove the result, which explains why we did not use the result to define the integral of a bounded function. Now, what if the convergence is not uniform? The Riemann integral does not have much to do with passing the limit inside the integral. The Lebesgue integral, on the other hand, has other options if the convergence is pointwise. More precisely, if the convergence is pointwise, then it has the following alternative options: to be bounded, to be monotonic, and to be dominated by another integrable function. It gradually turns out that the Lebesgue integral does not require uniform convergence to handle the limit-integral interchanging. There is, however, a price to pay. The sequence of functions has to be either uniformly bounded, increasing, or dominated by another integrable function. Here are some of the fundamental convergence theorems we possibly need.

3.4.2 Bounded Convergence Theorem

Theorem 3.4.3 (Bounded Convergence Theorem (BCT)) *Let $\{f_n\}$ be a sequence of measurable functions on A, with $\mu(A) < \infty$ and $f_n \longrightarrow f$ p.w. on A. If there exists $M > 0$ such that $|f_n| \leq M$, then*

$$\lim \int_A f_n = \int_A f.$$

Proof Let $\epsilon > 0$. By Egoroff Theorem, there exists a measurable set $E \subset A$ such that $\mu(E) < \epsilon_1 < \epsilon$ and $f_n \longrightarrow f$ uniformly on $A \setminus E$, i.e., there exists $N \in \mathbb{N}$ such that

$$|f_n - f| < \epsilon_2 < \epsilon$$

for all $n \geq N$. Note that in A, hence E we have

$$|f_n - f| \leq 2M.$$

Then

$$\begin{aligned}\left|\int_A f_n - \int_A f\right| &\leq \int_A |f_n - f| \\ &= \int_{A\setminus E} |f_n - f| + \int_E |f_n - f| \\ &\leq \epsilon_2 \mu(A \setminus E) + \epsilon_1 2M.\end{aligned}$$

Now, let $\epsilon_1 = \dfrac{\epsilon}{4M}$ and $\epsilon_2 = \dfrac{\epsilon}{2\mu(A \setminus E)}$, and the result follows. □

It is worth noting that the BCT is not valid for Riemann integral. For example, let $f_n = \chi_{s_n}(x)$, where $s_n = \{r_i\}_{i=1}^n$ and r_i is the enumeration of all rationals in $[0, 1]$. Then

$$\int_0^1 f_n = 0,$$

but

$$f_n \longrightarrow f = \chi_{\mathbb{Q}}(x),$$

which is not Riemann integrable.

3.4.3 Monotone Convergence Theorem

The following theorem generalizes Theorem 3.3.2 to measurable functions instead of simple functions and with relaxed conditions. The conditions of being nonnegative and nondecreasing are still required, but the condition of uniform boundedness of the sequence is no longer required.

Theorem 3.4.4 (Monotone Convergence Theorem (MCT)) *Let $\{f_n\}$ be a sequence of measurable functions on E which is nonnegative and increasing. If* $\lim f_n = f$ *a.e., then*

$$\lim \int_E f_n = \int_E f.$$

Proof Note that f is the limit of measurable increasing functions, so $\int_E f_n$ and $\int_E f$ exist (why?). Since $f_n \leq f$, by monotonicity we have

$$\int_E f_n \leq \int_E f.$$

Taking the limit, we get

$$\lim \int_E f_n \leq \int_E f. \tag{3.4.1}$$

For the other direction, we use SAT to find an increasing sequence of simple functions $\{\varphi_n\}$ that converge to $\{f_n\}$ for every n. Hence for every n we can find a simple function $0 \leq \varphi_n \leq f_n(x)$ that can approximate f_n, such that

$$f_n(x) - \frac{1}{n} < \varphi_n(x) \leq f_n(x)$$

for all n. Note that $\varphi_n \nearrow f$, hence by Theorem 3.3.2, $\lim \int_E \varphi_n = \int_E f$. We also have, $\varphi_n \leq f_n(x)$, so by monotonicity of integrals and taking limit gives

$$\int_E f = \lim \int \varphi_n \leq \lim \int f_n. \tag{3.4.2}$$

The result follows by (3.4.1) and (3.4.2). □

The necessary condition in MCT is essential. To see this, let

$$f_n = \begin{cases} \frac{1}{n} & 0 < x \leq n \\ 0 & x > n \end{cases}.$$

Then it's clear that $f_n \longrightarrow 0$. However, $\int_0^\infty f_n = \frac{1}{n} \cdot n = 1$.

3.4.4 Fatou's Lemma

The following lemma is one of the fundamental results in analysis and has wide applications in real and functional analysis and probability theory.

Lemma 3.4.5 (Fatou's Lemma) *Let $\{f_n\}$ be a sequence of measurable functions, $f_n \geq 0$, and $f_n \longrightarrow f$ a.e. on a set E. Then*

$$\int_E f \leq \underline{\lim} \int_E f_n.$$

Proof Define the sequence

$$h_n = \inf_{k \geq n} f_k(x).$$

Then $h_n \leq f_n$, and by monotonicity

$$\int_E h_n \leq \int_E f_n,$$

for all n, so

$$\lim \int_E h_n \leq \underline{\lim} \int_E f_n. \tag{3.4.3}$$

But $\lim h_n = \underline{\lim} f_n$. Note that the sequence $\{h_n\}$ by its construction is increasing, so by MCT we obtain

$$\lim \int_E h_n = \int_E \underline{\lim} f_n. \tag{3.4.4}$$

The result follows from (3.4.3) and (3.4.4). □

We have the following remarks:

(1) The term $\underline{\lim} \int_E f_n$ may take the value ∞, and in this case no conclusion can be drawn about the integrability of f. If, however, $\underline{\lim} \int_E f_n < \infty$, then f is integrable on A.
(2) The inequality can be strict. For example,

$$f_n(x) = \chi_{[n,n+1]} \to 0.$$

However,

$$\int f_n = 1 \quad \text{for all } n.$$

The lemma does not assume anything besides the nonnegativity of the functions. The result provides an upper bound for the integral of a limit of this sequence. The simplicity of the condition and its importance make this lemma a fascinating and powerful tool.

3.4.5 Dominated Convergence Theorem

One of Fatou's lemma's advantages is the following great theorem, which is regarded as one of the most important results in analysis.

Theorem 3.4.6 (Dominated Convergence Theorem (DCT)) *Let $\{f_n\}$ be a sequence of measurable functions on E. If $f_n \to f$ p.w. a.e., and is dominated by some integrable function g over E, that is, $|f_n| \leq g$ for all n and for almost all x in E, then f is integrable and*

$$\lim \int_E f_n = \int f.$$

Proof Notice that $h_n = g - f_n \geq 0$, and $h_n \longrightarrow h = g - f$. Applying Fatou's lemma,

$$\int_E h \leq \underline{\lim} \int_E h_n. \tag{3.4.5}$$

Notice also that $|f| \leq g$ (why?). So f is integrable. Applying linearity gives

$$\int_E g - f = \int_E g - \int_E f \leq \underline{\lim} \int_E g - f_n = \int_E g - \overline{\lim} \int_E f_n.$$

Hence

$$\overline{\lim} \int_E f_n \leq \int_E f. \tag{3.4.6}$$

Now let $k_n = g + f_n$, and proceeding the same as before, we obtain

$$\int_E f \leq \underline{\lim} \int_E f_n. \tag{3.4.7}$$

The result follows as a conclusion from (3.4.6) and (3.4.7). □

The following results are some consequences that can be concluded from the MCT or DCT.

Corollary 3.4.7 *Let $\{f_n\}$ be a sequence of nonnegative measurable functions on a measurable set E and let $\sum_{n=1}^{\infty} f_n = f$. Then*

$$\sum_{n=1}^{\infty} \int_E f_n = \int_E f_n.$$

Proof Note that

$$g_m = \sum_{k=1}^{m} f_k$$

is an increasing sequence of nonnegative functions, and $g_n \nearrow f$. Apply MCT and linearity of the integral to obtain the result. □

Proposition 3.4.8 *Let $f \geq 0$ be integrable on E. For every set $A \subseteq E$, if $\mu(A) \to 0$ then $\int_A f \to 0$.*

Proof The result is clear if f is bounded. Suppose f is unbounded. Define

$$f_n(x) = \min\{f(x), n\}.$$

Then $\{f_n\}$ is increasing and bounded for every n and $f_n \longrightarrow f$. Now, use MCT. Details are left to the reader. □

Theorem 3.4.9 (Generalized DCT) *Let $\{f\}$ be a sequence of measurable functions and $f_n \longrightarrow f$ a.e. Moreover, let $\{g_n\}$ be a sequence of integrable functions and $g_n \longrightarrow g$ a.e. If $|f_n| \leq g_n$, and $\lim \int g_n = \int g$, then*

$$\lim \int f_n = \int f.$$

Proof Use DCT and Fatou's lemma. The details are left as an exercise (see Problem 39). □

One of the important results that can be concluded from convergence theorems is that Lebesgue integrals are absolutely continuous, as shown by the following theorem.

Theorem 3.4.10 (Absolute Continuity of the Integral) *Let f be integrable on a set A. For every $\epsilon > 0$, there exists $\delta > 0$ such that if $E \subseteq A$, and $m(E) < \delta$ then $\int_E |f| < \epsilon$.*

Proof Define the following sequence:

$$f_n(x) = \begin{cases} |f| & |f| \leq n \\ n & |f| > n \end{cases}.$$

Then $f_n \geq 0$ and $f_n \nearrow |f|$. Using MCT on a set E, we have

$$\int_E |f - f_n| \to 0.$$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that for all $n \geq N$, we obtain

$$\int_E |f - f_n| < \frac{\epsilon}{2}.$$

So

$$\int_E |f| < \int_E |f - f_n| + \int_E |f_n|$$

$$< \frac{\epsilon}{2} + Nm(A).$$

Choose

$$m(E) < \delta = \frac{\epsilon}{2N}.$$

This completes the proof. □

3.4.6 Historical Remark

In 1885, Arzela formulated a version of the Bounded Convergence Theorem for the Riemann integral. In 1971, W. Luxemburg established the Dominated Convergence Theorem for the Riemann integral. In 2010, B. Thomas established the Monotone Convergence Theorem for the Riemann integral. Unlike the case for Lebesgue's version, in all these results, the sequence of functions $\{f_n\}$ and the limit function f are required to be Riemann integrable in Riemann Theory, but this is not the case in Lebesgue Theory as seen from the statement of DCT. The integrability of the limit function f is a conclusion of the DCT in the Lebesgue integral rather than a condition, and this is one reason why the Lebesgue integral is more potent than the Riemann integral.

3.5 Lebesgue Integrability

3.5.1 Riemann Integrability

Recall that the integrand function is assumed to be bounded in Riemann integral, and the integral is taken over a bounded set, say $[a, b]$. Then the definition can be applied to obtain the upper and lower Riemann sums.

$$U(f; P) = \sum_{k=1}^{n} M_k(x_k - x_{k-1}), \quad L(f; P) = \sum_{k=1}^{n} m_k(x_k - x_{k-1}),$$

where M_k and m_k are the maxima and minima of f throughout the intervals (x_{k-1}, x_k) in a partition $P = \{x_0, x_1, \ldots, x_n\}$. Taking the infimum of $U(f; P)$ over all partitions P and the supremum of $L(f; P)$ over all partitions P gives

$$U(f) = \inf_P U(f; P) = \overline{\int_a^b} f, \tag{3.5.1}$$

$$L(f) = \sup_P L(f; P = \underline{\int_a^b} f. \tag{3.5.2}$$

Note that $U(f; P)$ is bounded from below by $m(b-a)$ where $m = \inf_{[a,b]} f$, and $L(f; P)$ is bounded from above by $M(b-a)$ where $M = \sup_{[a,b]} f$. Hence the infimum and supremum that are taken in (3.5.1) and (3.5.2) are finite, and therefore the integrals are well defined. So we have the following definition

Definition 3.5.1 (*Riemann Integrable*) Let f be a real-valued bounded function defined on $[a, b]$, and consider $U(f)$ and $L(f)$ defined in (3.5.1) and (3.5.2). Then f is *Riemann integrable* on $[a, b]$ if $U(f) = L(f)$. The integral of f on $[a, b]$ is denoted by $\int_a^b f$.

3.5.2 Improper Riemann Integral

Based on the definition above, it turns out that a function f may not be Riemann integrable on an interval I if one (or more) of the following situations occurs:

(1) The function is unbounded on I. This means that there exists $c \in I$ such that $f(x) \longrightarrow \pm\infty$ as $x \longrightarrow c^+$, or c^-. In this case, either $M = \infty$ or $m = -\infty$, so in any chosen partition P, the function attains infinity in one of the subintervals (x_{k-1}, x_k) resulting in an infinite sum. In this case, $U(f)$ or $L(f)$ does not exist.
(2) The interval I is unbounded, say $I = [a, \infty)$ or $I = (-\infty, b]$. In this case, any finite partition P of I will contain an unbounded subinterval, and any infinite partition of form $I_k = [k, k+1]$ results in an infinite summation instead of the finite Riemann sum. In this case, $U(f)$ or $L(f)$ does not exist.
(3) The function is bounded on a bounded interval of the form $I = [a, b]$, but the oscillating or the variation of the values of the function throughout this interval is infinitely large that in any possible subinterval I_k, we have

$$m_k = r_1 < r < M_k = r_2$$

for some $r \in \mathbb{R}$. Then for any partition P, we have $L(f; P) = r_1(b-a)$ and $U(f; P) = r_2(b-a)$. In his case, we have

$$L(f) = r_1(b-a) < r_2(b-a) = U(f).$$

Dirichlet's function is a typical example of the third situation. Letting

$$f(x) = \begin{cases} 1 & x \in [a,b] \cap \mathbb{Q} \\ 0 & x \in [a,b] \setminus \mathbb{Q} \end{cases}.$$

Then

$$L(f) = 0 < U(f) = 1.$$

This deficiency illustrated in the third situation above has no treatment. The first two situations can be treated using limits. Suppose f approaches ∞ as $x \to b^-$, and consider the interval $I_r = [a, r]$. Then f is integrable on I_r. Take $r \to b^-$. If the limit exists, say L, then the integral $\int_a^b f$ converges to L. The case when f approaches ∞ as $x \to a^+$ can be handled similarly. The value L was not obtained as the Riemann sum $U(f) = L(f)$, but as a limit of these sums. This is called *Improper Integral*, and a function f that can be integrated in this way (without using the definition) is said to be *Improperly Riemann Integrable*.

3.5.3 Lebesgue Integrability

As we saw in Sect. 3.2, the way the Lebesgue integral is constructed enables us to integrate an unbounded function over unbounded intervals. The Lebesgue integral, in this case, is defined as the supremum over all simple functions. This supremum may attain infinity, and since $\pm\infty$ have been included in the extended real number system, the integral becomes well defined even if it equals ∞. This implies the following:

(1) Unlike the Riemann integral case, there exists no improper integral in the Lebesgue Theory. The Lebesgue integral of an unbounded function or a bounded function over an unbounded interval can be evaluated and

$$\int_A f < \infty \quad \text{or} \int_A f = \infty.$$

In the former case, we say that f is integrable on A, and in the latter case, f is not integrable on A, though the integral is well defined and equals ∞. It turns out that the integral is not improper. This is considered a dramatic shift in the theory of integration which gives the Lebesgue's approach more power and freedom in dealing with integral complications.

(2) The only case the Lebesgue integral becomes undefined is when

$$\int f^+ = \int f^- = \infty,$$

since this will give us $\int f = \infty - \infty$. If either f^+ or f^- is integrable on a set A, the integral of f on A exists and well defined and may equal any value from $-\infty$ to ∞. However, f is said to be integrable on A if $\int_A f < \infty$.

(3) Since $|f| = f^+ + f^-$ and $f = f^+ - f^-$, we immediately conclude the following nice result:

Proposition 3.5.2 *Let f be measurable on a set E. Then f is integral on E if and only if $|f|$ is integrable on E.*

This property does not hold in the Riemann Theory. Take for example

$$f(x) = \begin{cases} -1 & x \in [a,b] \cap \mathbb{Q} \\ 1 & x \in [a,b] \setminus \mathbb{Q} \end{cases}.$$

As discussed above, the function is not integrable on $[-1, 1]$ since

$$L(f) = -1 < U(f) = 1.$$

But

$$\int_{[-1,1]} |f| = \int_{-1}^{1} 1 = 2.$$

The idea of integrability of $|f|$ but not f is unfortunate and seems odd in the Riemann Theory, but this is no longer possible in the Lebesgue Theory.

The following result demonstrates that the Lebesgue Theory is rich enough to deal with comparisons between functions and in a more effective way than the Riemann Theory. The previous result gives more flexibility to compare between two functions. In Riemann Theory, both functions are required to be nonnegative. However, this is no longer necessary and crucial in the Lebesgue Theory.

Proposition 3.5.3 (Comparison Test for L-Integrability) *If $|f| \leq g$ a.e. on E and g is integrable on E, then so is f.*

Proof This is clear since $f^+ \leq g$ and $f^- \leq g$, so by monotonicity on E, both f^+ and f^-, and hence f, are integrable on E. □

3.5.4 Improper Riemann Integral Versus Lebesgue Integral

It has been demonstrated in Theorem 3.2.5 that if a function f is R-integrable on a set A, then it is L-integrable and the two integrals are equal. The question that arises is: Can we extend the result to improper integrals? The following example shows that the answer is, unfortunately, no.

Example 3.5.4 Consider the sinc function[3]

$$\text{sinc}x = \frac{\sin x}{x}.$$

Its integral over $(0, \infty)$ is well known to be equal to $\frac{\pi}{2}$. There are many ways to verify that

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

One can use the Laplace transform to obtain

$$\mathcal{L}\left\{\frac{\sin x}{x}\right\} = \int_s^\infty \mathcal{L}\{\sin x\} = \int_s^\infty \frac{ds}{s^2+1}.$$

One can also use double integration techniques to obtain

$$\int_0^\infty \frac{\sin x}{x} dx = \int_0^\infty \int_0^\infty \sin x e^{-xt} dt dx.$$

We can also use the theory of complex integration using the function $f(x) = \frac{e^{iz}}{z}$. So the sinc function is improperly integrable on $(0, \infty)$ and the integral converges to $\frac{\pi}{2}$. However, the integral is not absolutely convergent. One way to see this is to write the integral as

$$\begin{aligned}\int_0^\infty \left|\frac{\sin x}{x}\right| dx &= \sum_{k=0}^\infty \int_{k\pi}^{(k+1)\pi} \left|\frac{\sin x}{x}\right| dx \\ &\geq \sum_{k=0}^\infty \int_{k\pi}^{(k+1)\pi} \frac{|\sin x|}{(k+\frac{1}{2})\pi} dx \\ &= \frac{2}{\pi} \sum_{k=0}^\infty \frac{1}{k+\frac{1}{2}}.\end{aligned}$$

Since the infinite series $\sum_{k=0}^\infty \frac{1}{k+\frac{1}{2}}$ diverges, so is the integral. Hence we have

[3] This is a well-known function in Fourier analysis and signal processing.

$$\int_0^\infty \left|\frac{\sin x}{x}\right| dx = \infty,$$

and we conclude that sinc$x$ is not L-integrable on $(0, \infty)$. This is a classic example of a function that is not L-integrable but is improperly R-integrable. Nevertheless, the following theorem provides sufficient conditions for an improperly R-integrable function to be L-integrable.

Proposition 3.5.5 *Let f be a measurable nonnegative function. Then the following holds:*

(1) If f is R-integrable on $[a, r]$ for every $r > a$ and is improperly R-integrable on (a, ∞) then f is L-integrable on (a, ∞).
(2) Let $f \to \infty$ as $x \to a^+$. If f is R-integrable on $(a + \epsilon, b)$ for each $\epsilon > 0$ and is improperly R-integrable on (a, b) then f is L-integrable on (a, b).

Proof For (1), we define the following sequence:

$$f_n(x) = f(x)\chi_{[a,n]}(x). \tag{3.5.3}$$

Then it is easy to see that the sequence is nonnegative, increasing, and converging to f. The result follows from the MCT.

For (2), define the following sequence:

$$f_n(x) = \begin{cases} f(x) & x \in [a + \frac{1}{n}, b] \\ 0 & x \in (a, a + \frac{1}{n}] \end{cases}. \tag{3.5.4}$$

Then f_n is nonnegative, increasing, and converging to f. The result again follows from the MCT. □

It turns out that the MCT is an efficient tool to determine the integrability of a function. The way the two sequences of functions are constructed in the proof of the previous proposition can be applied and used to evaluate the L-Integral of a function. The following example of *p-Integrals* illustrates this idea.

Example 3.5.6 In calculus courses, it was demonstrated that

$$\int_0^1 \frac{1}{x^p} dx = \begin{cases} \frac{1}{1-p} & p < 1 \\ \infty & p \geq 1 \end{cases},$$

and

$$\int_1^\infty \frac{1}{x^p} dx = \begin{cases} \frac{1}{p-1} & p > 1 \\ \infty & p \leq 1 \end{cases}.$$

We verify that the same results hold for L-integrals as well. For the first integral, the function in lights of Lebesgue Theory takes the form

$$f(x) = \begin{cases} \dfrac{1}{x^p} & 0 < x \leq 1 \\ \infty & x = 0 \end{cases}.$$

Define the following sequence:

$$f_n(x) = \begin{cases} x^{-p} & \frac{1}{n} \leq x \leq 1 \\ n^p & 0 \leq x < \frac{1}{n} \end{cases}.$$

The sequence is increasing and converging to f. Hence we use MCT to obtain

$$\int_0^1 f = \lim \int_0^1 f_n = \lim_n \left(\int_0^{1/n} n^p + \int_{1/n}^1 x^{-p} \right) = 0 + \frac{1}{1-p}.$$

The second integral can be treated similarly using the sequence in (3.5.4).

Proposition 3.5.5 and the previous example demonstrate the efficiency of MCT to deal with integrals that are treated improperly in Riemann Theory. The result is the same, with the exception that the MCT gives the "value" of the integral, rather than the "limit" of the integral as in Riemann Theory. The MCT can also be used to show that a function is not L-integrable. The following example illustrates this idea.

Example 3.5.7 Let $f(x) = e^x$ on $[0, \infty)$. Define

$$f_n(x) = \begin{cases} e^x & 0 \leq x \leq n \\ 0 & x > n \end{cases}.$$

Clearly $\{f_n\}$ is nonnegative, increasing, and converging to f, that is, $0 \leq f_n \nearrow f$. Using the MCT, we obtain

$$\begin{aligned} \int_0^\infty e^x d\mu &= \lim \int_0^\infty f_n d\mu \\ &= \lim \left(\int_0^n e^x + \int_n^\infty 0 \right) \\ &= \lim(e^n - 1) = \infty. \end{aligned}$$

Hence e^x is not L-integrable on $[0, \infty)$, although the integral itself exists and equals ∞.

Now, we can use convergence theorems to evaluate the limits of integrals.

Example 3.5.8 To evaluate the limit

$$\lim \int_0^n \left(1 + \frac{x}{n}\right)^n e^{-2x} d\mu,$$

and consider

$$f_n = (1 + \frac{x}{n})^n e^{-2x}.$$

Then we see that $0 \leq f_n \leq f_{n+1}$ and

$$f_{n+1} \nearrow f(x) = e^x e^{-2x} = e^{-x}.$$

Hence by MCT

$$\lim \int_0^n (1 + \frac{x}{n})^n e^{-2x} d\mu = \int_0^\infty e^{-x} d\mu. \tag{3.5.5}$$

But the integral in the RHS of (3.5.5) can also be computed with the aid of Proposition 3.5.5, and we find that e^{-x} is clearly L-integrable on $[0, \infty)$ and the integral is equal to 1. Hence we obtain

$$\lim \int_0^n \left(1 + \frac{x}{n}\right)^n e^{-2x} d\mu = 1.$$

The next example uses DCT instead of MCT.

Example 3.5.9 To evaluate

$$\lim \int_0^n \frac{\cos\left(\frac{x}{n}\right)}{nx^2 + 1} d\mu,$$

we set

$$f_n(x) = \frac{\cos(\frac{x}{n})}{nx^2 + 1}.$$

Note that $\{f_n\}$ are measurable functions, and $f_n \longrightarrow 0$. Moreover,

$$\left|\frac{\cos(\frac{x}{n})}{nx^2 + 1}\right| \leq \frac{1}{nx^2 + 1} < \frac{1}{x^2}.$$

In light of Example 3.5.6, the function $\dfrac{1}{x^2}$ is integrable, so we use the DCT to obtain

$$\lim \int_0^n \frac{\cos(\frac{x}{n})}{nx^2+1} d\mu = \int_0^\infty 0 = 0.$$

3.5.5 Riemann–Lebesgue Lemma

We conclude the section with the following famous lemma, which discusses a significantly important integral limit with broad applications in analysis and applied mathematics, known as "*Riemann–Lebesgue Lemma*".

Theorem 3.5.10 (Riemann–Lebesgue Lemma) *Let f be Lebesgue integrable on $I = [a, b]$. Then*

$$\lim_{n\to\infty} \int_I f(x) \sin nx dx = \lim_{n\to\infty} \int_I f(x) \cos nx dx = 0.$$

Proof Since f is integrable on $I = [a, b]$, given $\epsilon > 0$ we can find a partition

$$\{a = x_0, x_1, \ldots, x_n = b\}$$

and a step function s such that

$$s(x) = \sum_{j=0}^{n} c_j \chi_{[x_j, x_{j+1}]},$$

and

$$\int_I |f - s| < \epsilon.$$

Writing the integrand as

$$f(x) \cos nx = f(x) \cos nx - s(x) \cos nx + s(x) \cos nx,$$

and using triangle inequality, we get

$$\left| \int_a^b f(x) \cos nx dx \right| \leq \int_a^b |f(x) - s(x)|\, dx + \int_a^b |s(x) \cos nx|\, dx$$
$$\leq \epsilon + \sum_{j=0}^{n} |c_j| \int_{x_j}^{x_{j+1}} |\cos nx|\, dx.$$

But

$$\int_{x_j}^{x_{j+1}} |\cos nx|\, dx = \frac{1}{n} \left|\sin nx_{j+1} - \sin nx_j\right| \leq \frac{2}{n} \longrightarrow 0.$$

This completes the proof. □

3.6 Lebesgue Fundamental Theorem of Calculus

3.6.1 The Recovering Problem

Another important and significant aspect of Riemann Theory is the ability to recover a function from its derivative. This can be achieved by the famous Fundamental Theorem of Calculus (FTC), which states the following:

Theorem 3.6.1 (Fundamental Theorem of Calculus (FTC) for Riemann Integral) *Let f be continuous on $[a, b]$. Then*

(1) $\int_a^b \frac{d}{dx}(f(x))dx = f(b) - f(a)$.
(2) $\frac{d}{dx}\int_a^x f(x)dx = f(x)$.

The goal of the present section is to show that this result extends to Lebesgue Integral. We have seen that the Lebesgue Differentiation Theorem states that every monotone function on a set is differentiable a.e. on that set. Given that a monotone function need not be continuous, we do not expect to obtain Theorem 3.6.1 in general. Even if the function is continuous and differentiable, its derivative may not be integrable. Consider for example the function

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x^2} & x \neq 0 \\ 0 & x = 0 \end{cases}.$$

However, we can obtain an inequality that uses the quantity $f(b) - f(a)$ in FTC(1) as an upper bound.

Proposition 3.6.2 *Let f be increasing on $[a, b]$. Then,*

$$\int_a^b f'(x)dx \leq f(b) - f(a).$$

Proof Let f be increasing on $[a, b]$, and $f(x) = f(b)$ for all $x \geq b$. Define the sequence

$$f_n = n\left[f\left(x + \frac{1}{n}\right) - f(x)\right].$$

It can be easily shown that $f_n \geq 0$ and $f_n \longrightarrow f'$ a.e. Then, by Fatou's Lemma

$$\begin{aligned}\int_a^b f' &\leq \underline{\lim} \int_a^b f_n \\ &= \underline{\lim} \left[n \int_{a+\frac{1}{n}}^{b+\frac{1}{n}} f(x)dx - n \int_a^b f(x)dx \right] \\ &= \underline{\lim} \left[n \int_b^{b+\frac{1}{n}} f(x)dx - n \int_a^{a+\frac{1}{n}} f(x)dx \right].\end{aligned}$$

Using the fact that f increases on $[a, b]$, and $f(b + \frac{1}{n}) = f(b)$, we see that $f(x) = f(b)$ on $[b, b + \frac{1}{n}]$ and $f(a)$ is minimum on $[a, a + \frac{1}{n}]$. Substituting above gives

$$\int_a^b f' \leq \underline{\lim}(nf(b))\left[b + \frac{1}{n} - b\right] - (nf(a))\left[a + \frac{1}{n} - a\right] = f(b) - f(a). \qquad \square$$

3.6.2 *Integrability of The Derivative*

The previous proposition, together with Jordan Decomposition Theorem (Theorem 2.6.7), leads to the following important result:

Proposition 3.6.3 *If f is of bounded variation on $[a, b]$, then f' is integrable on $[a, b]$.*

Proposition 3.6.2, though interesting, does not give sufficient information about the recovering problem. The inequality sign needs to be replaced with equality, and the indefinite integral of f' must recover f. It becomes apparent that the bounded variation property is not sufficient and so we need a stronger condition. This brings up the notion of absolute continuity to play a dominant role here. The following result presents the class of absolutely continuous functions as an important and efficient tool that can be used to prove the Fundamental Theorem of Calculus for Lebesgue integrals.

Proposition 3.6.4 *If f is integrable on $[a, b]$, and $F(x) = \int_a^x f(t)dt$ for $x \leq b$, then $F \in AC[a, b]$.*

Proof This follows from Theorem 3.4.10. Indeed, let $\epsilon > 0$, then there exists $\delta > 0$ such that if $A = [a, b]$ and $m(A) < \delta$, then

$$\int_A |f| < \epsilon.$$

Let $[a_k, b_k]$, $k = 1, \ldots n$ be pairwise disjoint intervals in $[a, b]$ such that

$$\sum_{k=1}^{n} (b_k - a_k) < \delta.$$

Let $E = \bigcup_k [a_k, b_k]$. Then

$$\sum_{k=1}^{n} |F(b_k) - F(a_k)| = \sum_{k=1}^{n} \left| \int_{a_k}^{b_k} f \right| \leq \sum_{k=1}^{n} \int_{a_k}^{b_k} |f| \leq \int_E |f| < \epsilon.$$

□

3.6.3 Differentiation of Integral

Now we are ready to solve the recovering problem for Lebesgue integrals. We will prove two versions of the FTC for Lebesgue integrals. The first version deals with recovering the function after differentiating its indefinite integral, while the second version deals with recovering the function after integrating its derivative. We begin by the following useful lemma.

Lemma 3.6.5 *Let f be integrable on $[a, b]$. If*

$$\int_a^x f = 0$$

for all $x \in [a, b]$, then

$$f = 0. \text{ a.e.}$$

Proof Suppose not. Then WLOG we may assume that $f > 0$ on a set E, such that $\mu(E) > 0$. Then there exists a closed set $F \subset E$, such that $\mu(F) > 0$ and $\int_F f > 0$. Let $O = [a, b] \setminus F$ be an open set. Then

$$0 = \int_a^b f = \int_O f + \int_F f,$$

which implies that

$$\int_O f = -\int_F f < 0.$$

Moreover, O can be written as a countable collection of open intervals

$$O = \bigcup (a_n, b_n),$$

hence

$$\int\limits_{\bigcup (a_n, b_n)} f = \sum_n \int\limits_{a_n}^{b_n} f < 0.$$

But this means that for some $j \leq n$ we have $\int_{a_j}^{bj} f < 0$, and this contradicts the fact that $\int_a^{bj} f = 0$. □

Now we prove the first version of the Fundamental Theorem of Calculus for Lebesgue integrals.

Theorem 3.6.6 (Fundamental Theorem of Calculus (Version I)) *Let f be integrable on $[a, b]$ and*

$$F(x) = \int\limits_a^x f(t)dt.$$

Then

$$F'(x) = f(x)$$

a.e.

Proof Suppose first that f is bounded measurable on $[a, b]$, i.e., $|f| < M < \infty$ on $[a, b]$. Let

$$F(x) = \int\limits_a^x f(t)dt.$$

Then by Proposition 3.6.4, $F \in AC[a, b]$, so by Proposition 2.7.5(1) F' exists a.e., that is, if

$$g_n = n\left[F\left(x + \frac{1}{n}\right) - F(x)\right],$$

then

$$\lim n\left[F\left(x + \frac{1}{n}\right) - F(x)\right] = F'(x).$$

Note that

$$|g_n| = \left|n\left[F\left(x + \frac{1}{n}\right) - F(x)\right]\right| = \left|n \int\limits_x^{x+\frac{1}{n}} f(t)dt\right| \leq M.$$

By the BCT, $\lim \int g_n = \int g$ on $[a, b]$, so for all $x \in [a, b]$, we have

$$\lim \int_a^x \left| n\left[F\left(x+\frac{1}{n}\right) - F(x) \right] \right| dt = \int_a^x F'(t)dt. \tag{3.6.1}$$

On the other hand,

$$\int_a^x n\left[F\left(x+\frac{1}{n}\right) - F(x) \right] dt = n\left[\int_{a+\frac{1}{n}}^{x+\frac{1}{n}} F(t)dt - \int_a^x F(t)dt \right]$$
$$= n\left[\int_x^{x+\frac{1}{n}} F(u)du - \int_a^{a+\frac{1}{n}} F(u)du \right].$$

Taking $n \longrightarrow \infty$, and using Mean Value Theorem for integrals since $F \in AC[a,b]$, gives

$$\lim \int_a^x \left| n\left[F\left(x+\frac{1}{n}\right) - F(x) \right] \right| dt = F(x) - F(a).$$

Combining the result with (3.6.1), taking into account that $F(a) = 0$, gives

$$F(x) - F(a) = F(x) = \int_a^x F'(t)dt.$$

Thus

$$\int_a^x f(t)dt = \int_a^x F'(t)dt$$

which implies that

$$\int_a^x [F'(t) - f(t)]dt = 0 \ \ \forall x \in [a,b].$$

By Lemma 3.6.5, we conclude that

$$F'(t) = f(t) \ \ a.e.$$

Now, assume that f is integrable on $[a,b]$. WLOG, we can assume that $f \geq 0$. Then we can write $f = f^+ - f^-$. Define the sequence

$$f_n = \min\{f, n\}.$$

Then for each n, $f_n \geq 0$ and is bounded measurable function. Define the following sequence of functions:

$$h_n(x) = \int_a^x (f(t) - f_n(t))\, dt. \tag{3.6.2}$$

It is easy to see that $\{h_n\}$ is increasing and differentiable a.e. on $[a, b]$, and $h_n'(x) \geq 0$. From (3.6.2) we have

$$F(x) = \int_a^x f(t)dt = \int_a^x f_n(t)dt. + h_n(x).$$

Applying the result for the bounded case in the previous step gives $F'(x) \geq f_n(x)$ a.e. Taking the limit gives

$$F'(x) \geq f(x) \ a.e. \tag{3.6.3}$$

and integrating on $[a, b]$ gives

$$\int_a^b F'(x) \geq \int_a^b f(x). \tag{3.6.4}$$

On the other hand, by Proposition 3.6.2,

$$\int_a^b F'(x)dx \leq F(b) - F(a) = F(b) = \int_a^b f(x)dx.$$

Combining with (3.6.4) gives

$$\int_a^b \big(F'(x) - f(x)\big)\, dx = 0.$$

But from (3.6.3), the integrand $F'(x) - f(x) \geq 0$. It follows by Proposition 3.3.4(5) that

$$F'(x) = f(x). \qquad \square$$

3.6.4 Indefinite Integral of Derivative

The first version of the FTC deals with the differentiation of an integral, and it shows that differentiating a function after integrating it will recover the function (i.e., give it back). The second version deals with the reverse order.

Theorem 3.6.7 (Fundamental Theorem of Calculus (version II)) *Let $F \in AC[a, b]$, and $F'(x) = f(x)$ a.e. on $[a, b]$. Then*

$$F(x) = \int_a^x f(t)dt + F(a),$$

and

$$\int_a^b f(x)dx = F(b) - F(a).$$

Proof Let F be absolute continuous on $[a, b]$, and let $F' = f$. Assume

$$G(x) = \int_a^x f(t)dt.$$

Then by Proposition 3.6.4, $G \in AC[a, b]$. Define the function

$$h(x) = G(x) - F(x).$$

Then by Proposition 2.7.2(4) $h \in AC[a, b]$, so h' exists a.e., and using FTC (version I) gives

$$h' = G' - F' = f - f = 0.$$

By Proposition 2.7.5(2) $h = c$, so $G(x) - c = F(x)$. Substituting with $x = a$ gives $c = -F(a)$. Therefore

$$G(x) + F(a) = F(x) = \int_a^x f(t)dt + F(a).$$

The other conclusion of the theorem can immediately be obtained by substituting $x = b$. □

We end the section with the following observation: One of the consequences of the Fundamental Theorem of Calculus (version II) is that every absolutely continuous function is the indefinite integral of its derivative. On the other hand, Proposition 3.6.4 implies that every indefinite integral is absolutely continuous, which is our observation's reverse order. Thus we proved:

Theorem 3.6.8 *A function F is absolutely continuous on $[a, b]$ if and only if it is the indefinite integral of its derivative.*

3.7 Lebesgue Double Integral

3.7.1 The Double Integral

The idea of Lebesgue integral can be extended to integrating multivariable measurable functions. We will be focusing on two variables, so basically, our extension will be the space $\mathbb{R}^2$, and the procedure can be routinely extended to $\mathbb{R}^n$. It is well known from calculus that if a function is continuous on the rectangle $R = [a, b] \times [c, d]$ then

$$\int_R f(x, y)dA = \int_z^b \left(\int_c^d f(x, y)dy \right) dx = \int_c^d \left(\int_a^b f(x, y)dx \right) dy.$$

Our main motive behind this section is to extend this fact to Lebesgue integral. This is called the $Fubini\ Theorem$. The Lebesgue product measure on $\mathbb{R}^2$ studied in Section 1.7 is to be used in this regard. All results related to the Lebesgue integral can be easily established for the double Lebesgue integral. A basic property to consider is Theorem 1.7.4, which states that

$$\mu^*(A \times B) = \mu^*(A) \cdot \mu^*(B).$$

For simple function $\varphi(x, y)$ defined on $A \subset \mathbb{R}^2$, write $A = \bigcup_{k=1}^n R_k$, where $R_k = I_k \times J_k$ for intervals I_k and J_k. Then φ is defined as

$$\varphi(x, y) = \sum_{k=1}^n a_k \chi_{R_k}(x, y).$$

3.7.2 Sections of Sets and Functions

As discussed earlier in Sect. 3.3, the existence of a sequence of simple monotone functions has played a central role in establishing the definition of Lebesgue integral. Unfortunately, the monotone property has no sensible and standard definition for this property on $\mathbb{R}^2$ because it is not totally ordered. One way to overcome this obstacle is to introduce the "$section$" of a function. This concept requires a product set, so we need to define sections for sets before moving to functions.

Definition 3.7.1 (*Section of a Set*) Let E be a set of the product space $X \times Y$. Then, by fixing a point $x \in X$, the set

$$E_x = \{(y \in Y : \ (x, y) \in E\}$$

is called the *section* of E determined by x. Similarly, given a point $y \in Y$, the set

$$E^y = \{x \in X : \ (x, y) \in E\}$$

is called the *section* of E determined by y.

Keep in mind that the definition implies that E_x is a subset of Y, whereas E^y is a subset of X, and neither is a subset of E. Writing the variables x and y as a subscript and a superscript is merely a matter of terminology.

One can easily deduce the following basic properties of sections.

Proposition 3.7.2 *Let A and B be subsets of the product space $X \times Y$. Then*

(1) $(A \cup B)_x = A_x \cup B_x$.
(2) $(A \cap B)_x = A_x \cap B_x$.
(3) $(A \setminus B)_x = A_x \setminus B_x$.

The result can be easily extended to countable union or countable intersection of sets. The idea of the section of a set can be conveyed to functions via the following definition.

Definition 3.7.3 (*Section of a Function*) Let $f : X \times Y \to \mathbb{R}$ be a real-valued function defined on the product space $X \times Y$. Then the *section* of f determined by $x \in X$ is given by $f_x(y) = f(x, y)$. Similarly, the *section* of f determined by $y \in Y$ is given by $f^y(x) = f(x, y)$.

The following proposition is useful in proving some integral results.

Proposition 3.7.4 *Let $E \subset \mathbb{R}^2$ and $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$. Then*

(1) If E is measurable in $\mathbb{R}^2$, then E_x and E^y are measurable in $\mathbb{R}$.
(2) If f is measurable with respect to $\mathbb{R}^2$, then f_x and f^y are measurable with respect to $\mathbb{R}$.

Proof For (1), it suffices to prove the result for E_x, since the measurability of E^y can be proved similarly. If E is a measurable rectangle, then we can write E as $E = A \times B$ for some $A \in X$, and $B \in Y$, and so we have

$$E_x = \begin{cases} B & x \in A \\ \emptyset & x \notin A \end{cases},$$

which is clearly measurable in Y. Otherwise, let $E \subset R_\sigma$ be the union of the almost disjoint rectangles E_i, and WLOG we can assume these subsets $\{E_i\}$ to be measurable in $X \times Y$ and $E = \bigcup E_i$. Fix $y \in Y$. Then

$$\chi_{E_x}(y) = \chi_E(x, y) = \sup_i \chi_{E_i}(x, y) = \sup_i \chi_{(E_i)_x}(y). \tag{3.7.1}$$

If E_i is measurable, then χ_{E_i} is measurable function. Hence χ_{E_x} is measurable, which implies that E_x is measurable.

Now, let $E \subset R_{\sigma\delta}$. Then, we can write $E = \bigcap E_i$, where $E_i \in R_\sigma$. It follows that

$$\chi_{E_x}(y) = \chi_E(x, y) = \inf_i \chi_{E_i}(x, y) = \inf_i \chi_{(E_i)_x}(y).$$

It is easy to conclude the result using Proposition 3.7.2 and the previous argument.

For (2), since f is measurable, $E = \{(x, y) : f(x, y) > \alpha\}$ is measurable. By (1), $E_x = \{y : f(x, y) > \alpha\}$ is measurable, hence f_x is measurable. □

3.7.3 *Fubini–Tonelli Theorems*

In what follows, we establish two lemmas that will help us in proving our main result.

Lemma 3.7.5 *Let $f(x, y)$ be integrable on $X \times Y \subseteq \mathbb{R}^2$ with respect to two-dimensional Lebesgue product measure $\mu_2 = \mu_x \times \mu_y$. Then $f_x(y)$ is integrable on Y for a.e. x and $f^y(x)$ is integrable on X for a.e. y.*

Remark Here, the sets X and Y are usually the set $\mathbb{R}$, but the only reason we used this notation is to distinguish between integration with respect to x and with respect to y.

Proof It suffices to prove the result for f_x since the same procedure can be followed for f_y. Assume

$$f(x, y) = \chi_E(x, y)$$

for some measurable $E \subseteq X \times Y$ with $\mu_2(E) < \infty$. Then

$$\mu_2(E) = \int_{X \times Y} f(x, y) d(x, y).$$

Let

$$f(x, y) = f_x(y) = \chi_{E_x}(y).$$

Integrating both sides over Y, we get

$$\int_Y f_x(y) dy = \int_Y \chi_{E_x}(y) dy = \mu(E_x) < \infty.$$

Hence $f_x(y) = \chi_{E_x}(y)$ is integrable on Y. This result extends to simple functions $f(x, y) = \varphi(x, y)$, that is, if $\varphi(x, y)$ is an integrable simple function on Y then φ_x is integrable on Y.

Now let $f \geq 0$. Fix $x \in X$ and write $f(x, y) = f_x(y)$. Then there exists a sequence of simple functions

$$\varphi_n(x, y) = (\varphi_n)_x(y)$$

such that

$$0 \leq (\varphi_n)_x \nearrow f_x$$

and since f is integrable, φ_n is integrable for each n, hence by previous argument $(\varphi_n)_x$ is also integrable for each n, and by MCT

$$\int_Y f_x(y)dy = \lim \int_Y (\varphi_n)_x(y)dy = \lim \int_Y \varphi_n(x, y)dy < \infty.$$

□

Lemma 3.7.6 *Let $E \subset \mathbb{R}^2$ be a measurable set of a finite measure, and let $g(x) = \mu(E_x)$. Then g is a measurable function and*

$$\int_{\mathbb{R}} g(x)dx = \mu_2(E).$$

Proof Let $E = A \times B \subset \mathbb{R}^2$, for some measurable sets $A, B \subset \mathbb{R}$. Write g as

$$g(x) = \int_{\mathbb{R}} \chi_{E_x}(y)dy.$$

This gives

$$\begin{aligned} \int_{\mathbb{R}} g(x)dx &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \chi_{E_x}(y)dy \right) dx \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \chi_E(x, y)dy \right) dx \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \chi_{A\times B}(x, y)dy \right) dx. \end{aligned} \tag{3.7.2}$$

Since

$$\chi_{A\times B}(x, y) = \chi_A(x) \cdot \chi_B(y),$$

Equation (3.7.2) can be written as

$$\begin{aligned}\int_{\mathbb{R}} g(x)dx &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \chi_A(x) \cdot \chi_B(y)dy \right) dx \\ &= \left(\int_{\mathbb{R}} \chi_A(x)dx \right) \left(\int_{\mathbb{R}} \chi_B(y)dy \right) \\ &= \mu(A)\mu(B) = \mu_2(E).\end{aligned}$$

Let $E \in R_\sigma$ be of a finite measure. Write $E = \bigcup E_i$, where each E_i is a measurable rectangle. Let

$$g_k(x) = \mu((E_k)_x).$$

Then for all k, $g_k \geq 0$ is measurable by the previous argument, and $g(x) = \sum g_k$, which implies that g is measurable. By Corollary 3.4.7

$$\int_{\mathbb{R}} g(x)dx = \sum \int_{\mathbb{R}} g_k(x)dx = \mu_2(E).$$

Let $E \in R_{\sigma\delta}$ be of finite measure. Write this as $E = \bigcap E_k$ where $E_{k+1} \subseteq E_k$ and $\mu_2(E_1) < \infty$. Let

$$g_k(x) = \mu((E_k)_x).$$

Then by the previous argument

$$\int_{\mathbb{R}} g_k(x)dx = \mu_2(E_k). \tag{3.7.3}$$

We also have

$$\int_{\mathbb{R}} g_1(x)dx = \mu_2(E) < \infty,$$

hence g_1 is integrable on $\mathbb{R}$. Using the definition of sections and Proposition 3.7.2, it is easy to show that $E_x = \bigcap (E_k)_x$. Then using the property of continuity of the measure from above (Theorem 1.7.5(3))

$$\lim g_k(x) = \lim \mu((E_k)_x) = \mu(E_x) = g(x),$$

and

$$\lim \mu_2(E_k) = \mu_2(E). \tag{3.7.4}$$

Then by DCT, and using (3.7.3) and (3.7.4)

$$\int_{\mathbb{R}} g(x)dx = \lim \int_{\mathbb{R}} g_k(x)dx = \lim \mu_2(E_k) = \mu_2(E).$$

Now let E be measurable in $\mathbb{R}^2$ of finite measure. By Proposition 1.7.3(3) exists $G \in R_{\sigma\delta}$ such that $\mu_2(G \setminus E) = 0$, or $\mu_2(E) = \mu_2(G)$, so for almost all x, $\mu(E_x) = \mu(G_x)$. From previous argument about $R_{\sigma\delta}$, we have

$$g(x) = \mu(G_x) = \mu(E_x),$$

therefore

$$\int_{\mathbb{R}} g(x)dx = \mu_2(G) = \mu_2(E).$$

□

The Fubini Theorem was introduced by the Italian mathematician G. Fubini in 1907. Another variant of double Lebesgue integral has been introduced later and independently by another Italian mathematician, L. Tonelli, who published his result in 1909. While the Fubini Theorem requires the function to be integrable over its domain, the Tonelli Theorem only requires the function to be nonnegative measurable, without necessarily be integrable. We will combine the two theorems into one because the proof will follow the same procedure.[4]

Theorem 3.7.7 (Fubini–Tonelli Theorems)

1. ***Tonelli's Theorem***. *If $f \geq 0$ is measurable on $\mathbb{R}^2$ then*
   a. *The functions*

$$F(x) = \int_{\mathbb{R}} f(x, y)dy \;\&\; F(y) = \int_{\mathbb{R}} f(x, y)dx$$

   are measurable on $\mathbb{R}$, and:
   b. *The following equality holds*

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f dy \right) dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f dx \right) dy = \int_{\mathbb{R}^2} f d(x, y) \leq \infty.$$

2. ***Fubini's Theorem***. *If f is integrable over $\mathbb{R}^2$ then*
   a. *The functions*

$$F(x) = \int_{\mathbb{R}} f(x, y)dy \;\&\; F(y) = \int_{\mathbb{R}} f(x, y)dx$$

[4] The reader, however, needs to understand that the theorems and their proofs were completely different when their authors published them independently.

are integrable over $\mathbb{R}$*, and:*

b. The following equality holds

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f dy \right) dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f dx \right) dy = \int_{\mathbb{R}^2} f d(x, y) < \infty.$$

Proof To prove Tonelli Theorem, let $f \geq 0$. Assume $f(x, y) = \chi_E(x, y)$ for some measurable $E \subseteq \mathbb{R}^2$. Then

$$\mu_2(E) = \int_{\mathbb{R}^2} f(x, y) d(x, y). \tag{3.7.5}$$

On the other hand, fixing x gives

$$f(x, y) = f_x(y) = \chi_{E_x}(y).$$

Integrating both sides over $\mathbb{R}$

$$\int_{\mathbb{R}} f_x(y) dy = \mu(E_x).$$

Integrating again, and using Lemma 3.7.6, gives

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) dy \right) dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f_x(y) dy \right) dx = \int_{\mathbb{R}} \mu(E_x) dx = \mu_2(E). \tag{3.7.6}$$

Combining (3.7.5) and (3.7.6) gives

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) dy \right) dx = \int_{\mathbb{R}^2} f(x, y) d(x, y). \tag{3.7.7}$$

Repeating the same procedure on $f_y(x) = \chi_{E_y}(x)$ gives

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) dx \right) dy = \int_{\mathbb{R}^2} f(x, y) d(x, y). \tag{3.7.8}$$

Now, (1) follows from (3.7.7) and (3.7.8) for the case $f(x, y) = \chi_E(x, y)$. Using linearity property, this result can immediately be extended to simple functions. Fix $x \in X$, then there exists a sequence of simple functions $\varphi_n(x, y) = (\varphi_n)_x(y)$ such that $(\varphi_n)_x \nearrow f_x$. By MCT

$$\lim \int_{\mathbb{R}} (\varphi_n)_x dy = \int_{\mathbb{R}} f_x(y) dy.$$

Now letting

$$\Phi_n(x) = \int_{\mathbb{R}} (\varphi_n)_x dy$$

and

$$F(x) = \int_{\mathbb{R}} f_x(y) dy.$$

Note that $\{\Phi_n\}$ is measurable, and since the sequence converges to F, we conclude that $F(x)$ is measurable in x and nonnegative, and similar argument shows the same conclusions for $F(y)$ as well, which proves 1(a,b).

Since $(\varphi_n)_x \nearrow f_x$, we have $\Phi_n \nearrow F$. Applying MCT again

$$\lim \int_{\mathbb{R}} \Phi_n(x) dx = \int_{\mathbb{R}} F(x) dx,$$

and similar argument shows that

$$\lim \int_{\mathbb{R}} \Phi_n(y) dy = \int_{\mathbb{R}} F(y) dy.$$

Therefore

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) dy \right) dx = \lim \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \varphi_n(x, y) dy \right) dx = \int_{\mathbb{R}^2} f(x, y) d(x, y),$$

Since f may or may not be integrable, the values of the integrals are evaluated in the extended real line, which proves 1(a,b).

To prove Fubini Theorem, let f be integrable. Then we can write $f = f^+ - f^-$ where both functions f^+ and f^- are nonnegative and integrable. So WLOG we can assume $f \geq 0$ and hence use the same argument of the proof of (1). Since f is integrable and nonnegative, f_x is also integrable. Assume $f(x, y) = \chi_E(x, y)$. Then

$$\int_{\mathbb{R}} f_x(y) dy = \mu(E_x) < \infty.$$

Then, equality (3.7.6) above becomes

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y)dy \right) dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f_x(y)dy \right) dx = \mu_2(E) < \infty.$$

This proves 2(a,b) for characteristic functions, and the result extends by linearity to simple functions. Let $f \geq 0$ be integrable over $\mathbb{R}$, which implies by Lemma 3.7.5 that f_x is integrable over $\mathbb{R}$ for almost all x. Using the same argument above for Tonelli, we have

$$\lim \int_{\mathbb{R}} (\varphi_n)_x dy = \int_{\mathbb{R}} f_x(y)dy < \infty.$$

Performing MCT again as above, taking into account that f is integrable over $\mathbb{R}^2$, gives

$$\begin{aligned} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y)dy \right) dx &= \lim \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \varphi_n(x, y)dy \right) dx \\ &= \lim \int_{\mathbb{R}^2} \varphi_n(x, y)d(x, y) \\ &= \int_{\mathbb{R}^2} f(x, y)d(x, y) < \infty. \end{aligned}$$

This proves 2(a,b). This also shows that $F(x)$ (and similarly) $F(y)$ are integrable over $\mathbb{R}$, which proves 2(a,b). This completes the proof. □

Fubini–Tonelli Theorems provide a powerful tool for computing double integrals by interchanging the order of integration. One of their immediate consequences is the following important corollary.

Corollary 3.7.8 *Let $f(x, y) = g(x)h(y)$ be integrable over $A \times B \subseteq \mathbb{R}^2$. Then*

$$\int_{A \times B} f d(x, y)d(x, y) = \left(\int_A g(x)dx \right) \cdot \left(\int_B h(y)dy \right).$$

Example 3.7.9 To integrate the Gaussian function $f(x) = e^{-x^2}$ over $\mathbb{R}^2$, we consider the following function:

$$f(x, y) = e^{-(x^2+y^2)}.$$

Then, we change to polar coordinates

$$\int_{\mathbb{R}^2} e^{-(x^2+y^2)}dxdy = \lim_k \int_0^k 2\pi r e^{-r^2}dr = \pi.$$

Using Corollary 3.7.8,

$$\begin{aligned}\int_{\mathbb{R}^2} e^{-x^2-y^2}dxdy &= \int_{\mathbb{R}^2} e^{-x^2}e^{-y^2}dydx \\ &= \left(\int_{\mathbb{R}} e^{-x^2}dx\right)\left(\int_{\mathbb{R}} e^{-y^2}dy\right) \\ &= \left(\int_{\mathbb{R}} e^{-x^2}dx\right)^2.\end{aligned}$$

Therefore,

$$\int_{\mathbb{R}} e^{-x^2}dx = \sqrt{\pi}.$$

3.7.4 Convolution

Another important application of the Fubini–Tonelli Theorems is what so called the *convolution* of two measurable functions.

Definition 3.7.10 (*Convolution*) Let f, g be measurable functions. The convolution of f and g is defined by

$$(f * g)(x) = \int_{\mathbb{R}} f(x-y)g(y)dy.$$

Remark The definition and the subsequent results can all be extended to $\mathbb{R}^n$.

We conclude the section by the following result which shows that this convolution is integrable over $\mathbb{R}^2(\mathbb{R}^{2n})$ if both f and g are integrable over $\mathbb{R}$ $(\mathbb{R}^n)$.

Proposition 3.7.11 *If f, g are measurable and integrable functions over $\mathbb{R}$, then $f * g$ is integrable over $\mathbb{R}^2$. The result also holds for $\mathbb{R}^n$ and $\mathbb{R}^{2n}$.*

Proof If $f(x)$ is measurable in $\mathbb{R}$ then we can write f as a composition with the linear transformation $T(x, y) = x - y$ (which is continuous, and thus measurable), so $f(x-y)$ is measurable (see Problem 48 for an alternative argument). Hence $f(x-y)g(y)$ is measurable. Then we can integrate the convolution over $\mathbb{R}$ using Tonelli's Theorem on $|f(x-y)g(y)|$ to obtain

$$\begin{aligned}\int_{\mathbb{R}} |f * g|\, dy &= \int_{\mathbb{R}} \left| \int_{\mathbb{R}} f(x-y)g(y)dx \right| dy \\ &\leq \int_{\mathbb{R}} \int_{\mathbb{R}} |f(x-y)g(y)dx|\, dy \\ &= \int_{\mathbb{R}^2} |f(x-y)|\, |g(y)|\, dxdy \\ &= \left(\int_{\mathbb{R}} |f(x)|\, dx \right) \left(\int_{\mathbb{R}} |g(y)|\, dy \right) \\ &< \infty.\end{aligned}$$

□

3.8 Problems

(1) Let $f(x) \geq 0$ on a measurable set A. If $\mu(A) = 0$, show that $\int_A f = 0$.
(2) Let $\varphi \geq 0$. Show that

$$\nu(A) = \int_A \varphi$$

defines a measure.
(3) Show that for a general integrable function f and $c < 0$ we have $\int cf = c \int f$.
(4) Consider the collection of sequences $f : \mathbb{N} \longrightarrow [0, \infty)$, $f(x) = a_k$.
(a) Find a sequence of simple functions $f_n(k) \geq 0$ such that $f_n \leq f_{n+1}$ and $f_n \longrightarrow f$ p.w.
(b) Show that

$$\lim \int_{\mathbb{N}} f_n d\mu = \sum_{k=0}^{\infty} a_k.$$

Deduce that f is integrable with respect to the counting measure μ iff $\sum a_k$ is absolutely convergent.
(c) Generalize the result for any $f : \mathbb{N} \longrightarrow \mathbb{R}$.
(5) Show that if f is integrable on a set A, then $|f| < \infty$ a.e. on A.
(6) **Countable additivity of integrals**: Let E be measurable set and $\{E_n\}_{n=1}^{\infty}$ be a partition of E and f is integrable over E. Show that

$$\int_{\overset{\infty}{\cup} E_n} f = \sum^{\infty} \int_{E_n} f.$$

(7) Consider the definition in (3.3.5):

$$\int_E f = \lim \int_E \varphi_n\,, \quad \{0 \le \varphi_n \le f, \ \ \varphi_n \text{ is a simple function}\}.$$

(a) Show that the definition is well defined.
(b) Show that the definition is equivalent to Definition 3.3.1.

(8) Show that the UCT for Lebesgue Integrals does not hold for A with $\mu(A) = \infty$.

(9) Does BCT works on unbounded intervals, i.e., $\mu(A) = \infty$? Discuss your answer.

(10) Prove the Bounded Convergence Theorem from the Dominated Convergence Theorem.

(11) Let $\{f_n\}$ be a sequence of measurable functions on a set A. If $|f_n| \le g$ on A and g is integrable on A, show that

$$\int_A \underline{\lim} f_n \le \underline{\lim} \int_A f_n \le \overline{\lim} \int_A f_n \le \int_A \overline{\lim} f_n.$$

(12) Prove or disprove: If $\{f_n\}$ is a sequence of integrable functions on A and $f_n \longrightarrow f$ p.w., then f is integrable on A.

(13) Give an example to make the inequality in Fatou's Lemma strict.

(14) Let $f_n \ge 0$ be a sequence of measurable functions on A, such that $f_n \longrightarrow f$ p.w. on A. If $f_n \le f$ on A for all n, show that

$$\lim \int_A f_n = \int_A f.$$

(15) Let $f_n \ge 0$ be a sequence of functions and $f_n \searrow f$ on A. If f_1 is integrable on A, show that

$$\lim \int_A f_n = \int_A f.$$

(16) If $\int_A |f| = 0$, prove that $f = 0$ a.e. on A.

(17) Let $h(x)$ be a bounded measurable function such that $\mu(\{x : h(x) \ne 0\}) < \infty$. Let $f \ge 0$ be a measurable function on a measurable set E . Show that

$$\int_E f = \sup_{0 \le h \le f} \int_E h.$$

(18) Using the previous problem,
(a) Verify the linearity property: $\int f + g = \int f + \int g$.
(b) Prove Fatou's Lemma.

(19) Prove the MCT from Fatou's Lemma.

(20) Let $f \geq 0$ and $A_n = \{x : f(x) \geq n\}$. Show that $\lim \int_{A_n} f = 0$.

(21) Let $f \geq 0$ over E. Show that there exists a sequence of bounded measurable functions $\{f_n\}$ such that

$$\int_E f = \lim \int_E f_n.$$

(22) Use DCT to prove the absolute continuity of the integral.

(23) Let $f_n \longrightarrow f$ p.w. on A. If $|f_n| \leq g$ and g is integrable on A, then show that

$$\int_A |f_n - f| \longrightarrow 0.$$

(24) Find a sequence $f_n \longrightarrow 0$ on $I = [0, 1]$ but $\int_I f_n \longrightarrow 1$ a.e.

(25) **Beppo Levi Theorem**: Let $\{f_n\}$ be an increasing sequence of integrable functions on E and

$$\sup \int_E f_n = \lim \int_E f_n < \infty.$$

Show that $f_n \nearrow f$ for some integrable function f, and

$$\lim \int_E f_n = \int_E f.$$

(26) Let f_n be a sequence of integrable functions on A and $f_n \longrightarrow f$. If f_n, f are integrable on A, show that $\int_A |f_n - f| \longrightarrow 0$ iff $\int_A |f_n| \longrightarrow \int_A |f|$.

(27) Let f be a function of two variables and

$$\lim_{t \to 0} f(x, t) = f(x).$$

If $|f(x, t)| \leq g(x)$ and g is integrable on A, show that

$$\lim_{t \to 0} \int_A f(x, t) = \int_A f(x).$$

(28) Using the techniques of Theorem 3.3.2 to prove the Monotone Convergence Theorem directly from the definition.

(29) In Theorem 3.3.2, show that if f is bounded, then the convergence of the simple functions φ_n to f is uniform.

(30) Show that if f and g are measurable, and f^2 and g^2 are integrable, then $f.g$ is integrable on A.

(31) If f is Lebesgue integrable on E, determine whether f^2 is Lebesgue integrable on E.

(32) Let f be integrable on $\mathbb{R}$. Show that

$$\lim_{n\to\infty}\int_{\mathbb{R}} f(x)\sin nx dx = \lim_{n\to\infty}\int_{\mathbb{R}} f(x)\cos nx dx = 0.$$

(33) Let f be Lebesgue integrable on a measurable set A.
(a) If $\{A_n\}$ is an ascending countable collection of subsets of A, then

$$\int_{\cup A_n} f = \lim \int_{A_n} f.$$

(b) If $\{A_n\}$ is an descending countable collection of subsets of A, then

$$\int_{\cap A_n} f = \lim \int_{A_n} f.$$

(34) If f is integrable on a set A, show that $|f|$ is integrable on A and

$$\left|\int_A f\right| \le \int_A |f|.$$

(35) Show that the L-Integral of $\sin x$ on $\mathbb{R}$ is not defined.

(36) Show that if$|f|$ be improperly R-integrable on $\mathbb{R}$ then $|f|$ is L-integrable.

(37) Show that the following functions are improperly R-integrable on $[0, 1]$ but not L-integrable.
(1) $f(x) = \frac{1}{x}\sin(\frac{1}{x})$.
(2) $f(x) = \frac{1}{x}\sin(\frac{1}{x}) + \cos(\frac{1}{x})$.
(3) $f(x) = 2x\sin(\frac{1}{x^2}) - \frac{2}{x}\cos(\frac{1}{x^2})$.

(38) Let $f(x) = \frac{1}{\sqrt{x}}$. Find a sequence f_n such that $f_n \longrightarrow f$ and

$$\lim \int_0^1 f_n = \int_0^1 f.$$

(39) Prove the Generalized Dominated Convergence Theorem 3.4.9.

(40) Evaluate the following Lebesgue Integrals:

(1) $\int_1^\infty \frac{e^{1/x}}{x^2} d\mu.$

(2) $\int_0^1 \frac{e^{\sqrt{x}}}{\sqrt{x}} d\mu.$

(3) $\int_1^2 \frac{1}{\sqrt[3]{x-1}} d\mu.$

(4) $\int_{-\infty}^\infty e^{-|x|} d\mu.$

(41) Evaluate the following integral limits as $n \to \infty$:

(1) $\lim \int_0^1 \cos(x^n).$

(2) $\lim \int_0^1 \sin(\frac{x}{n}) e^{x/n}.$

(3) $\lim \int_0^1 \frac{x}{nx^2+1} d\mu.$

(4) $\lim \int_1^\infty \frac{\sqrt{x-1}}{1+\sqrt{n}x^3} d\mu.$

(5) $\lim \int_1^\infty \frac{n}{n^2x^2+1} d\mu.$

(6) $\lim \int_0^\infty (1-\frac{x}{n})^n e^{x/2} d\mu.$

(7) $\lim \int_0^\infty \frac{n \sin(x/n)}{xe^x} d\mu.$

(8) $\lim \int_0^\infty \frac{\sin(x/n)}{(1+\frac{x}{n})^n} d\mu.$

(9) $\lim \int_0^\infty (1+\frac{x}{n})^{-n} \sin(\frac{x}{n}) d\mu.$

(10) $\lim \int_0^\infty \frac{1+nx^2}{(1+x^2)^n} d\mu.$

(11) $\lim \int_0^\infty \frac{1+nx}{(1+x^2)^n} d\mu.$

(12) $\lim \int_0^\infty \frac{1}{2+\cos x+nx^2} d\mu.$

(42) Give an example of a function F that is the indefinite integral of some integrable function f but $F' \neq f$.

(43) Give an example of an indefinite integral which is not differentiable on $[a, b]$.

(44) Show that any absolutely continuous function can be written as the difference of two absolutely continuous functions.

(45) Show that if $f \in AC[\epsilon, 1]$ and f is continuous at $x = 0$ and $f \in BV[0, 1]$, then $f \in AC[0, 1]$.

(46) Prove Proposition 3.7.2.

(47) Show that if E is a set of zero measure in $\mathbb{R}^2$, then E_x is a set of zero measure in $\mathbb{R}$.

(48) (a) If $E \subseteq \mathbb{R}$ is measurable, show that $\hat{E} = \{(x, y) : x - y \in E\}$ is measurable in $\mathbb{R}^2$.
(b) If f is measurable on $\mathbb{R}$, show that $g(x, y) = f(x - y)$ is measurable on $\mathbb{R}^2$.

(49) Use Fubini–Tonelli Theorem to evaluate

(1) $\int_0^\infty \frac{\sin x}{x} dx$.

(2) $\int_0^{2\pi} \int_0^\infty y \cos y e^{-xy} dx dy$ for some $a > 0$.

(3) $\int_0^\infty \int_0^\infty (\arctan x)(\sin y)(e^{-xy}) dx dy$.

(50) Show that
(a) $f * g = g * f$.

(b) $(f * g) * h = f * (g * h)$.

Chapter 4
Lebesgue Spaces



4.1 Norms and Linear Spaces

4.1.1 Finite-Dimensional Linear Spaces

Recall a space X is called linear space (or vector space), if it is closed under addition and scalar multiplication, that is, if for every $x, y \in X$, $c \in F$, $cx + y \in X$. Other properties of these spaces are

1. If $v_1, v_2 \in V$ and $c_1, c_2 \in \mathbb{R}$ then $c_1 v_1 + c_2 v_2 \in V$.
2. Commutative and associative laws of addition hold.
3. Distributivity of scalar multiplication with respect to field and vector additions. So there exist $v_1, v_2 \in V$ and $a, b \in F$ such that $a(v_1 + v_2) = av_1 + av_2$, and $(a + b)v = av + bv$.
4. Additive and multiplicative identities exist. There exists $0 \in V$ such that $v + 0 = v$ for all $v \in V$, and there is $1 \in V$ such that $1.v = v$ for all $v \in V$.
5. Scalar multiplication is associative; that is $a(bv) = (ab)v$ for all $v \in V$.
6. There exists an additive inverse $-v$ such that $v + (-v) = 0$.

The most well-known example of a linear space is the Euclidean spaces $\mathbb{R}^n$ for all $n \geq 1$. These spaces consists of vectors of the form $(x_1, x_2, \ldots x_n)$. Addition of vectors takes the following form

$$(x_1, x_2, \ldots x_n) + (y_1, y_2, \ldots y_n) = (x_1 + y_1, x_2 + y_n, \ldots x_n + y_n),$$

and scalar multiplication takes the form

$$c \cdot (x_1, x_2, \ldots x_n) = (cx_1, cx_2, \ldots cx_n).$$

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The distance between any two vectors is given by

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_n)^2 + \cdots + (x_n - y_n)^2}. \tag{4.1.1}$$

If $y = 0$, then $d(x, 0)$ represents the length of the vector $x = (x_1, x_2, \ldots x_n)$.

In general, a linear space may not be $\mathbb{R}^n$. They could be spaces of sequences, spaces of functions, or spaces of matrices. For spaces of functions, the algebraic operations take the form

$$f(x) + g(x) = (f + g)(x),$$

$$c \cdot f(x) = (cf)(x).$$

One central difference between these spaces and the Euclidean spaces $\mathbb{R}^n$ is that the latter spaces has only finite number of coordinates, while the former spaces possess uncountably infinite coordinates if the domain of the variable x is uncountable set, so that $f(x)$ will take uncountably infinite values $(f(x_r))$ where $r \in [a, b]$, or any uncountable set, and as a result, the space will have an infinite number of dimensions. Many geometric and topological properties of the finite-dimensional spaces fail to satisfy in infinite-dimensional spaces. In particular, the notion of distance and length will be unclear when our elements are functions or sequences, so one has to generalize these notions to adapt the new environment of spaces with infinite dimensions.

4.1.2 Definition of Norm

An important notion related to linear spaces is what so-called the *norm*. A *norm* on a space X is a nonnegative real-valued function defined on X with the following axioms:

1. $\|x\| \geq 0$ for every $x \in X$.
2. $\|x\| = 0$ if and only if $x = 0$.
3. $\|cx\| = |c| \, . \, \|x\|$ for every $x \in X$ and $a \in \mathbb{R}$.
4. $\|x + y\| \leq \|x\| + \|y\|$, for every $x, y \in X$.

If the function satisfies all but axiom (2), it is called *seminorm*. Writing the norm of x, as a function $f(x) = \|x\|$, it is clear that $f(0) = 0$. Using axiom 4 above,

$$\|x\| = \|x - y + y\| \leq \|x - y\| + \|y\| \, .$$

This gives

$$|\|x\| - \|y\|| \leq \|x - y\| \, ,$$

or

$$|f(x) - f(y)| \leq |f(x - y)|,$$

which implies that the norm is continuous.

4.1.3 The $p-$Norm

Consider a vector $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$. The 2-norm

$$\|x\|_2 = \left(|x_1|^2 + |x_2|^2 + \cdots + |x_n|^2\right)^{\frac{1}{2}}.$$

This is also known as the *Euclidean norm*. This is basically the distance and length (4.1.1). When we extend the number of coordinates to infinity, then our vectors become sequences, and the 2-norm will take the form

$$\|x\|_2 = \left(\sum_{i=1}^{\infty} |x_i|^2\right)^{1/2}.$$

If the coordinates become uncountably infinite, the sequences convert to functions $f(x)$, and the 2-norm will take the form of integral

$$\|f\|_2 = \left(\int |f|^2\right)^{1/2}.$$

Furthermore, the 2-norm can be generalized to any number p, to become

$$\|f\|_p = \left(\int |f|^p\right)^{1/p}. \tag{4.1.2}$$

Here, the absolute value is taken to satisfy axiom 1, and the power of $1/p$ is taken to satisfy axiom 3. It should be noted, however, that the formula in (4.1.2) is not always guaranteed to be a norm for any class of functions, as it may easily fail to satisfy any of the axioms, especially axiom 2. Thus, extra care should be taken when dealing with this type of norms for function spaces, and these functions should be subject to suitable conditions. This $p-$norm is the norm that will be adopted in our new Lebesgue spaces which will be introduced in the next section. We shall see that equipping these function spaces with the $p-$norm gives a nice geometric structure in the spaces which will allow us to introduce the concept of length, convergence compactness, and many other topological properties that have already been studied for the finite-dimensional spaces.

4.2 Basic Theory of L^p Spaces

4.2.1 The Space of Lebesgue Integrable Functions

Consider the collection of all Lebesgue integrable functions on $[a, b]$. As can be easily checked, this set clearly defines a linear space. Let us denote this space by $L[a, b]$. We also have

$$\int_a^b |f(x)|\, dx < \infty$$

for every Lebesgue integrable function f. The question arises is: Can we define this quantity as a norm for the space? To answer this question, define the following:

$$\int_a^b |f(x)|\, dx = \|f\|.$$

Note that the integral is always nonnegative, hence $\|f\| \geq 0$. Let f be a Lebesgue integrable function. Then

$$\|af\| = \int_a^b |af(x)|\, dx = |a| \int_a^b |f(x)|\, dx.$$

Hence $\|af\| = |a| \,.\, \|f\|$. If f, g are two Lebesgue integrable functions, then

$$\begin{aligned} \int_a^b |f(x) + g(x)|\, dx &\leq \int_a^b |f(x)| + |g(x)|\, dx \\ &= \int_a^b |f(x)|\, dx + \int_a^b |g(x)|\, dx. \end{aligned}$$

Hence, we have

$$\|f + g\| \leq \|f\| + \|g\|.$$

So the norm $\|\cdot\|$ satisfies axioms 1, 3, and 4 above for the norm. It remains to check axiom 2. If $\|f\| = 0$, then

$$\int_a^b |f(x)|\, dx = 0.$$

But this only means that $f = 0$ a.e. as discussed in Chap. 3. So one can choose $f(x) = 0$ on $[a, b]$ except at finitely many points, then $\|f\| = 0$ although $f \neq 0$, and this violates axiom 2 of norms. On the other hand, recall that two functions f and g are the same if $f = g$ except on a set of measure zero. Since Lebesgue integration ignores sets of measure zero, these functions fall under the same class, which means we can view the two functions as being the same. This allows us to form the following equivalence classes:

$$f \sim g \Longleftrightarrow f = g \ a.e. \tag{4.2.1}$$

We can then view the elements of the space as if they are equivalence classes of functions rather than functions, and the space is considered to be the space of these functions, quotiented by the equivalence relation in (4.2.1). Therefore, the function $\|\cdot\|$ is a norm, and the space $L[a, b],\ \|\cdot\|_1)$ is a normed space. This can also be defined as $L(\Omega)$ where $\Omega \subseteq \mathbb{R}^n$.

4.2.2 Definition of L^p Space

The space $L[a, b]$ can be also generalized to $L^p[a, b]$, the space of all functions such that $|f|^p$ is integrable (in Lebesgue sense) on $[a, b]$ for every $f \in L^p[a, b]$, where $1 \leq p < \infty$, together with the norm

$$\|f\|_p = \left(\int_a^b |f(x)|^p \, dx \right)^{1/p}.$$

Remark We still describe the members of the space as functions, not classes of functions, and we will continue to describe it this way. Although this is not rigorous as illustrated above, it is more convenient.

It is easy to verify that the space L^p is a linear space. Let $f, g \in L^p$. Then $|f|^p$ and $|g|^p$ are integrable, and

$$|f + g|^p \leq 2^p \max\{|f|^p, |g|^p\} \leq 2^p(|f|^p + |g|^p).$$

This implies that

$$f + g \in L^p.$$

Moreover, $af \in L^p$ for every $a \in \mathbb{R}$. Hence L^p is a subspace of the linear space of all functions defined on $[a, b]$, which implies L^p is itself a linear space. To show $\|\cdot\|_p$ is a norm, note that the axioms 1, 2, and 3 are easy to establish. Indeed,

$$|f(x)|^p \geq 0$$

for all x, then so is $\|f\|_p$. Moreover, as argued in the L case, axiom 2 holds if we think of the elements of the space as equivalence classes of functions. Let $c \in \mathbb{R}$, then

$$\|c.f\|_p = \left(\int_a^b |cf(x)|^p \, dx \right)^{1/p} = \left(|c|^p \int_a^b |f(x)|^p \, dx \right)^{1/p} = |c| \, \|f\|_p .$$

It remains to establish the triangle inequality, so we need the Minkowski Inequality which will be introduced in the next section.

4.2.3 L^∞ Spaces

The question that arises now is: What happens if p keeps increasing to infinity? Well, if we need to have

$$\int_a^b |f(x)|^p \, dx < \infty,$$

then taking $p \longrightarrow \infty$, this requires the function to be bounded, or almost everywhere bounded. That is, f is bounded on $[a, b]$, except maybe on a set of measure zero. This gives rise to a new space which will be called the L^∞ space. Recall that a function f is bounded on a set A if there exists $M > 0$ such that $|f| \leq M$ for all $x \in A$. The following definition takes into account the exception of a set of measure zero.

Definition 4.2.1 (*Essential Supremum*) Let $f : \Omega \longrightarrow \mathbb{R}$. Then, the *essential supremum* $\operatorname{ess\,sup}(f) = \inf\{M : |f(x)| \leq M, \text{ for almost every } x \in \Omega\}$.

In other words, we say that

$$\operatorname{ess\,sup}(f) = \inf\{M : \mu(x \in \Omega : |f(x)| > M) = 0\}.$$

So the essential supremum of a function is the smallest number M such that the set $\{x : f(x) > M\}$ is of measure zero. As it appears in its definition, the essential supremum disregards sets of measure zero. This feature enables us to define the "*supremum norm*", or the "*infinity norm*", denoted $\|\cdot\|_\infty$, in the following sense: Define $L^\infty(\Omega)$ to be the space of all essentially bounded measurable functions, and define the norm

$$\|f\|_\infty = \operatorname{ess\,sup} |f| .$$

We have the following observations:

1. A function that is essentially bounded isn't necessarily bounded. Consider

$$f(x) = \begin{cases} x & : x \in \mathbb{Q} \\ 1 & : x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}.$$

2. The supremum of a function is not necessarily equal to the essential bound. Consider

$$f(x) = \begin{cases} 3 & : x = 1 \\ 2 & : x \neq 1 \end{cases}.$$

   Then, $\|f\|_\infty = 2$, but $\sup\{f\} = 3$.
3. If $\|f\|_\infty = 0$, then $f = 0$ a.e.. This demonstrates the advantage of using the essential supremum over the supremum, which allows $\|f\|_\infty$ to fulfill the axioms of the norm. This, in turn, will help define a norm on the L^∞ space.

Now, consider the space $L^\infty[a, b]$ of all measurable and essentially bounded function on $[a, b]$. Define the norm to be

$$\|f\|_\infty = \operatorname{ess} \sup_{x\in[a,b]} |f(x)|.$$

Now, let $f, g \in L^\infty[a, b]$. Then

$$\begin{aligned} \|f + g\|_\infty &= \operatorname{ess} \sup_{x\in[a,b]} |f(x) + g(x)| \\ &\leq \operatorname{ess} \sup_{x\in[a,b]} |f(x)| + \operatorname{ess} \sup_{x\in[a,b]} |g(x)| \\ &= \|f\|_\infty + \|g\|_\infty. \end{aligned}$$

So, $\|f\|_\infty$ satisfies all axioms of the norm, and therefore the $(L^\infty[a, b],\ \|\cdot\|_\infty)$ is a normed space.

4.2.4 Inclusions of L^p Spaces

The next result illustrates some relation between Lebesgue spaces of different $p-$norms.

Proposition 4.2.2 *Let $1 \leq p < r < \infty$ and $\Omega \subseteq \mathbb{R}^n$, with $\mu(E) < \infty$. Then*

$$L^\infty(\Omega) \subset L^r(\Omega) \subset L^p(\Omega).$$

Proof Let $f \in L^\infty(\Omega)$, and choose $r \geq 1$. Then

$$\int_\Omega |f|^r \, d\mu \leq M^r \cdot \mu(\Omega) < \infty,$$

so $f \in L^r(\Omega)$.

Let $f \in L^r(\Omega)$ and $E \subseteq \Omega$ defined by $E = \{x \in \Omega : |f| > 1\}$. Then

$$\begin{aligned} \int_\Omega |f|^p &= \int_E |f|^p + \int_{E^c} |f|^p \\ &\leq \int_E |f|^r + \mu(E) \\ &\leq \infty. \end{aligned}$$

□

Remark We note the following two comments to Proposition 4.2.2.

1. The inclusion is proper, i.e., there are functions in $L^r(\Omega)$ that are not in $L^p(\Omega)$. As an example, let $f(x) = x^{-1/r}$ and $\Omega = [0, 1]$. Then

$$\int_\Omega f^r = \int_{[0.1]} \frac{1}{x}$$

diverges, so $f \notin L^r([0, 1])$, but

$$\int_\Omega f^p < \infty$$

since $\frac{p}{r} < 1$, so $f \in L^p([0, 1])$

2. The condition $\mu(E) < \infty$ is essential and cannot be removed. As an example, let $f(x) = x^{-1/p}$ and $\Omega = [1, \infty)$. Then

$$\int_1^\infty x^{-r/p} < \infty.$$

Hence $f \in L^r(\mathbb{R})$. But $\int_1^\infty f^p$ diverges, which means $f \notin L^r(\mathbb{R})$.

4.3 Fundamental Inequalities

4.3.1 Young's Inequality

The inequalities discussed in this section are of utmost importance in the theory of L^p spaces. The first inequality was found by Holder in 1889, and it will be used to prove the $Minkowski\ Inequality$. Before we state $Holder\ Inequality$, we have the following definition:

Definition 4.3.1 (*Conjugates*) Two numbers p and q are said to be conjugates of each other if $1 \leq p, q \leq \infty$ and $\dfrac{1}{p} + \dfrac{1}{q} = 1$.

The conjugate of 1 is given as ∞, and

$$\frac{1}{\infty} = 0.$$

From now on, we will refer to conjugate numbers by p and q. We begin by the following fundamental inequality:

Lemma 4.3.2 (Young's Inequality) *If* $a, b \geq 0$, *and* $1 < p$, *then*

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof Notice that $f(x) = -\ln x$ is concave up, so the line $t \ln x + (1-t)\ln y$ must be above the graph of f between x and y for all $t \in (0, 1)$. Then we have

$$\ln(tx + (1-t)y) \geq t \ln x + (1-t)\ln y.$$

Now substitute with $x = a^p$ and $y = b^q$ and take the exponential of both sides of the equation. □

4.3.2 Holder Inequality

Theorem 4.3.3 (Holder Inequality) *Let* $f \in L^p(\Omega)$ *and* $g \in L^q(\Omega)$, *for* $1 \leq p, q \leq \infty$, *p and q are conjugates. Then* $fg \in L^1(\Omega)$, *i.e., Lebesgue integrable, and*

$$\int_\Omega |fg| \leq \|f\|_p \cdot \|g\|_q .$$

Proof If $p = 1$ and $q = \infty$, then it is immediate that

$$\int_{\Omega} |fg| \leq \|g\|_{\infty} \|f\|_1 .$$

Consider $p > 1$. Let $F = \dfrac{f}{\|f\|_p}$ and $G = \dfrac{g}{\|g\|_q}$. Clearly

$$\|F\|_p = \|G\|_q = 1.$$

Using Young's inequality and then integrate over Ω, we obtain

$$\int_{\Omega} |FG| \leq \int_{\Omega} \frac{F^p}{p} + \int_{\Omega} \frac{G^q}{q} = \frac{1}{p} + \frac{1}{q} = 1. \tag{4.3.1}$$

Writing F and G back in terms of f and g and substitute in (4.3.1), the result follows. □

Note that the Holder inequality is viewed as a generalization to Cauchy-Schwartz inequality for the case $p = q = 2$.

4.3.3 Minkowski Inequality

The next inequality is a generalization of the triangle inequality in $p-$norm.

Theorem 4.3.4 (Minkowski Inequality) *Let $f, g \in L_p$ for $1 \leq p \leq \infty$. Then*

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p .$$

Proof We have

$$|f + g|^p = |f + g|^{p-1} |f + g|^1 \leq |f| |f + g|^{p-1} + |g| |f + g|^{p-1} . \tag{4.3.2}$$

Since $p = (p - 1)q$ (verify), we have $(f + g)^{p-1} \in L^q$. Now, apply Holder Inequality to both terms in the RHS of (4.3.2), gives

$$\begin{aligned} \int |f + g|^p \leq & \left(\int |f|^p \right)^{1/p} \cdot \left(\int |f + g|^{(p-1)q} \right)^{1/q} \\ & + \left(\int |g|^p \right)^{1/p} \cdot \left(\int |f + g|^{(p-1)q} \right)^{1/q}, \end{aligned}$$

so we obtain

$$\int |f + g|^p \leq \left[\left(\int |f|^p \right)^{1/p} + \left(\int |g|^p \right)^{1/p} \right] \left(\int |f + g|^p \right)^{1/q},$$

and writing the inequality in terms of $p-$norm gives

$$\|f+g\|_p^p \leq \|f+g\|_p^{p/q}\,(\|f\|_p + \|g\|_p). \tag{4.3.3}$$

Dividing both sides of (4.3.3) by $\|f+g\|_p^{p/q}$,taking into account that $\dfrac{1}{p}+\dfrac{1}{q}=1$, gives the result.

For $p=\infty$, let $f, g \in L^\infty$. Then

$$|f(x)+g(x)| \leq |fx)| + |g(x)| \leq \|f\|_\infty + \|g\|_\infty\,.$$

Taking the supremum will extend the result to L^∞. □

This inequality leads to axiom 4 of norms, and thus we have the following important result:

Corollary 4.3.5 *L^p are normed linear spaces for $1 \leq p \leq \infty$.*

4.4 Further Spaces of Order p

Recall a Lebesgue space is the normed space consisting of measurable functions that are pth power Lebesgue integrable equipped with the $p-$norm. Associated with this space, there are other linear spaces that have the same property of pth power of the elements. These spaces shall be called "*spaces of order p*". We introduce three classes of spaces of order p that can be derived from Lebesgue spaces: the first class in which the values of p are changed to $0 < p < 1$ rather than $p \geq 1$. The second class in which the Lebesgue integrable functions are replaced with the Riemann integrable functions, and the third class in which the Lebesgue measure is replaced by the counting measure and $\mathbb{R}$ is replaced by $\mathbb{N}$, The purpose of this section is to introduce these spaces which have properties similar to those of the Lebesgue spaces, but are not considered Lebesgue spaces since the first two spaces are not normed spaces and the third space is not a function space, although it is a normed space and possesses most of the properties that L^p have.

4.4.1 L^p Spaces, $0 < p < 1$

The first class of spaces is considering Lebesgue spaces when $0 < p < 1$. The collection $L^p(E)$ consists of all measurable functions $f : E \longrightarrow \mathbb{R}$ such that

$$\int_E |f(x)|^p\,dx < \infty.$$

This space is a linear space. Similar to the argument for the case $1 \leq p < \infty$, let $f, g \in L^p$. Then $|f|^p$ and $|g|^p$ are integrable, and

$$0 \leq |f+g|^p \leq 2^p \max\{|f|^p, |g|^p\} \leq 2^p(|f|^p + |g|^p).$$

So, $f+g \in L^p(E)$. To select a norm, we set the function

$$\|f\|_p = \left(\int_E |f(x)|^p \, dx \right)^{1/p}. \tag{4.4.1}$$

The following inequalities hold for the case $0 < p < 1$.

Proposition 4.4.1 (Reverse Holder Inequality) *Let $0 < p < 1$, and q be the conjugate of p. Let $f \in L^p$ and $g \in L^q$. Then*

$$\int |fg| \geq \|f\|_p \, \|g\|_q .$$

Proof Since $0 < p < 1$, $q < 0$. Let $r = \dfrac{1}{p}$ and $t = -\dfrac{q}{p}$. Then r, t are conjugates (verify), and $1 < r < \infty$. Moreover, $(fg)^p \in L^r$ and $g^{-p} \in L^t$. Using Holder Inequality (Theorem 4.3.3)

$$\begin{aligned} \int_\Omega \left| |fg|^p \, |g|^{-p} \right| &\leq \left(\int_E \left(|fg|^p \right)^r dx \right)^{1/r} \cdot \left(\int_E \left(|g|^{-p} \right)^t dx \right)^{1/t} \\ &= \left(\int |fg| \right)^p \left(\int |g|^q \right)^{-p/q}. \end{aligned}$$

Hence,

$$\left(\int |f|^p \right)^{1/p} \leq \left(\int |fg| \right) \left(\int |g|^q \right)^{-1/q}.$$

Multiplying both sides by $\left(\int |g|^q \right)^{1/q}$ gives the result. □

Since Holder's inequality was reversed, we expect the same case will hold for Minkowski Inequality. Indeed, this is the case as the next result shows.

Proposition 4.4.2 (Reverse Minkowski Inequality) *Let $f, g \in L_p$ for $1 \leq p \leq \infty$. Then*

$$\|f\|_p + \|g\|_p \leq \|f+g\|_p .$$

Proof This is similar to the proof of the original inequality. We write $h = |f| + |g|$. Then

$$|h|^p = ||f| + |g|| \cdot ||f| + |g||^{p-1}.$$

Note that

$$||f| + |g||^{p-1} |f + g|^1 = |f| |f + g|^{p-1} + |g| |f + g|^{p-1}, \tag{4.4.2}$$

and $|f|, |g| \in L^p$ while $|f + g|^{p-1} \in L^q$ where $q(p-1) = p$, hence q is the conjugate of p. Then we integrate both sides, then apply Holder Inequality on the RHS of (4.4.2) to obtain the inequality. □

The most important consequence of the inequality is that the function in (4.4.1) is not a norm, and thus $(L^p, \|\cdot\|_p)$ is not a normed space.

4.4.2 ℓ^p Spaces

The second type of spaces can be produced from the Lebesgue L^p as follows: Let the L^p measure space be $(\mathbb{N}, \Sigma, \upsilon)$, where $\mathbb{N}$ is the set of natural number, Σ is a σ-algebra of subsets of $\mathbb{N}$, and ν is the counting measure. Then $f \in L^p(\mathbb{N})$ will be of the form $f = (f_1, f_2, \ldots,)$ such that

$$\|f\|_p = \left(\sum_{j=1}^{\infty} |f_j|^p \right)^{1/p} < \infty.$$

We obtain a space of sequences instead of functions, and these sequences converge in p-norm. Let $\{c_n\}$ be a sequence, and define

$$f(x) = \sum c_n \chi_{[n,n+1]}(x).$$

Then

$$\|f\|_p = \left(\int \sum c_n \chi_{[n,n+1]}(x) dx \right)^{1/p} = \left(\sum c_n^p \right)^{1/p} = \|c_n\|_p.$$

Hence $\|f\|_p < \infty$ if and only if $\|c_n\|_p < \infty$. It turns out that the new sequence space is a special class of L^p spaces on $X = \mathbb{N}$ when the underlying measure is the counting measure. We will therefore call the new sequence version of the L^p space the $\ell^p = \ell^p(\mathbb{N}, \Sigma, \upsilon)$ spaces, where υ is the counting measure.

If $p = \infty$, then

$$\begin{aligned}\|f_n\|_\infty &= \inf_n\{M:\ \mu(n\in\mathbb{N}:\ |f(n)|>M)=0\}\\ &= \inf_n\{M\geq 0:\ \ |c_n|\leq M\}\\ &= \sup_n\{|c_n|\}.\end{aligned}$$

Here is the following definition of these spaces.

Definition 4.4.3 (ℓ^p *Spaces*) Let $1\leq p<\infty$. The space $(\ell^p, \|\cdot\|_p)$ is the space of all infinite sequences with finite p-norm, i.e., for any $x_n\in(\ell^p, \|\cdot\|_p)$ we have

$$\|x_n\|_p=\left(\sum|x_n|^p\right)^{1/p}<\infty.$$

If $p=\infty$, then for any $x_n\in\ell^\infty$ we have

$$\|x_n\|_\infty=\sup_n|x_n|<\infty.$$

The spaces $(\ell^p, \|\cdot\|_p)$ and $(\ell^\infty, \|\cdot\|_\infty)$ are linear vector spaces, where the elements of the former space are $p-$summable infinite sequences, and the elements of the latter space are bounded sequences. Indeed, for $x_n, y_n\in\ell^p$ we have

$$\sum_n|x_n+y_n|^p\leq\sum_n\left(2\max(|x_n|,|y_n|)^p\leq 2^p\sum_n\left(|x_n|^p+|y_n|^p\right)<\infty,\right.$$

so it is a linear space.

Since ℓ^p is a special case of the space L^p, we expect it will be normed space for $1\leq p\leq\infty$. Clearly, $\|x_n\|\geq 0$ with $\|x_n\|=0$ iff $x_n=0$. We also have for a scalar c

$$\|cx_n\|=\left(\sum|cx_n|^p\right)^{1/p}=|c|\left(\sum|x_n|^p\right)^{1/p}=|c|\,\|x_n\|.$$

For the triangle inequality, the following theorem predicts that Holder and Minkowski Inequalities still hold.

Theorem 4.4.4 *Let $x_n\in\ell^p$ and $y_n\in\ell^q$ then*

1. **Holder Inequality for sequences**: $x_n\cdot y_n\in\ell^1$ and

$$\|x_ny_n\|_1\leq\|x_n\|_p\cdot\|y_n\|_q$$

for $p>1$.

2. **Minkowski Inequality for sequences**: If $q=p$, then

$$\|x_n+y_n\|_p\leq\|x_n\|_p+\|y_n\|_p$$

for $p\geq 1$.

Proof The proof is very similar to the case for L^p.

(1) For Holder Inequality, assume the special case:

$$\sum |x_1|^p = \sum |y_i|^q = 1.$$

Then using Young Inequality,

$$\begin{aligned}\sum_{k=1}^{\infty} |x_k y_k| &= \sum_{k=1}^{\infty} |x_k| \,.\, |y_k| \\ &\leq \sum \left(\frac{1}{p} |x_n|^p + \frac{1}{q} |y_n|^q \right) \\ &= \frac{1}{p} + \frac{1}{q}.\end{aligned}$$

Hence

$$\sum_{k=1}^{\infty} |x_k y_k| \leq 1. \tag{4.4.3}$$

This can be used to prove the inequality in the general case where $\sum |x_i|^p \neq 1$, $\sum |y_i|^q \neq 1$ by letting

$$\tilde{x_n} = \frac{x_n}{\|x_n\|_p}, \quad \tilde{y_n} = \frac{y_n}{\|y_n\|_p},$$

then

$$\|\tilde{x_n}\|_p = \|\tilde{y_n}\|_q = 1,$$

and we can substitute them in (4.4.3).

(2) For Minkowski Inequality, we have

$$\sum |x_n + y_n|^p \leq \sum (|x_n| + |y_n|) \, |x_n + y_n|^{p-1} \,. \tag{4.4.4}$$

Note that $(p-1)q = p$. So

$$|x_n + y_n|^{p-1} = |x_n + y_n|^{p/q} \in \ell^q.$$

Using Holder's inequality in the RHS of (4.4.4), and letting $K = \sum |x_n + y_n|^p$ we obtain

$$\sum (|x_n| + |y_n|) \, |x_n + y_n|^{p-1} \leq K^{1/q} \left[\left(\sum |x_n|^p \right)^{1/p} + \left(\sum |y_n|^p \right)^{1/p} \right], \tag{4.4.5}$$

Combining (4.4.4) and (4.4.5) then multiplying both sides by $K^{-1/q}$ gives

$$\left(\sum |x_n + y_n|^p\right)^{1/p} \leq \left(\sum |x_n|^p\right)^{1/p} + \left(\sum |y_n|^p\right)^{1/p}.$$

The proof is complete. □

Since every $x_n \in \ell^p$ is necessarily bounded, $\ell^p \subseteq \ell^\infty$. For the relations between ℓ^p for $1 \leq p < \infty$ we have the following

Proposition 4.4.5 *If* $1 \leq p < r < \infty$ *then*

$$\ell^p \subset \ell^r \subset \ell^\infty.$$

Proof The inequality $\ell^r \subset \ell^\infty$ was already established. Let $x_n \in \ell^p$, so $x_n \to 0$, i.e., there exists $N \in \mathbb{N}$ such that $|x_n| < 1$ for all $n \geq N$. Hence

$$\begin{aligned}\sum |x_n|^r &= \sum_{|x_n|>1} |x_n|^r + \sum_{|x_n|<1} |x_n|^r \\ &\leq \sum_{|x_n|>1} |x_n|^r + \sum |x_n|^r \\ &< \infty.\end{aligned}$$

so $x_n \in \ell^r$. □

In comparison with Proposition 4.2.2, we see that the inclusion relations for the two spaces are completely the opposite. If $1 \leq p < r < \infty$ then we have

$$L^\infty \subset L^r \subset L^p \subset L^1$$

$$\ell^1 \subset \ell^p \subset \ell^r \subset \ell^\infty.$$

4.4.3 $\mathcal{R}^p$ Spaces

The third type of spaces of pth power is the space of all Riemann integrable functions of pth power. This space consists of all pth power Riemann integrable functions on $\Omega \in \mathbb{R}^n$ such that

$$\int_\Omega |f|^p < \infty.$$

As we see, the formulation of the space is similar to L^p. We denote this space by $\mathcal{R}^p(\Omega)$. For the norm, we choose the $p-$ norm

$$\|f\|_p = \int_\Omega |f|^p.$$

It is clear that this function satisfies all axioms except axiom 2. Indeed, there are nonzero Riemann integrable functions such that

$$\int_{\Omega} |f|^p = 0.$$

Note that this situation doesn't cause any issue in Lebesgue spaces since they deal with Lebesgue integrable functions, and we have seen that in Lebesgue theory, and according to Definition 2.2.6, functions that are equal almost everywhere are considered equal. This allows us to form the equivalent classes in (4.2.1), and consequently, the $p-$norm satisfies axiom 2 based on these settings. This is no longer valid in Riemann theory since Riemann integrable functions don't ignore sets of measure zero, so we cannot form these equivalence classes. The function $\|f\|_p$ is, therefore, a seminorm on this space, and so the space $(\mathcal{R}^p(\Omega), \|f\|_p)$ is not a normed space. This is another reason why it is more convenient to deal with Lebesgue integrable functions rather than Riemann integrable functions.

4.5 Convergence in L^p

4.5.1 Convergence in $p-$Norm

Recall that the norm is a generalization of the idea of absolute value and can be used as a tool to measure distances between functions of the space. Thus, in L^p the convergence is characterized by the p-norm as follows:

Definition 4.5.1 (*Convergence in $p-$ Norm*) Let $(f_n) \in L^p(\Omega)$. Then f_n converges to f in $p-$ norm (or in the pth mean) if

$$\|f_n - f\|_p \to 0.$$

The convergence is called: $p - norm\ convergence$.

According to the above definition, $f_n \to f$ in the $p-$norm for $f_n, f \in L^p$ if and only if

$$\int_{\Omega} |f_n - f|^p \longrightarrow 0.$$

This is another reason to justify why considering equivalence classes rather than functions is essential.

4.5.2 Comparison Between Types of Convergence

What relations do we have between the three types of convergence: the uniform convergence, the pointwise convergence, and the $p-$norm convergence? In Chap. 2, we demonstrated that uniform convergence implies pointwise convergence, but the converse is not true. It should be an easy task (using supremum) to show that uniform convergence implies $p-$norm convergence, provided that Ω is bounded, but the converse is not necessarily true. Moreover, no general implication between $p-$norm convergence and pointwise convergence, but we can always impose particular conditions to validate the implications. The following two propositions establish the relation between $p-$norm convergence and the other two types.

Proposition 4.5.2 *Let $f_n \in L^p$. If $f_n \longrightarrow f$ uniformly on a set Ω of finite measure, then $f_n \longrightarrow f$ in $p-$norm.*

Proof Note that

$$\int_{\Omega} |f_n - f|^p \leq (\mu(\Omega)) \max |f_n - f|$$

Since $f_n \longrightarrow f$ uniformly, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have

$$\max |f_n - f|^p < \epsilon.$$

The result follows since $\mu(\Omega) < \infty$. □

The result does not hold if $\mu(\Omega) = \infty$. The reader should be able to provide a counterexample.

Another feature of these spaces is that they are closed (complete) in the sense that if $f_n \in L^p$ and $f_n \longrightarrow f$ in the p- norm, then

$$\|f\|_p \leq \|f - f_n\|_p + \|f_n\|_p$$

and so $f \in L^P$. Now, if $f_n \in L^p$ and $f_n \longrightarrow f$ a.e, then two questions arise:
1. Is $f_n \longrightarrow f$ in L^p?
2. Is $f \in L^p$? The following example shows that the answer to the first question is No, and a counterexample for the first question is left to the reader. Let

$$f_n(x) = n^2 \chi_{(0,\frac{1}{n})}(x).$$

Then, it's clear that $f_n \in L^p$ and $f_n \longrightarrow 0$ p.w.a.e. (verify). But

$$\int |f_n - 0|^p = \int_0^{1/n} n^{2p} \longrightarrow \infty.$$

Hence $f_n \nrightarrow f$ in L^p.

4.5.3 $p-$Norm Dominated Convergence Theorem

However, we can impose some conditions to guarantee the desired convergence. Recall the DCT deals with the convergence of Lebesgue integrable functions, i.e., functions in L^1. The following theorem generalizes DCT and can be extended to any L^p for $p \geq 1$ and gives an affirmative answer to the above two questions.

Theorem 4.5.3 ($p-$ Norm Dominated Convergence Theorem) *Let $\{f_n\} \in L^p(\Omega)$, for $1 \leq p < \infty$. If $f_n \to f$ a.e., and f_n is dominated by some $g \in L^p(\Omega)$, i.e., $|f_n| \leq g$ for all n, and for almost all x in Ω, then $f \in L^p(\Omega)$, and $f_n \longrightarrow f$ in $p-$norm.*

Proof Note that

$$|f_n - f| \leq |f_n| + |f| \leq 2g.$$

So

$$|f_n - f|^p \leq 2^p g^p \in L^p.$$

Apply the DCT to $|f_n - f|^p$ gives

$$\lim \int_{\Omega} |f_n - f|^p = 0.$$

□

This can also be used to impose other conditions to obtain a new interesting result, as explained in the following proposition.

Proposition 4.5.4 *Let $f_n \to f$ a.e., and f_n, $f \in L^p$, for $1 \leq p < \infty$. Then $f_n \longrightarrow f$ in $p-$norm iff $\|f_n\|_p \longrightarrow \|f\|_p$ a.e.*

Proof Let $f_n \longrightarrow f$ in $p-$norm, that is, $\|f_n - f\|_p \longrightarrow 0$. By Minkowski Inequality, we have

$$\|f\|_p \leq \|f_n - f\|_p + \|f_n\|_p .$$

So

$$\left| \|f\|_p - \|f_n\|_p \right| \leq \|f_n - f\|_p \to 0.$$

To prove the other direction, let $\|f_n\|_p \to \|f\|_p$ a.e., so $|f_n|^p \longrightarrow |f|^p$ a.e. Since

$$|f_n - f|^p \leq 2^p \left(|f_n|^p + |f|^p \right),$$

write

$$h_n = |f_n - f|^p \ \ \& \ g_n = 2^p (|f_n|^p + |f|^p).$$

Then, clearly $g_n \in L^1$, and

$$g_n \longrightarrow g = 2^{p+1} |f|^p \in L^1.$$

Applying GDCT (Theorem 3.4.9) on h_n and g_n gives

$$\int |f_n - f|^p \to 0.$$

□

4.5.4 Inclusion Relations

Now, we study mean convergence for different values of p. Recall Proposition 4.2.2 gives the inclusion relation as

$$L^\infty(\Omega) \subset L^r(\Omega) \subset L^p(\Omega).$$

for $1 \leq p < r < \infty$ and $\Omega \subseteq \mathbb{R}^n$, with $\mu(E) < \infty$. A consequence of Proposition 4.2.2 is

Proposition 4.5.5 *Let $1 \leq p < r$. Let $\mu(E) < \infty$, and $f_n \in L^r(E)$. If f_n converges in $L^r(E)$, then it converges in $L^p(E)$.*

Proof If $\{f_n\}$ converges to f in $L^r(E)$, then $\|f_n - f\|_r \to 0$, so $|f_n - f|^r \to 0$. Note that

$$\int_E \left(|f_n - f|^p\right)^{r/p} = \int_E |f_n - f|^r < \infty,$$

so

$$|f_n - f|^p \in L^{r/p}(E).$$

Note that $\dfrac{r}{p} > 1$. Let $s = \dfrac{r}{p}$ and let $t = \dfrac{r}{r-p}$ be the conjugate of s. Define

$$g(x) = \chi_E(x).$$

Then, by Proposition 4.2.2. $g \in L^t(E)$, and

$$\int_E |f_n - f|^p = \int_E |f_n - f|^p \, g.$$

Applying Holder's inequality on $|f_n - f|^p$ and g

$$\begin{aligned}\int_E |f_n - f|^p &\leq \|f_n - f\|_s^r \left(\int_E |g|^t \right)^{1/t} \\ &\leq (\mu(E))^{1/t} \|f_n - f\|_s^r \\ &= c \left(\int_E |f_n - f|^r \right)^{1/s} \longrightarrow 0,\end{aligned}$$

where $c = (\mu(E))^{1/t}$, and the result follows. □

4.5.5 Convergence in L^∞

As in the p case, we say that a sequence $f_n \in L^\infty(\Omega)$ converges to $f \in L^\infty(\Omega)$ in L^∞ if

$$\|f_n - f\|_\infty \longrightarrow 0.$$

But this implies

$$\operatorname{ess\,sup} |f_n - f| = 0.$$

Since the supremum gives rise to uniform convergence, we see that the essential supremum gives almost uniform convergence. In other words,

$$f_n \longrightarrow f \text{ in } L^\infty,$$

if there exists $I \subset \Omega$ with $\mu(E) = 0$ such that $f_n \longrightarrow f$ uniformly on $\Omega \setminus E$. In view of Proposition 4.5.2, this implies the following result

Proposition 4.5.6 *Let $f_n \in L^\infty(\Omega)$ for $\mu(\Omega) < \infty$. If f_n converges in $L^\infty(\Omega)$, then it converges in $L^p(\Omega)$, $1 \leq p < \infty$.*

Proof Use Proposition 4.5.2. □

4.6 Approximations in L^p

4.6.1 Dense Subspaces

Let A and B be two sets such that $A \subseteq B \subseteq \mathbb{R}$. Then the set A is called a *dense* subset of B if whenever $x \in B$, either $x \in A$ or x is a limit point of A. In other words, every element in B is either in A or on the boundary of A, which means we can find a sequence $\{a_k\}$ in A such that $\lim a_k = x$. Hence for every $\varepsilon > 0$, we can

find $N > 0$ such that $|a_n - x| < \varepsilon$ for $n \geq N$, and so we can get the sequence as close as we wish to x. This, in turn, requires the notion of a distance. In the case of general linear spaces, say X, we need a general notion of distance, which can be expressed by the metric, so if X is a normed space, then one can define denseness and arbitrary closeness.

Let X be a normed space, and let $Y \subseteq X$ be a subspace of X. Then Y is a *dense subspace* of X if for every $f \in X$ there exists a sequence $f_n \in Y$ such that f_n converges to f in the norm of X, i.e.,

$$\|f_n - f\|_X \to 0.$$

The most famous example is the *Weierstrass Approximation Theorem*, which states that the collection of polynomials in $[a, b]$ is dense in the collection of continuous functions in $[a, b]$. The advantage of this concept is that one can approximate the functions defined on some space by other functions defined on a subspace. Approximating functions by functions of more regularity is of great importance to the theory of partial differential equations and some areas of applied mathematics.

As observed in Chap. 2, measurable function are "nearly" continuous. This is the idea of Lusin Theorem. Due to the continuity of the integral and integrability of measurable functions, we established the result that Lebesgue integrable functions could be approximated by continuous functions. In the terminology of this chapter, this says that functions in L^1 can be approximated by continuous functions.

Proposition 4.6.1 *The collection of simple functions in L^p is dense in L^p for all $1 \leq p$.*

Proof Let $f \in L^p(\Omega)$ for $1 \leq p < \infty$. WLOG, we can assume $f \geq 0$ since a general f can be written as $f = f^+ - f^-$. Now since f is measurable, by SAT there exists a sequence of simple functions $\{\varphi_n\} \in L^p(\Omega)$ such that $\varphi_n \nearrow f$ a.e. and $|\varphi_n| \leq f$ on Ω for all n. So we have

$$|\varphi_n - f|^p \leq 2^{p+1} |f|^p .$$

Applying DCT, we conclude that $\varphi_n \longrightarrow f$ in L^p. This completes the proof. □

4.6.2 Density Results in L^p

Theorem 2.4.3 asserts that simple functions can be approximated by continuous functions. Moreover, since simple functions are linear combinations of characteristic functions defined on measurable sets, and since measurable sets can be approximated by open and closed sets, we infer that simple functions can be approximated by step functions in L^p. Indeed, let

$$\varphi(x) = \sum_{j=1}^{n} a_j \chi_{A_j}(x)$$

be a simple function, where A_j are collection of measurable sets. Then, by the First Littlewood Principle, there exists a finite disjoint collection $\{I_j\}_{j=1,\ldots,n}$ of open intervals such that

$$U = \bigcup_{j=1}^{n} I_j$$

and

$$\mu(A \setminus U) + \mu(U \setminus A) < \epsilon^p.$$

For simplicity, let

$$\varphi(x) = \chi_A(x),$$

and consider the following step function

$$s(x) = \chi_U(x).$$

We see that

$$\|\varphi - s\|_p = \left(\int |\chi_U(x) - \chi_A(x)|^p \right)^{\frac{1}{p}} = (\mu(A \setminus U) + \mu(U \setminus A))^{\frac{1}{p}} < \epsilon.$$

So simple functions can be approximated by step functions. Therefore, we obtain the following:

Corollary 4.6.2 *Step functions in $L^p(\Omega)$ are dense in $L^p(\Omega)$ for $\Omega \subseteq \mathbb{R}^n, \mu(\Omega) < \infty$ and $1 \leq p < \infty$.*

Now, we use Lusin Theorem to obtain a continuous function $g(x)$ that linearize $s(x)$ in each I_j. Geometrically, this can be achieved by connecting polygonal lines between values of a step function

$$s(x) = \sum_{j=1}^{n} a_j \chi_{II}(x).$$

such that

$$\mu(E) < \left(\frac{\epsilon}{2M} \right)^p,$$

where

$$E = \{x : g(x) \neq s(x)\}$$

and $M = \max(s(x))$. But

$$|g - s|^p \leq [2^p M]^{\frac{1}{p}} = 2M^{\frac{1}{p}}.$$

Hence

$$\|s - g\|_p < \epsilon.$$

Therefore, we conclude that the space of continuous functions on Ω is dense in $L^p(\Omega)$.

But we have a little more to add here. Consider the space of all continuous functions f on $\mathbb{R}$, denoted by $C(\mathbb{R})$, and the space of all continuous functions on $\mathbb{R}$ that have compact support, denoted by $C_c(\mathbb{R})$. It is clear that

$$C_c(\mathbb{R}) \subset C(\mathbb{R});$$

nevertheless we have the following result.

Theorem 4.6.3 *The space $C_c(\Omega)$ is dense in $L^p(\Omega)$ for $\Omega \subseteq \mathbb{R}^n$, and $1 \leq p < \infty$.*

Proof The previous theorem asserts that simple functions in L^p are dense in L^p. Note here that for every $\varphi \in L^p$,

$$\mu(\{x \in \Omega : \varphi(x) \neq 0\}) < \infty.$$

If not, then $\varphi(x) = c \neq 0$ on some set $E \subset \Omega$, $\mu(E) = \infty$, so φ is not in L^1. Therefore, every simple function in L^p is of compact support, say E. Using Lusin Theorem, we can find a continuous function g and a closed set $F \subseteq E$, hence compact, such that

$$\mu(E \setminus F) < \epsilon$$

and $\varphi = g$ on F. Let

$$|g| \leq \|\varphi\|_\infty < \infty.$$

It follows that $g \in C_c$, and for $\epsilon > 0$

$$\|g - \varphi\|_p^p = \int\limits_{E\setminus F} |g - \varphi|^p + \int\limits_F |g - \varphi|^p \leq \int\limits_{E\setminus F} 2^p \|\varphi\|_\infty^p + 0 \leq 2^p \|\varphi\|_\infty^p \epsilon.$$

Rescaling ϵ, we conclude that φ is approximated by g, and therefore $g \in C_c \cap L^p$. □

The theorem could have been proved using a different approach. If f is unbounded in Ω, we need to truncate f by the following sequence

$$f_n = \begin{cases} f(x) & |f(x)| \leq n \\ 0 & |f(x)| > n \end{cases}.$$

Then, the sequence $\{f_n\}$ is bounded for all n and thus we can approximate it by a continuous function g using Lusin Theorem. If f is of infinite support, i.e.,

$$\mu(\{x \in \Omega : f(x) \neq 0\}) = \infty,$$

then, we need to support f by the following sequence

$$f_n = \begin{cases} f(x) & \|x\| \leq n \\ 0 & \|x\| > n \end{cases}.$$

In $\mathbb{R}$, the sequence can be written as

$$f_n = f(x) \cdot \chi_{[-n,n]}(x).$$

Note that the functions $\{f_n\}$ are of compact support for all n, so by Lusin Theorem we can approximate f by a continuous function with compact support.

The result in Theorem 4.6.3 simply says that we can always approximate L^p functions by continuous functions with compact support. This is a very interesting result since it provides us with more flexibility in dealing with complicated problems regarding L^P functions and clarifies many aspects of the topological structure of L^p spaces. The previous theorem implies that functions in L^p can be approximated by simple functions in L^p, which in turn, can be approximated by continuous functions with compact support, thanks to Lusin Theorem.

4.6.3 Density Results in L^∞

For L^∞ spaces, since all functions in L^∞ are essentially bounded, the simple functions play a dominant role as in the following result.

Proposition 4.6.4 *Simple functions in $L^\infty(\Omega)$ are dense in $L^\infty(\Omega)$, for $\mu(\Omega) < \infty$.*

Proof Let $f \in L^\infty(\Omega)$. Then there exists $M > 0$ such that $|f| \leq M$ on $\Omega \setminus E$ for some null set E, $\mu(E) = 0$. By SAT, we can find a sequence of simple functions $\varphi_n \longrightarrow f$ uniformly on $\Omega \setminus E$, hence $\varphi_n \longrightarrow f$ in L^∞, i,e. $\|f - \varphi_n\|_\infty \longrightarrow 0$. So, for every $\epsilon > 0$, we can find φ_N such that $\|f - \varphi_N\|_\infty < \epsilon$. □

Recall that Corollary 4.6.2 concluded that step functions are dense in L^p because they are dense in the collection of simple functions with respect to the $p-$norm. In the $\infty-$ norm, the situation is quite different. We cannot approximate a simple function by a step function in a set of measure less than ϵ. The value of f at this set could be large so that the supremum will not be arbitrary small in the domain. Similarly, in Lusin Theorem, the statement is not talking about a set of measure zero, but a set of measure $\epsilon > 0$, hence, if we can find a continuous function g such that

$f = g$ on a set of measure less than ϵ, then the two functions are not equal in L^p spaces and the supremum of $|f - g|$ may not be arbitrarily small. We discussed this idea when indicating the difference between almost everywhere property and the "nearly" concept. It turns out that, unlike the $p-$norm where the size of sets can be taken as small as needed over which the integrals can be made small, the $\infty-$ norm is too restrictive to obtain small values when dealing with supremum. Indeed, let $a < r < b$ and

$$f(x) = \chi_{[a,r]}(x) \in L^{\infty}[0, 1].$$

Let $\epsilon > 0$, then

$$\left\| f - \chi_{[a,r+\epsilon]}(x) \right\|_{\infty} = 1. \tag{4.6.1}$$

Another reason is that in L^{∞} the convergence is uniform, and a continuous function on a compact set is uniformly continuous. The sequence of uniformly continuous functions converges uniformly to a continuous function, but not all functions in L^{∞} are continuous. So, for arbitrary $f \in L^{\infty}[a, b]$, we cannot guarantee a sequence of continuous function that converges to f. Therefore, the collection of continuous functions cannot be dense in $L^{\infty}[a, b]$.

4.7 Bounded Linear Functionals on L^p

4.7.1 Notion of Functional

The following definition provides a generalization for functions.

Definition 4.7.1 (*Functional*) Let X and Y be two normed vector spaces. Then, a mapping $T : X \longrightarrow Y$ is said to be an operator. If $Y = \mathbb{R}$ then T is called $functional$.

Moreover, T is said to be a $linear$ if the following holds:

1. $T(cx) = cT(x)$.
2. $T(x + y) = T(x) + T(y)$.

Definition 4.7.2 1. **Continuous Linear Operator**: A linear operator T is said to be $continuous$ at $x_0 \in X = Dom(T)$ if for every $\epsilon > 0$, there exists $\delta > 0$ such that if

$$\|x - x_0\| \leq \delta,$$

then

$$\|T(x) - T(x_0)\| \leq \epsilon.$$

2. **Bounded Linear Operator**: Let $T : X \longrightarrow Y$ be a linear operator, and X and Y are two normed spaces with norms $\|\cdot\|_X$ and $\|\cdot\|_Y$, respectively. Then T is called

bounded linear operator if there exists $M \in \mathbb{R}$, such that for all $x \in X$,

$$\|Tx\|_Y \leq M \|x\|_X .$$

The definition of continuity seems natural and extends the definition of a real-valued function on $\mathbb{R}^n$. The definition of boundedness can be written as

$$\frac{\|Tx\|}{\|x\|} \leq M. \tag{4.7.1}$$

If there is no such M to satisfy (4.7.1), then we say that the operator is unbounded. The smallest possible number satisfying (4.7.1) is the supremum

$$\|T\| = \sup_{\|x\| \neq 0} \frac{\|Tx\|}{\|x\|}.$$

This can also be written as

$$\|T\| = \sup_{\|x\|=1} \|T(x)\| .$$

Clearly $\|T\| \geq 0$, and $\|T\| = 0$ if $T(x) = 0$. Moreover,

$$\|cT\| = \sup \|c.T(x)\| = |c| \, \|T(x)\| ,$$

and

$$\|T_1 + T_2\| = \sup \|T_1(x) + T_2(x)\| \leq \sup \|T_1(x)\| + \sup \|T_2(x)\| = \|T_1\| + \|T_2\| .$$

The next result shows that, surprisingly, these two concepts, i.e., continuity and boundedness, are equivalent for linear functionals.

Theorem 4.7.3 *Let $T : Dom(T) = X \longrightarrow Y$ be a linear operator from a normed space X to a normed space Y. Then, T is bounded if and only if T is continuous at all $x \in X$.*

Proof Let $T : Dom(T) = X \longrightarrow Y$ be linear. Suppose it is bounded, and let $x_0 \in X$. Then for every $\epsilon > 0$

$$\|T(x) - T(x_0)\| = \|T(x - x_0)\| \leq \|T\| \, \|x - x_0\| . \tag{4.7.2}$$

Since T is bounded, $\|T\| = M$. So let $\delta = \dfrac{\epsilon}{M}$, and continuity holds. For the reverse direction, let T be continuous for all $x \in X$, in particular at $x = 0$. Then, there exists $\delta > 0$ such that if $\|x\| \leq \delta$, then $\|Tx\| \leq 1$. So, for any $x \in X$ we have

$$\left\| T\left(\frac{\delta x}{\|x\|} \right) \right\| \leq 1.$$

Therefore

$$\|Tx\| \leq \frac{1}{\delta} \|x\|.$$

Hence T is bounded. □

4.7.2 Integral Functional on L^p

Let us discuss the following functional on L^p.

Example 4.7.4 (*Integral Functional*) Let $g \in L^q(E)$ for measurable $E \subseteq \mathbb{R}$. Let $T : L^p \longrightarrow \mathbb{R}$, $1 \leq p < \infty$ be a functional defined by

$$T(f) = \int_E fg.$$

The functional is clearly linear. Moreover, $fg \in L^1$ by Holder's inequality, so the functional T is well defined. Then we have

$$|T(f)| \leq \int_E |f|\,|g| \leq \|f\|_p \|g\|_q.$$

Since $g \in L^q(E)$, $\|g\|_q < \infty$. We see that T is bounded and

$$\|T\| \leq \|g\|_q.$$

Now, consider the function

$$f = |g|^{q/p} \operatorname{sgn}(g).$$

Then

$$|f|^p = |g|^q \in L^1,$$

i.e.,

$$\int_E |f|^p = \int_E |g|^q.$$

This can be written as

$$\|f\|_p = (\|g\|_q)^{q/p}.$$

On the other hand, since $\frac{q}{p} + 1 = q$ (verify) and

$$|g| = g \cdot \operatorname{sgn}(g),$$

we have

$$fg = |g|^{\frac{q}{p}+1} = |g|^q \in L^1.$$

Hence

$$T(f) = \int |g|^q = (\|g\|_q)^q = (\|g\|_q)^{q/p} \|g\|_q = \|f\|_p \|g\|_q .$$

Therefore, we conclude that T is bounded and

$$\|T\| = \|g\|_q .$$

4.7.3 Riesz Representation Theorem

Example 4.7.4 shows that for a space L^p, $1 \leq p < \infty$, for every $g \in L^q$ we can find a bounded integral functional T on L^p such that $\|T\| = \|g\|$, or in other words, there are as many bounded linear functionals on L^p as there are functions in the conjugate space L^q, which reflects the abundance of the bounded linear functionals on L^p spaces. But what about the converse? how many functions does L^q have? In an astonishing theorem, Riesz proved that the converse is true as well, and there are as many functions $g \in L^q$ as there are bounded linear functionals on L^p of the same form of the functional in Example 4.7.4. This shows that the integral functional in the example is the only bounded linear functional we can have on L^p and it defines a one-one correspondence between the bounded linear functionals on L^p and the functions on the conjugate space L^q. So basically one can identify members of L^p with their representations in L^q. Before establishing this great result, we need the following lemma, taking into account that whenever we refer to q, we mean the conjugate number of p such that $p^{-1} + q^{-1} = 1$.

Lemma 4.7.5 *Let $g \in L^1$ and $\int g\varphi \leq M \|\varphi\|_p$ for $1 \leq p < \infty$ for every simple function $\varphi \in L^p$. Then $g \in L^q$ and $\|g\|_q \leq M$.*

Proof Since $|g|$ is a nonnegative measurable, by SAT (Theorem 2.4.4), there exists a sequence of simple functions $\varphi_n \longrightarrow |g|$, hence

$$\varphi_n^q \longrightarrow |g|^q . \tag{4.7.3}$$

Define

$$h_n(x) = \operatorname{sgn}(g)\varphi_n^{q-1}.$$

Then $h_n \in L^p$ and

$$\int_E \varphi_n^q \leq \int_E g f_n \leq M \|h_n\|_p = M \left(\int_E \varphi_n^q \right)^{1/p}.$$

Dividing both sides by $\left(\int_E \varphi_n^q\right)^{1/p}$ gives

$$\|\varphi_n\|_q \leq M,$$

or

$$\int_E |\varphi_n|^q \leq M^q. \tag{4.7.4}$$

Given (2.4.1), we apply MCT or Fatou's Lemma (Lemma 3.4.5) in (4.7.4) and the result follows. □

Now, we state and prove the Riesz Representation Theorem.

Theorem 4.7.6 (Riesz Representation Theorem for L^p Spaces) *For every bounded linear functional T on $L^p(E)$, $E \subseteq \mathbb{R}$, $1 \leq p < \infty$, there exists a unique $g \in L^q(E)$ such that*

$$T(f) = \int_E fg, \quad \|T\| = \|g\|_q.$$

Proof We begin with the characteristic function to define T, then we extend T to simple functions, then we define it for $f \in L^p$. Assume first that $\mu(E) < \infty$. In order to express any $T \in (L^p(E))^*$ in integral form, we use the Fundamental Theorem of Calculus (Theorem 3.6.8), which states that a function is absolutely continuous on $[a, b]$ if and only if it is the indefinite integral of its derivative. It suffices to prove the result for $E = [a, b]$ since we can extend the result to a Borel set B such that $\mu(B \setminus E) = 0$. Let $T \in (L^p(E))^*$. So we define

$$f(x) = \chi_{[a,b]}(x).$$

Then $T(f)$ is well defined since $f \in L^p$. Now, consider the function $\chi_{[a,x)}$ for $x < b$. Let

$$T(\chi_{[a,x)}) = h(x)$$

We consider a partition of disjoint subintervals $\{(a_k, b_k)\}$ of $[a, b]$. Then it is easy to see that T is absolutely continuous on $[a, b]$, hence by Theorem 3.6.8 we can write it as an integral of the form

$$T(\chi_{[a,x)}) = h(x) = \int_a^x h',$$

or

$$T(\chi_E) = \int_E h' = \int_E g\chi_E$$

for some $g \in L^q$. Now, linearity of T gives

$$T(\varphi) = \int_E \varphi g$$

for all simple functions φ. Since T is bounded, we have

$$|T(\varphi)| \leq \|T\| \, \|\varphi\|_p .$$

By Lemma 4.7.5, $g \in L^q$. Let $f \in L^p$. Then by SAT, there exists $\varphi_n \nearrow f$, and $\|\varphi_n - f\|_p \longrightarrow 0$. Hence by DCT

$$T(f) = \lim T(\varphi_n) = \lim \int_E \varphi_n g = \int_E fg.$$

Let $f \in L^p$, then we can apply the previous result to obtain

$$T(f\chi_E) = \int_E f\chi_E g.$$

Now, let $\mu(E) = \infty$. Write $E = \bigcup E_k$ where $E_k \subseteq E_{k+1}$, and each $\mu(E_k) < \infty$. Then, on each χ_{E_k} we write

$$T(f\chi_{E_k}) = \int_E f\chi_{E_k} g_k,$$

where $\|g_k\|_q \leq \|T\|$, and $g_k \in L^q(E)$. Note that from the construction of g_k in the previous argument, we see that $g_k(x)$ vanishes outside E_k and $g_k(x) = g_{k+1}(x)$ for almost all $x \in E_k$. So we can define a measurable function g such that $g(x) = g_k(x)$ for every k and almost all $x \in E_k$. Letting $k \longrightarrow \infty$, then Fatou's Lemma yields

$$\|g\|_q = \lim \|g_k\|_q \leq \|T\| .$$

Hence, $g \in L^q$. Note that $f\chi_{E_k} \longrightarrow f$ a.e., so we have

$$\left\| f\chi_{E_k} - f \right\|_p \longrightarrow 0.$$

Consequently,

$$T(f) = \lim T(f\chi_{E_k}) = \lim \int_E f\chi_{E_k} g_k = \lim \int_{E_k} fg = \int_E fg.$$

Uniqueness of g arises from the fact that

$$\int f(g_1 - g_2) = 0$$

implies $g_1 = g_2$ a.e. For $p > 1$, let

$$f = |g_1 - g_2|^{q-2}(g_1 - g_2).$$

For $p = 1$, let $f = \operatorname{sgn}(g_1 - g_2)$. □

4.7.4 The Case $p = \infty$

It is worth noting that the theorem doesn't hold for the case $p = \infty$. This amounts to saying that there exists a bounded linear functional on L^∞ such that it cannot be represented as an integral functional of the form of Example 4.7.4. Consider the following functional $F : L^\infty[0, 1] \to \mathbb{R}$ defined by

$$F(y) = y(0), \tag{4.7.5}$$

for every continuous function y, where $y \in L^\infty[0, 1]$, taking into account that $L^\infty[0, 1]$ is endowed with the supremum norm $\|\cdot\|_\infty$. The functional is bounded and linear (verify). Let (y_n) be a sequence of continuous functions on $[0, 1]$ defined as $y_n(0) = 1$ and $y_n(x) \searrow 0$ on $[0, 1]$. Then $|y_n| \leq 1$, and

$$F(y_n) = 1. \tag{4.7.6}$$

Now, suppose there exists $g \in L^1$ such that

$$F(y) = \int_0^1 y_n(x)g(x)dx.$$

Note that $y_n(x)g(x) \to 0$ and

$$|g(x)y_n(x)| \leq g(x) \in L^1[0, 1].$$

So by DCT we get

$$\int_0^1 y_n(x)g(x)dx \to 0.$$

But this contradicts with (4.7.6). This implies that there is no such g.

In Sect. 4.6, we also observed that the space of bounded continuous functions is dense in L^p for all $1 \leq p$, but not in L^∞. This basically indicates that the space L^∞ has a deep structure, and the representation of its members is a rather complicated process that can't be identified using one-one correspondence.

4.8 Problems

1. Prove the generalized Holder's inequality: Let

$$\frac{1}{q_1} + \frac{1}{q_2} + \cdots + \frac{1}{q_n} = 1,$$

where $1 \leq q_i \leq \infty$. If $g_i \in L^{q_i}$ then $g_1 \ldots g_n \in L^1$ and

$$\|g_1 \ldots g_n\| \leq \prod_{i=1}^{n} \|g_i\|_{q_i}.$$

2. Show that Minkowski's Inequality holds in infinite case:

$$\left\| \sum_{j=1}^{\infty} f_j \right\|_p \leq \sum_{j=1}^{\infty} \|f_j\|_p.$$

3. If f is bounded and measurable function on $[0, 1]$, show that

$$\lim_{p\to\infty} \|f\|_p = \|f\|_\infty.$$

4. If $f, g \in L^\infty$, show that

$$\|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty.$$

5. Let f and g be bounded measurable functions on $[0, 1]$. Show that

$$\|fg\|_\infty \leq \|f\|_\infty \cdot \|g\|_\infty.$$

6. Show that the Minkowski Inequality turns into equality, i.e.,

$$\|f+g\|_p = \|f\|_p + \|g\|_p\,,$$

if and only if $f = cg$ for $c \in \mathbb{R}$.

7. Let $f \in L^p[a,b]$ and $1 \le p = s.t.$ Prove that $f^t \in L^s[a,b]$ and $f^s \in L^t[a,b]$.
8. If $1 \le p < q$. and $f \in L^p(\mathbb{R}) \cap L^q(\mathbb{R})$, show that $f \in L^r(\mathbb{R})$ for $p < r < q$.
9. Let $f \in L^r(E)$ for $\mu(E) < \infty$. If $1 \le p < r$, show that

$$\|f\|_p \le (\mu(E))^{\frac{1}{p}-\frac{1}{r}}\,\|f\|_r\,.$$

10. Give an example of a sequence $\{f_n\}$ that converges uniformly to f but not in L^p, and an example of a sequence $\{g_n\}$ that converge to f p.w. but not in L^p.
11. (a) Give an example to show that none of the spaces $L^r(\mathbb{R})$ and $L^s(\mathbb{R})$ for $1 \le r < s$ is included in the other one.
(b) Give an example to show that $L^\infty(\mathbb{R})$ is not included in $L^p(\mathbb{R})$ for any $1 \le p < \infty$.
12. If $f^2, g^2 \in L^1(\mathbb{R})$, show that $fg \in L^1(\mathbb{R})$.
13. For every $1 \le p < q$, find a function $f \in L^p[0,1]$ but not in $L^q[0,1]$.
14. Let $f \in L^r$ and $g \in L^s$, where $\dfrac{1}{s} + \dfrac{1}{r} = \dfrac{1}{t}$. Show that $fg \in L^t$ and

$$\|fg\|_t \le \|f\|_r \cdot \|g\|_s\,.$$

15. For each of the following functions, determine the value of p for which $f \in L^p[0,1]$.
(a) $f(x) = x^{-\frac{1}{k}}\chi_{[a,b]}(x)$.
(b) $f(x) = \ln x$.
16. Let $T : \ell^p \longrightarrow \ell^1$, $T(x_n) = n^r x_n$. Determine all values of r for which T is bounded.
17. If $\mu(A) < \infty$, show that every square-integrable function on A is integrable on A.
18. In the previous problem, verify by an example that the condition $\mu(A) < \infty$ is essential.
19. Give an example of a sequence $\{x_n\}$ such that $x_n \in \ell^3$ but $x_n \notin \ell^2$.
20. Determine the values of p such that the following sequence become in ℓ^p.
(a) $x_n = \dfrac{1}{\sqrt{n}\log n}$.
(b) $x_n = \dfrac{1}{\sqrt[4]{n^3+1}}$.
(c) $x_n = e^{-n}$.
21. (a) Show that

$$\bigcap_{p>1} \ell^p \neq \ell^1.$$

(b) Show that

$$\bigcup_{p>1} L^p[0,1] = L^1[0,1].$$

22. Let $f_n \in L^2[a,b]$ and $\lim f_n = 0$ a.e. on $[a,b]$. Show that

$$\lim \int_{[a,b]} f_n = 0.$$

23. Let f and g be two nonnegative and integrable functions on $I = [0,1]$. If $fg \geq 1$, prove that

$$(\int_I f)(\int_I g) \geq 1.$$

24. Let $f_n, f \in L^p$ with $0 < p < 1$, and $f_n \longrightarrow f$ in L^p. Show that

$$|f_n|^p \longrightarrow |f|^p$$

in L^1.

25. Let $f_n, f \in L^p$ and $f_n \longrightarrow f$ in L^p, $p \geq 1$. Show that

$$|f_n|^p \longrightarrow |f|^p$$

in L^1.

26. **Brezis-Lieb Lemma**: Show that

$$\|f_n\|_2^2 - \|f_n - f\|_2^2 \longrightarrow \|f\|_2 .$$

27. Let $f \in L^1$. Show that there exists $g \in L^\infty$ such that $\int fg = \|f\|_1$.
28. Let $f_n \in L^p$ and $f_n \longrightarrow f$ a.e.. If $\|f_n\|_p \leq M$, show that

$$\int f_n g \longrightarrow \int fg$$

for all $g \in L^q$.

29. Let $A \subseteq \mathbb{R}$ with $\mu(A) < \infty$.
(b) Show that for $1 \leq r < s \leq \infty$,

$$\|f\|_r \leq (\mu(A))^{\frac{1}{r}-\frac{1}{s}} \|f\|_s .$$

(c) Show that the condition $\mu(A) < \infty$ is necessary.

30. (a) Prove the *Fatou's Lemma of* $p-$ *norm*: Let $\{f_n\}$ be a sequence of nonnegative measurable functions, and $f_n \longrightarrow f$ a.e. on a set E. Then

$$\|f\|_p \leq \underline{\lim} \|f_n\|_p .$$

(b) Apply the above Lemma to prove Proposition 4.5.4.

31. Let g be integrable function on $[0, 1]$. If

$$\left| \int fg \right| \leq M \|f\|_p$$

for some $M > 0$, $1 < p < \infty$, and for every bounded function f, show that $g \in L^q$ and $\|g\|_q \leq M$.

32. (a) Let $f_n \in L^p$ for $p \geq 1$, and $f_n \longrightarrow f$ p.w. on A, $\mu(A) < \infty$. If $\|f_n\| \leq M$, show that $f \in L^p$.
(b) Show that the result in (a) also holds for $p = \infty$.

33. Let $f_n \in L^p$ for $p \geq 1$. If $\sup |f_n| \in L^p$, and $f_n \longrightarrow f$ a.e., show that $f \in L^p$ and $f_n \xrightarrow{p} f$.

34. Let $f_n \in L^p(A)$, $\mu(A) < \infty$ and let $\sum^{\infty} \|f_n\|_p < \infty$. Show that

$$\sum^{\infty} |f_n(x)| < \infty.$$

35. In the previous problem, show that if $f(x) = \sum^{\infty} f_j(x)$ a.e., then $f \in L^p$ and

$$\|f\|_p \leq \sum \|f_n\|_p .$$

36. Let $f_n \xrightarrow{p} f$ for $1 \leq p < \infty$, and let $|g_n| \leq M$ for some sequence of measurable functions g_n and $g_n \longrightarrow g$ p.w. Show that

$$f_n g_n \longrightarrow fg \text{ in} L^p.$$

37. **Convergence in Measure**. A sequence $\{f_n\}$ is said to converge to f in measure on A, denoted $f_n \xrightarrow{\mu} f$, if for every $\epsilon > 0$,

$$\mu(\{x \in A : |f_n(x) - f(x)| > \epsilon\} \longrightarrow 0.$$

(a) Show that if $f_n \xrightarrow{p} f$ on A then $f_n \xrightarrow{\mu} f$ on A.
(b) Let $f_n \in L^p$ and $f_n \xrightarrow{\mu} f$. If $|f_n| \leq g \in L^p$, show that $f_n \xrightarrow{p} f \in L^p$.
(c) Let $f_n \in L^P$ and $f_n \xrightarrow{\mu} f$. If $\|f_n\|_p \longrightarrow \|f\|_p$, show that $f_n \xrightarrow{p} f \in L^p$.

38. Let $f_n, f \in L^p$ and $g_n, g \in L^q$. If $f_n \xrightarrow{p} f$ and $g_n \xrightarrow{q} fg$ in L^q, show that $f_n g_n \longrightarrow fg$ in L^1.

39. Let $f_n \in L^P$ and $g_n \in L^q$. If $f_n \xrightarrow{p} f$ and $g_n \xrightarrow{q} g$, show that

$$f_n g_n \xrightarrow{\mu} fg.$$

40. Let g be integrable function on $[0, 1]$. If $\left|\int fg\right| \leq M \|f\|_1$ for some $M > 0$, and for every bounded function f, show that $g \in L^\infty[0, 1]$ and $\|g\|_\infty \leq M$.
41. Let $g \in L^1$ and $\int g\varphi \leq M \|\varphi\|_p$ for $1 \leq p < \infty$ for every simple function $\varphi \in L^p$. Show that $g \in L^q$ and $\|g\|_q \leq M$.
42. Let $s(x) = \chi_{[a,b]}(x)$. Construct a sequence of continuous functions on $\mathbb{R}$ and converging to $s(x)$ in $L^p(\mathbb{R})$.
43. Let $f \in L^p$. Show that f can be approximated by a bounded measurable function in L^p.
44. Show that every bounded measurable function in L^p can be approximated by a step function in L^p.
45. Show that L^∞ is dense in L^p for $1 \leq p < \infty$.
46. Show that the functional defined in (4.7.5) is bounded and linear.
47. Show that T is bounded and find $\|T\|$.
    (1) $T : \ell^p \longrightarrow \ell^p$, $T(x_1, x_2, x_3, \ldots) = (x_2, x_3, x_4, \ldots)$.
    (2) $T : \ell^1 \longrightarrow \ell^1$, $T(x_n) = \dfrac{x_n}{2^n}$.
48. Let $a < r < s < \infty$. Define the linear operator $T : L^s[a, b] \longrightarrow L^r[a, b]$, $T(f) = f$. Show that T is bounded.
49. Let $y \in \ell^\infty$ and let $f(x) = \sum x_k y_k$.
    (a) Show that $f \in (\ell^1)^*$.
    (b) Show that $|f| = \|y\|_\infty$.

Chapter 5
Abstract Measure Theory



5.1 Generalization of Measure Theory

5.1.1 Introduction

The objective of this chapter is to extend measure theory by generalizing the concept of measure to include more general sets other than sets in $\mathbb{R}^n$. Of course, this newly defined measure won't be Lebesgue measure, so in this general measure theory, we are generalizing the measure function and the measurable sets. Since we will be dealing with abstract spaces with no clear idea of the identity of elements, it would be hard to employ a definition of measure without making sure the axioms of the measure will be satisfied. In this view, it is customary to deal with a nonempty set X and then consider the σ-algebra on X. This has been introduced in Definition 1.3.6 as the collection of subsets of X that is closed under countable unions and set complement, and includes the empty set $\emptyset$. This σ-algebra of X shall be denoted by $\Sigma(X)$, or simply Σ. We say that a collection Σ of subsets of a nonempty set X is said to be a σ-algebra on X if

(1) $\emptyset \in \Sigma$,
(2) If $E \in \Sigma$ then $X \setminus E \in \Sigma$.
(3) If $\{E_k\}$ is a countable collection of sets in Σ, then $\bigcup_k E_k \in \Sigma$.

$\{X, \emptyset\}$ is clearly a σ-algebra on X, and the collection of power set $\mathcal{P}(X)$ consisting of all subsets of X is also a σ-algebra on X. We also saw that the collection of all Lebesgue measurable subsets of X forms a σ-algebra on X.

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5.1.2 Measurable and Measure Spaces

Definition 5.1.1 Let X be a nonempty set, and its σ-algebra is Σ. Then the pair (X, Σ) is called a *measurable space*. A subset A of X is called *measurable* if $A \in \Sigma$.

Now that we identified our measurable sets, we can use it to define our measure function.

Definition 5.1.2 Let (X, Σ) be a measurable space. A function $\mu : \Sigma \to [0, \infty]$ is called a *measure* if

(1) $\mu(\emptyset) = 0$,
(2) For any countable collection $\{E_j\}$ of pairwise disjoint sets in $\mathcal{M}$ we have $\mu(\bigcup_j E_j) = \sum_j \mu(E_j)$.

The triple (X, Σ, μ) is called a *measure space*. The classical example that we already have and discussed is $(\mathbb{R}, \mathcal{L}, \mu)$, where $\mathcal{L}$ is the collection of all Lebesgue measurable sets in $\mathbb{R}$ and μ is the Lebesgue measure. If $\mu(X) < \infty$ then μ is a finite measure, while if $\mu(X) = \infty$, but

$$X = \bigcup_{i=1} A_i,$$

and $\mu(A_i) < \infty$ for every i then μ is called a σ-*finite measure*.

Note that the Lebesgue measure μ on $X = [0, 1]$ is finite since every member of Σ, including X, is of finite measure. If $X = \mathbb{R}$, then $\mu(\mathbb{R}) = \infty$, but we can write $\mathbb{R}$ as

$$\mathbb{R} = \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} [-n, n],$$

where $\mu(A_n) < \infty$ for all n. Hence μ is not a finite measure on $\mathbb{R}$ but it is σ-finite measure. On the other hand, if we choose the measure μ to be the counting measure (defined in Problem 1.8.4), then $\mu(\mathbb{R}) = \infty$, and $\mathbb{R}$ can't be written as the union of countable union of sets of finite measure. Therefore, we say that μ is an infinite measure on $\mathbb{R}$.

Almost all properties of a measure μ are expected to be the same to those for the Lebesgue measure studied in Chap. 1. Moreover, the properties of measurable function in Chap. 2 extends to measurable functions on abstract measure spaces, and the results obtained for the Lebesgue integration on the real line extends to general integration on abstract measure spaces, and their proofs are almost the same, so we shall not repeat the results here.

We give a definition for the measurable function on X.

5.1.3 Measurable Functions

Definition 5.1.3 (*Measurable Function*) Let (X, Σ, μ) be a measure space. A function $f : X \longrightarrow \overline{\mathbb{R}}$ is called *measurable* with respect to μ, if given any open set $O \subseteq \mathbb{R}$, $f^{-1}(O) \in \Sigma$ (i.e., measurable).

Proposition 5.1.4 *Let f be an extended real-valued function on a measurable space on (X, Σ). Then the following are equivalent*

(1) f is measurable.
(2) $f^{-1}(F)$ is measurable set for every closed set F.
(3) $f^{-1}([-\infty, r)) = \{x \in \Omega : f(x) < r\}$ is measurable set for every $r \in \mathbb{R}$.
(4) $f^{-1}([-\infty, r]) = \{x \in \Omega : f(x) \leq r\}$ is measurable set for every $r \in \mathbb{R}$.
(5) $f^{-1}((r, \infty]) = \{x \in \Omega : f(x) > r\}$ is measurable set for every $r \in \mathbb{R}$.
(6) $f^{-1}([r, \infty]) = \{x \in \Omega : f(x) \geq r\}$ is measurable set for every $r \in \mathbb{R}$.

Proposition 5.1.5 *Let f, g, f_n be extended real-valued measurable functions on a measurable space (X, Σ). Then*

(1) $cf + g$, $f \cdot g$, $\dfrac{f}{g}$, $f \vee g$, $f \wedge g$ are all measurable.
(2) $\inf_n f_n$, $\limsup_{n\to\infty} f_n$, and $\liminf_{n\to\infty} f_n$ are measurable.
(3) $\{f_n\}$ converges to f p.w. if and only if $\limsup f_n = \liminf f_n = f$.

Theorem 5.1.6 (Simple Approximation Theorem (SAT)) *Let f be a measurable function on a set $E \in \Sigma$ in a measurable space (X, Σ). Then there exists a sequence of simple functions $\{\varphi_n\}$ such that $|\varphi_n| \leq |f|$ for all n and $\varphi_n \longrightarrow f$ p.w. on E. If f is bounded on E then $\varphi_n \longrightarrow f$ uniformly.*

Theorem 5.1.7 (Lusin Theorem) *Let f be a measurable function on a set $E \in \Sigma$ in a measure space (X, Σ, μ). Then for every $\epsilon > 0$, there exists a continuous function g and a closed set $F \subseteq E$ such that $\mu(E \setminus F) < \epsilon$ in which $f = g$ on F.*

Theorem 5.1.8 (Egoroff Theorem) *Let $\{f_n\}$ be a sequence of measurable functions and $f_n \to f$ a.e. on a set E, $\mu(E) < \infty$ in a measure space (X, Σ, μ). Then for every $\epsilon > 0$, there exists a set A, $\mu(A) < \epsilon$ such that $f_n \longrightarrow f$ uniformly on $E \setminus A$.*

5.1.4 Integration Over Abstract Measurable Spaces

Definition 5.1.9 Let (X, Σ, μ) be a measure space. Then

(1) **Integral of Simple Function**. Let $\varphi : \Sigma \longrightarrow \mathbb{R}$,

$$\varphi(x) = n \sum_{i=1}^{n} c_i \chi_{A_i}$$

be a nonnegative simple function defined on a set A where $\{A_i\}$ is a measurable partition of a set $E \in \Sigma$ in a measure space (X, Σ, μ). Then

$$\int_E \varphi(x)dx = \sum_{i=1}^{n} c_i \mu(A_i).$$

(2) **Integral of Bounded Function**: If f is bounded and measurable defined on a measurable set $E \in \Sigma$ with finite measure (i.e., $\mu(E) < \infty$) in a measure space (X, Σ, μ). Then we define the integral of f over E as

$$\int_E f = \sup_{\varphi \le f} \int_E \varphi.$$

(3) **Integral of Nonnegative Function**: Let $f : E \longrightarrow [0, \infty]$ be a nonnegative measurable function defined on a measurable set $E \in \Sigma$ in a measure space (X, Σ, μ). Then the integral of f over E is defined as

$$\int_E f = \sup \int_E \varphi,$$

where the supremum is taken over all possible simple functions φ with $0 \le \varphi \le f$. The function f is said to be $integrable$ over E if

$$\int_E f < \infty.$$

(4) **Lebesgue Integrable Function**: Let f be a measurable function defined on a measurable set E. If f^+ and f^- both integrable over E, then f is $integrable$.

Theorem 5.1.10 *Let $f \ge 0$ be a measurable function defined on a set $E \in \Sigma$ in a measure space (X, Σ, μ). If $\{\varphi_n\}$ is an increasing sequence of nonnegative simple functions and $\varphi_n \nearrow f$, then*

$$\int_E f d\mu = \lim \int_E \varphi_n d\mu.$$

Proposition 5.1.11 *Let f and g be $\mu-$integrable over $E \in \Sigma$ in a measure space (X, Σ, μ). Then the following properties hold:*

(1) If $f \le g$ a.e. then $\int_E f d\mu \le \int_E g d\mu$.
(2) $\int_E (af + g)d\mu = a \int_E f d\mu + \int_E g d\mu$ for $a, b \in \mathbb{R}$.
(3) If $E = A \cup B$ where $A \cap B = \emptyset$, then $\int_E f = \int_A f d\mu + \int_B f d\mu$.
(4) If $f = g$ a.e. on E and f is integrable on E, then g is integrable on E and $\int_E f d\mu = \int_E g d\mu$.

(5) If $f \geq 0$ and $\int_E f\,d\mu = 0$, then $f(x) = 0$ a.e. on E.
(6) If $\mu(E) = 0$ then $\int_E f\,d\mu = 0$.

Theorem 5.1.12 (Bounded Convergence Theorem) *Let $\{f_n\}$ be a sequence of measurable functions on a set $E \in \Sigma$ in a measure space (X, Σ, μ), with $\mu(E) < \infty$ and $f_n \longrightarrow f$ p.w. on A. If there exists $M > 0$ such that $|f_n| \leq M$, then*

$$\lim \int_E f_n = \int_E f.$$

Theorem 5.1.13 (Fatou's Lemma) *Let $\{f_n\}$ be a sequence of measurable functions in a measure space (X, Σ, μ), $f_n \geq 0$, and $f_n \longrightarrow f$ a.e. on a set $E \in \Sigma$. Then*

$$\int_E f\,d\mu \leq \underline{\lim} \int_E f_n d\mu.$$

Theorem 5.1.14 (Monotone Convergence Theorem) *Let $\{f_n\}$ be a sequence of nonnegative and increasing measurable functions on $E \in \Sigma$ in a measure space (X, Σ, μ). If $\lim f_n = f$ a.e., then*

$$\lim \int_E f_n d\mu = \int_E f\,d\mu.$$

Theorem 5.1.15 (Dominated Convergence Theorem) *Let $\{f_n\}$ be a sequence of measurable functions on $E \in \Sigma$ in a measure space (X, Σ, μ). If $f_n \to f$ p.w. a.e., and is dominated by some $\mu-$integrable function g over E, that is; $|f_n| \leq g$ for all n and for almost all x in E, then f is $\mu-$integrable and*

$$\lim \int_E f_n d\mu = \int f\,d\mu.$$

The formulation of the product measure on $\mathbb{R}^n$ extends to abstract measure spaces. Let (X, Σ) and (Y, Γ) be two measurable spaces. Then the collection

$$\Sigma \otimes \Gamma = \{A \times B : \ A \in \Sigma, \ B \in \Gamma\}$$

is a σ-algebra on $X \times Y$. Now, let (X, Σ, μ) and (Y, Γ, λ) be two measure spaces. Then $\mu \times \lambda$ is a measure on the product space $X \times Y$ with $\Sigma \otimes \Gamma$. Consequently, one has the following general version of Fubini–Tonelli Theorems on abstract measure spaces.

Theorem 5.1.16 (Fubini–Tonelli Theorems) *Let* (X, Σ, μ) *and* (Y, Γ, λ) *be two* σ*-finite measure spaces.*

(1) ***Tonelli's Theorem****. If* f *is a nonegative measurable on* $(X \times Y, \mu \times \lambda)$*, then*

(a) $F(x) = \int_Y f(x, y) d\Gamma_y$ *and* $F(y) = \int_X f(x, y) d\mu_x$ *are measurable functions on* (X, μ) *and* (Y, Γ)*, respectively, and*
(b) $\int_X \left(\int_Y f d\Gamma_y\right) d\mu_x = \int_Y \left(\int_X f d\mu_x\right) dy = \int_{X\times Y} f d(\mu \times \lambda) \leq \infty$.

(2) ***Fubini's Theorem****. If* f *is* $\mu \times \lambda-$*integrable over* $X \times Y$*, then*

(1) $F(x) = \int_Y f(x, y) d\Gamma_y$ *and* $F(y) = \int_X f(x, y) d\mu_x$ *are integrable functions over* X *and* Y*, respectively, and*
(2) $\int_X \left(\int_Y f dy\right) d\mu_x = \int_Y \left(\int_X f d\mu_x\right) d\Gamma_y = \int_{X\times Y} f d(\mu \times \lambda) < \infty$.

We conclude the section with the abstract Lebesgue space $L^p(X) = L^p(X, \Sigma, \mu)$*, where* X *is a space,* Σ *is a* σ*-algebra on* X *and* μ *is a measure on* (X, Σ)*. This space is defined as the space of all* μ*-measurable functions* $f : X \to \mathbb{R}$ *such that*

$$\int_X |f|^p \, d\mu < \infty.$$

5.2 Signed Measure

5.2.1 Notion of Signed Measure

One of the main features of abstract measure theory is that measures are sometimes accepted as negative values in addition to positive values. We begin with this definition:

Definition 5.2.1 Let (X, Σ) be a measurable space. A signed measure on (X, Σ) is a function $\nu : \Sigma \longrightarrow [-\infty\infty]$ such that:

(a) ν assumes at most one of the values $+\infty$, $-\infty$.
(b) $\nu(\emptyset) = 0$.
(c) If $\{E_j\}$ is any countable collection of pairwise disjoint measurable sets then

$$\nu\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \nu(E_j),$$

where the series converges absolutely when $\nu(\bigcup_{j=1}^{\infty} E_j)$ is finite.

The first condition appears in order to avoid obtaining the unfortunate result $\infty - \infty$. In view of the above definition, we observe that every measure is a signed measure, and the difference of two measures is a signed measure. The following is the classic example of a signed measure.

Example 5.2.2 Let μ be a measure defined on a measurable space (X, Σ). Let $f : X \longrightarrow \overline{\mathbb{R}}$, a measurable with respect to the measure μ. Define the set function

$$\nu(E) = \int_E f d\mu, \quad E \in \Sigma.$$

Then it is easy to see that ν is a signed measure. Indeed, $\nu(\emptyset) = 0$, and

$$\nu\left(\bigcup_{j=1}^{\infty} E_j\right) = \int_{\bigcup E_j} f d\mu = \sum_{j=1}^{\infty} \int_{E_j} f d\mu = \sum_{j=1}^{\infty} \nu(E_j).$$

It is understood that if $f \in L^1(\mu)$ then ν is a finite measure.

One interesting point about the example above is that it shows that sign measures can be represented by integrals. In fact, we will see later that every signed measure can be represented by an integral provided some conditions hold.

It is important to observe that signed measure don't satisfy the monotone property, i.e., if $A \subseteq B$ then we may not have $\nu(A) \leq \nu(B)$. This should be understood from the fact that ν may accept negative values on subsets of B.

Example 5.2.3 Let $f(x) = 1 - x$ be defined on $\mathbb{R}$ with the Lebesgue measure, and let $A = [0, 1]$ and $B = [0, 3]$, then $\nu(A) \geq \nu(A)$ although $A \subseteq B$. Moreover, if $C = [0, 2]$ then it is easy to see that $\nu(C) = 0$ although the set C is uncountable in $\mathbb{R}$.

Example 5.2.3 raises the question about the difference between null sets and sets of measure zero. In fact, the set C is of measure zero but it is not a null set. This motivates the following classification of measurable sets with respect to signed measures.

5.2.2 Positive and Negative Sets

Definition 5.2.4 Let ν be a signed measure on a measurable space (X, Σ) and let $A \in \Sigma$. Then:

1. **Positive Set**: If $\nu(E) \geq 0$ for every $E \in \Sigma$ and $E \subseteq A$, then we say that A is positive with respect to ν. We will denote a positive set by P.
2. **Negative Set**: If $\nu(E) \leq 0$ for every $E \in \Sigma$ and $E \subseteq A$, then we say that A is negative with respect to ν. We will denote a negative set by N.
3. **Null Set**: If $\nu(E) = 0$ for every $E \in \Sigma$ and $E \subseteq A$, then we say that A is null with respect to ν.

Remark We have the following observations from the definition above.

(1) The empty set is an example of all of the three cases.

(2) Every positive set is a set of positive measure but the converse is not necessarily true. Similarly, every negative set is a set of negative measure but the converse is not necessarily true, and every null set is a set of measure zero but the converse is not necessarily true.
(3) If a set A is both positive and negative with respect to a signed measure ν, then A is null with respect to ν.
(4) A set can be none of the three cases above. Consider the set B in Example 5.2.3.

Proposition 5.2.5 *Let ν be a signed measure on a measurable space (X, Σ).* Then

(1) Every measurable subset of a positive set in Σ is positive, and the countable union of positive sets in Σ is positive.
(2) very measurable subset of a negative set in Σ is negative, and the countable union of negative sets in Σ is negative.
(3) Every measurable subset of a null set in Σ is null, and the countable union of null sets in Σ is null.

Proof We will only prove the first statement since the other two statements can be proved similarly. The first part of the first assertion is immediate from definition. Let $\{P_n\}$ be a countable family of positive sets in Σ, and set

$$A = \bigcup_i P_i.$$

To show that A is positive let $E \subseteq A$ be measurable. Define $E_1 = E \cap P_1$, and

$$E_n = E \cap (P_n \setminus P_{n-1}).$$

Note that $\bigcup_n E_n = E$, and since $E_n \subseteq P_n$, $\{E_n\}$ is also a countable family of pairwise disjoint positive sets in Σ, i.e., $\nu(E_n) \geq 0$ for all n. Therefore we have

$$\nu(E) = \nu\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \nu(E_j) \geq 0.$$

The result follows since E is arbitrary. □

5.2.3 Hahn's Lemma

Recall that we have made a remark that every positive set is a set of positive measure but the converse is not necessarily true. However, the next result (sometimes it is called: Hahn's Lemma) states that a set of positive measure contains a positive measurable set.

Lemma 5.2.6 (Hahn's Lemma) *Let ν be a signed measure on a measurable space (X, Σ) and let $E \in \Sigma$ such that $0 < \nu(E) < \infty$. Then there exists a measurable subset of E that is positive and of positive measure.*

Proof We will assume that E is not positive since there is nothing to proof otherwise. Then E contains a set of negative measure. Let n_1 be the smallest integer for which there exists a set $E_1 \subset E$ such that

$$\nu(E_1) < \frac{-1}{n_1}.$$

If $E \setminus E_1$ is positive we are done. Otherwise, we repeat the argument for $E \setminus E_1$. Let n_2 be the smallest integer for which there exists a set $E_2 \subset E \setminus E_1$ such that

$$\nu(E_2) < \frac{-1}{n_2}.$$

We repeat the argument for $E \setminus (E_1 \cup E_2)$. If this process stops after k steps then $E \setminus (\bigcup_{j=1}^{k} E_j)$ is positive and is subset of E and we are done. Otherwise continue by induction to define the set

$$P = E \setminus \left(\bigcup_{j=1}^{\infty} E_j \right)$$

where $\{E_j\}$ is a pairwise disjoint subsets of E such that

$$\nu(E_k) < \frac{-1}{n_k}.$$

This will also produce an infinite sequence $n_1, n_2, \ldots$. Then

$$\nu(P) = \nu(E) - \sum_{k=1}^{\infty} \nu(E_k).$$

Since $\nu(P) < \infty$, we conclude that the series $\sum_{k=1}^{\infty} \frac{1}{n_k}$ absolutely converges, so $\frac{1}{n_k} \to 0$. This implies that

$$\nu(P) \geq \nu(E) + \sum_{k=1}^{\infty} \frac{1}{n_k} > \nu(E) > 0.$$

We have proved that P is a set of positive measure. To show that P is positive, let $F \subseteq P$. Then for each k we have

$$F \subseteq P \subseteq E \setminus \left(\bigcup_{j=1}^{k-1} E_j \right).$$

This implies that

$$\nu(F) \geq -\frac{1}{n_k - 1}.$$

The result follows since $n_k \to \infty$. □

5.3 Decomposition of Measures

5.3.1 Hahn Decomposition Theorem

This section deals with decomposition theorems. The first theorem deals with decomposing a measurable space using a signed measure. The second theorem deals with decomposing the signed measure into two measures. Recall that a decomposition of a set X is a pair of subsets of X, say A and B, such that $X = A \cup B$ and $A \cap B = \emptyset$.

Theorem 5.3.1 (Hahn Decomposition Theorem (HDT)) *Let ν be a signed measure on a measurable space (X, Σ).* Then there exist a positive set P and a negative set N such that $X = P \bigcup N$ and $P \bigcap N = \emptyset$. Moreover, this decomposition is unique up to null sets.

Proof In view of axiom 1 of Definition 5.2.1, we can assume without loss of generality that $-\infty \leq \nu < \infty$, i.e., ν does not take the value $+\infty$. Let $M = \sup\{\nu(A)$, where A is positive set with respect to $\nu\}$. Clearly $0 < M < \infty$. Let $\{P_k\}$ be a sequence of positive sets such that $\nu(P_n) \to M$. Let

$$P = \bigcup P_n.$$

Then by Proposition 5.2.5, A is positive, and

$$\nu(P) = \nu(P - P_k) + \nu(P_k) \geq \nu(P_k).$$

So

$$M = \lim \nu(P_k) \leq \nu(P) \leq M, \tag{5.3.1}$$

and consequently,

$$\nu(P) = M.$$

Let $N = X \setminus P$. We want to show that N is negative. If not, then there exists a set $A \subseteq N$ of positive measure. By Lemma 5.2.6, there is a positive set $A_1 \subseteq A$. Then $A_1 \cap P = \phi$ and $A_1 \cup P$ is positive, and so

$$\nu(A_1 \cup P) \geq \nu(P) = M.$$

This contradicts (5.3.1), which implies that N contains no sets of positive measure. This establishes the existence of the decomposition $\{P, N\}$ of X. Let $\{P', N'\}$ be another Hahn decomposition of X. Then note that

$$P \setminus P' \subset P, \quad P \setminus P' \subset N'.$$

Hence $P \setminus P'$ is both negative and positive set, and therefore it is null set, and so is $P' \setminus P$, and thus $P = P'$ a.e. The same argument can be applied for N and N'. □

The result gives the first important consequence of a signed measure, as it allows to decompose the space into a positive set and a negative set that are disjoint.

5.3.2 *Jordan Decomposition Theorem*

Another important consequence of the HDT is that it allows identifying the positive and negative parts of the measure. Let $X = P \bigcup N$, and let $E \subseteq X$ be measurable. Then

$$\nu(E) = \nu(E \cap P) + \nu(E \cap N).$$

Since this can be negative, we write

$$\nu^+(E) = \nu(E \cap P), \tag{5.3.2}$$

and

$$\nu^-(E) = \nu(E \cap N). \tag{5.3.3}$$

Then we can write our signed measure as

$$\nu(E) = \nu^+(E) - \nu^-(E).$$

Thus, we constructed the following decomposition.

Theorem 5.3.2 (Jordan Decomposition Theorem (JDT)) *Let ν be a signed measure on a measurable space (X, Σ).* Then, there exist unique positive measures ν^+, ν^- *such that*

$$\nu = \nu^+ - \nu^-.$$

Proof The existence of the decomposition has been constructed. To prove uniqueness of the decomposition, let $\nu = \mu^+ - \mu^-$. In view of (5.3.2) and (5.3.3) we must have another Hahn decomposition $\{P', N'\}$ of X such that

$$\mu^+(E) = \nu(E \cap P'), \quad \mu^-(E) = \nu(E \cap N').$$

Then by uniqueness of Hahn decomposition, $P' \setminus P$ and $N \setminus N'$ are null sets. Therefore,

$$\mu^+(E) = \nu(E \cap P') = \nu(E \cap P) = \nu^+(E),$$

$$\mu^-(E) = \nu(E \cap N') = \nu(E \cap N) = \nu^-(E). \qquad \square$$

As we see in the proof of the theorem, the uniqueness of the Hahn decomposition leads to the uniqueness of the Jordan decomposition. According to the theorem, any signed measure ν can be written as the difference of two positive measures ν^+ and ν^-, which are called the positive and negative parts of ν, respectively. Since ν cannot take both values $\pm\infty$, one of the two parts must be finite. If both are finite then ν is a finite signed measure.

The decomposition reminds us of the Jordan Decomposition Theorem of functions of bounded variations that were discussed in Theorem 2.6.7. Likewise, one can also define what is called the total variation of the measure v by

$$|\nu|(E) = \nu^+(E) + \nu^-(E), \quad E \in \Sigma.$$

5.4 Absolute Continuity of Measures

5.4.1 Motivating Problem

In Example 5.2.2, we provided an example of a signed measure as follows:

$$\nu(E) = \int_E f d\mu. \tag{5.4.1}$$

The measure defined in (5.4.1) has also the property that for all measurable sets E such that $\mu(E) = 0$, then $\nu(E) = 0$.

The question arises is whether any signed measure can be represented by an integral of a measurable function with respect to a measure. The Radon–Nikodym Theorem came to discuss this problem, and the answer is yes, provided some conditions should be imposed on the measures and the function. This section is devoted to discuss the result of the Radon–Nikodym Theorem, which is one of the most significant results in measure theory that has many important applications in probability theory and stochastic analysis.

To get an answer to the problem above, we note the following two observations:

(1) It is natural to think of ν and μ as two measures related in such a way that if μ vanishes at a set, then ν should also vanishes at the same set, otherwise the equality (5.4.1) no longer holds.
(2) Moreover, Theorem 3.6.8 states that a function is absolutely continuous on $[a, b]$ if and only if it is the indefinite integral of its derivative.

5.4.2 Definitions

The two observations remind us of the notion of absolute continuity defined in Sect. 2.6, were we define an absolute continuous function on $[a, b]$ if given $\epsilon > 0$, there exists $\delta > 0$ such that if $\{[x_i, y_i] : i = 1, 2, \ldots, n\}$ is a finite collection of pairwise disjoint subintervals in $[a, b]$ and

$$\sum_{i=1}^{n} |y_i - x_i| < \delta$$

then

$$\sum_{i=1}^{n} |f(y_i) - f(x_i)| < \epsilon.$$

We have also seen that such a function is of bounded variation. This motivates us to extend the concept somehow to measures. So it seems that we need to invoke the idea of absolute continuity for measures if we want (5.4.1) to work conversely. Analogous to Theorem 3.6.8, we also need that ν is a finite measure. Here is the definition:

Definition 5.4.1 (*Absolutely Continuous Finite Measure*) Let ν be a finite measure on a measure space (X, Σ, μ), and $E \in \Sigma$. We say that ν is *absolutely continuous* with respect to μ if for every $\epsilon > 0$, there exists $\delta > 0$ such that if $\mu(E) < \delta$ then $\nu(E) < \epsilon$.

The above definition is restricted to finite signed measures, and since we will be dealing with σ-finite signed measure ν, we may need to extend Definition 5.4.1 to accept σ-finite measures rather than finite measures. The following definition fulfills our needs.

Definition 5.4.2 (*Absolutely Continuous Measure*) Let ν be a measure on a measure space (X, Σ, μ), and $E \in \Sigma$. We say that ν is *absolutely continuous* with respect to μ if $\nu(E) = 0$ whenever $\mu(E) = 0$. This is denoted by $\nu \ll \mu$.

It is easy to see that the $\epsilon - \delta$ definition implies the other definition (Definition 5.4.2) regardless of whether ν is finite or not. The next proposition shows that Definition 5.4.2 implies Definition 5.4.1 when ν is finite.

Proposition 5.4.3 *Let ν be a finite signed measure on a measure space (X, Σ, μ).* If $E \in \Sigma$ and $\nu(E) = 0$ whenever $\mu(E) = 0$, then for every $\epsilon > 0$, *there exists $\delta > 0$ such that if $\mu(E) < \delta$ then $\nu(E) < \epsilon$.*

Proof Suppose not, then there is $\epsilon > 0$ and $E_k \in \Sigma$ such that

$$\mu(E_k) < \frac{1}{2^k} \ \& \ \nu(E_k) \geq \epsilon.$$

Define

$$F_m = \bigcup_{k=m}^{\infty} E_k.$$

Note that F_m is decreasing sequence, and

$$\mu(F_m) \leq \sum_{k=m} \frac{1}{2^k} = \frac{1}{2^{m-1}}.$$

Now, let

$$F = \bigcap_{m=1}^{\infty} F_m.$$

Clearly $F \in \Sigma$. Then

$$\mu(F) \leq \lim_{m\to\infty} \frac{1}{2^{m-1}} = 0.$$

But on the other hand, we have

$$\nu(F_1) \leq \nu(X) < \infty,$$

and $\nu(F_m) \geq \epsilon$ for all m, hence $\nu(F) \geq \epsilon$, which is a contradiction. □

The next proposition states some properties of absolute continuity.

Proposition 5.4.4 *(HDT) Let ν be a signed measure and μ is a measure. Then $\nu \ll \mu$ if and only if $\nu^+ \ll \mu$ and $\nu^- \ll \mu$.*

Proof Left to the reader as an exercise. □

The following Lemma will be used to prove the Radon-Nikodym Theorem.

Lemma 5.4.5 *Let ν & μ be finite measure on a measurable space (X, Σ), and $\nu \ll \mu$. Then there exists $A \in \Sigma$ and $\epsilon > 0$ such that $\mu(A) > 0$ and $\nu(E) \geq \epsilon\mu(E)$ for all $E \subseteq A$.*

Proof Define the following signed measure on (X, Σ)

$$\lambda_n = \nu - \frac{\mu}{n}. \tag{5.4.2}$$

For each n, write $X = P_n \bigcup N_n$. Let

$$P = \bigcup_n P_n \ \& \ N = \bigcap_n N_n.$$

Clearly $N = X \setminus P$, so $\lambda_n(N) \leq 0$, and using (5.4.2), for all n we have

$$0 \leq \nu(N) \leq \frac{1}{n}\mu(N) \leq \frac{1}{n}\mu(N_n) \leq \frac{1}{n}\mu(X) < \infty. \tag{5.4.3}$$

Taking $n \to \infty$ gives $\nu(N) = 0$. Now, $\mu(P) \geq 0$ since μ is a measure on (X, Σ). If $\mu(P) = 0$, then $\nu(P) = 0$ by absolute continuity of ν with respect to μ, and so

$\nu(P_n) = 0$ for all n, hence $\nu(E) = 0$ for all $E \subseteq P_n$. But from (5.4.3) we also have $\nu(E) = 0$ for all $E \subseteq N$, and therefore $\nu(E) = 0$ for all $E \in \Sigma$, i.e., $\nu \equiv 0$ is the zero measure, which is a contradiction. Thus, $\mu(P) > 0$, and so $\mu(P_{n_0}) > 0$ for some n_0. Let $A = P_{n_0}$. Then A is positive for λ_{n_0}, and hence

$$\lambda_{n_0}(P_{n_0}) > 0.$$

Use (5.4.2) and let $\epsilon = \dfrac{1}{n_0}$. Then $\lambda_{n_0}(E) \geq 0$ for all $E \subseteq A$. □

5.5 Radon–Nikodym Theorem

5.5.1 Radon–Nikodym Theorem for Finite Measures

Now, we come to our main result of this section.

Theorem 5.5.1 (Radon–Nikodym Theorem (RNT) for finite measures) *Let ν & μ be finite measure on a measurable space (X, Σ).* If $\nu \ll \mu$ then there exists a unique (up to a.e.) nonnegative $\mu-$integrable f on X such that for all $E \in \Sigma$,

$$\nu(E) = \int_E f d\mu.$$

Proof Define the following family

$$\mathcal{F} = \{f : X \to [0, \infty] : \int_E f d\mu \leq \nu(E) \ E \in \Sigma\}.$$

Using the proof in Lemma 5.4.5, define

$$f(x) = \frac{1}{n}\chi_{P_n}(x).$$

Then for any $E \in \Sigma$, and using singular measure in (5.4.2), we have

$$\int_E f d\mu = \frac{1}{n}\mu(P_n \cap E) \leq \nu(P_n \cap E) \leq \nu(E).$$

Hence, $\mathcal{F}$ is nonempty. Let

$$M = \sup_{f \in \mathcal{F}} \int_X f d\mu.$$

Since μ is finite, we have

$$M \leq \mu(X) < \infty.$$

Let $f_1, f_2 \in \mathcal{F}$, then

$$f_1 \vee f_2 = \max\{f_1, f_2\} \in \mathcal{F}$$

(check). Now, let $\{f_n\}$ be a sequence in $\mathcal{F}$ such that

$$\int_X f_n d\mu \to M. \tag{5.5.1}$$

Define the increasing sequence $h_n = \max\{f_1, \ldots, f_n\}$. It is left to the reader to show that $h_n \in \mathcal{F}$, and $\int_X h_n d\mu \nearrow M$. It is clear that $\lim h_n = \sup f_n = f$, so by MCT (Theorem 5.1.14), we have

$$\int_X f d\mu = \lim \int_X h_n d\mu \leq M, \tag{5.5.2}$$

But from (5.5.1), and given the fact that $f_n \leq h_n$ for all n

$$M = \lim \int_X f_n d\mu \leq \lim \int_X g_n d\mu = \int_X f d\mu. \tag{5.5.3}$$

From (5.5.2) and (5.5.3) we obtain

$$\int_X f d\mu = M, \tag{5.5.4}$$

and

$$\int_E f d\mu = \lim \int_E h_n d\mu \leq \nu(E),$$

that is, $f \in \mathcal{F}$. Now define

$$\lambda(E) = \nu(E) - \int_E f d\mu.$$

Clearly λ is a finite measure and $\lambda \ll \mu$ since $\nu \ll \mu$ (verify). Suppose $\lambda \neq 0$. Then by Lemma 5.4.5, there exists $E \in \Sigma$ and $\epsilon > 0$ such that $\mu(E) > 0$ and

$$\lambda(E) \geq \epsilon \mu(E),$$

i.e.,

$$\nu(E) - \int_E f d\mu \geq \epsilon \mu(E),$$

or

$$\nu(E) \geq \int_E (f(x) + \epsilon \chi_E(x)) d\mu.$$

Thus, $f + \epsilon \chi_E \in \mathcal{F}$, and by (5.5.4)

$$\int_X (f + \epsilon \chi_E) d\mu \geq \int_X f d\mu = M.$$

This is a contradiction that implies that $\lambda = 0$, and therefore

$$\nu(E) = \int_E f d\mu.$$

To prove uniqueness, let g be another function satisfying the result. Then by linearity of the integral we have

$$\int_E (f - g) d\mu = \int_E f d\mu - \int_E g d\mu = \nu(E) - \nu(E) = 0,$$

for all $E \in \Sigma$. Thus $f = g$ a.e. with respect to μ. □

According to the theorem any measure ν can be expressed by a measurable function on a measure space (X, Σ, μ), provided that $\nu \ll \mu$. Representing measures by integrals is indeed a breakthrough result since it opens the doors to investigate more results about the properties of measures. The theorem was proved first by Johann Radon in 1913 in $\mathbb{R}^n$ with the Lebesgue measure, then the result was extended by Otton Nikodym in 1930 to general sigma-finite measure spaces.

5.5.2 *Extended Radon-Nikodym Theorem*

It is worth noting that the theorem can be extended to more general settings. The first type of extension is to assume ν to be signed measure. In this case, we can apply the Jordan decomposition $\nu = \nu^+ - \nu^-$ and use the argument of the proof on ν^+, ν^- to obtain

$$\begin{aligned}\nu(E) &= \nu^+(E) - \nu^-(E)\\ &= \int_E f_1 d\mu - \int_E f_2 d\mu\\ &= \int_E (f_1 - f_2) d\mu,\end{aligned}$$

where f_1 is the function predicted by RNT for ν^+ and μ while f_2 is the function predicted by RNT for ν^- and μ. Now, set $f = f_1 - f_2$, which will be our function predicted by RNT.

The second type of extension is to assume ν, μ are σ-finite measures. In this case, we write

$$X = \bigcup_{k=1}^{\infty} A_k$$

where $\mu(A_k), \nu(A_k) < \infty$ for all i. Then for each $E \in \Sigma$ we define

$$\begin{aligned}\mu_k(E) &= \mu(E \cap A_k)\\ \nu_k(E) &= \nu(E \cap A_k).\end{aligned}$$

Then we apply RNT for this finite case and define

$$\nu(E \cap A_k) = \int_{E \cap A_k} f_k d\mu,$$

where $\{f_k\}$ are the functions predicted by RNT for each finite case in k. Then we set

$$f(x) = f_k(x), \ \ x \in A_k. \tag{5.5.5}$$

It is easy to show that f is measurable. Note that

$$E = E \cap \left(\bigcup_k A_k \right) = \bigcup_k (E \cap A_k).$$

Using (5.5.5), it follows that

$$\int_E f d\mu = \sum_k \int_{E \cap A_k} f_k d\mu = \sum_k \mu(E \cap A_k) = \mu(E). \tag{5.5.6}$$

We leave the details for the reader, and thus we state the following generalization of the RNT.

Theorem 5.5.2 (Radon–Nikodym Theorem (RNT)) *Let ν be a σ-finite signed measure & μ be σ-finite measure on a measurable space (X, Σ). If $\nu \ll \mu$ then there exists a unique (up to a.e.) $\mu-$integrable f on X such that for all $E \in \Sigma$,*

$$\nu(E) = \int_E f d\mu.$$

5.6 Radon–Nikodym Derivative

The function f predicted by the theorem is referred to as the Radon–Nikodym derivative of ν with respect to μ, and is denoted by

$$f = \frac{d\nu}{d\mu}, \ d\nu = f d\mu. \tag{5.6.1}$$

The uniqueness of the function f implies the uniqueness of the RN derivative, and this allows us to provide some calculus on RN derivative (5.6.1).

5.6.1 Fundamental Theorem of Calculus for RN Derivatives

We begin by the following result which is following analogous to Theorem 3.6.8.

Theorem 5.6.1 (Fundamental Theorem of Calculus for RN derivatives) *Let ν be a σ-finite signed measure and μ be σ-finite measure on a measurable space (X, Σ). Then $\nu \ll \mu$ if and only if $\frac{d\nu}{d\mu}$ exists and*

$$\nu(E) = \int_E \frac{d\nu}{d\mu} d\mu.$$

Proof The ($\Rightarrow$) is already proved by Theorem 5.5.2, and the reverse direction follows immediately from the definition of absolute continuity and Proposition 5.1.11(6). □

5.6.2 Calculus of RN Derivatives

Proposition 5.6.2 (Calculus of RN Derivative) *Let ν, λ be two σ-finite signed measures and μ is a σ-finite measure on a measurable space (X, Σ). Then*

(1) If $\frac{d\nu}{d\mu}$ *exists and* $c \in \mathbb{R}$, *then*

$$\frac{dc\nu}{d\mu} = c\frac{d\nu}{d\mu}$$

a.e. on X.

(2) If $\nu + \lambda$ *is a defined measure on* (X, Σ) *such that* $\frac{d\nu}{d\mu}$ *and* $\frac{d\lambda}{d\mu}$ *both exist, then*

$$\frac{d(\nu + \lambda)}{d\mu} = \frac{d\nu}{d\mu} + \frac{d\lambda}{d\mu}$$

a.e. on X.

(3) If $\lambda \geq 0$ *(i.e., not signed) and* $\frac{d\lambda}{d\mu}$ *exists, then*

$$\frac{d\lambda}{d\mu} \geq 0.$$

Proof Left as an exercise. □

It is seen that if $\nu \ll \mu$, then

$$\nu(E) = \int_E \frac{d\nu}{d\mu} d\mu,$$

on the other hand we have

$$\nu(E) = \int_E d\nu.$$

So

$$\int_E d\nu = \int_E \frac{d\nu}{d\mu} d\mu.$$

This suggests the following important result

Theorem 5.6.3 *Let* ν *be a signed* σ*-finite measure and* μ *be a* σ*-finite measure on a measurable space* (X, Σ). *If* $\frac{d\nu}{d\mu}$ *exists, then for every* μ*-measurable function on* X *we have*

$$\int_X f d\nu = \int_X f \frac{d\nu}{d\mu} d\mu.$$

Proof First, we assume $f(x) = \chi_E(x)$. Then

$$\int_X f d\nu = \int_X \chi_E(x) d\nu = \nu(E) = \int_E \frac{d\nu}{d\mu} d\mu$$
$$= \int_X \chi_E(x) \frac{d\nu}{d\mu} d\mu = \int_X f(x) \frac{d\nu}{d\mu} d\mu.$$

Hence the formula holds for characteristic functions. Then, by linearity, it also holds for simple functions $\varphi(x) = \sum_j^n a_j \chi_{E_j}(x)$. Now, let $f \geq 0$ be μ-measurable function on X. Then there exists an increasing sequence of nonnegative simple functions $\{\varphi_n\}$ such that $\varphi_n \nearrow f$. We have

$$\int_X \varphi_n d\nu == \int_X \varphi_n \frac{d\nu}{d\mu} d\mu. \tag{5.6.2}$$

Taking the limit of both sides of (5.6.2), and using MCT,

$$\lim \int_X \varphi_n \frac{d\nu}{d\mu} d\mu = \int_X f \frac{d\nu}{d\mu} d\mu.$$

Thus, we have

$$\int_X f d\nu = \int_X f \frac{d\nu}{d\mu} d\mu. \tag{5.6.3}$$

Finally, let f be general μ-measurable function on X. Let $f = f^+ - f^-$,and apply (5.6.3) to both parts □

5.6.3 *Chain Rule of RN Derivative*

This formula is important for establishing chain rule for the RN derivative.

Proposition 5.6.4 (Chain Rule of RN Derivatives) *Let μ, ν, λ be σ-finite measures on a measurable space (X, Σ). If $\frac{d\nu}{d\mu}$, $\frac{d\mu}{d\lambda}$ exist on X for Then $\frac{d\nu}{d\lambda}$ exists and*

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \cdot \frac{d\mu}{d\lambda}.$$

Proof Let $E \in \Sigma$. Then we apply Theorem 5.6.3

$$\begin{aligned}\int_E \frac{d\nu}{d\mu}\cdot\frac{d\mu}{d\lambda}.d\lambda &= \int_X \chi_E \frac{d\nu}{d\mu}\cdot\frac{d\mu}{d\lambda}.d\lambda \\ &= \int_X \chi_E \frac{d\nu}{d\lambda}.d\lambda \\ &= \int_E \frac{d\nu}{d\lambda}.d\lambda \\ &= \mu(E).\end{aligned}$$

Hence, $\frac{d\nu}{d\mu}\cdot\frac{d\mu}{d\lambda}$ is the RN derivative of ν with respect to λ, but this is just $\frac{d\nu}{d\lambda}$, thus the result follows by uniqueness of RN derivative. □

5.7 Lebesgue Decomposition of Measures

One of the main consequences of the RNT is what so-called: *Lebesgue decomposition Theorem*. Before stating the theorem, we need to introduce the following definition.

5.7.1 Mutually Singular Measures

Definition 5.7.1 (*Mutually Singular Measures*) Let μ and λ be two measure on a measurable space (X, Σ). Then μ and λ are said to be mutually singular (denoted by $\nu\perp\lambda$) on (X, Σ) if there exist a decomposition $\{A, B\}$ such that

$$\mu(A) = \lambda(B) = 0.$$

Applying the definition to the Jordan decomposition of ν, we immediately obtain the following corollary.

Corollary 5.7.2 *Let ν be a signed measure on a measurable space (X, Σ).* Then the two Jordan decomposition measures of ν are mutually singular, i.e. $\nu^+\perp\nu^-$.

Proof Left to the reader. □

5.7.2 Lebesgue Decomposition Theorem

Now, we apply the concept of mutually singular measures to introduce the main result of this section.

Theorem 5.7.3 (Lebesgue Decomposition Theorem) *Let ν be a σ-finite measure on a measurable space (X, Σ, μ). Then, there exist unique measures λ, ρ such that $\nu = \lambda + \rho$, where $\lambda \ll \mu$ and $\rho \perp \mu$.*

Proof One can easily show that $\eta = \nu + \mu$ is a σ-finite measure, and both $\nu, \mu \ll \eta$. Then by RNT, there exist f, g η-measurable on X such that for all $E \in \Sigma$,

$$\nu(E) = \int_E f d\eta, \quad \mu(E) = \int_E g d\eta.$$

Let $A = \{x : g(x) > 0\}$, and define

$$\lambda(E) = \nu(E \cap A), \ \ \rho(E) = \nu(E \cap A^c). \tag{5.7.1}$$

It is clear that both λ, ρ define measures on (X, Σ) and $\nu = \lambda + \rho$. Note that $\rho(A) = 0$ and $\mu(A^c) = 0$, thus $\rho \perp \mu$. Now, let $\mu(E) = 0$. Then

$$0 = \int_E g d\eta = \int_X g \chi_E d\eta = 0,$$

which implies that

$$g\chi_E(x) = 0 \text{ on } X.$$

It follows that

$$g\chi_{E\cap A}(x) = g\chi_E(x) g\chi_A(x) = 0,$$

so we have

$$\int_{E\cap A} g d\eta = 0.$$

But from the construction of A it is clear that $g(x) > 0$ for all $x \in E \cap A$, hence $\eta(E \cap A) = 0$, which implies from the definition of η that $\nu(E \cap A) = 0$. But, from (5.6.1) this is just $\lambda(E)$. Hence $\lambda \ll \mu$.

To prove the uniqueness of the decomposition, we assume two sets A, A' such that $\rho(A) = \rho(A') = 0$, and $\mu(A^c) = \mu(A'^c) = 0$. Let

$$\nu = \lambda + \rho = \lambda' + \rho',$$

where $\lambda, \lambda' \ll \mu$ and $\rho, \rho' \perp \mu$. Let $E \in \Sigma$ and note that $E = E_1 + E_2$ where

$$E_1 = E \cap (A \cap A'), \quad E_2 = E \setminus (A \cap A').$$

If $\rho(E_1) = \rho'(E_1) = 0$, then $\mu(E_2) = 0$, so $\lambda(E_2) = \lambda'(E_2) = 0$. It follows that

$$\nu(E_1) = \lambda(E_1) = \lambda'(E_1)$$
$$\nu(E_2) = \rho(E_2) = \rho'(E_2).$$

Hence

$$\lambda(E) = \lambda(E_1) = \lambda'(E_1) = \lambda'(E)$$
$$\rho(E) = \rho(E_2) = \rho'(E_2) = \rho'(E).$$

This establishes the uniqueness. □

5.8 Problems

(1) Determine whether the following collections are σ-algebra.
(a) $\{A \subseteq \mathbb{R} : A \text{ is finite or } A^c \text{ is finite}\}$.
(b) $\{A \subseteq \mathbb{R} : A \text{ is countable or } A^c \text{ is countable}\}$.

(2) Let $\{\mu_j\}$, $j = 1, 2, \ldots$ be a sequence of measure on a measurable space (X, Σ). Let $c_j \geq 0$ be a sequence of nonnegative numbers. Define

$$\mu = \sum_{j=1}^{\infty} c_j \mu_j.$$

Show that (X, Σ, μ) is a measure space.

(3) Let μ be a σ-finite measure on (X, Σ). Show that there exists a sequence of disjoint sets $\{A_n\}$ in Σ such that $\bigcup_n A_n = X$ and $\mu(A_n) < \infty$ for all n.

(4) Let (X, Σ, μ) be a measure space. Suppose that $A \in \Sigma$. Define Σ_A as the collection of all sets in Σ that are contained in Y, and $\mu_A(E) = \mu(E)$ if $E \in \Sigma_A$. Show that (A, Σ_A, μ_A) is a measure space.

(5) Give an example of a set of measure zero but not null.

(6) Let μ be a measure on (X, Σ) defined as

$$\mu(E) = \begin{cases} 0 & E \text{ infinite.} \\ \infty & E \text{ finite.} \end{cases} \text{ for } E \in \Sigma.$$

Show that X must be uncountable.

(7) Let μ be a finite measure in a measure space (X, Σ, μ). Let $\mathcal{F}$ be a collection of disjoint sets in Σ such that $\mu(A) > 0$ for every $A \in \mathcal{F}$. Show that $\mathcal{F}$ is at most countable.

(8) Let μ be a finite measure in (X, Σ, μ) and . Let f be μ-measurable defined on a set $A \in \Sigma$. Show that for every $\epsilon > 0$ there exists $E \in \Sigma$ such that $E \in A$, $\mu(E) < \epsilon$ and f is bounded in $A \setminus E$. (Strictly speaking, every function measurable on a finite measure space is nearly bounded on its domain).

(9) Prove the following version of Hahn's Lemma 5.2.6. Let ν be a signed measure on a measurable space (X, Σ) and let $E \in \Sigma$ such that $-\infty < \nu(E) < 0$. Then there exists a measurable subset of E that is negative and of negative measure.

(10) In the proof of Theorem 5.3.1 (HDT), it was assumed that ν does not take the value ∞. Prove the theorem in case $-\infty < \nu \leq \infty$.

(11) Let ν be a signed measure on a measurable space (X, Σ). Show that

$$\nu^+(E) = \sup\{\nu(B) : B \subseteq E\}$$

and

$$\nu^-(E) = -\inf\{\nu(B) : B \subseteq E\}.$$

(12) Find a Hahn decomposition for $\mathbb{R}$ with respect to the signed measure

$$\nu(E) = \int_E f d\mu,$$

and find the Jordan decomposition for ν.

(13) If

$$\nu(E) = \int_E f d\mu,$$

for some signed measure ν, show that

$$|\nu|(E) = \int_E |f| \, d\mu.$$

(14) Let f be integrable on (X, Σ) with respect to a signed measure μ. Show that

$$\left| \int_E f d\mu \right| \leq \int_E |f| \, d\mu.$$

(15) Show that a measurable set is null with respect to ν if and only if $|\nu|(E) = 0$.

(16) Show that

$$|\nu(E)| \leq |\nu|(E)$$

for every $E \in \Sigma$.

(17) Let ν be a signed measure on a measurable space (X, Σ). Show that

$$|\nu|(E) = \sup\{\left| \int_E f d\nu \right| : |f| \leq 1\}.$$

(18) Let ν, λ be two signed measures. Show that

$$|\nu + \lambda| \leq |\nu| + |\lambda|.$$

(19) Prove Proposition 5.4.4: Let ν be a signed measure and $\mu$is a measure. Then $\nu \ll \mu$ if and only if $\nu^+ \ll \mu$ and $\nu^- \ll \mu$.
(20) Show that $\nu \ll \mu$ if and only if $|\nu| \ll \mu$.
(21) Show that if $\nu_1 \ll \mu$ and $\nu_2 \ll \mu$ then $\nu_1 + \nu_2 \ll \mu$.
(22) In the proof of RNT (Theorem 5.5.1), show that $f_1, f_2 \in \mathcal{F}$, then

$$f_1 \vee f_2 = \max\{f_1, f_2\} \in \mathcal{F}.$$

(23) Let μ, λ be two finite measures on (X, Σ). Show that there exists a μ-measurable function f on (X, Σ, μ) such that

$$(\mu \wedge \lambda)(E) = \int_E g d\mu.$$

(24) Explain why f in (5.5.5) is well defined.
(25) Verify (5.5.6).
(26) Show that the sigma-finite condition in RNT (Theorem 5.5.2) is essential.
(27) Prove Proposition 5.6.2.
(28) Let $\mu \ll \nu \ll \lambda$. Show that $\dfrac{d\mu}{d\lambda}$ exists and

$$\frac{d\mu}{d\lambda} = \frac{d\mu}{d\nu} \cdot \frac{d\nu}{d\lambda}$$

without using the chain rule.
(29) Let μ, ν be two σ-finite measures on (X, Σ) such that $\nu \ll \mu$. If $\dfrac{d\nu}{d\mu} > 0$, show that $\mu \ll \nu$. Deduce that $\dfrac{d\mu}{d\nu}$ exists. In the case (a) above, show that

$$\frac{d\nu}{d\mu} = \frac{1}{\dfrac{d\nu}{d\mu}}.$$

(30) Let ν be a σ-finite measure on a σ-finite measure space (X, Σ, μ), such that $\nu \ll \mu$. Show that $\dfrac{d\nu^+}{d\mu}$ and $\dfrac{d\nu^-}{d\mu}$ both exist, and write formulas for them.
(31) Show that $\nu \perp \mu$ if and only if $|\nu| \perp \mu$.
(32) Show that if $\nu_1 \perp \mu$ and $\nu_2 \perp \mu$ then $\nu_1 + \nu_2 \perp \mu$.
(33) Let $\nu = \lambda - \mu$ for some σ-finite measure ν, λ and a finite measure μ. Show that $\lambda \geq \nu^+$ and $\mu \geq \nu^-$.
(34) Show that if $\nu \ll \mu$ and $\nu \perp \mu$ then $\nu = 0$.

Answer Key

Chapter 1

(1) Show that: $\sum_n \ell(I_i) = \sum_n (b_i - a_i) = b_n - a_1 + k$, $k > 0$.
(2) $\mu^*(A^c \cap B) + \mu^*(B) = \mu^*(A)$.
(3) Let $I \subseteq A$ be interval such that $I \cap E^c = \varnothing$. Then, $\mu(E^c \cap A) + \mu(I) \leq \mu(A)$.
(4) Use definition.
(5) Use definition.
(6) $\sup(\sum \lambda_n^*(A)) \leq \sum \sup(\lambda_n^*(A))$.
(7) Clearly $\mu(\varnothing) = 0$. Let A_n be pairwise disjoint, $\cup_\infty A_n = A$. Show:

$$\mu_i(A) \leq \sum \mu(A_n),$$

and take the supremum.
(8) μ^* is translation invariant, and since

$$((A \cap E) + x) = (A + x) \cap (E + x),$$

we have

$$\mu^*(A + x) = \mu^*(A + x) \cap (E + x) + \mu^*((A + x) \cap (E^c + x)).$$

Replace A with $A - x$ and prove:

$$\mu^*(A) = \mu^*(A \cap (E + x)) + \mu^*(A \cap (E^c + x)).$$

(9) (a) Let $G = \bigcap_\infty O_n$ where $O_n \supseteq E$, $\mu^*(O_n \setminus E) < \dfrac{1}{n}$.
(b) $\mu_*(E) \leq \mu_*(G) \leq \mu^*(G) = \mu(E)$.

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(10) $\mu(E_2 \setminus E_1) = 0$ and $A \setminus E_1 \subseteq E_2 \setminus E_1$.
(11) Use $\liminf A_n = \bigcup_{m=1}^{\infty} B_m$ where $B_m = \bigcap_{n=m}^{\infty} A_n$. Then, note that $B_m \subseteq A_m$ and $B_m \nearrow \bigcup_{m=1}^{\infty} B_m$.
(12) (a) Clear.
(b) Write

$$E = \bigcap_{n=1} \bigcup_{k \geq n} E_k.$$

Note that $\cup_{k \geq n} E_k$ decreasing.
(13) If E is measurable, then E^c is measurable so there exists open $O \supseteq E^c$ such that $\mu^*(O \setminus E^c) < \epsilon$, but $O \setminus E^c = E \setminus O^c$. Let $F = O^c$ be the closed set.
(14) Show that there exists closed $F \subseteq E_{n_0}$ for some n_0 sufficiently large such that $\mu(F \setminus F_n) < \dfrac{\epsilon}{2}$. Let $K = F \cap [-n_0, n_0]$.
(15) Let $\epsilon > 0$. Then, we can find open set O such that $K \subseteq O$ and

$$\mu(O) \leq \mu^*(K) + \epsilon.$$

Since $O \setminus K$ is open, we can write it as

$$O \setminus K = \bigcup_{j}^{\infty} I_j.$$

Choose $n \in \mathbb{N}$, then $\bigcup_j^n I_j = C$ is compact set, so $d(K, C) > 0$.
(16) $\mu(O) \leq \mu^*(A) + \dfrac{\epsilon}{2}$ and $\mu^*(A) \leq \mu(F) + \dfrac{\epsilon}{2}$. But $O = (O \setminus F) \cup F$ is measurable.
(17) Let A be set and E be measurable. For $\epsilon > 0$ $\mu^*(O \setminus F) < \epsilon$ for $F \subseteq E \subseteq O$. Let $U \supseteq A$ be open set. Show:

$$\mu(U \setminus F) \leq \mu(U \setminus O) + \mu(O \setminus F).$$

Conclude: $\mu^*(A \setminus E) + \mu^*(A \cap F) \leq \mu(U) + \epsilon$.
(18) Show that $\mu(A \setminus F) < \epsilon$ for some measurable $F \subseteq E$. For every n choose E_n such that $\mu(A \triangle E_n) < 2^{-n}\epsilon$. Let $E = \cap E_n$. Show that $\mu(E \setminus A) = 0$, then show that $\mu(A \setminus E) < \epsilon$.
(19) Find two disjoint sets which are G_δ and F_σ, respectively, and take the union.
(20) Note that the set

$$[0, 1] \setminus \mathbb{Q} = [0, 1] \setminus \bigcup \{q_n\} = \bigcap (0, 1) \setminus \{q_n\}.$$

(21) (b) $\varnothing$ is the only set of measure zero with respect to counting measure. (a) and (c) are clear.
(22) Write $E_1 \cup E_2 = (E_1 \setminus (E_1 \cap E_2)) \cup E_2$.
(23) Use Caratheodory criterion.

(24) Let $A \subseteq \mathbb{R}$, $\mu^*(A) < \infty$. Let $A_1 = A \cap (a, \infty)$ and $A_2 = A \cap (-\infty, a]$. For $\epsilon > 0$, we have $\sum \ell(I_n) < \mu^*(A) + \epsilon$. Consider $I'_n = I_n \cap (a, \infty)$ and $I''_n = I_n \cap (-\infty, a]$.

(25) If E is measurable, then $\mu^*(O \setminus E) = \mu^*(O) - \mu^*(E)$. Show that there exists $G = G_\delta$ such that $E \subseteq G$ and $\mu^*(G \setminus E) = 0$. Similarly, show that there exists $F \in F_\sigma$ such that $\mu^*(E \setminus F) = 0$.

(26) $A \cup B = ((A \cup B \cap A) \cup ((A \cup B) \cap A^c)$.

(27) Let $q_n \in \mathbb{Q} \cap [0, 1]$. For each $\epsilon > 0$ define the following set

$$A_\epsilon = \bigcup_{n=1}^{\infty} (q_n - \frac{\epsilon}{2^n}, q_n + \frac{\epsilon}{2^n}).$$

Show that A_ϵ is dense in $[0, 1]$. Then, show that A_ϵ is measurable and $0 < \mu(A_\epsilon) \leq 2\epsilon$.

(28) Let V be Vitali set. Then $I = [0, 1] = V \cup (I \setminus V)$. Show that

$$\mu^*(I \setminus V) = 1 - 0 = 1.$$

(29) Note that $\mu_*(E) \leq \mu_*(V)$.

(30) Consider $A = B = C$, the Cantor set.

(31) (a) Let $A = V \cup (1, \infty)$ where $V \subset [0, 1]$ is the Vitali set. Then $(1, \infty) \subset A$ hence $\mu_*(A) = \mu^*(A) = \infty$.
(b) Let $E = v \times \{0\}$ where V is the Vitali set.

(32) Note that if $A \subset R$ and every subset of A is Lebesgue measurable then $\mu(A) = 0$.

(33) Consider the mapping $\phi : C \longrightarrow [0, 1]$ defined by $\sum a_n 3^{-n} \longmapsto \sum b_n 2^{-n}$ where $b_n \in \{0, 1\}$.

(34) Let $A \in \mathbb{R}^n$. Let $E \subseteq O$. Show that

$$\mu^*(A) = \mu^*(A \cap O) + \mu^*(A \cap O^c) \geq \mu^*(A \cap E) + \mu^*(A \cap E^c).$$

(35) Similar to the proof of Theorem 1.4.9.

Chapter 2

(1) Let $f_n(x) = (1 + \frac{1}{n}) \cos n$.

(2) Clear from definition.

(3) For the first inequality, if $\epsilon > 0$ then $g_m \leq \underline{\lim}(g_n) + \epsilon$. So

$$f_m + g_m \leq f_n + \underline{\lim}(g_n) + \epsilon.$$

(4) (a) $f_n g_n \leq (\sup f_n)(\sup g_n)$. Take supremum then the limit.
(b) Note that for each $\epsilon > 0$,

$$\sup(f_n g_n) \geq (\overline{\lim}(f_n - \epsilon))(\overline{\lim}(g_n)).$$

(5) $(f^+)^{-1}((a, \infty]) = \mathbb{R}$ if $a < 0$ and $(f^+)^{-1}((a, \infty]) = f^{-1}((a, \infty])$ for $a \geq 0$. Similarly for f^-.
(6) Let $D = \text{Domain}(f)$. Show that

$$\{x \in D : \ f > r\} = \{x \in D \setminus E : \ f > r\} \cup \{x \in E : \ f > r\}.$$

(7) If $r \leq f$, then $r \leq |f|$ and if $r \geq |f|$, then $r \geq f$.
(8) (a) If n is odd, then $\{f^n > r\} = \{f > r^{1/n}\}$. If n is even and $r > 0$, then

$$\{f^n > r\} = \{f > r^{1/n}\} \cup \{f < -r^{1/n}\}.$$

(b) $\{|f|^p > r\} = \mathbb{R}$ if $r < 0$ and $\{|f|^p > r\} = \{|f| > r^{1/p}\}$.
(9) $f^{-1}(b) = \{f \geq b\} \cap \{f \leq b\}$.
(10) $\{x : \ f(x) = \infty\} = \bigcap_{n=1}^{\infty} \{x : \ f(x) \geq n\}$.
(11) $\{x : f \geq g\} = \bigcup_{q \in \mathbb{Q}} \{f \geq r\} \cap \{g \leq r\}$. Similarly for $\{x : f \leq g\}$.
(12) If A is dense, then there exists sequence $r_n \in A$ s.t. $r_n \longrightarrow r \in \mathbb{R}$. Let $r_n \nearrow r$. Then

$$\{x : f(x) \geq r\} = \bigcap_{n=1}^{\infty} \{x : f(x) \geq r_n\}.$$

(13) Consider $f(x) = \chi_V(x) - \chi_A(x)$ where $V \subseteq [0, 1]$ is the Vitali set and $A = [0, 1] \setminus V$.
(14) For every $M > 0$ $\exists N \in \mathbb{N}$ such that $f_n \geq \inf_{k \geq n} f_k > M$ for all $n \geq N$.
(15) $(g \circ f)^{-1}(-\infty, a) = f^{-1}(O)$ is measurable.
(16) Composite or product of measurable functions is measurable.
(17) Let $f = f^+ - f^-$.
(18) Use Propositions 2.3.1, and 2.3.2.
(19) Write $f' = \lim[n(f(x + \frac{1}{n}) - f(x)]$.
(20) Let $f_n(x) = \chi_{[n,n+1)}(x)$. Note that $|f_n - f| = \chi_{[n,n+1)} \nrightarrow 0$ uniformly.
(21) Let $E_n = E \cap [n - 1, n]$.
(22) For $r \in \mathbb{R}$, write

$$\{x : \ f(x) > r\} = \bigcup_n \{x \in F_n : \ f(x) > r\} \cup \{x \in E \setminus F : \ f(x) > r\}.$$

(23) (a) $D^+ f(0) = D_+ f(0) = 1$. $D^- f(0) = D_- f(0) = -1$.
(b) $D^+ f(0) = \limsup_{h \downarrow 0} \frac{f(h) - f(0)}{h} = \limsup_{h \downarrow 0} \sin(\frac{1}{h}) = \inf_{\delta} \sup_{0 < h < \delta} \sin(\frac{1}{h}) = 1$.
Similarly, $D^- f(0) = 1$ and $D_- f(0) = -1$.

(24) (a) $\limsup(-x_n) = -\liminf x_n$.
(b) $\limsup(x_n + y_n) \leq \limsup x_n + \limsup y_n$.

(25) Choose a partition $P = \{0, x_1, x_2, \ldots, x_n = 1\}$ where x_1 is irrational, x_2 is rational, and x_3 is irrational, and so on.

(26) $\dfrac{f(c) - f(c-h)}{h} < 0$ for $h > 0$. So $D^- f(c) < 0$. Similarly, $\dfrac{f(c+h) - f(c)}{h} > 0$ for $h > 0$.

(27) For every $x > a$, $|f(x) - f(a)| \leq V_a^x(f) = 0 \leq V_a^b(f) = 0$.

(28) Let P be a partition for $[a, b]$. Using mean value theorem,

$$\sum^{n} |f^{\alpha}(x_i) - f^{\alpha}(x_{i-1})| = \sum^{n} \alpha c^{\alpha-1} |f(x_i) - f(x_{i-1})|$$

for some $c \in (a, b)$.

(29) Note that $f \vee g = \dfrac{|f - g| + f + g}{2}$.

(30) (a) No.
(b) No.
(c) No.
(d) Yes.

(31) Let $f(x) = \begin{cases} 1 & x \in \mathbb{Q} \cap [a, b] \\ -1 & x \in [a, b] \setminus \mathbb{Q} \end{cases}$.

(32) For $y > x$, $g(y) - g(x) = g(x) = V_x^y(f) \geq f(y) - f(x)$.

(33) By Jordan decomposition, let $f = f_1 - f_2$. Let $g = f_1(x) + x$ and $h(x) = f_2(x) + x$.

(34) $\sum |f(g(x_i) - f(g(x_{i-1}))| \leq L \sum |(g(x_i) - (g(x_{i-1}))|$ where L is the Lipschitz constant of f.

(35) By Jordan Decomposition, $f = f_1 - f_2$. Let $c \in [a, b]$. Show that $\lim_{x \downarrow c} f$ exists.

(36) Let $f(x) = \sqrt{x}$ on $[0, 1]$ and

$$g(x) = \begin{cases} x^2 \left| \sin \dfrac{1}{x} \right| & 0 < x \leq 1 \\ 0 & x = 0 \end{cases}.$$

Let $h = f \circ g = x\sqrt{|\sin(1/x)|}$ and consider the partition $P = \{[\frac{2}{(2n+1)\pi}, \frac{2}{2n\pi}]\}$. Show that

$$v_a^b(h; P) \geq \frac{2}{\pi} \sum \frac{1}{(2n+1)}.$$

(37) If g is monotone, then $\{[g(y_i), g(x_i)]\}$ forms a collection of pairwise disjoint intervals. Since $f \in AC[a, b]$, for every $\epsilon > 0$ there exists $\delta > 0$ such that $\sum_n |f((y_i) - f(x_i)| < \epsilon$ whenever $\sum_n (y_i - x_i) < \delta$ for a disjoint collection of intervals $\{[x_i, y_i]\}$. Fix that value of δ, and show that $\sum_n |fg((y_i) - fg(x_i)| < \epsilon$.

(38) Since $f \in C[a, b]$, use definition of absolute continuity on f to find a partition in which $\sum_n |f((y_i) - f(x_i)| < \epsilon c^2$. Show that $\frac{1}{f} \in AC[a, b]$ then use the previous problem.

(39) (a) $\left|f'\right| \leq 2 + \frac{2}{\epsilon} < \infty$.

(b) Show that as $\epsilon \to 0$ $V_\epsilon^1(f) \nearrow \infty$.

(40) (a) Let $\left|f'\right| \leq M$. Then

$$|f(y) - f(x)| = \left|\frac{f(y) - f(x)}{y - x}\right| |y - x|.$$

Use mean value theorem.

(b) Let $f \ni AC[a, b]$ and is Lipschitz. Find a collection of pairwise disjoint intervals $\{[a_i, b_i], i = 1, \ldots, n\}\}$ of $[a, b]$. We have

$$|f(b_i) - f(a_i)| \leq M_i |b_i - a_i|$$

for some M_i. Define $M = \max\{M_1, \ldots M_n\}$.

(c) Let f be Lipschitz with a Lipschitz constant K. For $\epsilon > 0$ let $\delta = \frac{\epsilon}{K}$.

(d) $f'(x) = \frac{1}{2\sqrt{x}}$.

(41) (a) Note that f is continuous on a closed interval.

(b) Note that

$$f' = \alpha x^{\alpha-1} \sin(\frac{1}{x^\beta}) - \beta x^{\alpha-\beta-1} \cos(\frac{1}{x^\beta})$$

is integrable.

(42) Let $\epsilon > 0$, then there exists $\delta > 0$ satisfying the condition for absolute continuity. But, for every $\delta > 0$ there exists open set $O = \bigcup(x_i, y_i)$ such that $\mu(O) = \sum(y_i - x_i) < \delta$. Construct the collection of pairwise disjoint $\{[a_i, b_i]\}$ containing A such that $[a_i, b_i] \subset (x_i, y_i)$. Then clearly $\sum(b_i - a_i) < \delta$. Use absolute continuity, then show that

$$\mu(f(A)) \leq \sum |f(\beta_i) - f(\alpha_i)| < \epsilon.$$

(43) (a) Let $h = c_1 f_1 + c_2 f_2$. Note that

$$S(h, g; P) = c_1 S(f_1, g; P) + c_2 S(f_2, g; P).$$

(b) Note that

$$\left|\int_a^b f dg\right| \leq \max_{[a,b]} |f(t)| V(g),$$

where $V(g)$ is the total variation of g.

(44) (a) Note that $m(g(b) - g(a)) \leq S(f, g; P) \leq M(g(b) - g(a))$.
(b) Use (a) to conclude that for $x \neq y$, $F(y) - F(x) = c[g(y) - g(x)]$.
(c) Use (b).

Chapter 3

(1) Let $0 \leq \varphi \leq f$. Then $\int_A \varphi = 0$.
(2) $\nu(\varnothing) = 0$. Let $\varphi(x) = \sum_n a_k \chi_{E_k}(x)$ for some partition $\{E_k\}_{k=1}^n$ of a measurable set E. Show that $\nu(\cup E_j) = \sum \nu(E_j)$.
(3) $(cf)^+ = -cf^-$ and $(cf)^- = -cf^+$.
(4) (a) Let $f_n(k) = \sum_{k=0}^n f(k)\chi_{\{k\}}$.
(b) Use counting measure. Note that

$$\int_{\mathbb{N}} f_n d\mu = \sum_{k=0}^{n} a_k \mu(\{k\}) = \sum_{k=0}^{n} a_k.$$

Use MCT.
(c) Let $f(k) = f(k)^+ - f(k)^-$.
(5) Let $A_n = \{x : f(x) \geq n\}$. Show that $\int_E f \geq n\mu(A_n)$ and take $n \to \infty$.
(6) Let $\varphi(x) = \sum_n c_k \chi_{E_k}(x)$. Show that $\int_E \varphi = \sum_\infty \int_{E_n} \varphi$. Let $0 \leq \varphi \leq f$ and $\epsilon > 0$. There exists φ such that $\int_{E_n} \varphi \geq \int_{E_n} f - \epsilon$.
(7) (a) If $\varphi_n \nearrow f$, then there exists $N \in \mathbb{N}$ such that $\lim \varphi_n \geq \phi_N$. Show that $\lim \int_E \varphi_n \geq \lim \int_E \varphi_n$. Interchange φ_n and ϕ_n and repeat the work.
(b) For $\varphi_n \leq f$ we have $\lim \int \varphi_n \leq \sup \int \phi$ for $0 \leq \phi \leq f$.
(8) Let $A = [0, \infty)$ and $f_n(x) = \dfrac{n}{n^2 + x^2}$.
(9) No. The proof of BCT requires Egoroff Theorem which is not valid for unbounded intervals.
(10) If $h_n = f_n \chi_A$, then $h_n \longrightarrow h = f\chi_A$. Let $g = M\chi_A$ and apply DCT.
(11) Use Fatou's Lemma on $g_n = g - f_n$ and $h_n = g + f_n$.
(12) Let $A = \mathbb{R}$ and define $f_n(x) = \chi_{[-n,n]}(x)$.
(13) Let $f_n(x) = n^2 \chi_{[0,1.n]}(x)$.
(14) Use Fatou's Lemma to show that $\int_A f \leq \underline{\lim} \int_A f_n$.
(15) Let $g_n = f_1 - f_n$ and use MCT.
(16) Construct a sequence and use MCT.
(17) Combine Definitions 3.1.3 and 3.2.1.
(18) (a) Let $h \leq f + g$. Write $h = h_1 + h_2$ where $h_1 = \min(f, h)$. Then $h_1 \leq f$ and $h_2 \leq g$.
(b) Let h be as in the previous problem and define $h_n(x) = \min\{h, f_n\}$. Use BCT.
(19) By Fatou's Lemma $\int f \leq \underline{\lim} \int f_n$.

(20) Use Theorem 3.4.8.
(21) Define $f_n(x) = \min\{f, n\}$. Use MCT.
(22) Let $E_n = \{x : |f(x) > n|$. Let $g_n = f(x) \cdot \chi_{E_n}(x)$. Use DCT.
(23) Let $h_n = |f_n - f|$ and $h_n \leq |f_n| + |f| \leq 2g$. Use DCT.
(24) Let $f_n(x) = n\chi_{[0,\frac{1}{n}]}(x)$. Use MCT.
(25) Let $h_n = f_n - f_1$. Then $0 \leq h_n \nearrow f - f_1$. Use MCT.
(26) Use triangle inequality for one direction, and the generalized DCT for the other direction where $g_n = |f_n| + |f|$ and $h_n = f_n - f$.
(27) Let $g_n = f(x, t_n)$ where $t_n \to 0$. Apply DCT.
(28) Let $\varphi(x)$ be a simple function such that $0 \leq \varphi(x) \leq f(x)$ on A and define the set:

$$A_n = \{x \in A : f_n(x) \geq (1 - \epsilon)\varphi(x)\},$$

where $\epsilon > 0$. Show that $A_k \subseteq A_{k+1}$and$\cup_k A_k = A$, which implies lim $\int_{A_n} \varphi = \int_A \varphi$.
(29) Define

$$A_{n,k} = \{x : \frac{k-1}{n}M < f(x) < \frac{k}{n}M\}$$

and $\psi_n(x) = \sum(\frac{k}{n})\chi_{A_{n_k}}$ and $\varphi_n(x) = \sum(\frac{k-1}{n})\chi_{A_{n_k}}$. Then $\varphi_n \leq f \leq \psi_n$.
(30) $fg \leq \frac{1}{2}[f^2 + g^2]$.
(31) No. Let $f(x) = \begin{cases} 1/\sqrt{x} & x \in (0, 1] \\ 0 & x = 0 \end{cases}$.
(32) Show that there exists a step function $s(x) = \sum_{i=1}^{n} c_i \chi_{[a_i,b_i]}$ such that for every $\epsilon > 0$, $\int |f - s| < \epsilon$. Write

$$\left| \int_{\mathbb{R}} f(x) \cos nx dx \right| = \int |f - s\cos x + s\cos x| .$$

(33) Use countable additivity of integration. For (1), note that $B_n = A_{n+1} \setminus A_n$ are disjoint. For (2) define ascending sequence of measurable sets $B_n = A_1 \setminus A_{n+1}$.
(34) Note that $|f| = f^+ + f^-$, and $-|f| < f < |f|$.
(35) $\int f^+ = \int f^- = \infty$.
(36) Let $f_n = |f| \chi_{[-n,n]}(x)$. Use MCT.
(37) (1) Note that

$$\int_0^1 \left| \frac{1}{x} \sin(\frac{1}{x}) \right| d\mu = \lim \int_1^n \frac{|\sin x|}{x} d\mu = \infty,$$

and $\int_1^\infty \frac{\sin x}{x} d\mu = \frac{\pi}{2}$.

(2) $f = (x\cos(\frac{1}{x}))'$.

(3) $f = (x^2 \sin(\frac{1}{x^2}))'$.

(38) Let $f_n(x) = \dfrac{1}{\sqrt{x + \frac{1}{n}}}$. Use MCT.

(39) Similar argument of the DCT with $g_n - f_n$ and $g_n + f_n$. Using Fatou's Lemma on both functions.

(40) Use Proposition 3.5.5.

(41) (1) $|\cos(x^n)| \leq 1$.

(2) $\left|\sin(\frac{x}{n})e^{x/n}\right| \leq e$.

(3) $|f_n| \leq 1$.

(4) $|f_n| \leq 1/x^2$.

(5) $\dfrac{\sqrt{x-1}}{1+\sqrt{n}x^3} < \dfrac{\sqrt{x}}{x^3}$.

(6) $0 \leq f_n \nearrow e^{-x/2}$.

(7) $\left|\dfrac{n\sin(x/n)}{xe^x}\right| \leq e^{-x}$.

(8) Note that $|\sin(x/n)| \leq \dfrac{1}{n}$, and $(1 + \frac{x}{n})^n \geq 1 + x + \dfrac{x^2}{4}$.

(9) $|\sin(x/n)| \leq \dfrac{1}{n}$, and $(1 + \dfrac{x}{n})^n \geq 1 + x + \frac{x^2}{4}$.

(10) $f_n \longrightarrow \chi_{\{0\}}(x)$ and $|f_n| \leq 1$.

(11) $|f_n| \leq \dfrac{1+3x}{(1+x)^3} < \dfrac{1+3x}{x^3} = g$.

(12) $|f_n| \leq \dfrac{1}{1+x^2} = g$.

(42) Let $f(x) = \begin{cases} 1 & x \in \mathbb{Q} \cap [0,1] \\ 0 & x \in [0,1] \setminus \mathbb{Q}. \end{cases}$

(43) Let $F(x) = \begin{cases} x+1 & 0 \leq x \leq 1 \\ 2x & 1 < x \leq 2 \end{cases}$.

(44) Since f is AC, then it is of bounded variation and f' is integrable. Write $f = f^+ - f^-$, then $f = \int f' = \int f^{+\prime} - \int f^{-\prime}$.

(45) Let $f_n = f'\chi_{[\frac{1}{n},0]}$. Then, use DCT to show that

$$\int_{1/n}^{1} f' = f(1) - f(\frac{1}{n}).$$

Show that $\int_x^1 f' = f(1) - f(x)$.

(46) Use Definition 3.7.1.
(47) We can find a set G Borel in $\mathbb{R}^2$ such that $E \subseteq G$ and $\mu_2(G) = 0$.
(48) (a) Write $\hat{E}$ as

$$\hat{E} = \bigcap \{(x, y) : x - y \in O_i\} \setminus \{(x, y) : x - y \in G\},$$

where $\mu(G) = 0$.
(b) Note that $E = (f^{-1}(\alpha), \infty)$ for some $\alpha \in \mathbb{R}$ is measurable. Use (a).
(49) (a) Write $\dfrac{1}{x} = \int_0^\infty e^{-xt} dt$ for $x > 0$.
(b) Note that $\int_0^\infty e^{-xy} dx = \dfrac{1}{y}$.
(c) Note that $\int_0^\infty \sin y e^{-xy} dy = \dfrac{1}{x^2 + 1}$.
(50) (a) Let $z = x - y$, so $y = x - z$.
(b) $f * (g * h)(x) = \int \int f(x - y) g(y - z) h(z) dz dy$. Make the transformation $y \to y + z$.

Chapter 4

(1) Use induction on n. Note that

$$\sum_{j=2}^{n+1} \frac{q_1}{q_j(q_1 - 1)} = \frac{q_1}{q_1 - 1} \sum_{j=2}^{n+1} \frac{1}{q_j} = 1.$$

(2) Write $\left\| \sum_{j=1}^\infty f_j \right\|_p = \left\| \lim \sum_{j=1}^n f_j \right\|_p$.
(3) Let $\epsilon > 0$ and define

$$A = \{x : |f(x)| > \|f\|_\infty - \epsilon\}.$$

Then

$$\int_{[0,1]} |f|^p \geq \int_A |f|^p \geq \int_A (\|f\|_\infty - \epsilon)^p (\mu(A)).$$

(4) Use the previous problem.
(5) Show that

$$\|fg\|_p \leq \|f\|_\infty . \|g\|_\infty ,$$

and use previous problem.
(6) One direction is direct. For the other direction, use Minkowski Inequality.
(7) $\int \left| f^t \right|^s = \int |f^s|^t = \int |f|^{st}$.

(8) Let $A = \{x \in \mathbb{R} : |f(x)| > 1\}$.

$$\int_{\mathbb{R}} |f|^r = \int_A |f|^r + \int_{A^c} |f|^r .$$

But $\int_A |f|^r \leq \int_A |f|^q$ and $\int_{A^c} |f|^r \leq \int_{A^c} |f|^p$.

(9) Note that $|f|^p \in L^r(E)$. Let $g = |f|^p$, then $g \in L^{r/p}(E)$. Find s the conjugate of r/p, then use Holder Inequality on $g \in L^{r/p}(E)$ and $1 \in L^s(E)$.

(10) Let $f_n(x) = n^{-\frac{1}{p}} \chi_{[0,n]}(x)$, and $g_n(x) = n^{\frac{1}{p}} \chi_{[0,\frac{1}{n}]}(x)$.

(11) (a) Define $f(x) = x^{-1/s} \chi_{[0,1]}(x)$ and $g(x) = x^{-1/r} \chi_{[1,\infty)}(x)$.
(b) Let $f(x) = 1$.

(12) Note that $(f - g)^2 \geq 0$.

(13) Let $f(x) = x^{-1/q} \chi_{[0,1]}(x)$.

(14) Let $p = \dfrac{r}{t}$ and $q = \dfrac{s}{t}$.

(15) (a) $pk < 1$.
(b) $1 \leq p < \infty$.

(16) Note that $\|Tx\|_1 \leq \|x\|_p \left(\sum n^{rq}\right)^{1/q}$.

(17) Let $A_1 = \{x : |f| > 1\}$ and $A_2 = \{x : |f| \leq 1\}$. Then $|f|^2 > |f|$ on A_1 and $|f| \leq 1$ on A_2.

(18) Let $f(x) = \dfrac{1}{x}$ on $(1, \infty)$.

(19) Let $x_n = \dfrac{1}{\sqrt{n}}$.

(20) (a) $\sum \dfrac{1}{(n \log n)^{p/2}} \leq \sum \dfrac{1}{n^{p/2}}$.
(b) $\sum \dfrac{1}{(n^3 + 1)^{p/4}} \leq \sum \dfrac{1}{n^{3p/4}}$.
(c) $p > 0$.

(21) (a) Note that $\ell^1 \subset \bigcap_{p>1} \ell^p$.
(b) Note that $\bigcup_{p>1} L^p[0, 1] \subset L^1[0, 1]$.

(22) Since $f_n \longrightarrow 0$ a.e., by Egoroff Theorem, there exists $A \subset [a, b]$, $\mu(A) < \epsilon$ such that $f_n \longrightarrow 0$ uniformly (hence in norm) on $E = [a, b] \setminus A$. Use Holder's Inequality to obtain

$$\int_E f_n \leq (\int_E f_n^2)^{1/2} . (\mu(E))^{1/2} .$$

(23) Note that $f^{\frac{1}{2}}, g^{\frac{1}{2}} \in L^2(I)$. Use Holder's Inequality.

(24) Show that

$$\left| |f_n|^p - |f|^p \right| \leq |f_n - f|^p ,$$

then take the integral over E.

(25) Note that

$$\int \left| |f_n|^p - |f|^p \right| \leq p \int (|f_n| + |f|)^{p-1} |f_n - f| .$$

Use Holder's Inequality to obtain

$$\left\| |f_n|^p - |f|^p \right\|_1 \leq p(\|f_n\|_p + \|f\|_p)^{p/q} \|f_n - f\|_p .$$

(26) Use $|f_n - f|^2 = |f_n|^2 - 2ff_n + |f|^2$.
(27) Let $g(x) = \text{sign}(x)$.
(28) By Egoroff Theorem, for all $\epsilon > 0$ there exists $E \subset A$ s.t. $\mu(E) < \epsilon$ and $f_n \longrightarrow f$ uniformly on $A \setminus E$. Then

$$\left| \int f_n g - fg \right| \leq \int_E |f_n - f| |g| + \int_{A \setminus E} |f_n - f| |g| .$$

Use Holder's Inequality on $\int_E |f_n - f| |g|$.
(29) (a) Use Holder's Inequality with f^r and 1.
(b) Let $f(x) = 1$ on $[0, \infty)$.
(30) (a) Use Fatou's Lemma.
(b) Let $f_n \to f$ a.e. and $\|f_n\|_p \to \|f\|_p$ a.e. Then

$$2^p \left(|f_n|^p + |f|^p \right) - |f_n - f|^p \geq 0.$$

Apply (a) to obtain

$$\begin{aligned} \int 2^{p+1} |f|^p &\leq \underline{\lim} \int \left[2^p \left(|f_n|^p + |f|^p \right) - |f_n - f|^p \right] \\ &= \int 2^{p+1} |f|^p - \overline{\lim} \int |f_n - f|^p . \end{aligned}$$

The other direction is clear.
(31) Define $g_n(x) = g(x)$ if $|g(x)| \leq n$ and 0 if $|g(x)| > n$. Let

$$f_n = |g|^{q/p} \text{sgn}(g).$$

Then $f_n g_n = |g_n|^q$ and $\|f_n\|^p = \|g_n\|^q$.
(32) (a) Note that

$$|f_n|^p \longrightarrow |f|^p .$$

Use Fatou's Lemma.
(b) If $p = \infty$, then $|f_n(x)| \leq M$ hence $|f(x)| \leq M$. But $|f(x)| = \underline{\lim} |f_n(x)|$.

(33) Let $g = \sup_n |f_n|$. Then, $|f_n|^p \leq g^p$. Use GDCT to show $f_n \xrightarrow{p} f$ and use Fatou's Lemma to show that $f \in L^p$.

(34) Let $g_n(x) = \sum^n \left|f_j(x)\right|$ and $\sum \|f_n\|_p = M$. Show that $g_n \in L^p$ and

$$g_n(x) \longrightarrow g(x) = \sum^{\infty} \left|f_j(x)\right|.$$

Use Fatou's Lemma.

(35) Note that

$$\sum^{n} f_j(x) \leq g_n(x) \leq g(x).$$

Show that $f^p \in L^1$, hence $f \in L^p$. Also, note that $\|f\|_p \leq \sum_{\infty} \|f_n\|_p$.

(36) Note that

$$|f_n g_n - fg| \leq |f_n g_n - g_n f| + |g_n f - gf|$$

and

$$\int |f_n g_n - g_n f|^p \leq M^p \int |f_n - f|^p.$$

Moreover, define $h_n(x) = |g_n f - gf|^p$,and apply GDCT.

(37) (a) Let $A_n = \{x : |f_n(x) - f(x)|^p > \epsilon^p\}$. Then

$$\mu(A_n) \leq \int\limits_{A_n} |f_n(x) - f(x)|^p \, dx \leq \int\limits_{A} |f_n(x) - f(x)|^p \, dx.$$

(b) If not, then there exists subsequence f_{n_k} of f_n such that $\left\|f_{n_k} - f\right\|_p > \epsilon$. But $f_{n_k} \xrightarrow{\mu} f$.

(c) Similar to above.

(38) Use Holder's Inequality to show that $fg \in L^1$ and

$$\begin{aligned} \|f_n g_n - fg\|_1 &\leq |(f_n - f) \cdot g| + |f \cdot (g_n - g)| \\ &\leq \|f_n - f\|_p \|g_n\|_q + \|f\|_p \|g_n - g\|_q. \end{aligned}$$

(39) Follow the steps of the proof of the previous problem to obtain $\|f_n g_n - fg\|_1 \longrightarrow 0$. Let $A = \{x : |f_n g_n - fg| > \epsilon\}$. Show that

$$\mu(A) \leq \frac{1}{\epsilon} \int\limits_{A} |f_n g_n - fg|.$$

(40) Let $A = \{x : |g(x)| \geq M + \epsilon\} \subset [0, 1]$. Define $f = \chi_A(x)\text{sgn}(g)$. Note that

$$M \cdot \mu(A) = \left| \int fg \right| \geq (M + \epsilon)\mu(A).$$

(41) There exists a sequence of simple functions $\varphi_n \longrightarrow |g|$, hence $\varphi_n^q \longrightarrow |g|^q$. Let

$$f_n(x) = \operatorname{sgn}(g)\varphi_n^{q-1}.$$

Then $f_n \in L^p$ and

$$\int \varphi_n^q \leq \int g f_n \leq M \|f_n\|_p .$$

Show that $\|\varphi_n\|_q \leq M$ and use Fatou's Lemma.

(42) Define

$$g_n(x) = \begin{cases} 0 & : x < a \\ n(x-a) & : a \leq x \leq a + \frac{1}{n} \\ 1 & : a + \frac{1}{n} \leq x \leq b - \frac{1}{n} \\ n(b-x) & : b - \frac{1}{n} \leq x \leq b \\ 0 & : x > b. \end{cases}$$

Show that $0 \leq g_n(x) \leq s(x)$ and $g_n(x) \longrightarrow s(x)$. Then use DCT for p- norm.

(43) Let $f \geq 0$ and define $f_n(x) = \min\{f(x), n\}$. Show that $f_n \longrightarrow f$ p.w., and $|f - f_n|^p \longrightarrow 0$. Use DCT to obtain $\|f - f_n\|_p \longrightarrow 0$. For every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\|f - f_n\|_p < \epsilon$, and $n \geq N$.

(44) Let $|f| \leq M$ and $\epsilon > 0$. Choose N such that $M < N\epsilon$. Set

$$A_j = \{\frac{(j-1)M}{N} \leq f < \frac{jM}{N},\ j = -N+1, \ldots, N\}.$$

Define

$$\varphi(x) = \sum \frac{(j-1)M}{N}\chi_{A_j} \ \& \ \psi(x) = \sum \frac{jM}{N}\chi_{A_j}.$$

Show that $|f - \varphi| < \epsilon$. Let U_i be open sets such that $\mu(U_j \setminus A_j) + \mu(A_j \setminus U_j)$ is arbitrary small. Let $s(x) = \sum \frac{jM}{N}\chi_{U_j}$. Show that $s \in L^p$ and $|\varphi - s| < \epsilon$.

(45) Let $f \in L^p$. Then for $\epsilon > 0$ define

$$f_N = \begin{cases} N & N \leq f(x) \\ f(x) & |f| \leq N \\ -N & f(x) < -N \end{cases}.$$

Show that $f_N \in L^\infty$, and use DCT to obtain $\|f - f_N\|_p \longrightarrow 0$.

(46) Linearity of the functional is due to linearity of the integral, and for any continuous $y \in L^\infty$, $|y(0)| \leq \sup|y| = \|y\|_\infty$.

(47) (1) Notice that $\|Tx\|_p \leq \|x\|_p$. Let $x = (0, x_1, x_2, \ldots)$ and conclude $\|T\| = 1$.
(2) $\|T(x)\| \leq \|x_n\| \sum \frac{1}{2^n} = \|x_n\|$.

(48) Let $p = \dfrac{s}{r}$ and let q be the conjugate of p. Let $f \in L^s[a, b]$. Use Holder's Inequality with $|f|^{rp}$ and 1, then show that

$$\|T\| \leq (b-a)^{\frac{1}{r}-\frac{1}{s}}.$$

(49) (a) If $x = (x_n) \in \ell^1$ then $\sum |x_n| < \infty$ and if $y \in \ell^\infty$ then $\|y\|_\infty \leq M$. Show that $\|f\| \leq \|y\|_\infty$. Linearity is clear.
(b) Let $y = \delta_i$.

Chapter 5

(1) (a) NO.
(b) Yes.

(2) $\mu(\varnothing) = 0$, and $\mu \geq 0$ since $\mu_j \geq 0$ and $c_j \geq 0$ for all j. Let $\{E_n\} \in \Sigma$. Show that

$$\mu(\bigcup E_n) = \sum_n \mu(E_n).$$

Let $c_j \geq 0$ be a sequence of nonnegative numbers, and define $\mu = \sum_{j=1}^{\infty} c_j \mu_j$. Show that (X, Σ, μ) is a measure space.

(3) Let $\bigcup E_n = X$ such that $\mu(E_n) < \infty$ for all n. Let $A_1 = E_1$ and for $n \geq 2$, $A_n = E_n \setminus \bigcup_{j=1}^{n-1} E_j$.

(4) Show that Σ is a $\sigma-$algebra of X. Then show that μ_A is a measure on (X, Σ_A). $\mu_A(\varnothing) = 0$, and $\mu_A(\cup B_n) = \sum \mu_A(B_n)$.

(5) Let $E = [0, 1]$ and $f(x) = x - \frac{1}{2}$ with the signed measure $\nu(E) = \int_E f d\mu$.

(6) Suppose not. $X = \bigcup_n \{x_n\}$. Let $E_n = \{x_n\}$.

(7) Suppose $\mathcal{F}$ is an infinite collection of sets in Σ. Let $\mathcal{F} = \bigcup_k \mathcal{F}_k$, where $\mathcal{F}_k = \{A \in \mathcal{F} : \mu(A) \geq \dfrac{1}{k}\}$. Suppose $\mathcal{F}$ is uncountable. So, there is $\mathcal{F}_N$ for some $N \in \mathbb{N}$ such that $\mathcal{F}_m$ is infinite. Let $\{E_n\}$ be a sequence of disjoint sets in $\mathcal{F}_N$. Show that $\mu(\bigcup E_n) < \infty$, but $\sum \mu(E_n) \geq \infty$.

(8) Let $A_n = \{A : |f| \leq n\}$. Show that $\mu(A) = \lim \mu(A_n)$. Hence, for every $\epsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that $\mu(A) - \mu(A_n) < \epsilon$ for all $n > n_0$, so $\mu(A \setminus A_{n_0}) < \epsilon$.

(9) We will assume that E is not negative. Then E contains a set of positive measure. Let n_1 be the smallest integer for which there exists a set $E_1 \subset E$ such that $\nu(E_1) > \frac{1}{n_1}$. Proceed the same as in the proof.

(10) Let $m = \inf\{\nu(E) : E \text{ is negative}\}$. Let $\{N_k\}$ be a sequence of negative sets such that $\nu(N_k) \to m$. Let $N = \bigcup N_k$.

(11) Let $X = P \bigcup N$. Then $\nu^+(E) = \nu(E \cap P) \leq \sup\{\nu(B)\} : B \subseteq E\}$. On the other hand, let $B \subseteq E$ in Σ. Then $\nu(B) = \nu^+(B) - \nu^-(B) \leq \nu^+(B) \leq \nu^+(E)$. Note that $\nu^- = (-\nu)^+$.

(12) Let $P = \{x : f(x) \geq 0\}$ and $N = \{x : f(x) < 0\}$. Then

$$\nu^+(E) = \int\limits_{E \cap P} f d\mu$$

and

$$\nu^-(E) = -\int\limits_{E \cap N} f d\mu.$$

(13) Same as the previous problem.

(14) Note that $\mu = \mu^+ - \mu^-$. Write

$$\int\limits_E f d\mu = \int\limits_E f d\mu^+ - \int\limits_E f d\mu^-.$$

Use triangle inequality.

(15) Use Jordan decomposition.

(16) Note that

$$\nu(E) = \nu^+(E) - \nu^-(E) \leq \nu^+(E)$$

and

$$-\nu^-(E) \leq \nu^+(E) - \nu^-(E) = \nu(E),$$

so $-\nu^-(E) \leq \nu(E) \leq \nu^+(E)$.

(17) Let ν be a signed measure on a measurable space (X, Σ). Show that $|\nu|(E) = \sup\{\left|\int_E f d\nu\right| : |f| \leq 1\}$.

(18) Let ν, λ be two signed measures. Show that $|\nu + \lambda| \leq |\nu| + |\lambda|$. By Jordan decomposition, $\nu + \lambda = \mu^+ + \mu^-$. On the other hand,

$$\nu + \lambda = \nu^+ + \lambda^+ - (\nu^- + \lambda^-).$$

Use the previous problem.

(19) Let $X = P \cap N$ be HD for X. Let $\mu(E) = 0$. Then $\mu(E \cap P) = \mu(E \cap N) = 0$.

(20) Let $X = P \cap N$ be HD for X. Write $|\nu| = \nu^+ + \nu^-$ and use the previous problem.

(21) Clear from definition of absolute continuity.

(22) Let $A = \{x : f_1(x) > f_2(x)\}$. Then for all $E \in \Sigma$ we have

$$\int\limits_E f_1 \vee f_2 d\mu = \int\limits_{E \cap A} f_1 d\mu + \int\limits_{E \setminus A} f_2 d\mu.$$

(23) Show that $\mu \wedge \lambda \ll \mu$.
(24) This is because A_k are disjoint.
(25) Use MCT.
(26) Consider $[0, 1]$ with the Lebesgue measure μ, and λ the counting measure.
(27) For (1) use linearity of integrals. For (2) Note that $\nu + \lambda \ll \mu$. Use RNT and linearity of integrals. For (3), note that for every $E \in \Sigma$,

$$0 \leq \lambda(E) = \int_E f d\mu,$$

where $f = \dfrac{d\nu}{d\mu}$.
(28) Show that $\mu \ll \lambda$. Let $E \in \Sigma$ Show that

$$\mu(E) = \int_E \frac{d\mu}{d\nu}.\frac{d\nu}{d\lambda} d\lambda = \int_E \frac{d\mu}{d\lambda}.d\lambda,$$

then use the uniqueness of RN derivative.
(29) (a) Let $\nu(E) = 0$. Then $0 = \int_E \dfrac{d\nu}{d\mu} d\mu$, hence $\mu(E) = 0$.
(b) Note that $\nu \ll \mu \ll \nu$, and $1 = \dfrac{d\nu}{d\nu}$. Use chain rule.
(30) Since $\nu \ll \mu$, $\dfrac{d\nu}{d\mu}$ exists. Let $X = P \cup N$, and $\nu = \nu^+ - \nu^-$. So

$$\nu^+(E) = \nu(E \cap P) = \int_{E \cap P} \frac{d\nu}{d\mu} d\mu = \int_E \chi_P \frac{d\nu}{d\mu} d\mu,$$

and similarly for $\nu^-(E)$. Show that the RN derivatives of ν^+, ν^- are

$$\frac{d\nu^+}{d\mu} = \chi_P \frac{d\nu}{d\mu} \ \& \ \frac{d\nu^-}{d\mu} = -\chi_N \frac{d\nu}{d\mu}.$$

(31) Use Jordan decomposition and the definition of mutually singular measures.
(32) Let $X = A_1 \cup B_1 = A_2 \cup B_2$. We have $\nu_1(A_1) = \mu(B_2) = 0$ and $\nu_1(A_2) = \mu(B_2) = 0$. Consider $A = A_1 \cap A_2$ and $B = B_1 \cup B_2$.
(33) Let $X = P \cup N$. Since $\nu^+ \perp \nu^-$, $\nu^+(N) = \nu^-(P) = 0$. Show that $\nu^+(E) \leq \lambda(E)$ and $\nu^-(E) \leq \mu(E)$.
(34) Let E such that $\mu(E) = 0$, then $\nu(E) = 0$, but then $\mu(E^c) = 0$.

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Index

A
Absolute continuity, 72
Absolute continuity of measures, 189
Abstract Lebesgue space, 182
Algebra, 19
Almost everywhere property, 48

B
Beppo Levi theorem, 135
Borel–Cantelli Theorem, 40
Bounded convergence theorem, 101
Bounded variation, 66
Breizis-Lieb lemma, 173

C
Cantor Set, 31
Caratheodory criterion, 23, 36
Characteristic function, 44
Complete measure, 40
Convergence in p-norm, 155
Convergence in measure, 174
Counting measure, 39

D
Dense set, 159
Dini's derivative, 59
Dirac measure, 39
Dominated convergence theorem, 105

E
Egoroff's Theorem, 57
Essential supremum, 144
Euclidean norm, 141
Extended real numbers, 2

F
Fatou's lemma, 103
First Littlewood principle, 53
Fubini Theorem, 128
Fubini–Tonelli Theorems, 128
Functional, 164
Fundamental theorem of calculus—Riemann's case, 116
Fundamental theorem of calculus—version I, 119
Fundamental theorem of calculus—version II, 122

H
Hahn decomposition theorem, 186
Hahn's lemma, 184
Holder inequality, 147

I
Improperly Riemann integrable, 109
Inner measure, 9

J
Jordan decomposition, 71
Jordan decomposition theorem, 187

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L
Lebesgue decomposition theorem, 199
Lebesgue Differentiation Theorem, 64
Lebesgue integrable function, 97
Lebesgue measurability, 13
Linear space, 139
Littlewood principles, 53
ℓ^p spaces, 152
L^p spaces, 143
Lusin theorem, 58

M
Measurable function, 45
Measurable set, 19
Measurable space, 178
Measure, 2
Minkowski inequality, 148
Monotone convergence theorem, 102
Mutually singular measures, 198

N
Negative set, 183
Nonmeasurable Set, 30
Norm, 140

O
Outer measure, 3

P
Pointwise convergence, 52
Positive set, 183

R
Radon–Nikodym derivative, 195
Radon–Nikodym theorem, 195
Radon–Nikodym theorem for finite measures, 191
Riemann integrable, 108
Riemann–Lebesgue lemma, 115
Riesz representation theorem, 168

S
Second Littlewood principle, 58
Section of a function, 124
Section of a set, 124
Seminorm, 140
Signed measure, 182
Simple Approximation Theorem, 56
Simple function, 44
Supremum norm, 144

T
Third Littlewood principle, 57
Tonelli Theorem, 128
Total variation, 66

U
Uniform convergence, 52
Uniform convergence theorem, 100

V
Variation, 66
Vitali cover, 61
Vitali Covering Theorem, 61
Vitali Set, 28

Y
Young's inequality, 147